

畅销词汇书《托福核心词汇 21 天突破》作者李笑来最新力作！

新托福 iBT 词汇分类突破

李笑来 文

胖兔子粥粥 图

- ✔ 192 段分类专题，轻松记忆！
- ✔ 192 幅卡通场景，爆笑记忆！
- ✔ 192 段美音原声，强化记忆！



世界知识出版社

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前 言

真的只是词汇问题吗？

我前一本词汇书《TOEFL核心词汇21天突破》于2003年第一次付印之后，几经再版，多次加印，几年内畅销十几万册（盗版不计其数）。作为作者当然不免为之欣喜，但另一方面却又颇有些难以释怀。为什么呢？因为我总觉得自己编撰《TOEFL核心词汇21天突破》时尽管已力求做到完美，但仍自觉远远不够。起码我知道仅仅靠一本词汇书无法帮助学生解决所有问题。

这几年的教学经验不断验证我的看法，那就是：尽管词汇量的大小会影响学习者学习语言的效果，但除此之外一定存在更加隐蔽却又更加重要的问题需要解决。缺乏基础词汇量往往让学生举步维艰，但是好不容易跨过这个门槛的学生在筋疲力尽之余却发现自己的境遇不仅没有太大改善，甚至感觉更糟——每个词汇单独拿出来好像都认识，怎么放在一块却死活看不懂呢？如果仅靠词汇量就可以解决一切，又怎么会出现这种无法超越的尴尬？

如果我们把每个单词比喻成一块砖头，而把一篇文章比喻成一栋房子的话，那么就很容易明白文章并不只由词汇构成——除了砖头之外，还要有很多其他材料，最容易想到的起码还有钢筋和水泥。并不是你比别人拥有更多的砖头就意味着你肯定能够盖出比别人更大的房子；如果你只有砖头而没有钢筋水泥的话，你有的只不过是砖堆而已。实际上，我们都宁可砖头少一点，但一定要有足够的钢筋和水泥——这样的话我们就能造出房子。即便因为砖头没那么多因而房子小了一点，但是，毕竟还有房子可住，比既不中看又不中用的砖堆多多了吧？我真的很喜欢这个类比，它可以把事情讲得特别清楚：

An essay is written with words just as a house is built with bricks, but merely a collection of words cannot be called an essay any more than a pile of bricks can be called a house.

这就是我一直以来想解决的问题。传统词汇书（包括之前我写的那本）的局限在于只提供砖头，不提供钢筋和水泥。举例来

说，“habitat”这个单词，学生背过，知道是“栖息地”的意思；“destruction”这个学生单词早就认识，知道是“破坏”的意思；但是由于缺乏“水泥”，并不知道“habitat destruction”是“one of the most significant predicaments that endangered species are confronting with...”注意，我不是在说学生们看着“habitat destruction”猜不出“栖息地破坏”这个字面意思；我是在说，学生们看着“habitat destruction”往往并不能把这两个词拼起来之后再与“protecting endangered species”或者“biological diversity”等概念联系起来。

习惯的力量是惊人的。好的习惯需要很长时间积累才能发挥作用，然而坏习惯无需积累就可以时时暗自发挥作用。很多学生由于一直以来放眼“房子”的同时只关注“砖头”，而忽视“钢筋与水泥”，在每天遇到的尴尬数不胜数却又不明就里。有一个例子可以说明这个问题。“prey”这个单词，很多学生在词汇书中背过，知道是“名词，1) 被捕食的动物；2) 牺牲者、牺牲品”，然而，加上另外一个不可能不认识的词“bird”——“birds of prey”，很多学生就会“以为自己认识”。这个词组“birds of prey”是什么意思呢？很可能与他们想的（其实是猜的）恰恰相反，“birds of prey”的意思是，“猛禽、食肉鸟”。

我在讲座中经常提到这个例子，每次都会有人在台下惊呼“天哪！”每次我都会宽慰大家“相信我，你并不孤独……”并且我会接着说，“其实，你不知道并不可怕；真正可怕的是你并不知道‘你不知道’。”现在已经知道了自己的窘境来自于何处了，那摆脱困境就有了希望。在此之前，那种“并不知道自己不知道”的窘迫，曾经使得多少人好似玻璃瓶中的苍蝇，总是“看得见光明却找不到出路”……

前几年，我曾经一度尝试着使用“分类词汇”的方式帮助学生们扩大词汇量。尽管这么做在效果上来看显然要比“只提供单词列表”（就算加上例句）强出许多。但依然存在一些可以明显感受到的局限。现在，我可以用“砖头、钢筋、水泥和房子”的类比很生动地解释“分类词汇”的局限了：传统词汇书提供了“砖头”，分类词汇书不仅提供了“砖头”，还提供了“钢筋”（即，分类）——但在一定程度上，它并没有提供“水泥”，所以当然还有所欠缺。

近两年，我一直在尝试“细分专题+精简篇章+短词汇表”的模式帮助学生打造“扎实的基础词汇量”，效果非常明显——因为在这种模式中，砖头、钢筋、水泥、小房子一应俱全。很多学生往往在一个月內就可以基本掌握两千左右的核心词汇，并且掌握了词汇与词汇、概

念与概念之间的联系，了解词汇在不同语境中、概念在不同领域中的区别。

如果不通过篇章学习词汇，学生就很难把“L列表中的‘limb’”与“T列表中的‘twig’”联系起来——“limb”是“主干”，“twig”是“枝叉”（事实上，大多学生只知道“limb”是“人的四肢”，而从来没想到过这个单词与“树”有什么关系）——这是词汇与词汇之间的联系；很多学生不知道“rodent”（啮齿动物）必须不停地“gnawing”（磨牙）（很多学生甚至不知道“啮齿动物”是什么东西；而另外一些学生可能就算背过各种“高级词汇书”却只知道这是个名词：“不断的苦痛”）——这就是概念与概念之间的联系；“fault”在大多数语篇中是“错误”的意思，然而在涉及到“earthquake”（地震）的语篇中，往往更可能是“断层”的意思——这是词汇在不同语境中的区别；“adaption”（适应）这个概念在生物学中是一种竞争力的表现，但是在哲学中往往被当做是无力坚持而最终妥协的表现——这是概念在不同领域中的区别。

常识的重要性

在TOEFL/GRE/SAT培训中，流行一种在我看来比较奇怪的看法，而这种看法信奉者甚众：“不要用常识解题”。我很奇怪这些人怎么想的，常识是我们最有力的武器之一，为什么要缴械？每每仔细探究就会发现，但凡解题解错了，往往只有两种情况：1) 在正确常识的基础上逻辑推断错误；2) 逻辑推断正确却常识错误。逻辑推断非常重要，但要建立在合理的常识基础之上才可以。也许有人会反驳“那如果考试里的文章不符合科学常识怎么办？”答案是“在ETS的考试中这种情况不存在”——ETS的所有文章，都是符合现如今广泛接受的科学自然常识的。所以，我建议学生在考试的时候，大胆运用常识，前提有两个：1) 你的常识确实是正确的；2) 你所参加的考试是ETS设计的。

先看个例子：

科学家们很奇怪为什么在squirrel（松鼠）这种rodent（啮齿动物）身上竟然同时存在两种相反的品性：一方面它们“未雨绸缪”地accumulate（收集）acorns（橡果）以便度过寒冬，另外一方面它们又不停地squander（浪费、挥霍）——每个橡果它们都只吃掉1/3，而litter up（扔掉）2/3？经过调查之后，科学家们的解释是这样的：rodent（啮齿动物）的牙，外侧是坚硬的enamel（牙釉质），内侧却只有不停生长的dentine（牙质）。于是，它们必须不

停地gnawing（磨牙）才能保持incisors（门牙）keen（锋利）。所以说，实际上这些squirrel（松鼠）并没有squander（浪费），因为它们只要gnawing（磨牙）1/3的acorn（橡果）就使牙齿相当keen（锋利），那2/3当然要litter up（扔掉）——事实上（这是科学家们后来观察到的），当它们为了果腹而吃acorn（橡果）的时候，从来都是能吃多少就吃多少。

其实，上面的内容实际上是某次托福考试中的听力lecture的完整内容。

想象一下，当一考生被考到这样一个段子的时候，他究竟能听懂多少呢？乍一看来，好像他必须起码认识这些词汇：

□ accumulate

□ acorn

□ dentine

□ enamel

□ gnaw

□ incisor

□ keen

□ litter (up)

□ rodent

□ squander

□ squirrel

可事实上，就算他背过这些词汇，也不见得听懂。原因有很多，比如，其中一个原因是很多学生背单词的时候并不理会拼写，也不理会发音；于是，很多考生可能就算认得enamel，却不知道这个单词怎么读（重音到底在第几个音节上？），在这种情况下，录音里正确读出来的enamel会在这些考生的脑海里唤出正确的对应概念么？

然而，从另外一个角度来说，但凡有一点点科学自然常识的人，只要听得懂squirrel和gnawing，剩下的内容可能仅凭猜测也能八九

不离十地弄清楚。为什么呢？因为科学事实是几乎不变的，尤其不会因为ETS的托福考试改革了而发生变化的。很难想象一群小松鼠在开会，一个小松鼠对其它的松鼠说：

大家听好了！现在ETS的托福改革了，不再是pBT，也不是cBT了，而是iBT了……为了配合ETS的改革，我们从现在开始用橡果磨牙的时候不能再像以前那样用1/3扔2/3，而要统一改变行为，一定要用2/3扔1/3……

“细分专题+精简篇章+短词汇表”的模式在这面对学生有极大的帮助。不仅在记忆词汇的过程中通过上下文联系使学生能够更快更牢地记住单词，另外一方面还可以使学生用英文建立一个完整的、脉络清楚的、层次分明的概念体系，使他们即便在信息不尽完整的情况下同样的拥有更好的理解能力和更靠谱的推测能力。

（说明：关于“啮齿动物”这个专题，由于已经在这里当做例子讲解清楚了，所以，正文中不再重复收录。）

ETS在《新托福官方指南》中清楚地说明了TOEFL iBT考试所涉及的话题范围（见“附录一”）。所以，考生在背单词扩充词汇量的同时，一定要把相关的概念、常识，以及它们之间的必要背景弄清楚。ETS声称不会考“过分专业”的内容，事实上ETS也确实如此身体力行。然而，有些时候，所谓的“专业”是相对的。例如，从来不关注《生物》课程的我当初就确实不知道“啮齿动物”是什么东西，都包括哪些常见的动物，为什么它们被称为“啮齿动物”，它们的习性又是什么……还有另外一个因素就是由于地域的不同，很多概念可能并不在两种文化中完全重合：比如，在中国人的常识里，松鼠是吃松塔（pinecone）的；在美洲大陆常见的灰色小鸟是wren（鹪鹩），但亚洲大陆常见的灰色小鸟是sparrow（麻雀），乃至于TOEFL考试中出现过无数次的wren，却从来没有提过sparrow；我们知道的老鹰是eagle，而美国人熟悉的却是bald eagle和condor，连字典里的解释都是“（南美）秃鹰”……

这就是为什么当前这本词汇书非常有用的重要原因。因为这本书用192个小专题完整覆盖了ETS《新托福官方指南》中所给定的话题范围。书中所有的文字摘自精简改编的权威微软百科全书（Microsoft Encarta）以保持文字的地道，而且每个单词的释义经过精确的筛选而不是盲目收录。

上述三条中的最后一条尤为重要。很多学生只顾背词汇书而不在意语篇，其结果是往往只记住了“其实并不常用、并不常考的词

义”——其实，很多凭空编撰的词汇书对此也有不可推卸的责任。比如，很多学生对于cardinal这个单词，只记住了托福中不会考的名词词性含义“红衣主教”（因为托福几乎不涉及宗教）而不知道这个词在科学文章中常见的形容词词性含义是：“重要的、首要的”；对于radical这个单词，只记住了不会考到的“激进的”，而不知道常考的“剧烈的”（因为托福几乎不涉及政治）；关于canopy，只知道这个单词在字典里的第一个意思“天蓬、华盖”，却不知道常考的那个字典里的第二个释义“树冠”（因为托福考试中只有说明性文字，很少使用文艺性语言）；关于preoccupation这个单词，只背过“深入，全神贯注”，殊不知这个单词每次被考到的时候，意思从来都是“偏见”……

如何使用本书

1. 第一件事：“快速通读”

这个阶段建议在三天之内完成。这期间，最主要的任务并不是记忆单词，而是熟悉话题。一定要在最快的时间里了解托福考试所涉及的话题。因为一直以来，国内的种种英语教学材料的文本中——无论是体制内的教科书，还是体制外的“教辅书籍”，以及各种学习杂志，都有意无意地把话题更多地定位在“社科类”领域中——或许是因为很多人都觉得外语属于“文科”吧。最终，很多中国学生（哪怕是“理科生”）平时尽力读的英文内容也很少涉及自然科学。

然而，在TOEFL这个衡量考生英语实用能力的考试中，英语被定义为一个必要工具——到英语国家学习任何学科都必须使用的工具。所以，在托福考试中，自然科学和人文科学的话题几乎是平分天下，各占一半。于是在托福的考试内容中至少有一半会让很多考生“感觉”非常不适应。比如，关于地理地质（冰河、火山）的话题，或者关于微生物（浮游生物）古生物（恐龙）的话题。就算是一些在我们眼里常常被划分到“文科”的话题，实际上也不是中国学生经常阅读的，比如，关于考古学（化石）、人类学（古碑文）的话题。

所以，建议读者在最快的时间完成第一遍的“通读”。在通读的过程中，留意一下那些“一看就头晕”的话题，做出记号；在下一步制定计划的过程中，有意为这些话题多留一点时间。事实上，这些话题只是平时读得少而已，并没有那么可怕。相对其他原本就大致熟悉的话题来说，只不过需要多重复几遍就好。其实，背单词的诀窍只有一个：“重复”；对那些词汇中的“顽固分子”要做的更简单——“重复许多遍”。

2. 确定自我能力，制定合理计划。

这个阶段只需要一两个小时就够，但非常重要。认真核算一下每天能够用来学习词汇的时间，是两个小时，还是四个小时？你拥有的时间最终决定你可能完成的任务最多是多少。

在这里，先要弄清楚一个最重要的常识：“谁都无法百分之百

地高效率工作”。你肯定会被电话或者短信打断，如果你不像我一样永远把手机设定为“静音状态”；你肯定会在某个时刻不得不把书放下，因为有人敲门或者你自己需要去一趟洗手间……所以，制定计划的时候，一定要给自己的工作设置“冗余度”，给自己的时间设置“损耗值”。也就是说，如果你确定你每天能够花两个小时背单词的话，一定记得给自己分配的工作量最多应该在一个半小时之内就能完成。

确定自己每天能够用来做这项工作的时间之后，建议把每天背单词的时间划分为一次以上，比如两次——“早上一次，晚上一次”；或“下午一次，晚上一次”。因为这是一项特殊的工作，刚刚才提及到，背单词的诀窍其实只有一个，就是“重复”。

本书配有专题文本的MP3录音，建议每天重复收听多次。建议把零散的时间抽出来做这件事情，比如在路上的时候，比如在等车的时候，比如在健身的时候。听的时候不要只默默地听，要尽量跟读（哪怕读得不准，读得不顺）。为了能够通过跟读慢慢进步，最好只有一只耳朵戴着耳机，而空出另外一只耳朵才可以听到自己的声音，以便多多少少可以随时调整自己的发音。当然，还有个前提，相应的文本已经提前认真阅读过。

实施计划的最后一个建议是，整本书的第一遍记忆一定要在一个月之内完成。没有人能只用一遍就可以牢牢记住所有内容——大家都得靠重复。记住，第一遍不需要过分要求完美，不过一定要快，因为所有人的耐心都是有限的，第一遍拖太久，就很可能再也不想做第二遍了——那就注定要失败了。很多人做不好的原因其实只有一个，第一遍都没有完成，为什么呢？除了可能是因为本身的惰性之外，更可能的原因是因为他们第一遍就想完美，殊不知那是不可能的！但凡一个正常人（智商超过145的除外），基本上需要起码三遍以上的重复。

3. 最重要的单词记忆工具——拼读规则

“基础很重要”是经常地被忽视的老生常谈。其实，所谓的“基础差”只有一个原因——当初学习的时候缺乏耐心。缺乏耐心几乎是所有人的本性，也是绝大多数人最终未能矫正的习惯。很多人在生活的方方面面都因为缺乏耐心而处处吃亏却又从不自知；处处吃亏的结果是越来越急于求成，越来越缺乏耐心——于是一生都只能在这个死循环中左冲右撞而最终混个死因不详。

“英文单词的拼读规则”就是这样一个被大多数人忽视却非常

重要的基础知识。如果你从未忽视过这个重要的基础知识，那么恭喜你，可以不用接着读下去了，直接跳过本小节——不过，如果你至今还是有着那种“学英语也已经很多年了，怎么越学越差呢？”感觉的人，那么我将吐血推荐你耐下心来继续阅读下去——其实并不需要太多的时间。

很多人（这个比例很可能远远超过80%）记忆英文单词的时候是以字母为单位的，比如exorbitant这个单词，他们要总计记忆10个字母的组合（e-x-o-r-b-i-t-a-n-t）；而另外那些少数熟悉“英文单词的拼读规则”的人记忆这个单词，只须记忆三个音节（exor-bi-tant）。再比如ichthyosaur这个单词，其实一样只需要记忆三个音节（ich-thyo-saur）。

以音节（一个或者多个字母构成的组合）为单位而不是以字母为单位，会使背单词不知道轻松多少倍。尽管并非所有的英文单词都符合拼读规则，但是，根据Hanna、Hodges、和Rudorf（1966）的统计调查，保守估计超过96%以上的英文单词是符合基本的拼读规则的。有些人会反驳说，不符合拼读规则的单词多着呢！但那只是感觉，因为英文的词汇太多，所以，很小的比例也会让人“感觉很多”。

有关拼读规则，要切记以下两个原则：

1) 不能把这些规则当作推测单词发音的工具。

也就是说，你绝对不应该看着一个生词仅仅根据拼读规则就去猜测这个单词的发音，你应该做的是先搞清楚这个单词的发音，而后运用拼读规则进行辅助记忆。很多人都常常不经意地违背这个原则，所以才会经常读错单词。比如，很多人可能会读错以下五个单词中的至少三个：“cellist”、“facade”、“heir”、“niche”、“specific”。

2) 无需死记硬背所有的拼读规则。

花时间死记硬背这些拼读规则的意义其实并不大，其中一个原因是因为它只是辅助工具。就好像木匠在制作家具的时候，更多的时候他的注意力应该放在要制作的家具上，而不是全神贯注在诸如刨子之类制作家具所需要的工具上。最有效的学习方法是这样：先大致了解这些规则，而后再反复复习若干次，再然后要经常在新近习得的生词中反复“发现”这些规则的应用。用得多了，自然而然地就记住了，最终，你就会发现你记忆单词的时候早已经变得轻松许多。

另外，也没必要想一下子搞定所有的拼读规则。比如[ee]非常

常见，并且几乎只被读成/i:/，想记不住都难；[ea]往往要么被读做[i:]要么被读做/e/，也很容易记住。但另外一些规则不是那么常见。比如，/ieu/读成/e/的情况只在“lieutenant”中出现——并且还是在英式英语中，而美式英语中，这个单词中的音节“lieu”干脆被简化为读成/lu/。而像[æ/ée/ee/er/et/e/é]可能被读成/ei/，比如（vertebrae, café, épée, soiree, foyer, bouquet, dente, saké……），几乎全部是来自法文的词汇——基本可以暂时忽略，先把那些最常见的都搞定再说。

关于常见的元音拼法请参阅“附录二”，而常见的辅音拼法请参阅“附录三”。一般来说，想要一下子记住所有的拼读规则很难，其难度并不亚于想要一下子背下来一本词典。方法是先大致有个了解，而后反复查询参照——同样是通过“重复”巩固记忆。实际上，在阅读过程中，为了能够在下一次阅读的时候有效地提醒自己，通常情况下我最多标注的其实只有两个：1) 元音的长短；2) 重音所在位置。只要把握这两样东西，往往大部分问题都解决了。

比如之前提到过的单词enamel，第一个音节“e”究竟应该读做/e/，还是/i/呢？第一个音节如果是重音，那么应该是/e/，如果不是重音，那么就应该是/i/；/n/是没有什么其他读法的，同时因为重音是落在“na”这个音节上的，所以，这个音节应该被读做/næ/；而后面的“mel”只能是/mel/——这个单词读做/i'næmel/，于是，我只需要在第一个音节“e”和第二个音节“na”之间的上方标记上一个重音符号，就可以记住这个单词的发音了。再比如，遇到“beak”这个单词，究竟应该读/bi:k/呢，还是/bek/呢？因为“ea”可以读/i:/也可以读/e/。查过词典之后，我知道它应该读/bi:k/，于是，我会在“ea”的上方画一条横线，用来表示这是长元音——甚至没必要标记/i:/，因为也没有什么其他的读法了。这其实不难，用多了，自然就熟悉了。但是，一旦熟悉了拼读规则，背单词的效率就会大大提升。

4. 学以致用——如何快速提高阅读理解能力？

认识了一定的单词量之后，就一定要不停地用才可以真正把握。多读文章是扩充巩固词汇量的唯一有效手段。而读文章还是有一点点的方法的，考试（TOEFL/IELTS/SAT/GRE/GMAT）里的阅读文章更是如此。提高阅读理解的方法并不神秘，而真正有效的方法也往往非常简单。

当我们读进来第一句话（标识为S1）的时候，任务只有一个，“这句话在再说什么？”（What does S1 mean? 标识为M1）其实，有

些时候，这并不是一个简单任务。需要两样东西支撑：1)语法知识；2)概念体系。但是，很多人竟然以为只需要单词就够了。

然而，读进来第二句话（S2）的时候，任务就多了一个：不仅要搞清楚M2，还要搞清楚M1和M2之间的关系（标识为R1&2）——这是有很多人从来不做的事情。

M1和M2之间的关系，大抵上分为两种：

1) M1被M2支持（当然，也可能反过来，M1支持M2）。这时，M2往往可能从三个角度之一（或者三个角度组合使用）去解释M1——What?（举例、阐述）Why?（因果、比较、分类、目的）How?（方式、手段、步骤）

2) M1与M2共同支持另外一个句子。这时，M1与M2之间可能的关系分别是：并列、递进、转折。

如果，M1、M2、R1~2都已知，那么即意味着真正的“阅读理解”就已经全部完成。

然而，在考试中，考生往往遇到的情况是：

- (1). M1未知、M2已知、R1~2已知；
- (2). M1已知、M2未知、R1~2已知；
- (3). M1已知、M2已知、R1~2未知；

这其实与简单数学没什么两样（就好像“ $x+y=z$ ”）——即，一个方程式里面有三个变量，其中两个已知，现在需要推导出第三个变量的值。如果三个变量都是已知的，那也不算是什么考试了。

要知道，设计严谨科学的阅读理解考试中，是不会出现一个方程式里有三个变量，而其中却竟然有两个未知的情况的——那不叫考试，那叫“刁难”。这也是我为什么总是劝诫考生，轻易不要相信什么“模拟题”——反正，我研读过的所有非ETS官方的题目，大抵上都不严谨、不科学，不管作者和出版机构是多么地权威——不信，谁都可以用刚刚说的简单道理去自己作个判断。

同样的道理，段落与段落之间的关系也是如此。不仅要能够概括出第一段的大意（标记为MP1），然后还要再概括出第二段的大意（标记为MP2），最后还要弄清楚二段之间的关系（标记为RP1~2），后

又是解方程式了……

道理都清楚之后，就是平时练习的步骤了：

1) 搞清楚每一句话的确切含义。运用一切可以运用的手段——查词典，查语法书，甚至去查Google。“自己动手”比“花钱让别人替自己干活”（比如，报班上课听讲）在这方面不知道要有效多少倍。

2) 理解每句话之间、每个段落之间的关系。之后还有另外一个任务——概括。

3) 整理词汇。要知道，读完一篇文章之后，自己动手整理词汇，远比背单词书效率高出许多——可惜，大多数人并不相信。

4) 反复阅读若干遍。读着读着就可能发现很多第一次读并没有注意到的东西。

5) 复述文章。能够书面或者口头复述刚刚读过的文章，实际上需要很多综合能力：记忆力、逻辑能力、转述能力、重新组织能力、再理解能力等等。

6) 养成相隔若干天后复习的习惯。

其实，如此处理50篇左右的文章，就基本上所向披靡了。对于那些准备留学考试的学生来说，随便找来几套真题如法炮制就会有很大的提高。

5. 关于本书的配套录音

本书中总计有192段文字，每段文字都配有mp3录音文件。不妨在一段时间里（比如一个月）把这些录音作为mp3播放设备中的唯一内容，随时听，只要有可能就不要间断。

背单词的一个重要技巧就在于“尽量动用一切感官”——用眼睛看，用手写，用嘴说，用耳朵听……只有这样，大脑才可以真正活跃起来，精力才可能真正集中，效率才会真正提高。很多人背单词之所以没效率就是因为他们永远“默默地背”；只用眼睛看，从来不写，然后就在考场把“principal”看成了“principle”；从来不听也不说，然后就在考场上听到[tʃeləu]，全然没想到那是“cello”，想表达“specific”；却读成了[ˈspesifik]……最终吃了亏却也不知道究竟是怎么回事。

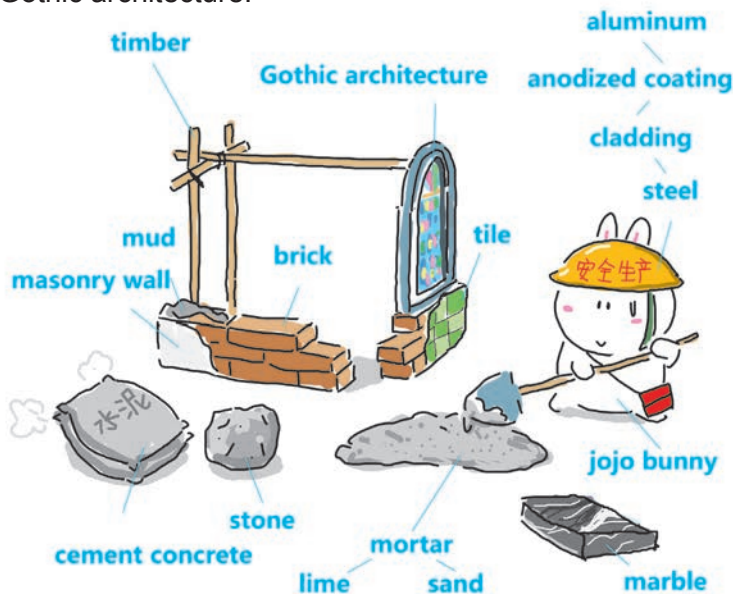
事实上，背句子很可能比背单词更为简单，尽管看起来好像要背得更多。因为在句子之中，词汇与词汇之间是有关联的，这种关联的存在会使大脑不由自主地以“有序”的方式存储这些词汇。本书的重大优势就在于所有的词汇都存在于文本之中，而非孤立存在。对大多数读者来说，大段背诵英文不仅是苦差事，并且也必然“感觉毫无必要”。不过，我们可以“退而求其次”——无需背诵，但必须“熟读”。最初的“快速通读”结束之后，就可以开始“反复跟读文本”，一来可以把生词以“有机、有序的方式”安全地存储在记忆之中，另一方面也可以顺带锻炼了自己的听力，打磨了自己的发音，可谓一举多得！

UNIT001—UNIT192

新 托 福 i B T 词 汇 分 类 突 破

UNIT001

Basic building materials include: timber, mud, stone, marble, brick, tile, steel, and cement concrete. Masonry walls are built with stones or bricks, cohered through the use of mortar, first composed of lime and sand. The profusion of aluminum and its anodized coatings provided cladding material that is lightweight and virtually maintenance free. Glass was known in prehistory and is celebrated for its contributions to Gothic architecture.



注释

timber ['tɪmbə] n. 木材

marble ['mɑ:bl] n. 大理石

tile [taɪl] n. 饰面砖, 瓷砖

cement concrete [si'ment-'kɒŋkri:t] n. 水泥混凝土

masonry wall ['meɪsənri-wɔ:l] n. 砖墙, 砖瓦墙

cohere [kəu'hiə] v. 粘合

mortar ['mɔ:tə] n. 砂浆, 灰浆

lime [laɪm] n. 石灰

profusion [prə'fju:ʒən] n. 大量 (a profusion of: 大量的)

aluminum [ə'lu:mənəm] n. 铝

anodized coating ['ænəudaɪzd-'keʊtɪŋ] n. 电镀层, 电镀膜 (anodize v. 阳极氧化: 用电解方式给 (金属面) 镀上一层保护性或装饰性的氧化物)

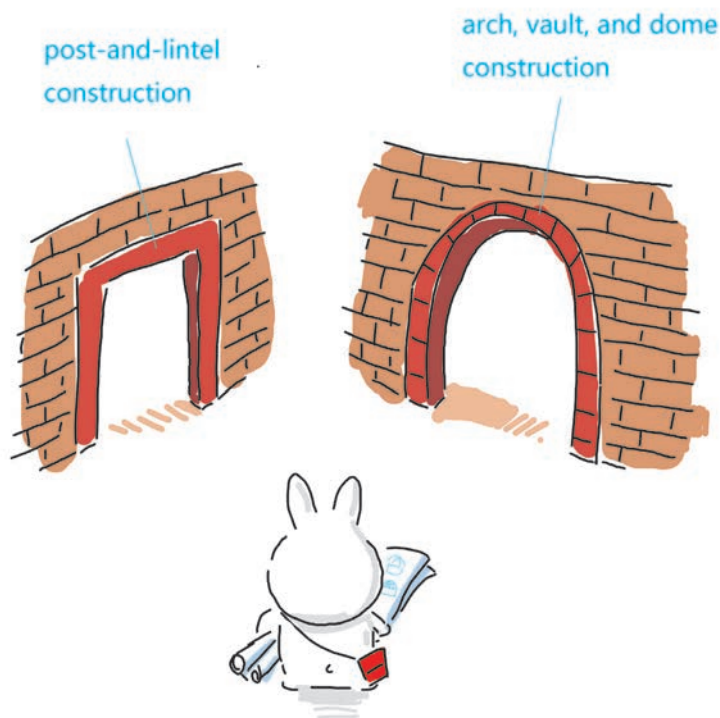
cladding ['klædɪŋ] n. 覆层 (在高温高压下结合到另一金属上的金属外层)

celebrated ['selɪbreɪtɪd] adj. 著名的

Gothic ['gɒθɪk] adj. 哥特式的

architecture ['ɑ:kitektʃə] n. 建筑, 建筑学

Two basic approaches to spanning over masonry walls are post-and-lintel construction and arch, vault, and dome construction.



注释

span [spæn] v. (桥、拱等) 横跨

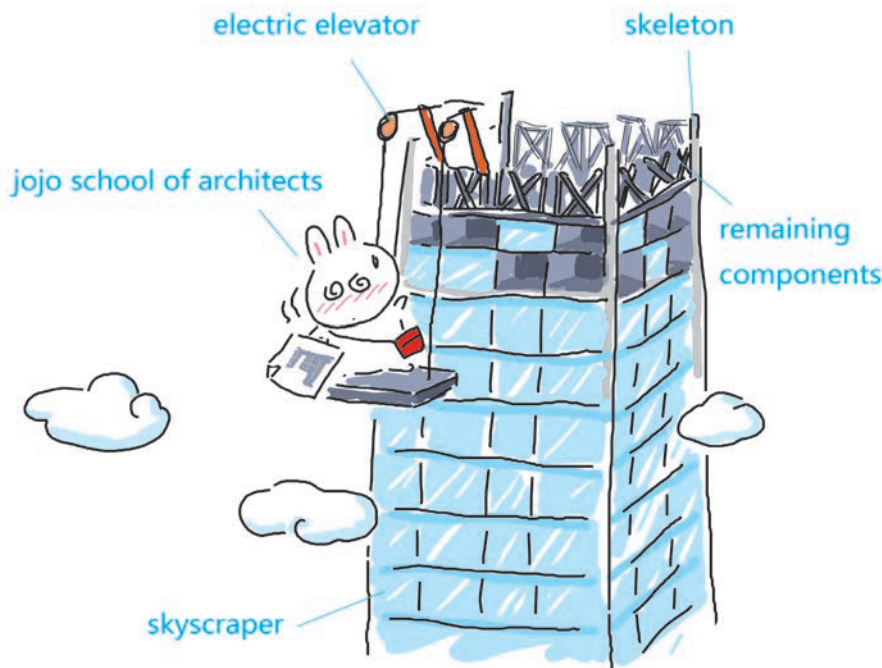
post-and-lintel ['pəust-ænd-'lɪntl] n. 连梁柱

arch [ɑ:tʃ] n. 拱 (连柱的)

vault [vo:lt] n. 拱 (不连柱的)

dome [dəʊm] n. 穹隆 (结构): 类似于穹隆 (弓形、有拱的) 或圆屋顶的结构

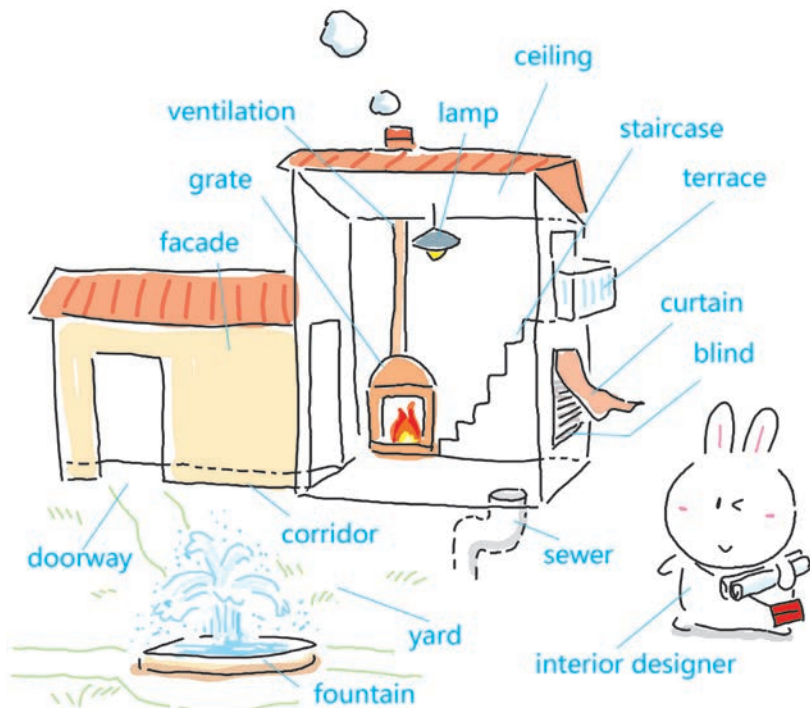
The Chicago architect Louis Sullivan's career converges with the so-called Chicago School of Architects, whose challenge was to invent the skyscraper or high-rise building, facilitated by the introduction of the electric elevator and the abundance of steel. The building's skeleton could be erected quickly and the remaining components hung on it to complete it, an immense advantage for high-rise buildings on busy city streets.



注释

- architect ['ɑ:kitekt] n. 建筑师
 converge [kən've:dʒ] v. 聚合于, 集中于一点
 school [sku:l] n. 学派
 skyscraper ['skaɪskreɪpə] n. 摩天大楼
 facilitate [fə'siilitet] v. 提供便利
 electric elevator [i'lektrik - 'eliveɪtə] n. 电梯
 abundance [ə'bʌndəns] n. 丰富
 skeleton ['skeliɪn] n. 框架结构整体框架
 erect [i'rekt] v. 竖立, 建立
 component [kəm'pəunənt] n. 零件, 构件
 immense [i'mens] adj. 极大的

Interior design of apartments and houses, that is, dwellings, usually involves both practical and aesthetic decisions. Choices have to be made for almost every element, ceilings and lamps, curtains and blinds, corridor and doorway, niches and ledges, grate and ventilations, terraces and staircases, even sewers or drainers. Sometimes such designs are extended to the facade of the building, and even to a fountain, or a gardening yard.



注释

interior design [in'tiəriə-di'zain] n. 室内设计

dwelling ['dwelɪŋ] n. 住处, 居住空间

involve [in'vɒlv] v. 包括

aesthetic [i:'sθetik] adj. 美学的

ceiling ['si:liŋ] n. 天花板

lamp [læmp] n. 灯

curtain ['kə:tn] n. 窗帘

blind [blaɪnd] n. 窗帘 (尤指固定在卷轴上能够拉下来的布质窗帘)

corridor ['kɒrɪdɔ:] n. 走廊

doorway ['dɔ:wei] n. 门口

niche [nitʃ] n. 壁橱

ledge [ledʒ] n. 壁架

grate [greɪt] n. 壁炉

ventilation [,ventɪ'leɪʃən] n. 通风设备

terrace ['terəs] n. 阳台

staircase ['steɪkɛɪsɪs] n. 楼梯

sewer [sjuə] n. 排水沟

drainer ['dreɪnə] n. 下水道装置, 滤水器

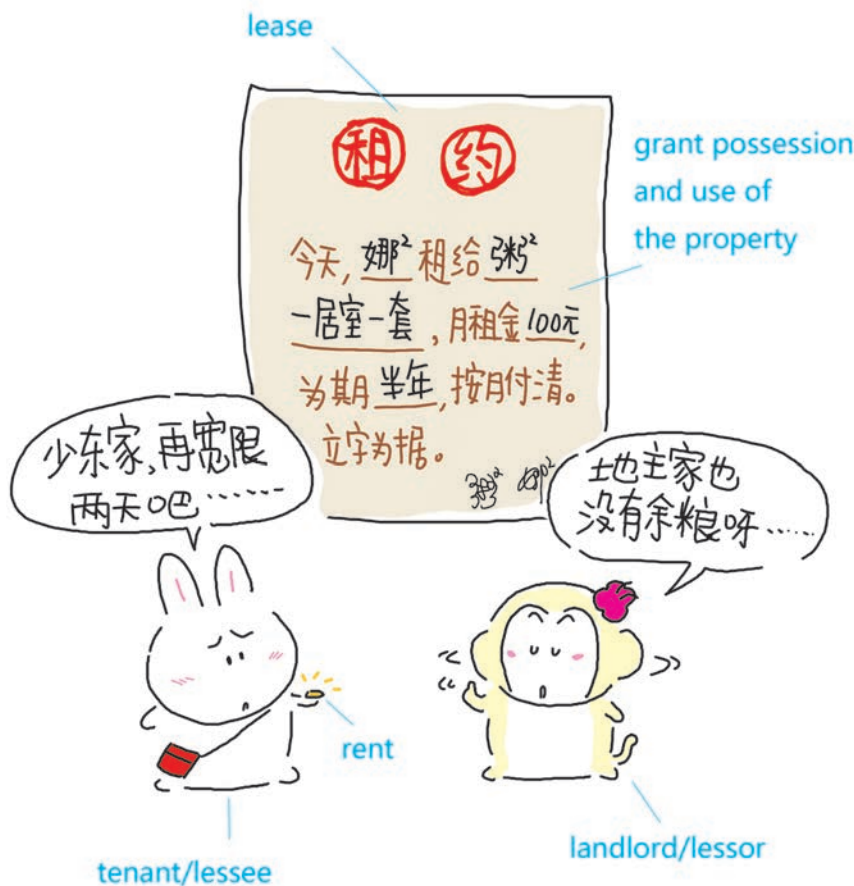
facade [fə'sɑ:dʒ] n. (建筑物的) 外立面, 尤指正面

fountain ['faʊntɪn] n. 喷泉

yard [jɑ:d] n. 庭院

UNIT005

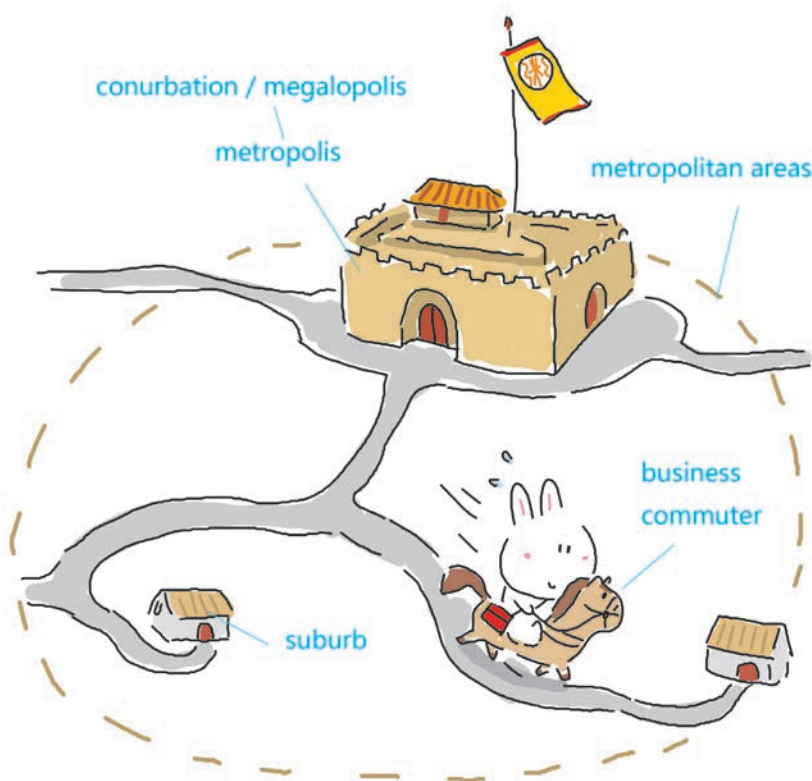
The lease is a contract of rent under which one party, called landlord or lessor, grants possession and use of the property for a limited term to the other party, who is called tenant or lessee.



注释

- lease [li:s] n. 租约
 rent [rent] n. 租金
 landlord ['lændlɔ:d] n. 房东
 lessor [le'sɔ:] n. 出租人
 grant [gra:nt] v. 授予 (权利); 让与 (财产)
 possession [pə'zeʃən] n. 占有, 拥有
 property ['prɒpəti] n. 财产, 所有物
 tenant ['tenənt] n. 房客
 lessee [le'si:] n. 承租人

A big city, or metropolis, is often accompanied by suburbs. Such cities are usually associated with metropolitan areas and urban sprawl, creating large amounts of business commuters. Once a city sprawls far enough to reach another city, this region can be deemed a conurbation or megalopolis.



注释

metropolis [mi'trɒpəlɪs] n. 大城市

suburb ['sʌbə:b] n. 城郊

be associated with [bi-ə'səʊʃieɪtɪd-wɪð] v. 与……相关

metropolitan [ˌmetrə'pɒlɪtən] adj. 大城市的

urban ['ə:bən] adj. 城市的

sprawl [sprɔ:l] n. (城市的) 无计划扩展

commuter [kə'mju:tə] n. 通勤者 (定期从一地到另一地旅行的人, 如从郊区到城市并回来)

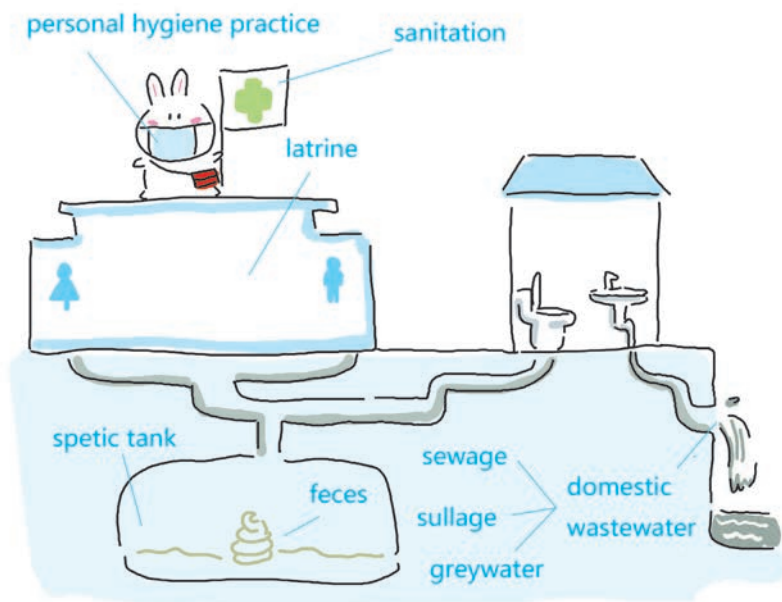
deem [di:m] v. 视为

conurbation [ˌkɒnə'beɪʃən] n. (连带卫星城镇和市郊的) 大都市

megalopolis [ˌmegə'lɒpəlɪs] n. 特大城市

UNIT007

Sanitation is the hygienic means of preventing human contact from the hazards of wastes, either physical, microbiological, biological or chemical agents of disease, which include human and animal feces, solid wastes, domestic wastewater (sewage, sullage, greywater), etc. Hygienic means of prevention can be practiced by using simple technologies (e. g. latrines, septic tanks), or by personal hygiene practices.



注释

sanitation [ˌsæni'teɪʃən] n. 公共卫生

hygienic [haɪ'dʒiːnɪk] adj. 卫生学的 (hygiene [ˈhaɪdʒiːn] 卫生学)

microbiological [ˈmaɪkrəʊˌbaɪə'lɒdʒɪkəl] adj. 微生物学的

agent [ˈeɪdʒənt] n. 媒介

feces [ˈfiːsɪz] n. 粪便

domestic [də'mestɪk] adj. 家庭的

sewage [ˈsjuɪdʒ] n. 污水

sullage [ˈsʌlɪdʒ] n. (房屋、街道等排出的) 污物

greywater [ˈɡreɪwɔːtə] n. 生活污水

latrine [lə'triːn] n. 公共厕所

septic tank [ˈseptɪk-tæŋk] n. 化粪池

personal hygiene practice [ˈpɜːsənəl-ˈhaɪdʒiːn-ˈpræktɪs] v. 个人卫生习惯

A public utility (usually just utility) is a company that maintains and provides the infrastructure for a public service. Although utilities can be privately owned or publicly owned, they often involve natural monopolies.



注释

public utility ['pʌblɪk-ju'tɪlɪti] n. 公用事业

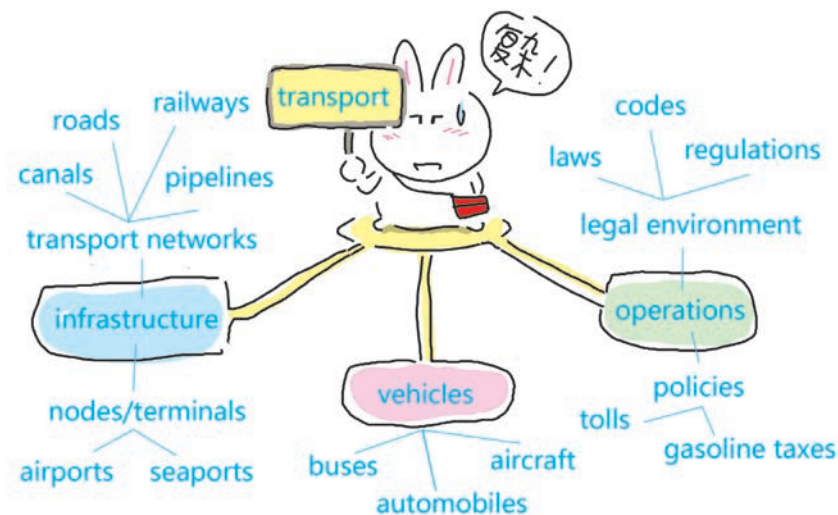
infrastructure ['ɪnfraˌstrʌktʃə] n. 基础设施

privately owned ['praɪvətli-əʊnd] adj. 私有的

publicly owned ['pʌblɪkli:-əʊnd] adj. 公有的

monopoly [mə'nɒpəli] n. 垄断

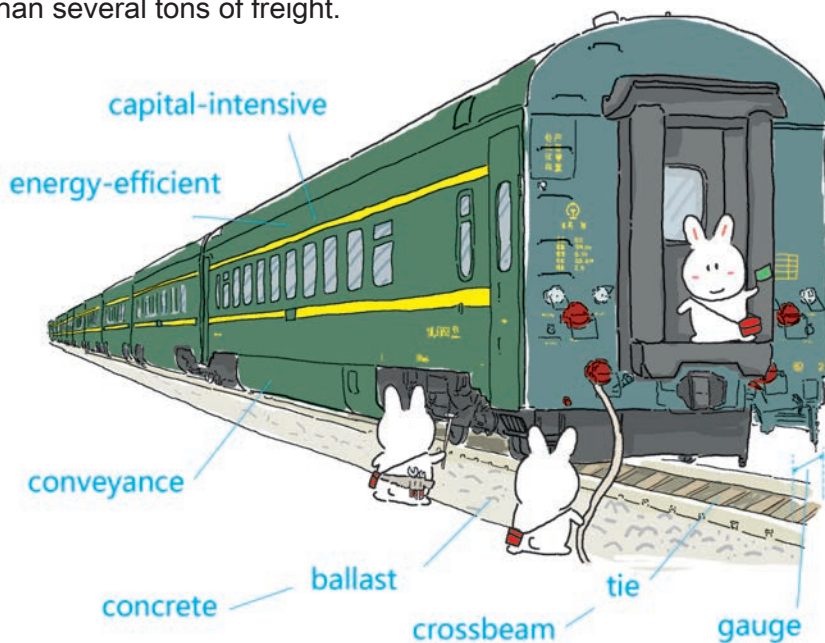
The field of transport has several aspects: infrastructure, vehicles, and operations. Infrastructure includes the transport networks (roads, railways, airways, waterways, canals, pipelines, etc.) that are used, as well as the nodes or terminals (such as airports, railway stations, bus stations and seaports). Vehicles travelling on the networks include automobiles, bicycles, buses, trains and aircraft. The operations deal with the way the vehicles are operated on the network, and the procedures set for this purpose including the legal environment (laws, codes, regulations, etc.). Policies, such as how to finance the system (for example, the use of tolls or gasoline taxes) may be considered part of the operations.



注释

- vehicle ['vi:ɪkl] n. 交通工具
 operation [ˌɒpə'reɪʃən] n. 运营
 canal [kə'neɪl] n. 运河
 pipeline ['paɪpleɪn] n. 管道, 输送管道
 node [nəʊd] n. 中转站
 terminal ['tɜ:mi:nəl] n. 终点站
 seaport ['si:pɔ:t] n. 港口
 law [lɔ:] n. 法律
 code [kəʊd] n. 法规
 regulation [ˌregjʊ'leɪʃən] n. 规章制度
 toll [təʊl] n. 通行费
 gasoline ['gæəsəli:n] n. 汽油

Rail transport is the conveyance of passengers and goods by means of wheeled vehicles specially designed to run along railways or railroads. Typical railway tracks consist of two parallel rails, normally made of steel, secured to crossbeams, termed ties. The ties maintain a constant distance between the two rails, a measurement known as the "gauge" of the track. To maintain the alignment of the track, it is either laid on a bed of ballast or else secured to a solid concrete foundation. Rail transport is an energy-efficient and capital-intensive component of logistics, in contrast, a traditional wagon can carry no more than several tons of freight.



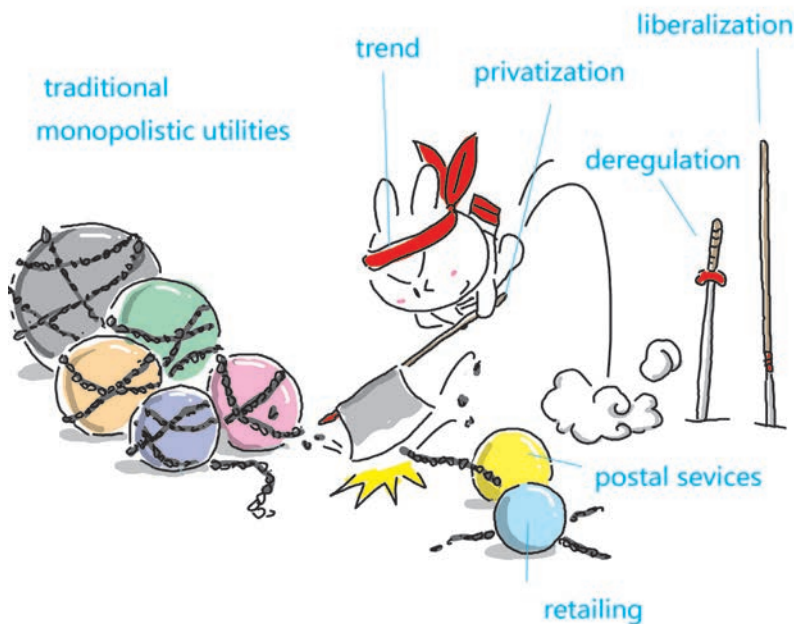
注释

rail transport [reɪl-trænspɔ:t] n. 铁路运输
 conveyance [kən'veɪəns] n. 运送
 by means of [baɪ-'mi:nz-əv] prep. 通过……方式
 consist of [kən'sɪst-əv] v. 由……组成
 secure to [sɪ'kjʊə-tu] v. 固定在……
 crossbeam ['krɒsbɪ:m] n. 横梁
 tie [taɪ] n. 枕木
 gauge [geɪdʒ] n. 轨距
 alignment [ə'laɪnmənt] n. 直线排列

ballast ['bæləst] n. 道碴
 concrete ['kɒkri:t] n. 混凝土
 energy-efficient ['enədʒi-'ɪfɪjənt] adj. 能源节约的
 capital-intensive [ˌkæpɪt-'ɪn-tensɪv] adj. 资本密集型的
 logistics [lə'dʒɪstɪks] n. 物流
 wagon ['wæɡən] n. 四轮马车
 freight [freɪt] n. 货物

UNIT011

Developments in technology have eroded some of the natural monopoly aspects of traditional public utilities. For instance, electricity generation, electricity retailing, telecommunication and postal services have become competitive in some countries, and the trend towards liberalization, deregulation and privatization of public utilities is growing, but the network infrastructure used to distribute most utility products and services has remained largely monopolistic.



注释

- erode [i'ru:əd] v. 侵蚀
 retailing ['ri:teɪlɪŋ] n. 零售业
 postal service ['pəʊstəl-'sɜ:vɪs] n. 邮电业
 competitive [kəm'petɪtɪv] adj. 竞争的
 trend [trend] n. 趋势
 liberalization [lɪbə'reɪl'aɪzɪʃən] n. 自由化
 deregulation [di:'regjʉ'leɪʃən] n. 缩小国家对经济干预的范围
 privatization [ˌpraɪvətə'zeɪʃən] n. 私有化
 infrastructure ['ɪnfə'strʌktʃə] n. 基础设施
 distribute [dɪ'strɪbjʉ:t] v. 分配
 monopolistic [mə'nɒpə'lɪstɪk] adj. 垄断的

An aspiring master would have to pass through the career chain from apprentice to journeyman before he/she could be elected to become a master craftsman. He/She would then have to produce a sum of money and a masterpiece before he/she could actually join the guild.



注释

aspiring [ə'spaɪərɪŋ] adj. 有志向的

master ['mɑ:stə] n. 大师

pass through ['pæs-θru:] v. 经过

apprentice [ə'prentɪs] n. 学徒

journeyman ['dʒə:nɪmən] n. 熟练工人

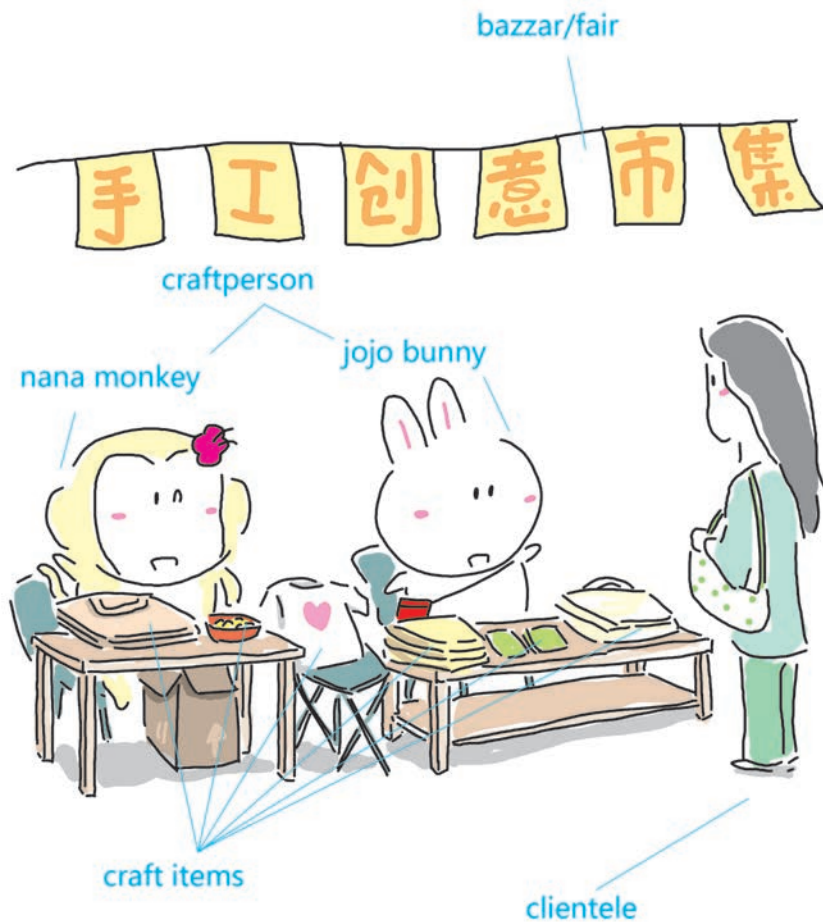
masterpiece ['mɑ:stəpi:s] n. 杰作

actually ['æktʃuəli] adv. 实际上

guild [gɪld] n. 协会

UNIT013

A craftsperson who perhaps has at first sold craft items only to friends or at local bazaars may find that increased demand leads to a wider clientele and sales by mail order, at crafts fairs, or through a shop.



注释

craft [kro:ft] n. 工艺

bazaar [bə'zɑ:] n. 集市

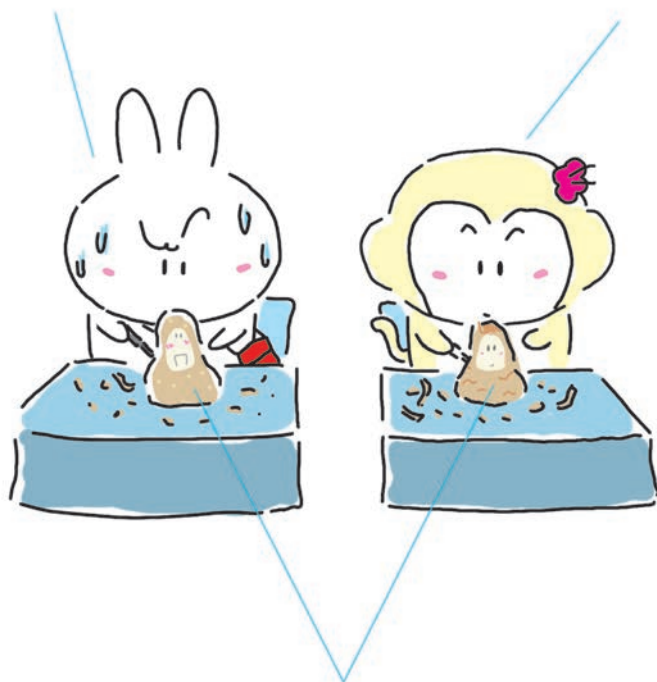
clientele [klaɪə'nɛl] n. 顾客

fair [fɛə] n. 集市

There is a fine line of distinction between crafts produced by amateur hobbyists for their personal satisfaction and those made by gifted artisans with a view toward the use and enjoyment of others.

amateur hobbyist

gifted artisan



a fine line of distinction

注释

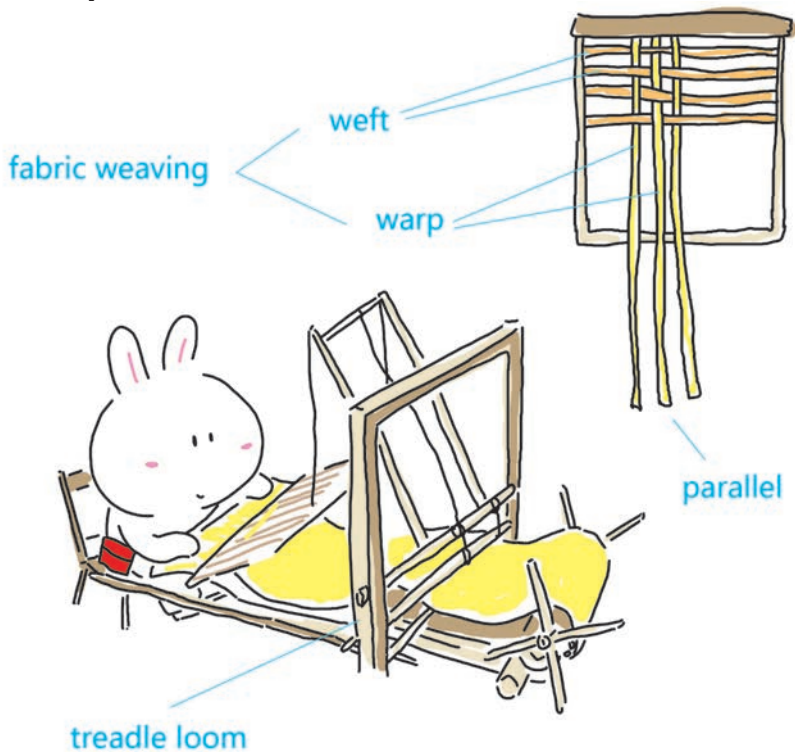
fine [fain] adj. (区别) 微小的 ; 细微的

distinction [dis'tɪŋkʃən] n. 区别

amateur hobbyist ['æmətə-'hɒbiɪst] n. 业余爱好者

artisan [ɑ:'ti:zən] n. 工匠

Fabric weaving involves interlacing warp and weft. The yarn threads of warp are arranged parallel to one another and held in tension on a loom, while the single thread wefts pass over or under the warp threads to create a solid or patterned piece of clothes. To create larger pieces, a treadle loom is usually needed.



注释

fabric weaving ['fæbrɪk - 'wi:vɪŋ] n. 纺织, 编织

involve [ɪn'vɒlv] v. 包括

interlace [ɪntə(:)'leɪs] v. (使) 交织

warp [wɔ:p] n. 织物上纵的方向的纱或线

weft [weft] n. 织物上横的方向的纱或线

yarn thread [jɑ:n-θred] n. 纺线丝

parallel ['pærəlel] adj. 平行的

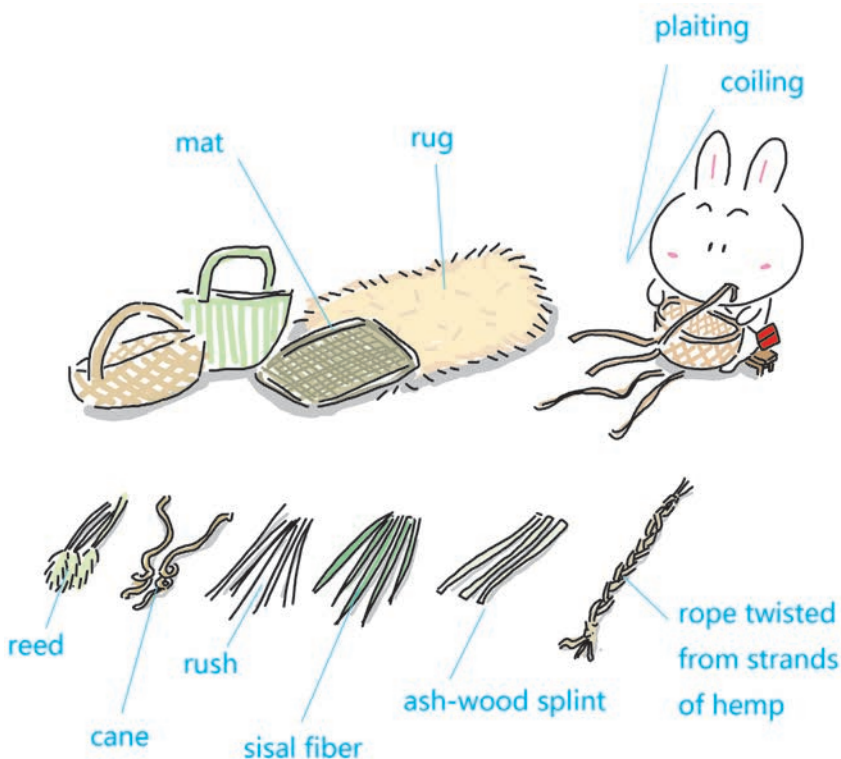
tension ['tenʃən] n. 绷紧状态

loom [lu:m] n. 织布机

solid ['sɒlɪd] adj. 结实的

treadle loom ['tredl-lu:m] n. 脚踏织机

Basketry is the craft of making baskets and bags, mats, rugs, and other items through weaving, plaiting, and coiling techniques, using materials such as reed, cane, rush, sisal fiber, ash-wood splints, and ropes twisted from strands of hemp.



注释

basketry ['bɑ:stkitrɪ] n. 编篮工艺

mat [mæt] n. 垫子

rug [rʌg] n. 地毯

plait [plæt] v. 把……打成辫

coil [kɔɪl] v. 盘绕

reed [ri:d] n. 芦苇

cane [keɪn] n. 藤条

rush [rʌʃ] n. 灯心草

sisal ['sɪsəl] n. 波罗麻：一种分布在墨西哥和中美洲的植物

fiber ['faɪbə] n. 纤维

ash-wood [æʃ-wu:d] adj. 柞木的

splint [splɪnt] n. 薄木条

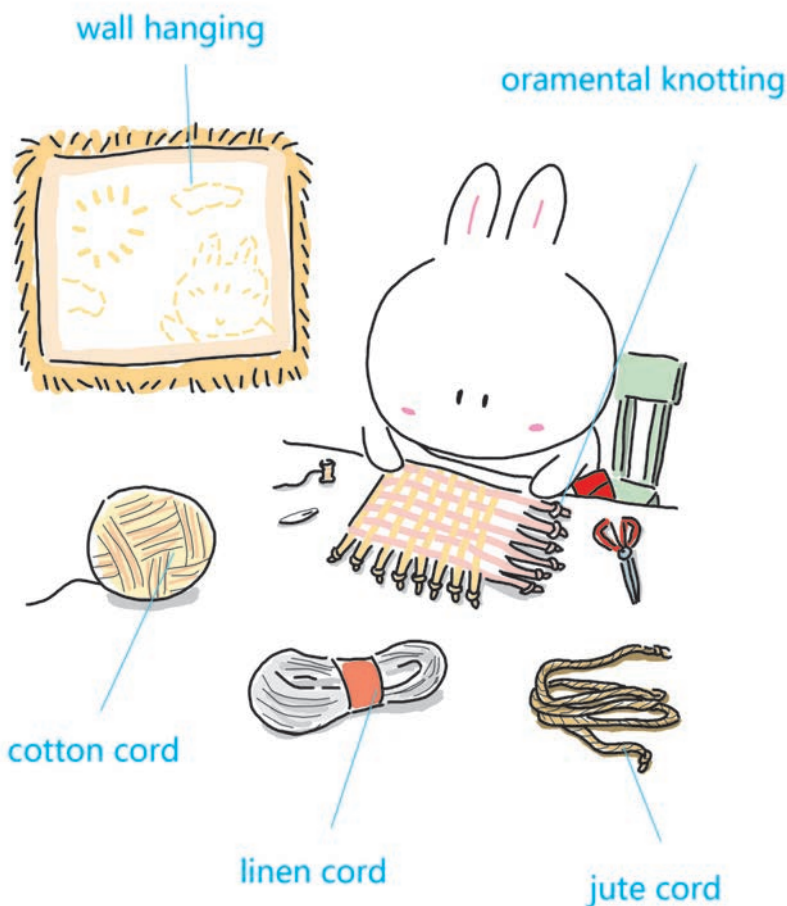
twist [twɪst] v. 捻

strand [strænd] n. 股

hemp [hemp] n. 大麻纤维

UNIT017

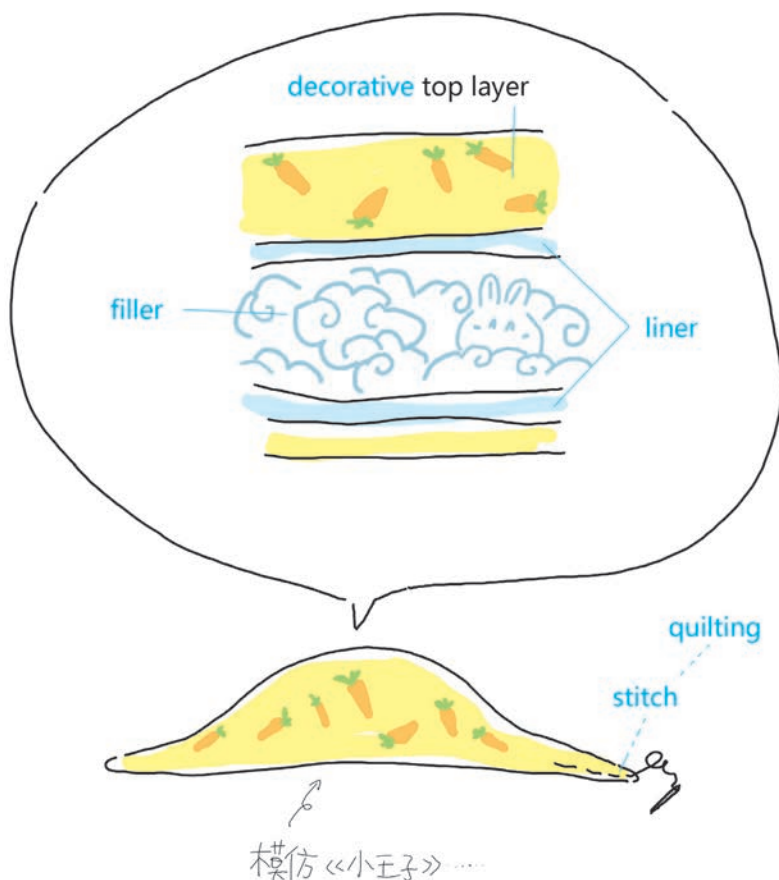
In ornamental knotting, cotton, linen, or jute cord is used to make bags, wall hangings, and containers.



注释

ornamental [ˌɔːnə'mentl] adj. 装饰性的
 knotting ['nɒtɪŋ] n. 结形花边 (流苏)
 cotton ['kɒtən] n. 棉布
 linen ['lɪnɪn] n. 亚麻布
 jute cord ['dʒuːt-kɔːd] n. 黄麻绳
 wall hanging ['wɔːl-'hæŋɪŋ] n. 壁毯

In quilting, three layers of fabric (a decorative top layer, filler, and a liner) are stitched or tied together.



注释

quilt [kwilt] n. 被子

quilting ['kwiltɪŋ] n. 被子面料

layer ['leɪə] n. 层

decorative ['dekəreɪv] adj. 装饰性的

filler ['fɪlə] n. 填充物

liner ['laɪnə] n. 衬里

stitch [stɪtʃ] v. 缝合

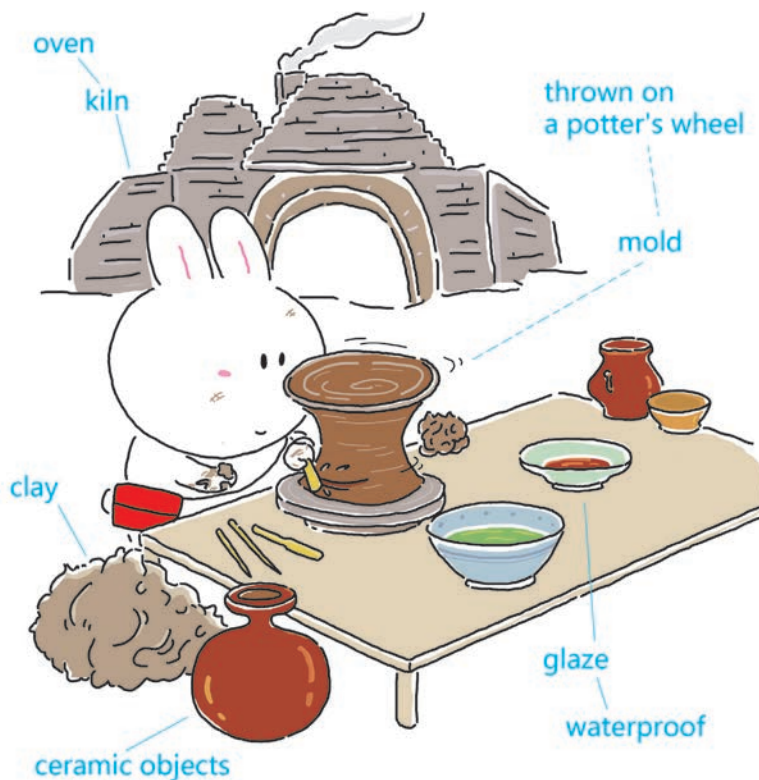
Many useful objects can be fashioned from leather, using such special tools as leather shears, punches for carving thick leather, thonging chisels, and lacing needles.



注释

- fashion ['fæʃən] v. 制造
 leather ['leðə] n. 皮革
 shears [ʃiəz] n. 剪刀
 punch [pʌntʃ] n. 打孔器
 carve [kɑ:v] v. 切割
 thong [θɒŋ] v. 给……装上皮带
 chisel ['tʃizl] n. 凿子
 lace [leis] v. 穿线
 needle ['ni:dl] n. 针

Ceramic objects can be molded completely by hand or thrown on a potter's wheel. When the clay hardens, it is fired in a high-temperature oven, or kiln, to strengthen it. To make the object waterproof, glazes may then be applied and the piece fired again.



注释

ceramic [si'ræmik] adj. 陶瓷的

mold [məʊld] v. 塑造

throw [θrəʊ] n. (在陶钧上)制(陶)坯

potter's wheel ['pɒtəz-wi:l] n. 陶轮

clay [kleɪ] n. 黏土

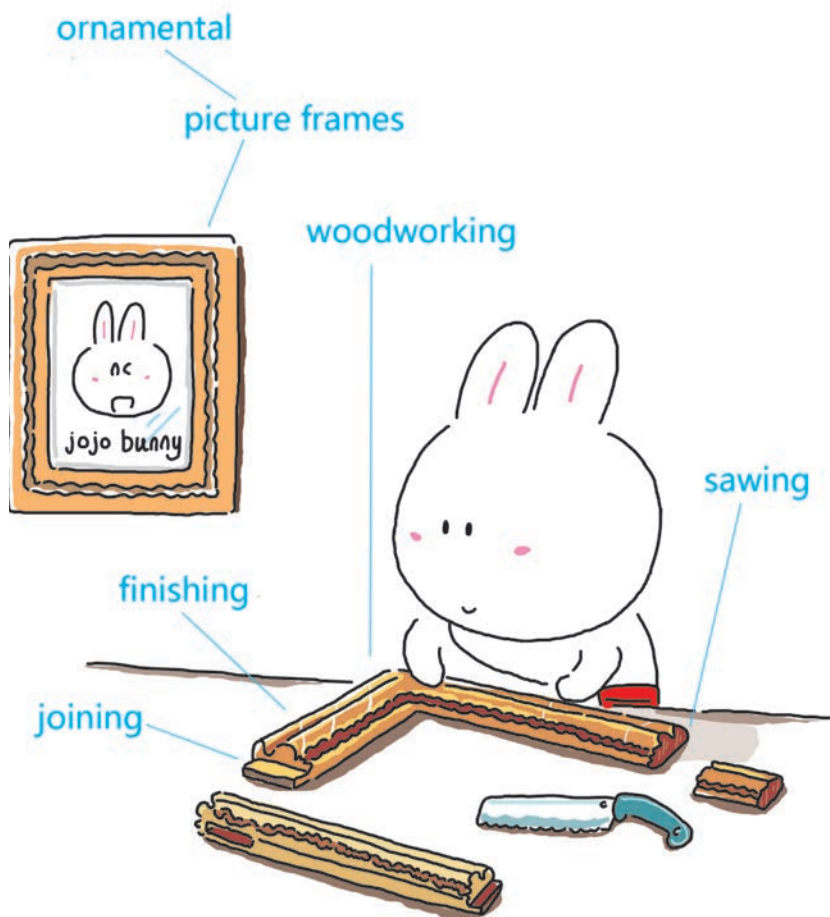
oven ['ʌvən] n. 烤箱

kiln [kɪln] n. 窑

waterproof ['wɔ:təpru:f] adj. 防水的

glaze [gleɪz] n. 釉料

Such basic woodworking techniques as sawing, joining, and finishing can be employed to make a wide variety of useful and ornamental objects, from jewelry boxes to picture frames.



注释

woodworking ['wʊd,wə:kin] n. 木工艺, 木工行业; adj. 木工的

sawing ['sɔ:ɪŋ] n. 锯

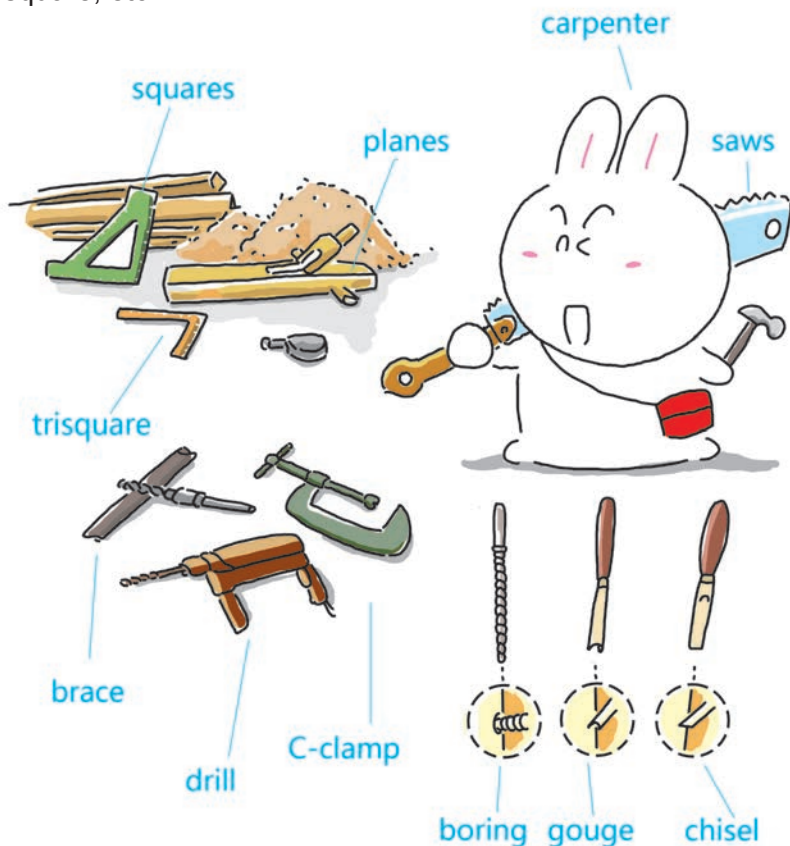
joining ['dʒɔɪnɪŋ] n. 拼接

finishing ['fɪnɪʃɪŋ] n. 抛光

ornamental [ɔ:nə'mentl] adj. 装饰性的

picture frame ['pɪktʃə-'freɪm] n. 相框

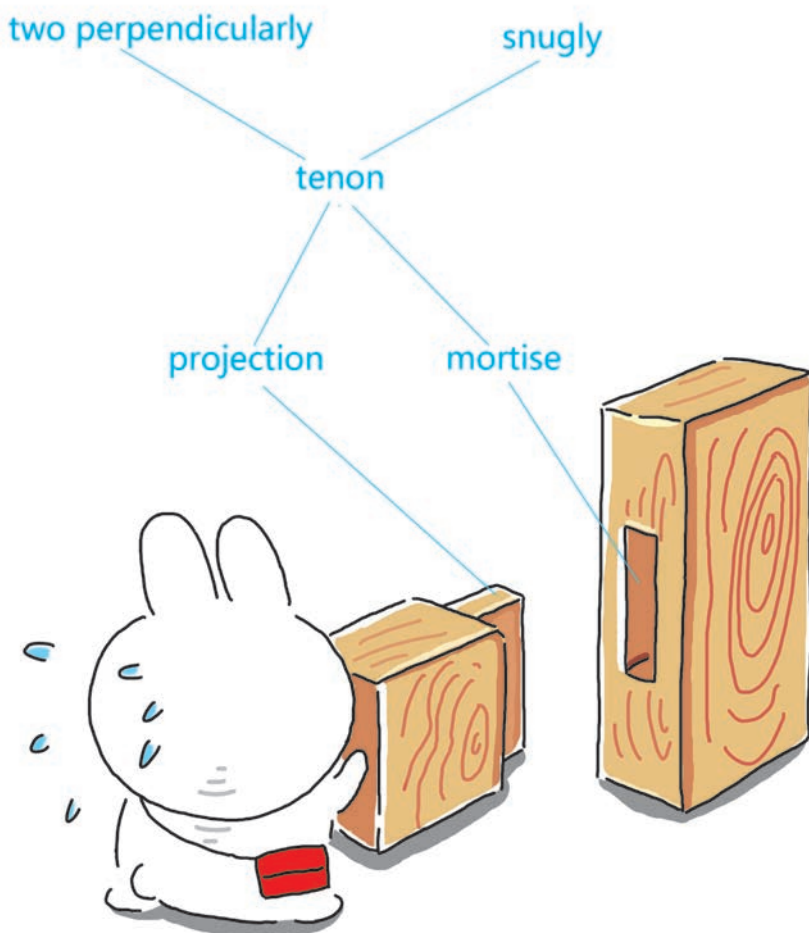
Carpenter tools include saws, planes, flat-edged chisels, semicircular gouge, boring tools such as brace, drills, and measuring tools such as steel rules, C-clamp, squares and trisquare, etc.



注释

- carpenter ['kɑ:pɪntə] n. 木匠
 saw [sɔ:] n. 锯
 plane [pleɪn] n. 刨子
 chisel ['tʃɪzl] n. 凿子
 gouge [gaʊdʒ] n. 半圆凿
 boring ['bɔ:rɪŋ] n. 钻 (孔)
 brace [breɪs] n. 手摇曲柄钻
 drill [drɪl] n. 钻孔机
 C-clamp ['si:klæmp] n. C型夹
 square [skwɛə] n. 直角尺
 trisquare ['traɪskwɛə] n. 曲尺

The tenon, a rectangular or square projection from the end of one member, fits snugly into the mortise cut in the second member, thus joining the two perpendicularly.



注释

tenon ['tenən] n. (木工的) 榫

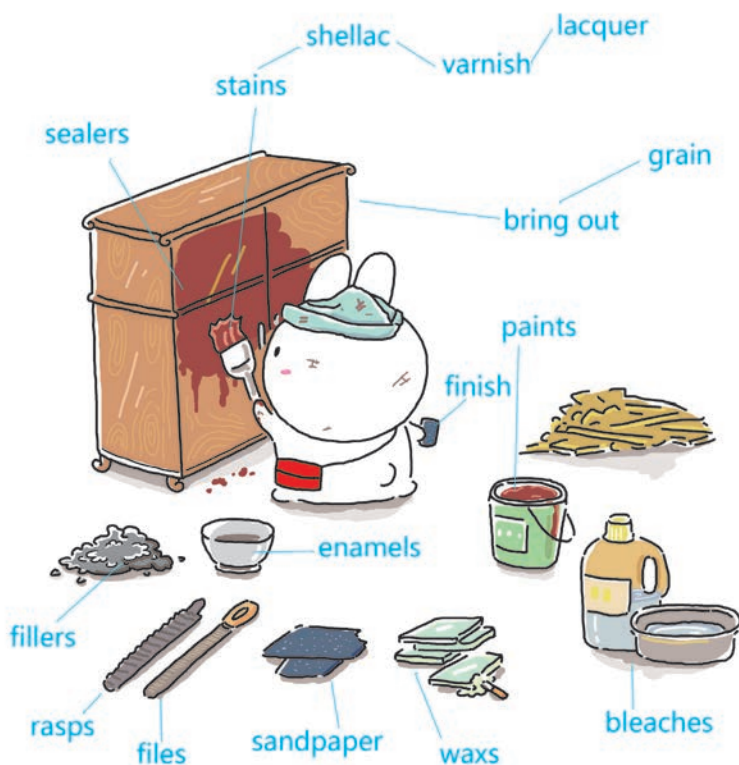
projection [prə'dʒekʃən] n. 凸出物

snugly ['snʌɡli] adv. 紧紧地

mortise ['mɔ:tis] n. 榫眼

perpendicularly [ˌpɜ:pən'dikjuləli] adv. 垂直地

Finishes serve to protect and preserve the wood and to bring out the beauty of the grain. Pieces are finished with rasps, files, and sandpaper. Common finishes include waxes, oils, bleaches, fillers, stains, shellac, varnish, lacquer, sealers, and paints, including enamels.



注释

bring out ['brɪŋ-aʊt] v. 显示出

grain [greɪn] n. 纹理图案

finish [fɪnɪʃ] n. 抛光、打磨

rasp [rɑːsp] n. 粗锉：一种有锋利、突出、尖锐凸起的钝锉

file [faɪl] n. 锉刀

sandpaper ['sænd,peɪpə] n. 砂纸；v. 用砂纸磨光

wax [wæks] n. 石蜡

bleach [bliːtʃ] n. 漂白剂

filler ['fɪlə] n. 填充料

stain [steɪn] n. 着色剂

shellac [ʃe'læk] n. 虫胶清漆：用于涂饰木料

varnish ['vɑːnɪʃ] n. 清漆：用于把表面涂成坚硬的、有光泽的透明的膜

lacquer ['lækə] n. 天然漆：一种用作表面涂层的光滑的树脂般的物质

sealer ['si:lə] n. 封底漆

paint [peɪnt] n. 油漆

enamel ['ɪnəməl] n. 瓷漆：一种烧制在金属、玻璃或瓷器上不透明的、保护性或装饰性的覆盖层

Paints are formed by mixing a pigment and a binder, a fluid vehicle, such as linseed oil, that solidifies when exposed to air. A varnish is a transparent solution that solidifies into a protective coating. Opaque and colored varnishes are called lacquers.



注释

pigment ['pɪgmənt] n. 色素

binder ['baɪndə] n. 粘合剂

vehicle ['vi:ɪkl] n. 调漆料

linseed oil ['linsi:d-oil] n. 亚麻油

solidify [sə'lidifai] v. (使)凝固

expose [iks'pəʊz] v. 暴露

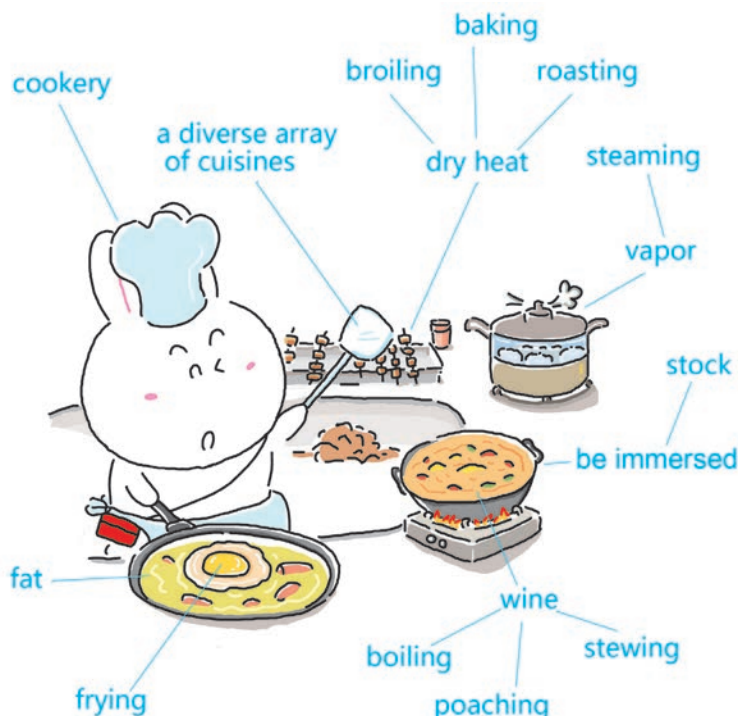
transparent [træns'pæərənt] adj.透明的

solution [sə'lu:ʃən] n. 溶液

coating ['kəʊtɪŋ] n. 涂层

opaque [əu'peik] adj. 不透明的

Cookery involves a diverse array of cuisines. Food may be immersed in liquids such as water, stock, or wine (boiling, poaching, stewing); immersed in fat or oil (frying); exposed to vapor (steaming and, to some extent, braising); exposed to dry heat (roasting, baking, broiling); and subjected to contact with hot fats (sautéing).



注释

cooking ['kʊkəri] n. 烹饪
diverse [dai'vəs] adj. 不同的
cuisine [kwi'zi:n] n. 烹饪法
immerse [i'mə:s] v. 浸没
stock [stɒk] n. 原汤
wine [wain] n. 葡萄酒
boil [bɔil] v. 煮
poach [pəʊtʃ] v. 水煮
stew [stju:] v. 炖: 用小火烧或慢慢地煮沸来烹饪
fat [fæt] n. 油脂
fry [frai] v. 油炸
vapor ['veipə] n. 蒸气

steam [sti:m] v. 蒸
braise [breiz] v. 炖: 先用油炒, 然后在有盖容器中炖
dry heat [drai-hi:t] n. 干热
roast [rəʊst] v. 烘烤
bake [beik] v. 烘焙
broil [brɔil] v. (用火) 烤
subject to ['sʌbdʒikt-tu:] v. 使遭受
contact ['kɒntækt] n. 关联
sauté [so:'teɪ] v. 炒

Essential modern kitchen equipment includes the following: a stove, or range; sink; work surface; various knives, pots and pans; such utensils as spatulas, whisks, specialized spoons, and rolling pins; a more highly specialized array of gear for producing pastries and other baked goods; and more recent sophisticated equipment such as blenders, food processors, and microwave ovens.

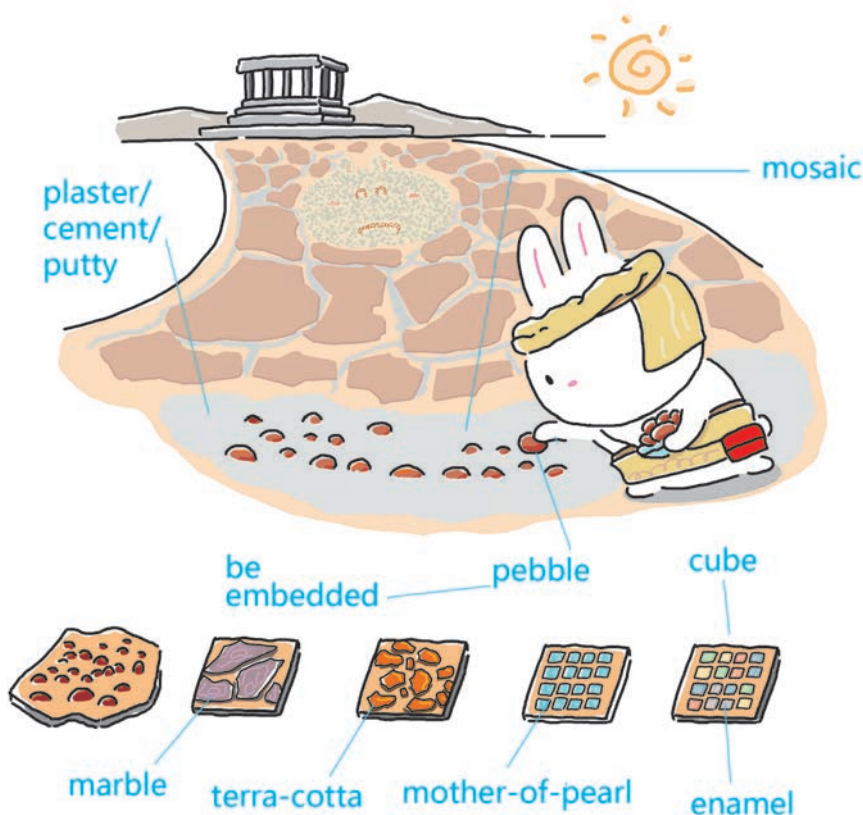


注释

kitchen ['kɪtʃɪn] n. 厨房
 stove [stəʊv] n. 炉子
 range [reɪndʒ] n. 火炉
 sink [sɪŋk] n. 水槽
 work surface ['wɜ:k-'sə:fəs] n. 工作台
 knife [naɪf] n. 刀
 pot [pɒt] n. 锅
 pan [pæn] n. 平底锅
 utensil [ju:'tensl] n. 器具

spatula ['spætjələ] n. (调拌用的) 刮铲
 whisk [hwɪsk] n. 搅拌器
 spoon [spu:n] n. 匙
 rolling pin ['rəʊlɪŋ-pɪn] n. 擀面杖
 gear [gɪə] n. 设备
 pastry ['peɪstri] n. 糕点
 sophisticated [sə'fɪstɪkeɪtɪd] adj. 复杂精巧的
 blender ['blendə] n. 搅拌机
 microwave oven ['maɪkrəweɪv-'ʌvən] n. 微波炉

In ancient times, mosaics were a form of floor decoration made of small pebbles and later of cut or shaped pieces of marble, hard stone, glass, terra-cotta, mother-of-pearl, and enamels. The shaped pieces, in the form of small cubes, are called tesserae. The tesserae are embedded in plaster, cement, or putty to hold them in place.



注释

mosaic [mo'zeiik] n. 镶嵌工艺
 pebble ['pebl] n. 鹅卵石
 marble ['mɑ:bl] n. 大理石
 terra-cotta ['terə-'kɒtə] n. 赤陶
 mother-of-pearl [ˌmʌðər-əv-'pɜ:l] n. 珍珠母
 cube [kju:b] n. 立方体

tesserae ['tesərə:] n. 镶嵌物
 (plural: tesserae ['tesəri:])
 embed [im'bed] v. 镶嵌
 plaster ['plɑ:stə] n. 石膏
 cement [si'ment] n. 水泥
 putty ['pʌti] n. (嵌装玻璃等用的) 油灰
 in place [in-'pleis] adv. 在适当的位置

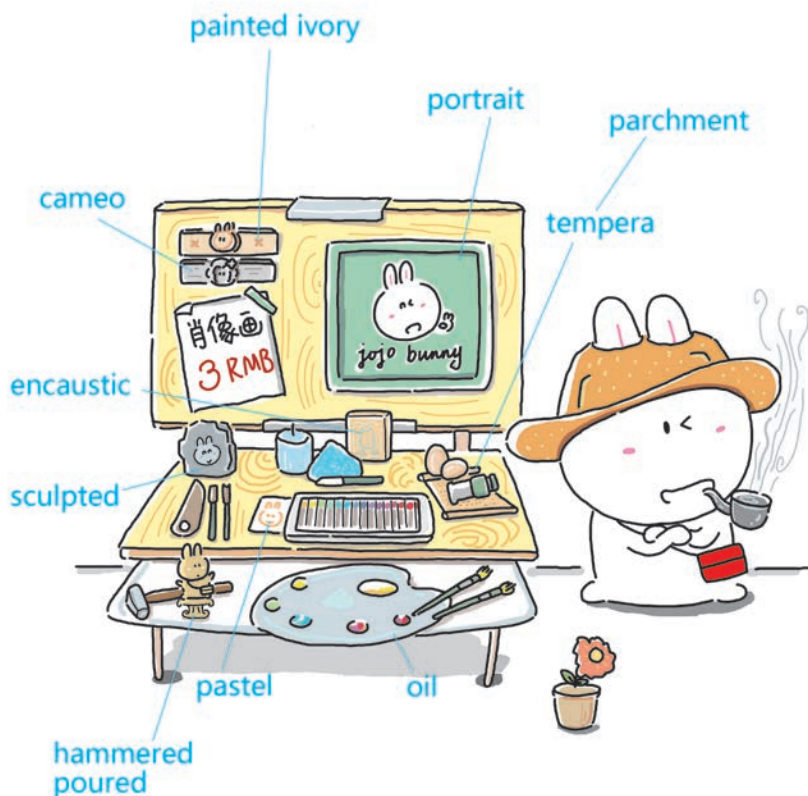
Molten glass was tinted in a wide range of colors with metal oxides and then poured on a flat surface such as a marble slab to form a disk of colored glass. This was scored with a sharp tool and broken into strips and cubes.



注释

molten ['məʊltən] adj. 熔化的; 熔融的
 tint [tɪnt] n. 带白的颜色, 淡色; v. 染色; 着色于……
 a wide range of [ə-'waɪd-reɪndʒ-əv] adj. 广泛的
 metal oxide ['metl-'ɒksaɪd] n. 金属氧化物
 pour on ['pɔːr-ɒn] v. 把……倒在……上
 flat [flæt] adj. 扁平的
 marble slab ['mɑːbl-'slæb] n. 大理石板
 score [skɔː] v. 打记号于
 strip [stri:p] n. 条

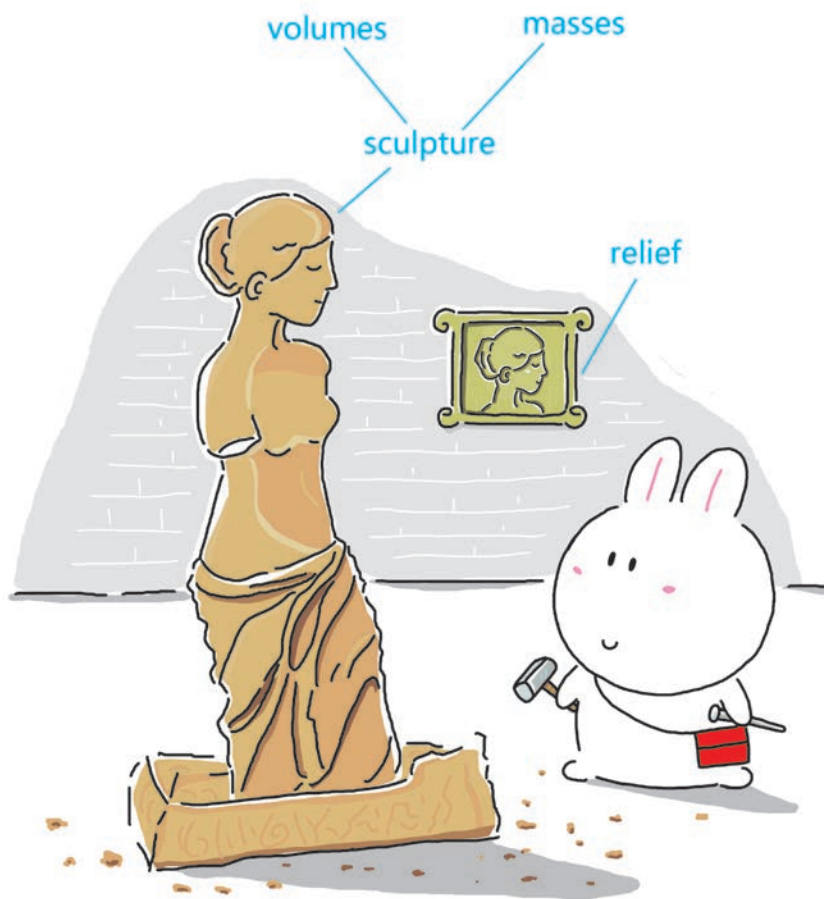
Portraits can be executed in any medium, including sculpted stone and wood, oil, painted ivory, pastel, encaustic on wood panel, tempera on parchment, carved cameo, and hammered or poured metal.



注释

- portrait ['pɔ:trɪt] n. 肖像
 sculpt [skʌlpt] v. 雕刻
 ivory ['aɪvəri] n. 象牙
 pastel [pæ'stel] n. 彩色蜡笔画
 encaustic [ɪn'kɔ:stɪk] n. 蜡画
 tempera ['tempərə] n. 蛋彩画
 parchment ['pɑ:tʃmənt] n. 羊皮纸
 cameo ['kæmiəu] n. 浮雕
 hammer ['hæmə] v. 锤打
 pour [pɔ:] v. 浇注

Sculpture (from Latin, "to carve"), three-dimensional art concerned with the organization of masses and volumes. The two principal types have traditionally been freestanding sculpture in the round and relief sculpture.



注释

sculpture ['skʌlptʃə] n.雕刻

mass [mæs] n.质量

volume ['vɒlju:m] n.体积

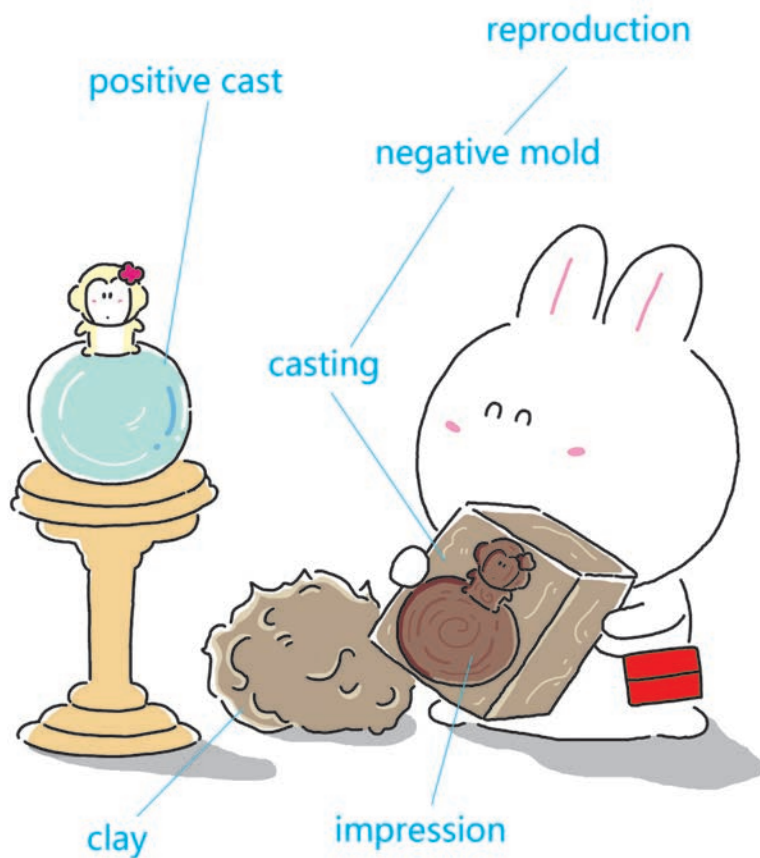
principal ['prɪnsəpəl] adj.主要的

freestanding ['fri:stændɪŋ] adj. (雕刻、建筑物等) 独立的, 自力撑持的

in the round [ɪn-ðə-'raʊnd] adj.雕刻成立体 (的), 栩栩如生的, 表现无余的

relief [ri'li:f] n.浮雕

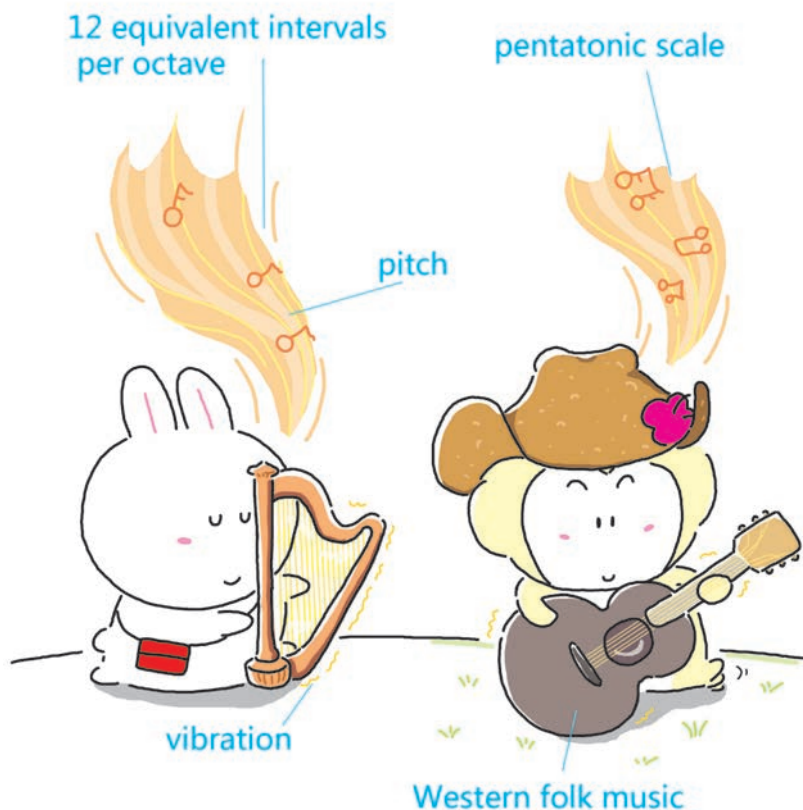
Casting is accomplished in two stages: an impression or negative mold is formed from the original — a clay model, for instance — and a positive cast or reproduction is made of the original work from the negative impression.



注释

- casting ['kɑ:stɪŋ] n. 铸造
 impression [im'preʃən] n. 印痕
 negative mold ['negətiv-məuld] n. 凹模
 clay [kleɪ] n. 黏土, 泥土
 positive cast ['pɒzitiv-kɑ:st] n. 正像
 reproduction [,ri:prə'dʌkʃən] n. 复制品

Pitch depends upon the rate of vibration, or frequency, of sound waves that produce a particular tone. Most Western music is based on 12 equivalent intervals per octave; however, a great deal of Western folk music conforms to pentatonic scale, the best-known form of which contains no half steps.

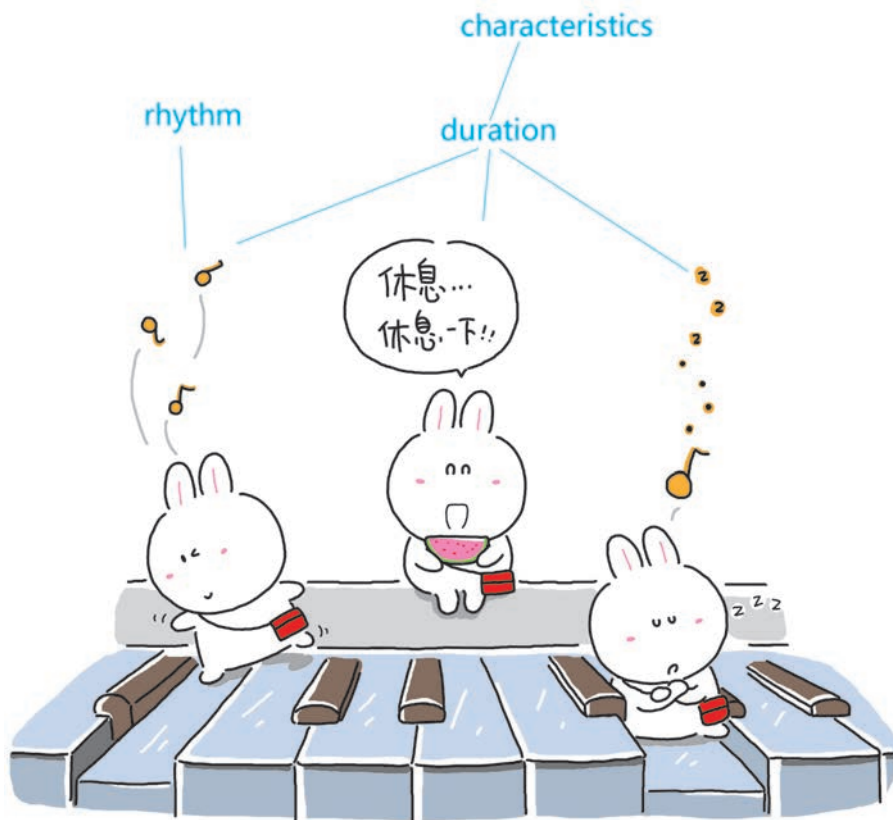


注释

pitch [pitʃ] n. 音高
 vibration [vai'breiʃən] n. 振动
 frequency ['fri:kwensi] n. 频率
 tone [təʊn] n. 音调
 equivalent interval [i'kwivələnt - 'intəvəl]
 n. 等间隔的音程
 octave ['ɒktiv] n. 八度音程 (音阶)

folk [fəʊk] adj. 民间的
 conform [kən'fɔ:m] v. (to, with) 遵照, 依照
 pentatonic [ˌpentə'tɒnik] adj. 五声音阶的
 scale [skeil] n. 音阶
 step [step] n. 音级

Rhythm is the way music uses time, which includes characteristics such as duration of tones and silences, and patterns of duration.



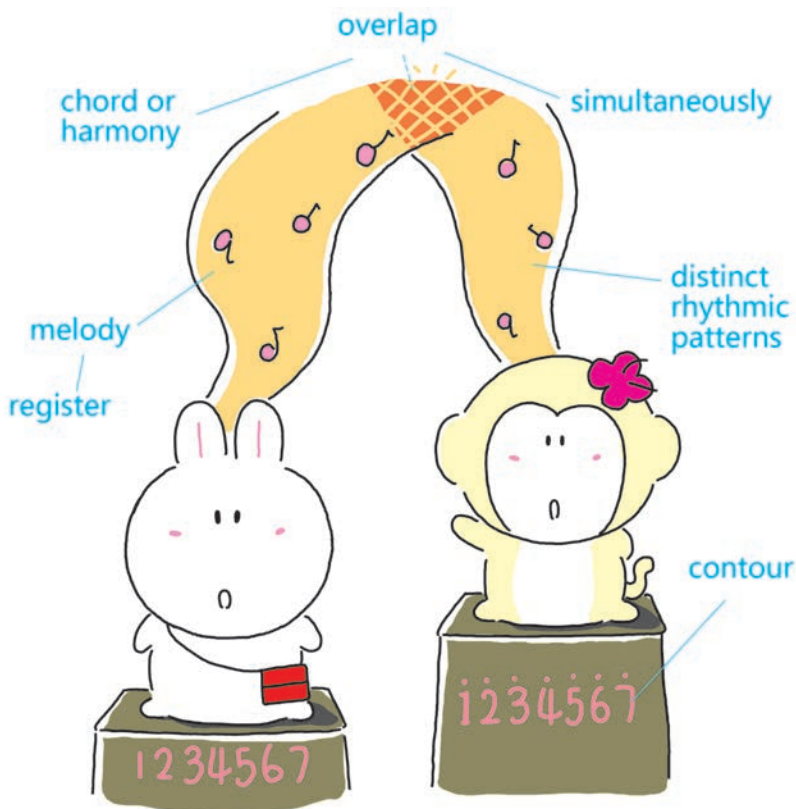
注释

rhythm ['rɪðəm] n. 节奏

characteristic [ˌkærɪktə'rɪstɪk] n. 特性

duration [dʒuə'reɪʃən] n. 持续时间

The most important and style-defining patterns are formed by pitches that overlap with one another in time, producing a chord, or harmony. Two or more tones heard simultaneously may belong to separate melodies that fit well together, but which occupy different octave registers, have distinct rhythmic patterns, or otherwise have different shapes or contours.



注释

style-defining [stail - di'fainiŋ] adj. 定义完整的, 仔细定义过的

overlap [əʊvə'læp] v. 与……重叠

chord [kɔ:d] n. 和弦

harmony ['hɑ:məni] n. 和声

simultaneously [siməl'teiniasli] adv. 同时地

melody ['melədi] n. 曲调

occupy ['ɒkjupaɪ] v. 占用, 填满

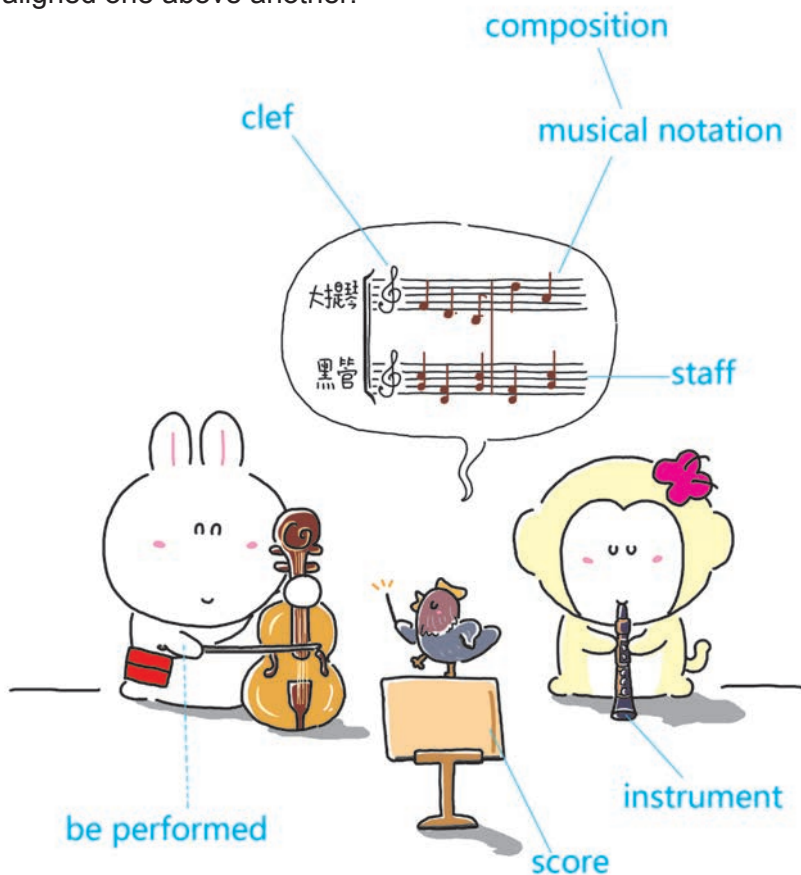
register ['redʒistə] n. 音域

distinct [dis'tɪŋkt] adj. 截然不同的

rhythmic patterns ['riðmik - 'pætənz] n. 有节奏的模式

contour ['kɒn.tʊə] n. (音调或声调的) 升降曲线; 变化方式

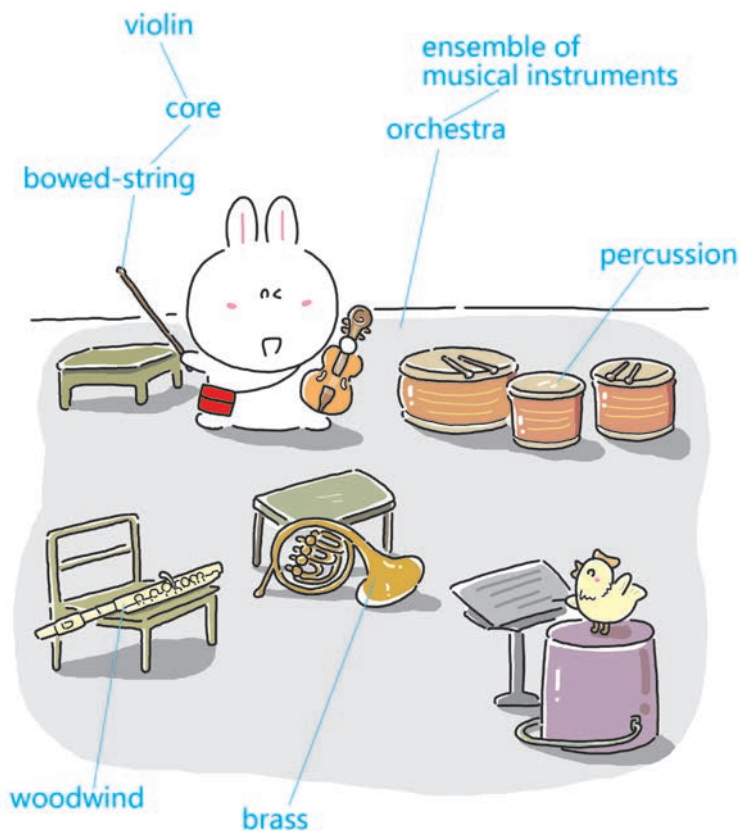
Score, musical notation for a multipart composition, in which the music to be performed by each voice or instrument, is written with clefs on a separate staff, all the staves being aligned one above another.



注释

- score [sko:] n. 乐谱
 notation [nəu'teɪʃən] n. 谱号 (musical notation: 音乐记谱)
 composition [,kɒmpə'zɪʃən] n. 乐曲
 perform [pə'fɔ:m] v. 演奏
 instrument ['ɪnstrəmənt] n. 乐器
 clef [klef] n. 谱号
 staff [sta:f] n. 五线谱 (plural: staves [sta:vz])
 align [ə'leɪn] v. 排列

Orchestra, in the narrowest sense, the characteristic ensemble of Western musical instruments, has as its core a group of bowed-string instruments of the violin family, augmented by woodwind, brass, and percussion instruments.



注释

orchestra [ˈɔ:kɪstrə] n. 管弦乐

ensemble [ɒŋ'sɒŋblə] n. 合奏

musical instrument ['mju:zɪkəl-'ɪnstrəmənt] n. 乐器

core [kɔ:] n. 中心

bowed-string [baʊd-'strɪŋ] adj. 演奏弦乐的

violin [ˌvaɪə'lɪn] n. 小提琴

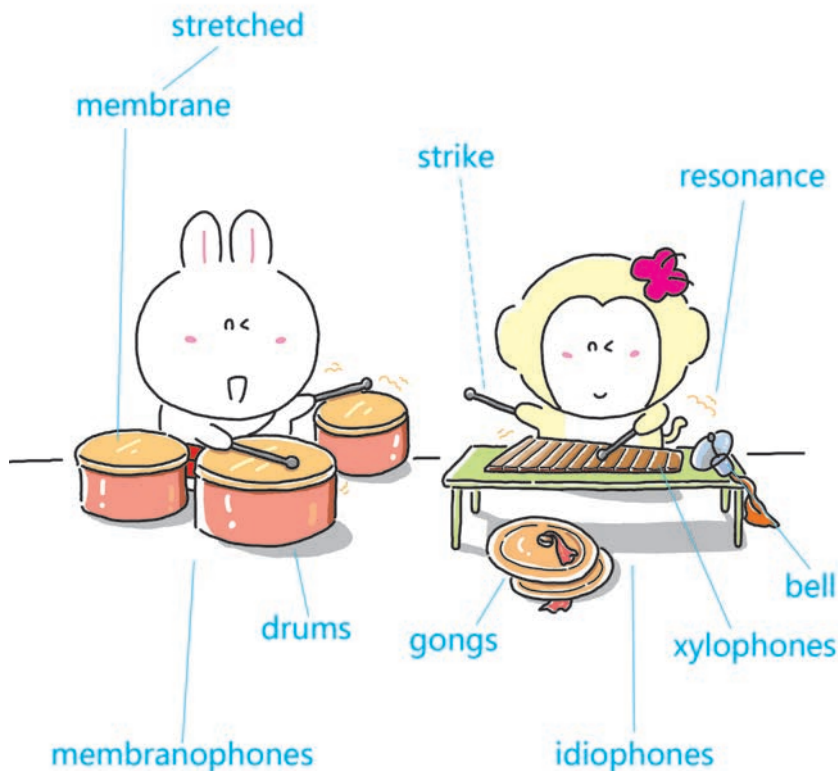
augment [ɔ:g'mənt] v. 增补, 扩充

woodwind ['wʊdwaɪnd] n. 木管乐器

brass [brɑ:s] n. 铜管乐器

percussion [pə:'kʌʃən] n. 打击乐器

Percussion instruments are referred to as membranophones if they produce sound through the vibrations of a stretched skin or other membrane. They are called idiophones if they produce sound through their natural resonance when struck, rubbed, plucked, or shaken. Drums are membranophones; hollowed logs, bells, gongs, xylophones, and pianos are examples of idiophones.

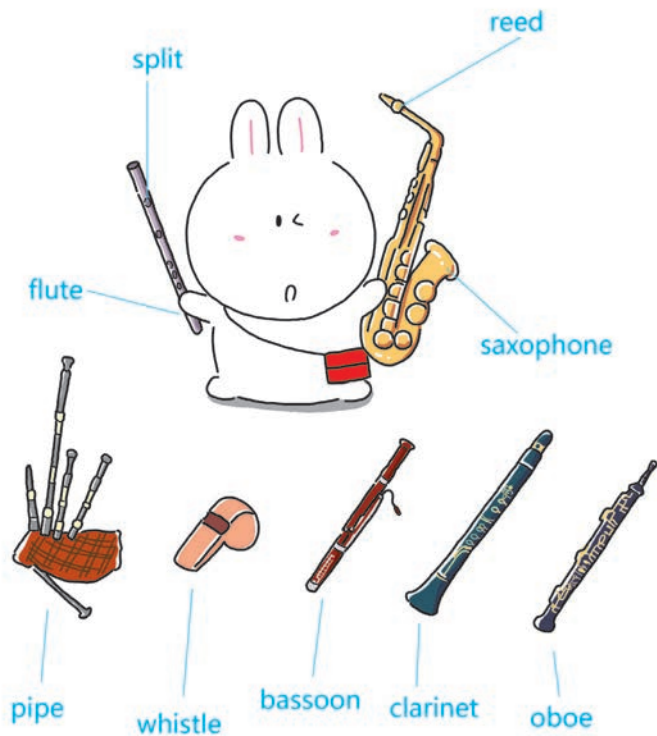


注释

membranophone [mem'breɪnəfəʊn] n. 膜鸣乐器
 stretched [stretʃ] adj. 有弹性的
 membrane ['membreɪn] n. (动物或植物体内的) 薄膜, 隔膜
 idiophone ['ɪdiəfəʊn] n. 体鸣乐器
 resonance ['rezənəns] n. 共振
 strike [straɪk] v. 敲击
 rub [rʌb] v. 摩擦

pluck [plʌk] v. 拨 (弦)
 shake [ʃeɪk] v. 摇晃
 drum [drʌm] n. 鼓
 hollowed log ['hɒləʊd-lɒɡ] n. 中空的原木
 bell [bel] n. 铃、钟、或钟琴
 gong [ɡɒŋ] n. 锣
 xylophone ['zaɪləfəʊn] n. 木琴
 piano [pi'æneɪ] n. 钢琴

Wind instruments, or aerophones, produce sound by vibration, which may be produced by performers' lips (brass), a column of air split across a sharp edge (flutes, pipes, whistles), or by one or two reeds, as with instruments such as the clarinet, saxophone, oboe and bassoon.

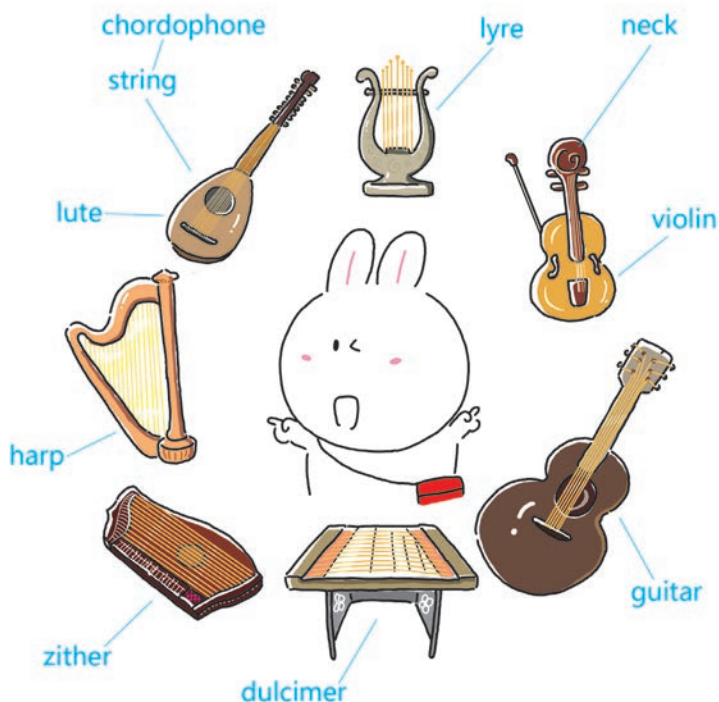


注释

wind instrument ['wind-'instrument] n. 管乐器
aerophone ['eərə'fəʊn] n. 管乐器
column ['kɒləm] n. 柱 (a column of air 气柱)
split [split] v. 劈开, 切开
edge [edʒ] n. 边缘
flute ['flu:t] n. 竖笛, 横笛
pipe [paɪp] n. 风笛

whistle ['hwɪsl] n. 口哨
reed [ri:d] n. 簧片
clarinet ['klæri'net] n. 单簧管, 黑管
saxophone ['sæksə'fəʊn] n. 萨克斯管
oboe ['əʊbəʊ, 'əʊbi] n. 双簧管
brass [bræs] n. 铜管 (乐器)
bassoon [bə'su:n] n. 低音管, 巴松

The string, or chordophone, family has several branches. In one branch, which includes the zither and dulcimer, strings are stretched across a flat body. In a second branch, each instrument has a neck, for example the lute, guitar, or violin. A third branch includes plucked instruments with multiple strings, such as the lyre or the harp, where each string produces only one pitch.



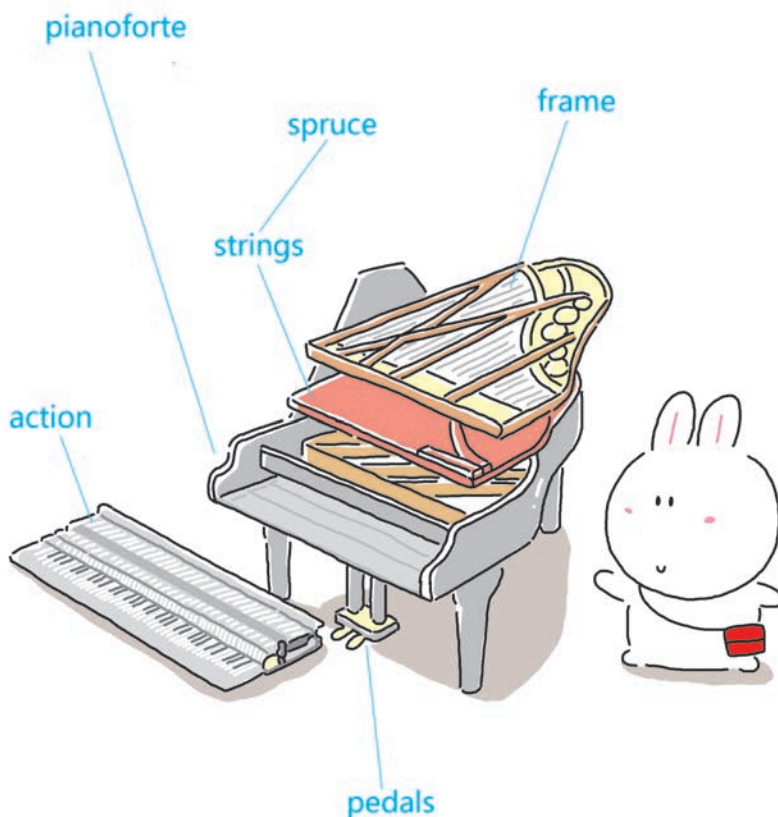
注释

- string [strɪŋ] n. 琴弦, 弦乐器
 chordophone ['kɔːdəʊfəʊn] n. 弦乐器
 zither ['zɪðə] n. 齐特琴 (与中国的“箏”相似)
 dulcimer ['dʌlsɪmə] n. 洋琴
 neck [nek] n. 琴颈
 lute [lu:t] n. 鲁特琴
 guitar [gɪ'tɑː] n. 吉他
 violin [ˌvaɪə'lɪn] n. 小提琴
 lyre ['laɪə] n. 里拉, 七弦竖琴 (古希腊的一种弦乐器)
 harp [hɑ:p] n. 竖琴

The modern pianoforte has six major parts: (1) The frame is usually made of iron. At the rear end is attached the string plate, into which the strings are fastened. In the front is the wrest plank, into which the tuning pins are set. Around these is wound the other end of the strings, and by turning these pins, the tension of the strings is regulated. (2) The soundboard, a thin piece of fine-grained spruce placed under the strings, reinforces the tone by means of sympathetic vibration. (3) The strings, made of steel wire, increase in length and thickness from the treble to the bass. The higher pitches are each given two or three strings tuned alike. The lower ones are single strings made heavier by being overspun — that is, wound around with a coil of thin copper wire. (4) The action is the entire mechanism required for propelling the hammers (wrapped with felt) against the strings. The most visible part of the action is the keyboard, a row of keys manipulated by the fingers. The keys corresponding to the natural tones are made of ivory or plastic; those corresponding to the chromatically altered tones, of ebony or plastic. (5) The pedals are levers pressed down by the feet. The damper, or loud pedal, raises all the dampers so that all the strings struck continue to vibrate even after the keys are released. The use of these pedals can produce subtle changes in tone quality. (6) According to the shape of the case, pianos are classified as grand, square, and upright. Grand pianos are built in various sizes, from the full concert grand, 2.69m long, to the parlor or baby grand, less than 1.8m long.

注释

- pianoforte [pi,ænou'fo:ti] n. 钢琴(piano的旧称)
 frame [freim] n. 骨架
 rear [riə] adj. 后面的
 attach [ə'tætʃ] v. 附加
 fasten ['fɑ:sən] v. 使固定; 加固
 plank [plæŋk] n. 支架 [wrest plank: 校准弦音的扭钥架]
 tuning pin ['tu:niŋ - pin] n. 调音弦轴
 wind [wind] v. 绕, 缠
 tension ['tenʃən] n. 压力, 张力
 regulate ['regjuleit] v. 调节, 校准
 fine-grained ['fain - 'greind] adj. 纹理细腻、平整的
 spruce [spru:s] n. 云杉质木材
 reinforce ['ri:in'fo:s] v. 加强



sympathetic vibration ['sɪmpə'tetɪk-vai'breɪʃən]

n. 共振

treble ['trebl] n. 高音部分

bass [beɪs] n. 低音部分

pitch [pɪtʃ] n. 音高

overspin wire ['əʊvəspɪn-'waɪə] n. 缠弦

coil [kɔɪl] n. (一)卷

action ['ækjən] n. 机械装置

mechanism ['mekənɪzəm] n. 机械装置

propel [prə'pel] v. 推动

hammer ['hæmə] n. 音锤

felt [felt] n. 毛毡

manipulate [mə'nɪpjuleɪt] v. (熟练地) 操作、使用 (机器等)

corresponding to [kərə'spɒndɪŋ-tu:]
adj. 与……相应的

ivory ['aɪvəri] n. 象牙

plastic ['plɑːstɪk] n. 塑料

chromatic [krəʊ'mætɪk] adj. 半音 (阶) 的, 含半音的

chromatically [krəʊ'mætɪkli] adv. 半音地

ebony ['ebəni] n. 乌木

pedal ['pedl] n. 踏板

lever ['li:və] n. 控制杆

dampener ['dæmpə] n. 制音器

release [rɪ'liːs] v. 释放

sustaining [səs'teɪnɪŋ] adj. 支持的

subtle ['sʌtl] adj. 微小的

grand [grænd] adj. 大的

square [skwɛə] adj. 正方形的

upright ['ʌpraɪt] adj. 立式的

various ['vɛəriəs] adj. 许多的, 多样的

parlor ['pɑːlə] adj. 客厅的

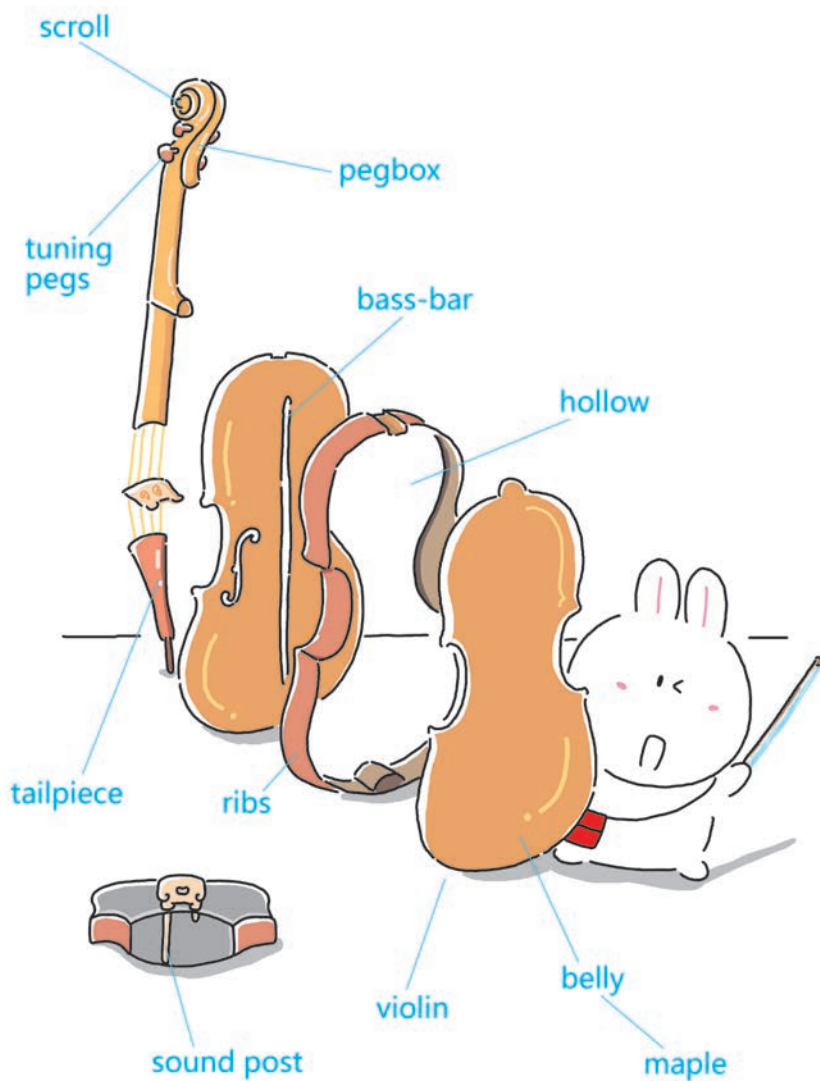
The main parts of the violin are the front, also called the belly, top, or soundboard, usually made of well-seasoned spruce; the back, usually made of well-seasoned maple; and the ribs, neck, fingerboard, pegbox, scroll, bridge, tailpiece, and f-holes, or soundholes. The front, back, and ribs are joined together to form a hollow sound box. The sound box contains the sound post, a thin, dowel-like stick of wood wedged inside underneath the right side of the bridge and connecting the front and back of the violin; and the bass-bar, a long strip of wood glued to the inside of the front under the left side of the bridge. The sound post and bass-bar are important for the transmission of sound, and they also give additional support to the construction. The strings are fastened to the tailpiece, rest on the bridge, and are suspended over the fingerboard, and run to the pegbox, where they are attached to tuning pegs that can be turned to change the pitch of the string. The strings are set in vibration and produce sound when the player draws the bow across them at a right angle near the bridge. Among the prized characteristics of the violin are its singing tone and its potential to play rapid, brilliant figurations as well as lyrical melodies. Violinists can also create special effects by means of the following techniques: pizzicato, plucking, tremolo, etc.

注释

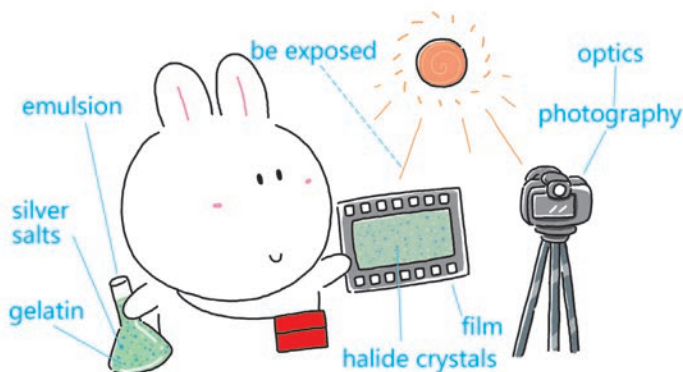
- violin [ˌvaɪəˈlɪn] n. 小提琴
 belly [ˈbeli] n. (弦乐器的) 面板
 well-seasoned [wel-ˈsi:znd] adj. 完全风干的
 spruce [spru:s] n. 云杉
 maple [ˈmeɪpl] n. 枫木
 rib [rib] n. 琴骨
 pegbox [ˈpeɡbɒks] n. (弦乐器的) 弦轴箱
 scroll [skrɔ:l] n. 涡卷形头: 提琴类乐器上的弯曲状饰头
 tailpiece [ˈteɪlpi:s] n. 系弦钮
 hollow [ˈhələʊ] adj. 中空的
 sound [saʊnd] n. 声音
 post [pəʊst] n. 杆
 dowel [ˈdaʊəl] n. 暗榫
 wedge [wedʒ] v. 楔入
 underneath [ˌʌndəˈni:θ] prep. 在……的下面
 bass bar [beɪs-bɑ:] n. 低音梁
 strip [stri:p] n. 条
 glue [glu:] v. 粘合
 transmission [trænzˈmɪʃən] n. 传播

suspend [sə'spend] v. 悬挂
 tuning peg ['tju:nɪŋ - peg] n. 调音弦轴
 pitch [pɪtʃ] n. 音质
 vibration [vai'breɪʃən] n. 振动
 draw [drɔ:] v. 拉
 bow [bəʊ] n. 琴弓
 prize [praɪz] v. 珍视

figuration [figju'reɪʃən] n. 用装饰音装饰
 lyrical ['lɪrɪkəl] adj. 抒情的
 melody ['melədi] n. 曲调
 pizzicato [pɪtsɪ'kɑ:təʊ] n. 拨奏乐曲
 pluck [plʌk] v. 拨
 tremolo ['tremələʊ] n. 颤音



Photography is based on principles of light, optics, and chemistry. For the purpose of producing a photograph, these silver salts are distributed in gelatin to make a mixture called an emulsion, which is applied to film or another supporting material in a thin layer. When the emulsion is exposed to light, the silver halide crystals undergo chemical changes and, after further processing, an image becomes visible. The stronger the light that strikes the crystals, the denser or more opaque that part of the film becomes. Most types of film produce a negative image, from which a positive final copy can be printed on sensitized paper. The dense (or dark) areas of the negative translate into light areas on the final photograph. Almost all modern photography relies on this negative-to-positive process.



注释

photography [fə'tɒɡrəfi] n. 摄影

principle ['prɪnsəpl] n. 原则

optics ['ɒptiks] n. 光学

silver salt ['sɪlvə-sɔ:lt] n. 银盐

distribute [dɪ'strɪbjʊ:t] v. 分布, 散布 (over)

gelatin ['dʒelə'tɪn, 'dʒelə'ti:n] n. 凝胶

emulsion [i'mʌljən] n. 感光乳剂

apply [ə'plai] v. 涂, 搽, 撒

film [fɪlm] n. 胶片

layer ['leɪə] n. 垫片, 层

expose [ɪks'pəʊz] v. 曝光, 暴露

halide ['hælaɪd] adj. 卤化物(的)

crystal ['krɪstəl] n. 晶体

undergo [ˌʌnde'ɡəʊ] v. 经历(变迁等); 遭受
(苦难等)

processing [prəʊ'sesɪŋ] n. 处理

image ['ɪmɪdʒ] n. 影像

strike [straɪk] v. 照在……上

opaque [əʊ'peɪk] adj. 不反射光线的

negative image [ˌneɡətɪv-'ɪmɪdʒ]

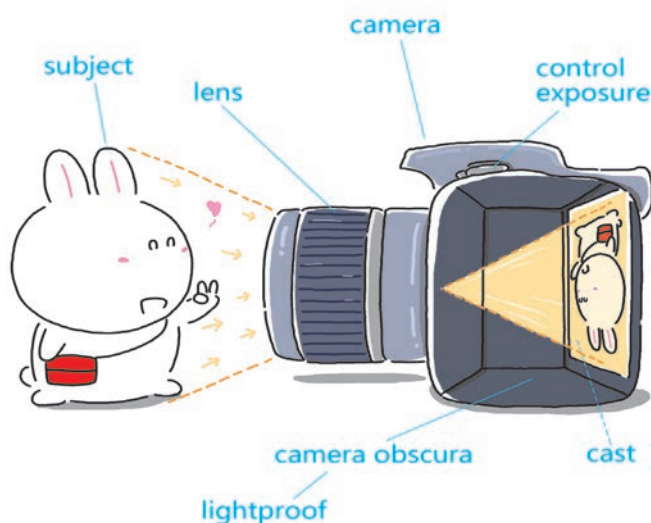
n. 底片(负像)

sensitized paper ['sensətaɪzd-'peɪpə] n. 感光纸

translate into [træns'leɪt-'ɪntu] v. 转化为

rely on [rɪ'laɪ-ɒn] v. 依赖, 依靠

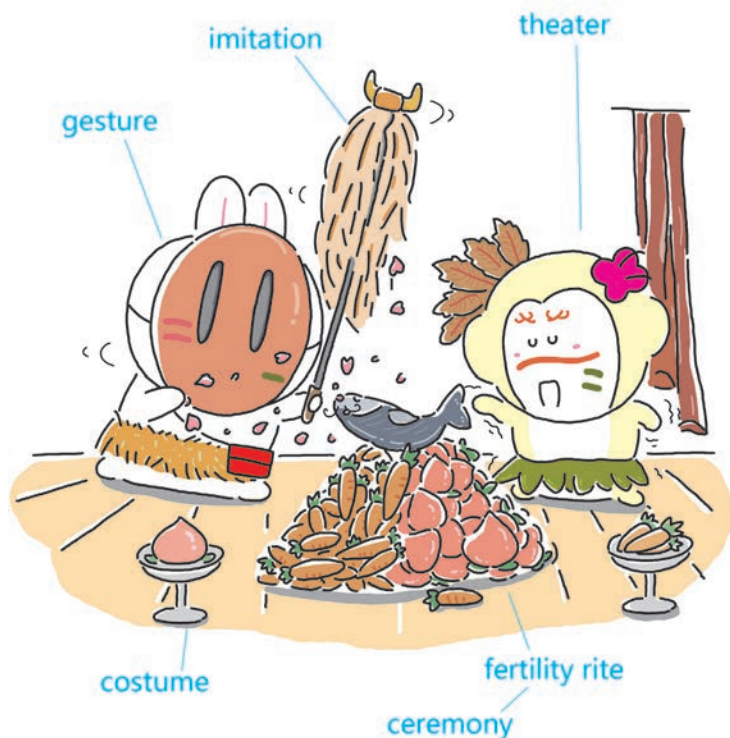
In most cases, the camera and its lens determine the appearance of the photographic image. Cameras work on the basic principle of the camera obscura. In both the camera obscura and the modern camera, light passes through a lens fitted into an otherwise lightproof box. Light passing through the lens casts an image of the camera's subject — the object, person, or scene in front of the camera — onto the inside of the box, which in a modern camera contains film. The camera and lens control how much light strikes the film, in what is called an exposure.



注释

- camera ['kæməɹə] n. 照相机
 lens [lenz] n. 镜头
 camera obscura ['kæməɹə-ɒb'skjuərə] n. 暗箱
 otherwise ['ʌðəwaɪz] adj. 别样的, 另外一种的
 lightproof ['laɪtpru:f] adj. 不透光
 cast [kɑ:st] v. 投射
 subject ['sʌbdʒɪkt] n. 对象
 scene [si:n] n. 景色
 exposure [iks'pəʊʒə] n. 曝光

Anthropologists and theater historians trace the origins of theater to myth and ritual found in dances and mimed performances by masked dancers during fertility rites and other ceremonies that marked important passages in life. Imitation, costumes, masks, makeup, gesture, dance, music, and pantomime were some of the theatrical elements found in early rituals.

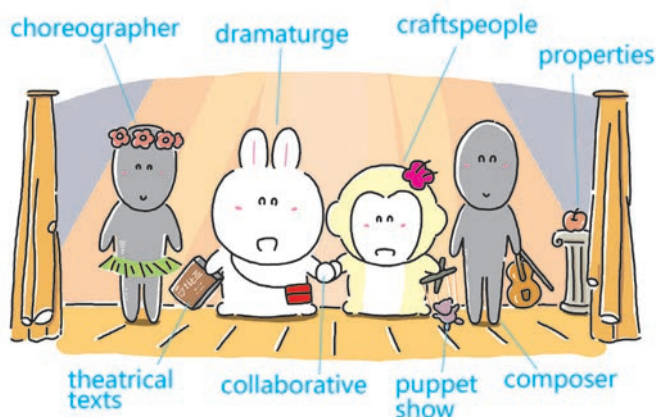


注释

anthropologist [ˌænθrəˈpɒlədʒɪst] n. 人类学家
 theater [ˈθiətə] n. 戏剧
 trace [treɪs] v. 追踪, 探索
 myth [miθ] n. 神话
 ritual [ˈrɪtʃuəl] n. (宗教) 仪式
 mimed performance [maɪmd-peɪfɔ:məns]
 n. 笑剧、闹剧表演
 fertility rite [fəˈtɪlɪti-raɪt] n. 丰收仪式

ceremony [ˈserɪməni] n. 仪式
 passage [ˈpæsidʒ] n. 转变
 imitation [ɪˈmɪteɪʃən] n. 道具
 costume [ˈkɒstju:m] n. 装束
 makeup [ˈmeɪkʌp] n. 化妆品
 gesture [ˈdʒestʃə] n. 造型
 pantomime [ˈpæntəmaɪm] n. 哑剧表演

In addition to the actor and the audience in a space, other elements of theater include a written or improvised text, costumes, scenery, lights, sound, and properties (props). Most theatrical performances require the collaborative efforts of many creative people, including dramaturge, craftspeople, designer, composer, or choreographer and managers working toward a common goal: the production. The performance is very often a play — a tragedy, comedy, or musical — but it need not be. Theater performances include vaudeville, puppet shows, mime, and other forms of entertainment. Theatrical texts, often referred to as drama, usually provide the vital framework of a performance. In Aristotle's famous definition, drama is an imitation of an action that is whole, complete, and of a certain magnitude or scope.

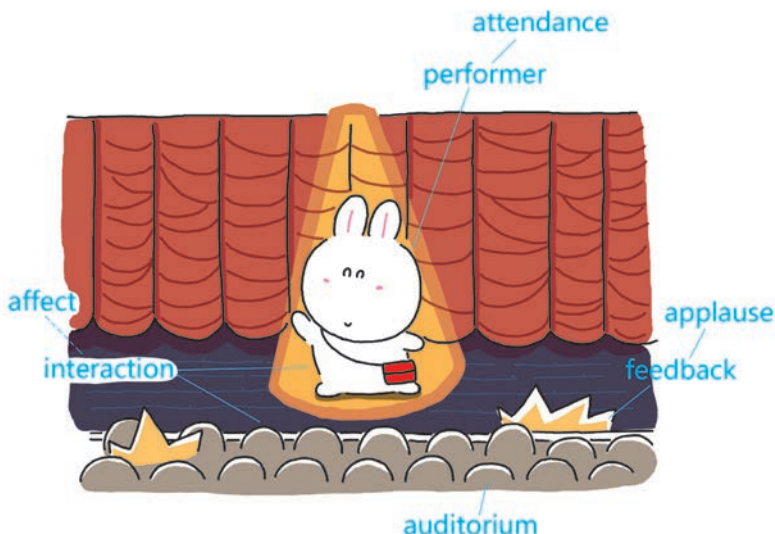


注释

audience ['ɔ: djəns] n. 观众
 improvised text ['ɪmpraɪvaɪzd-tekst] n. 同步字幕
 scenery ['si: nəri] n. 舞台布景
 property ['prɒpəti] n. 道具
 props [prɒps] n. 小道具
 collaborative [kə'ləbəreɪtɪv] adj. 协作的
 dramaturge ['dræmə,tə:dʒ] n. 剧作家
 craftspeople [kra:ftspi:pl] n. 道具师
 composer [kəm'pəuzə] n. 作曲家
 choreographer [kə'ri:ɔgrəfe] n. 舞蹈指导
 production [prə'dʌkʃən] n. 作品, 成品, 电影制片

tragedy ['trædzɪdi] n. 悲剧
 comedy ['kɒmɪdi] n. 喜剧
 vaudeville ['vɔ:deɪvɪl] n. 歌舞杂耍
 puppet show ['pʌpɪt-ʃəʊ] n. 木偶剧
 mime [maɪm] n. 笑剧
 theatrical text [θi'ætrɪkəl-tekst] n. 戏剧文本
 drama ['drɑ:mə] n. 戏剧
 framework ['freɪmwɜ:k] n. 结构
 magnitude ['mæɡnɪtju:d] n. 量值
 scope [skəʊp] n. 范围

Theater can serve many ends. It can be designed to entertain, instruct, motivate, persuade, and even shock. But whatever the intentions of the director, performers, and crew, the result depends on the interaction with an audience. The audience affects the performance by providing the performers with immediate feedback, such as laughter, tears, applause, or silence. Each night there is continuous interaction between the auditorium and the stage. Ultimately, audiences make their opinions known through their attendance or nonattendance. They support what appeals to them and generally fail to support what they find distasteful, offensive, or incomprehensible.



注释

serve [sə:v] v. 满足, 服务

end [end] n. 目的

entertain [ente'tein] v. 娱乐

instruct [in'strakt] v. 教导

motivate ['məuteiv] v. 激发

persuade [pe'sweid] v. 说服

shock [ʃɒk] v. 震撼

intention [in'tenʃən] n. 意图, 目的

director [di'rekte] n. 导演

performer [pe'fɔ:mə] n. 表演者

crew [kru:] n. 全体人员

interaction [intə'rækʃən] n. 相互作用

affect [ə'fekt] v. 影响

immediate [i'mi:diət] adj. 立即的

feedback ['fi:dbæk] n. 反馈

applause [ə'plo:z] n. 喝彩

auditorium [ɔ:'di'tɔ:riəm] n. 观众席

attendance [ə'tendəns] n. 出席

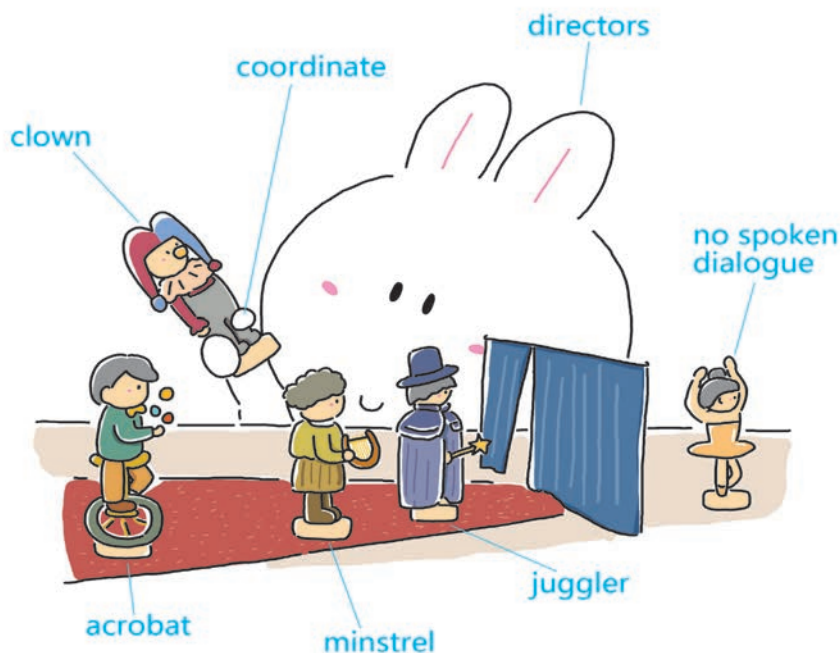
appeal [ə'pi:l] v. 对……有吸引力

distasteful [dis'teistful] adj. (令人) 不愉快的

offensive [ə'fensiv] adj. 冒犯的

incomprehensible [in,kompri'hensəbl] adj. 不能理解的

Directors assume responsibility for the overall interpretation of a script, and they have the authority to approve, control, and coordinate all the elements of a production. Throughout the ages, performers have been jugglers, mimes, minstrels, puppeteers, acrobats, clowns, singers, dancers, and amateur and professional actors/actresses. The first performers were most likely singers and dancers, as the first performances had no spoken dialogue.



注释

- assume [ə'sju:m] v. 承担
 interpretation [in'tɜ:pri'teɪʃən] n. 演绎
 script [skript] n. 剧本
 coordinate [kəu'ɔ:dineɪt] v. 调整
 juggler ['dʒʌglə] n. 玩杂耍的人
 mime [maɪm] n. 滑稽演员, 哑剧演员, 丑角
 minstrel ['mɪnstrel] n. 诗歌演唱者
 puppeteer [pʌpɪ'tiə] n. 演木偶戏的人
 acrobat ['ækɹəbæt] n. 杂技演员
 clown [klaʊn] n. 小丑
 amateur ['æmətə] n. 业余艺术家
 dialogue ['daɪələg] n. 对白

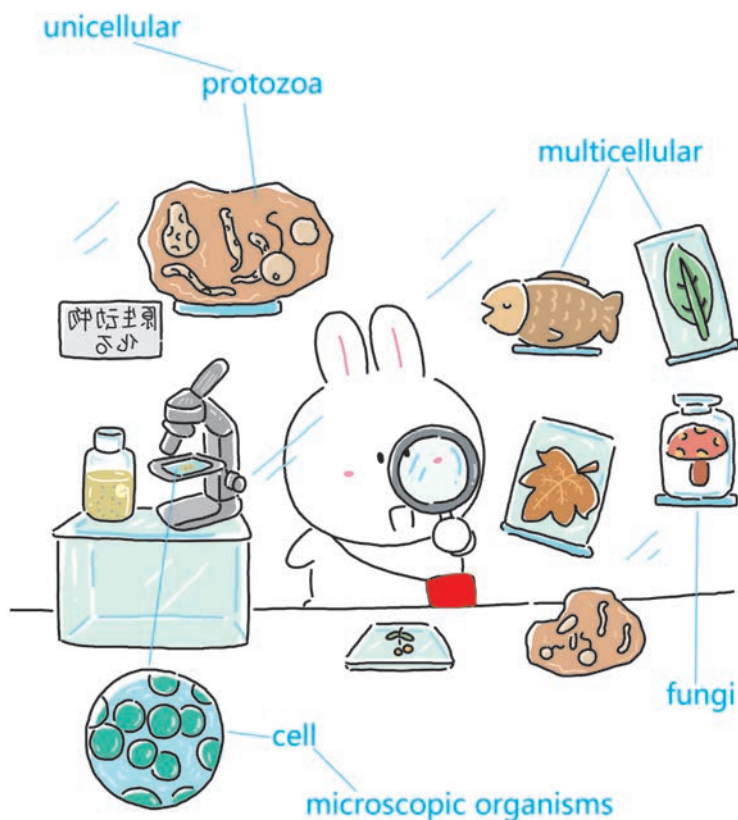
In commercial and nonprofit theaters, the producer is the person who puts together the financing, management staff, and the artistic team to produce the show. Usually, the producer works in tandem with a general manager and others to accomplish the daily running of the production, from rehearsals to closing. In this role, he or she selects a season of several plays, hires the artistic teams and technical staff, works with a casting director to audition and cast actors in various parts, controls the theater's funding, and acts as the final authority in all artistic and administrative operations.



注释

- commercial [kə'mɜːʃəl] adj. 商业的
 nonprofit [ˌnɒn'prɒfɪt] adj. 非赢利的
 producer [prə'djuːsə] n. 制片人
 put together ['put-tə'geðə] v. 组合
 staff [stɑːf] n. 全体职员
 in tandem with [in-'tændəm-wɪð] adv. 同……合作
 accomplish [ə'kɒmplɪʃ] v. 完成
 rehearsal [rɪ'hæ:səl] n. 彩排
 season ['siːzən] n. 适合某种活动的季节, 期间
 casting director ['kɑːstɪŋ-di'rektə] n. 负责挑选演员的人, 星探
 audition [ɔː'diʃən] v. 在试演中评估某一人物
 cast [kɑːst] v. 指定角色
 administrative [əd'mɪnɪstreɪtɪv] adj. 管理的

All living things are composed of cells. Some microscopic organisms, such as bacteria and protozoa, are unicellular, meaning they consist of a single cell. Plants, animals, and fungi are multicellular; that is, they are composed of a great many cells working in concert.



注释

cell [sel] n. 细胞

microscopic [ˌmaɪkrəˈskɒpɪk] adj. 极微小的

organism [ˈɔːɡənɪzəm] n. 有机生物

protozoan [ˌprəʊtəʊˈzəʊən] n. 原生动物 (plural: protozoa [ˌprəʊtəʊˈzəʊə])

unicellular [ˈjuːnɪˈseljʊlə] adj. 单细胞的

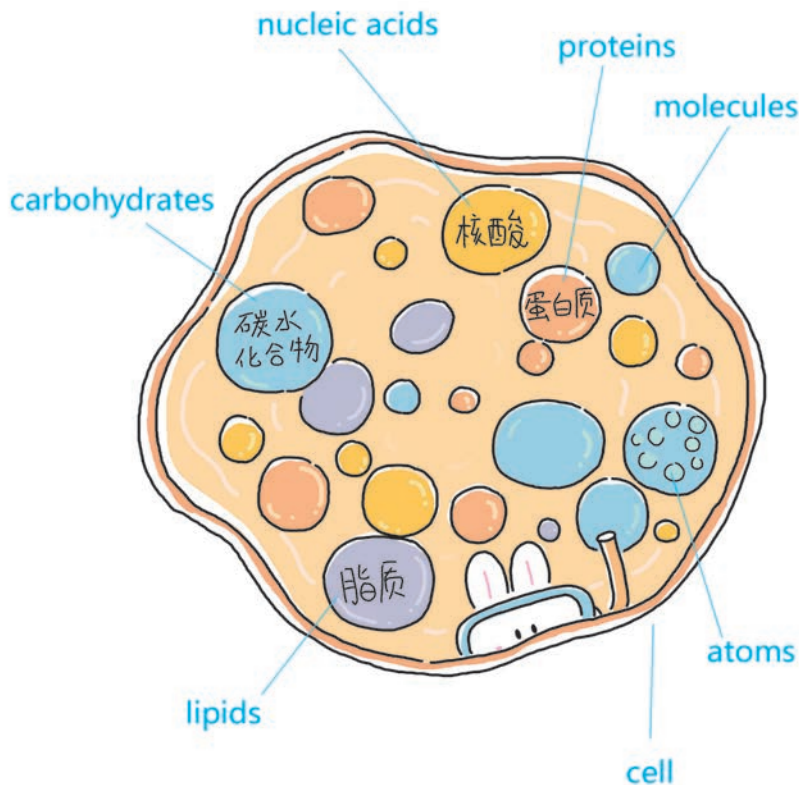
fungi [ˈfʌndʒaɪ, ˈfʌŋgaɪ] n. 真菌类

multicellular [ˌmʌltiˈseljʊlə] adj. 多细胞的

in concert [ɪn-ˈkɒnsət] adv. 同时

UNIT051

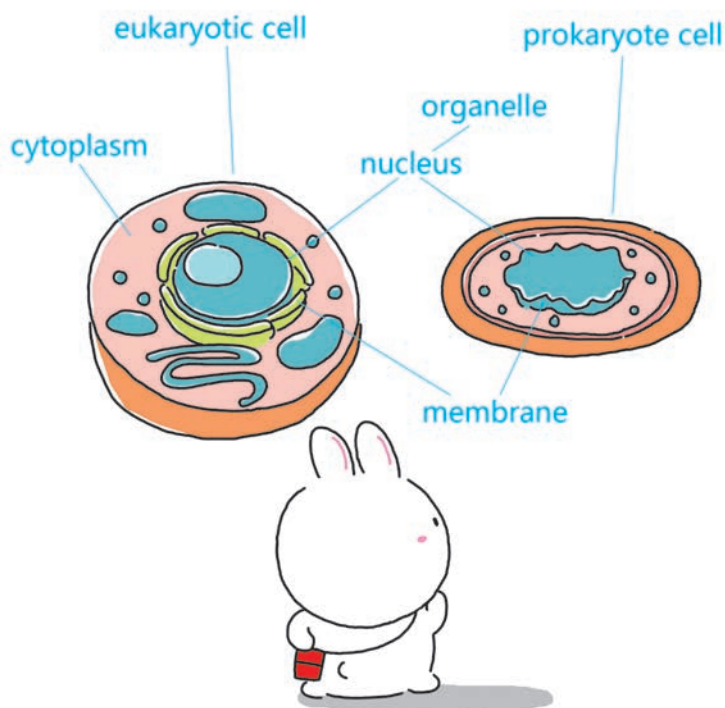
The components of cells are molecules, nonliving structures formed by the union of atoms. Small molecules serve as building blocks for larger molecules. Proteins, nucleic acids, carbohydrates, and lipids, which include fats and oils, are the four major molecules that underlie cell structure and also participate in cell functions.



注释

- molecule ['mɒlɪkjʊ:l] n. 分子
 union ['ju:njən] n. 结合
 atom ['ætəm] n. 原子
 building block ['bɪldɪŋ-'bɒk] n. 结构单元
 protein ['prəʊtɪn] n. 蛋白质
 nucleic acid [nju:'kli:k-'æsid] n. 核酸
 carbohydrate [kɑ:bəu'haidreit] n. 碳水化合物
 lipid ['lɪpɪd,'laɪpɪd] n. 脂质
 underlie [ˌʌndə'laɪ] v. 成为……的基础
 participate [pɑ:'tɪsɪpeɪt] v. 参与

The eukaryotic cell cytoplasm is similar to that of the prokaryote cell except for one major difference: eukaryotic cells house a nucleus and numerous other membrane-enclosed organelles. Like separate rooms of a house, these organelles enable specialized functions to be carried out efficiently. The building of proteins and lipids, for example, takes place in separate organelles where specialized enzymes geared for each job are located.

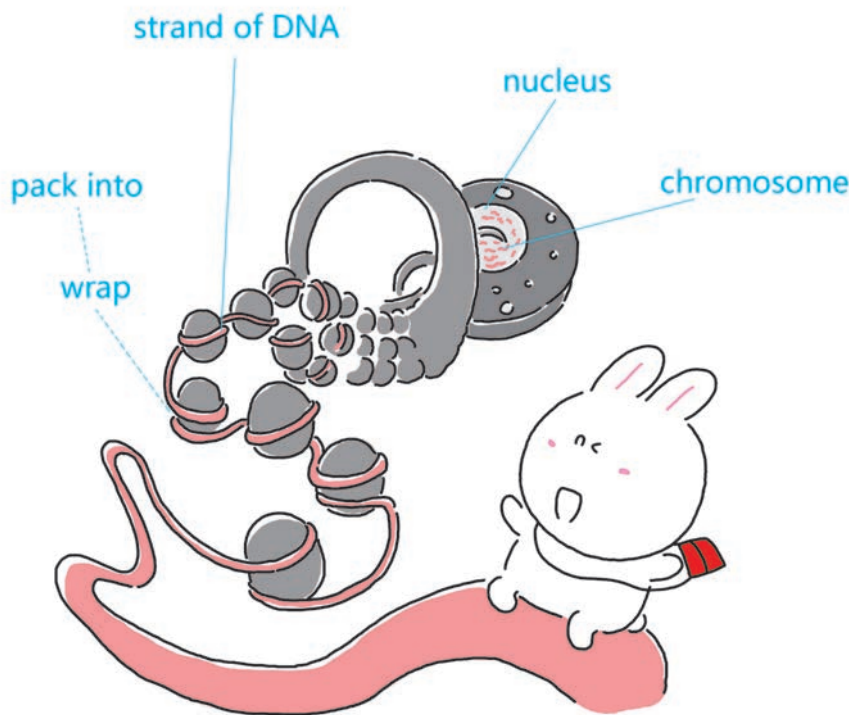


注释

- eukaryotic cell [ju'kæriəutik - sel] n. 真核细胞
 cytoplasm ['saɪtəplæzəm] n. 细胞质
 prokaryote cell [prəu'kæriəut - sel] n. 原核生物细胞
 house [haus] v. 包含
 nucleus ['nju:klɪəs] n. 细胞核
 membrane ['membrein] n. 膜
 organelle [ɔ:gə'nel] n. 细胞器
 enzyme ['enzaim] n. 酶
 gear [giə] v. 使适合于

UNIT053

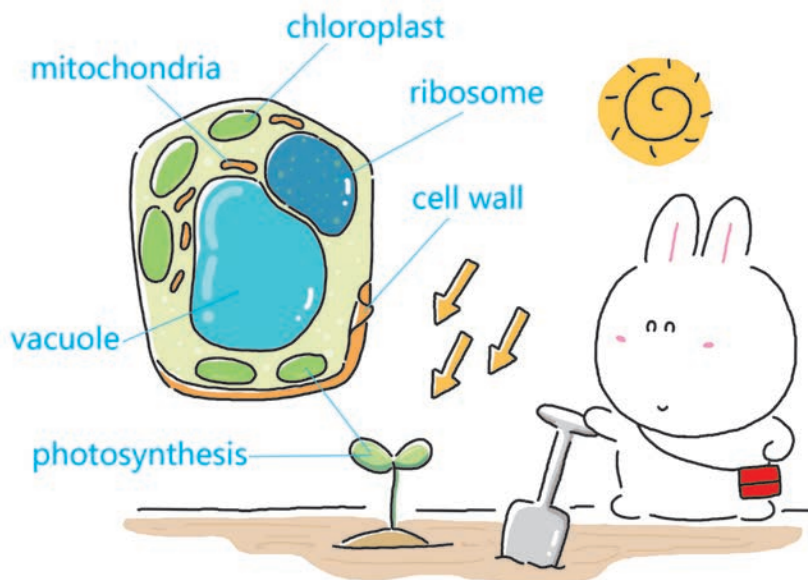
The nucleus is the largest organelle in an animal cell. It contains numerous strands of DNA, the length of each strand being many times the diameter of the cell. Unlike the circular prokaryotic DNA, long sections of eukaryotic DNA pack into the nucleus by wrapping around proteins. As a cell begins to divide, each DNA strand folds over onto itself several times, forming a rod-shaped chromosome.



注释

strand [strænd] n. 串, (绳子的) 股
 diameter [daɪ'æmɪtə] n. 直径
 pack into ['pæk-'intu] v. 将……装(塞, 挤)入
 wrap [ræp] v. 缠绕
 chromosome ['kræməsəʊm] n. 染色体

Plant cells have all the components of animal cells and several other added features, including chloroplasts, a central vacuole, and a cell wall. Chloroplasts convert light energy — typically from the Sun — into the sugar glucose, a form of chemical energy, in a process known as photosynthesis. Chloroplasts, like mitochondria, possess a circular chromosome and prokaryote like ribosomes, which manufacture the proteins that the chloroplasts typically need. The vacuole, a membranous bag, crowds the cytoplasm and organelles to the edges of the cell. The central vacuole stores water, salts, sugars, proteins, and other nutrients. In addition, it stores the blue, red, and purple pigments that give certain flowers their colors.



注释

component [kəm'pəʊnənt] n. 组成部分

chloroplast ['klɔːrəʊplæst] n. 叶绿体

vacuole ['vækjuəl] n. 液泡

cell wall [sel-wɔːl] n. 细胞壁

glucose ['gluːkəʊs] n. 葡萄糖

photosynthesis [ˌfəʊtəʊ'sɪnθəsis] n. 光合作用

mitochondria [ˌmaɪtə'kɒndrɪə] n. 线粒体

ribosome ['raɪbəseʊm] n. 核糖体

crowd ... to ... [kraʊd-tuː] v. 把……挤进……；把……塞进……

pigment ['pɪgmənt] n. 色素

Forests may be divided into the following eight general types on the basis of leaf characteristics and climate. (1) Deciduous forests of the temperate regions are the typical formation of the eastern United States. (2) Deciduous monsoon forests are characteristic of Bengal and Myanmar (formerly known as Burma) and common throughout Southeast Asia and India. (3) Tropical savanna forests are found in regions such as the campos of Brazil, where forest and grassland meet. (4) Northern coniferous forests form a worldwide belt in subarctic and alpine regions of the northern hemisphere. (5) Tropical rain forests are characteristic of central Africa and the Amazon watershed. (6) Temperate evergreen forests are found in the subtropical regions of North America and the Caribbean islands that have a warm maritime climate. (7) Temperate rain forests, with broad-leaved evergreen trees, are common on Mediterranean coasts. (8) Tropical scrub forests occur in regions of slight rainfall, bordering wetter forests.



注释

forest ['fɒrɪst] n. 森林

deciduous forest [di'sɪdʒuəs-'fɒrɪst] n. 落叶林

Deciduous monsoon forest [di'sɪdʒuəs-mən'su:n-'fɒrɪst]
n. 落叶季雨林

Bengal [ben'gɔ:l] n. 孟加拉

Myanmar ['mjænma:] n. 缅甸

Tropical savanna forest ['trɒpɪkl-sə'vænə-'fɒrɪst]
n. 热带稀树草原

campo ['kæmpəu] n. 南美草原

grassland ['grɑ:slænd] n. 草原

Northern coniferous forest ['nɔ:ð(ə)n-kəu'nɪfərəs-'fɒrɪst]
n. 北方针叶树林

worldwide belt ['wɜ:lɪdwaɪd-belt] n. 世界范围分布

subarctic ['sʌb'ɑ:ktɪk] adj. 亚寒带

alpine region ['ælpain-'rɪ:dʒən] n. 高山区

hemisphere ['hemi,sfiə] n. 半球

tropical rain forest ['trɒpɪkl-rein-'fɒrɪst] n. 热带雨林

watershed ['wɔ:təʃed] n. 流域

Temperate evergreen forest ['tempərɪt-evəɡri:n-'fɒrɪst]
n. 温带绿树林

subtropical ['sʌb'trɒpɪkəl] adj. 亚热带的

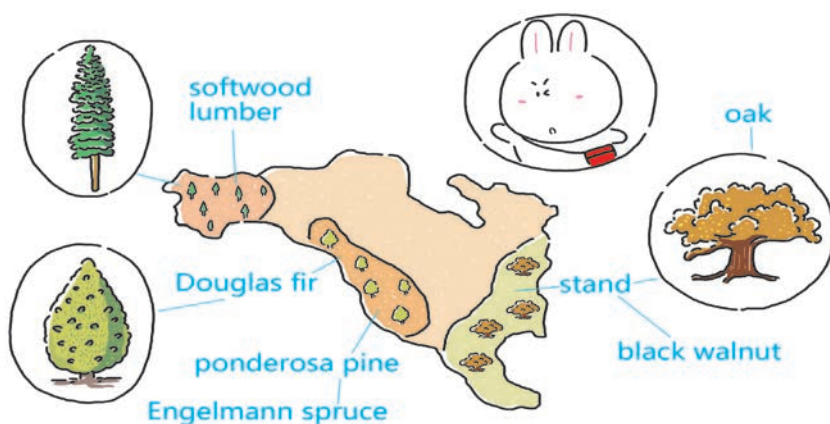
maritime ['mæritaim] adj. 海洋性的

Temperate rain forest ['tempərɪt-rein-'fɒrɪst]
n. 温带雨林

coast [kəʊst] n. 海岸

Tropical scrub forest ['trɒpɪkl-skrʌb-'fɒrɪst]
n. 热带灌丛林

Three major forest areas exist in the United States. The western forests of the Rocky Mountains and the Pacific coast are coniferous and contain Douglas fir, ponderosa pine, western white pine, Engelmann spruce, and white fir. More than half of the softwood lumber yield of the United States comes from the productive Douglas fir forests of the Pacific Northwest. The South Atlantic and Gulf states account for most of the remaining softwood lumber, chiefly from longleaf, shortleaf, loblolly, and slash pines. Hardwoods, yielding about one-fourth of the total production, are found in the eastern half of the United States, with particularly dense stands in the area surrounding the Mississippi and Ohio river valleys. Among the many hardwood species are oaks, black walnut, yellow poplar, and sugar maple.



注释

fir [fɜ:] n. 枞树, 冷杉

pine [paɪn] n. 松树

spruce [spru:s] n. 云杉

softwood ['sɒftwʊd] n. 针叶树

lumber ['lʌmbə] n. 木材

yield [ji:ld] n. 产量

account for [ə'kaunt-fɔ:] v. 占据

loblolly pine ['lɒblɒli-paɪn] n. 火炬松

slash pine [slæʃ-paɪn] n. 沼泽松

hardwood ['hɑ:dwʊd] n. 阔叶树

stand [stænd] n. 高大树丛

oak [əʊk] n. 橡树

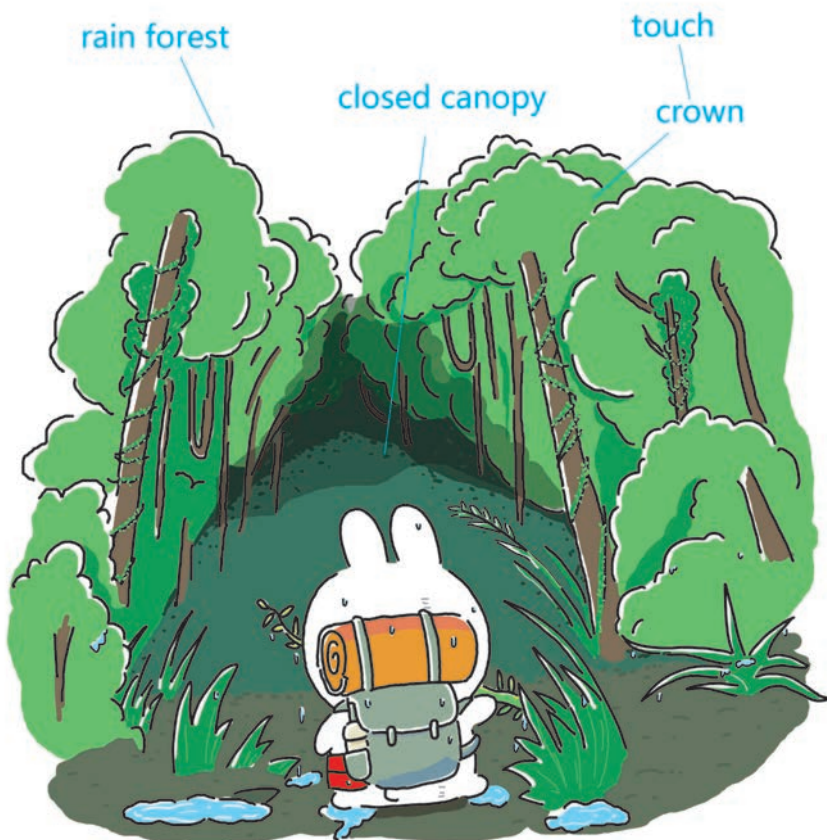
black walnut ['blæk-'wɔ:lənɪt] n. 黑胡桃

yellow poplar ['jeləu-'pɒplə]

n. 北美鹅掌楸 (poplar, 相当于中国人说的“杨树”)

sugar maple ['ʃʊgə-'meɪpl] n. 糖枫

To be classified as a rain forest, a forest must have a closed canopy, in which the treetops, or crowns, touch each other, creating a shaded forest interior. In addition, temperature and rainfall must be high and relatively even throughout the year. Forests that meet these criteria are found flanking the equator in South and Central America, Asia, Africa, and Australia.



注释

rain forest ['rein - 'fɒrɛst] n. 雨林

canopy ['kænəpi] n. 天篷: 树林中最上面一层

crown [kraun] n. 树冠

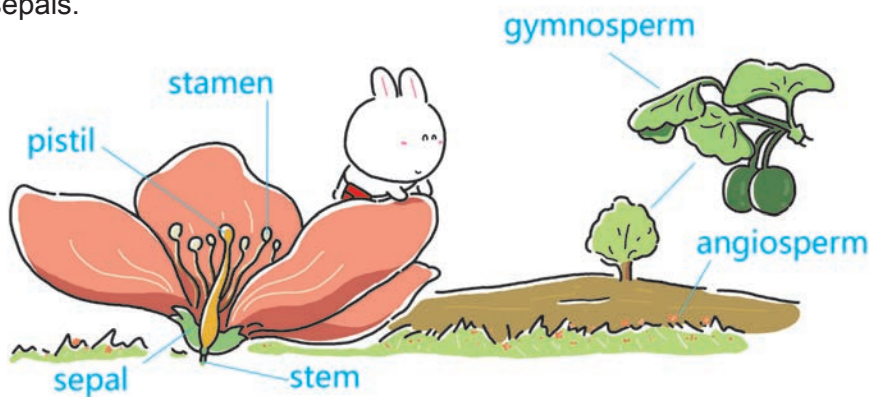
touch [tʌtʃ] v. 接触

criteria [krai'tiəriə] n. 标准

flank [flæŋk] v. 位于……的两侧

equator [i'kweɪtə] n. 赤道

Flower is the reproductive organ of most seed-bearing plants. Flowers carry out the multiple roles of sexual reproduction, seed development, and fruit production. Many plants — produce highly visible flowers that have a distinctive size, color, or fragrance. Almost everyone is familiar with beautiful flowers such as the blossoms of roses, orchids, and tulips. But many plants — including oaks', beeches, maples, and grasses — have small, green or gray flowers that typically go unnoticed. Whether eye-catching or inconspicuous, all flowers produce the male or female sex cells required for sexual reproduction. Botanists call them the cones, bearing plants gymnosperms, which means naked seeds. they refer to flowering plants as angiosperms, which means enclosed seeds. Flowers typically are composed of four parts, arranged in concentric rings attached to the tip of the stem. From innermost to outermost, these whorls are the pistil, stamens, petals, and sepals.



注释

reproductive organ ['ri:prə'daktiv-'ɔ:gen]

n. 生殖器官

fragrance ['freigrəns] n. 香气

blossom ['blɒsm] n. 花簇

orchid ['ɔ:kɪd] n. 兰花

tulip ['tju:lɪp] n. 郁金香

beech [bi:tʃ] n. 山毛榉

eye-catching ['ai-'kætfɪŋ] adj. 引人注目的

inconspicuous [ɪnkən'spɪkjʊəs] adj. 不显眼的

botanist ['bɒtənɪst] n. 植物学家

cone [kəʊn] n. 球果

gymnosperm ['dʒɪmnəʊspə:m] n. 裸子植物

refer to ['rɪfə:-tu:] v. 指的是……

angiosperm ['ændʒɪəspə:m] n. 被子植物

be composed of [bi:-kəm'pəʊzd-əv]

n. 由……组成

concentric [kən'sentrik] adj. 同一中心的

stem [stem] n. 茎

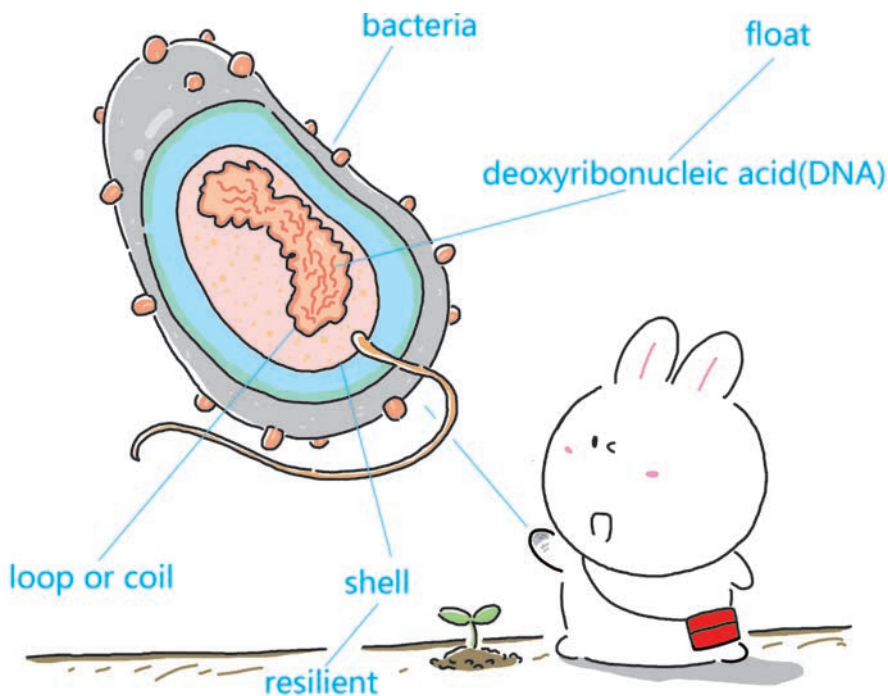
pistil ['pɪstɪl] n. 雌蕊

stamen ['steɪmen] n. 雄蕊

petal ['petl] n. 花瓣

sepal ['sepəl] n. 萼片

Bacteria lack a true nucleus, a feature that distinguishes them from plant and animal cells. In plants and animals the saclike nucleus carries genetic material in the form of deoxyribonucleic acid (DNA). Bacteria also have DNA but it floats within the cell, usually in a loop or coil. A tough but resilient protective shell surrounds the bacterial cell.



注释

bacteria [bæk'tiəriə] (pl.) n. 细菌 (single: bacterium [bæk'tiəriəm])

lack [læk] v. 缺乏

feature ['fi:tʃə] n. 特征

distinguish [dis'tɪŋɡwɪʃ] v. 区别

saclike ['sæklaɪk] adj. 囊状的

genetic material [dʒi'netɪk-mə'tiəriəl] n. 遗传物质

deoxyribonucleic acid [di:'ɒksi:raɪbəu'nju:kli:ik-'æsid] n. 脱氧核糖核酸

float [fləʊt] v. 漂浮

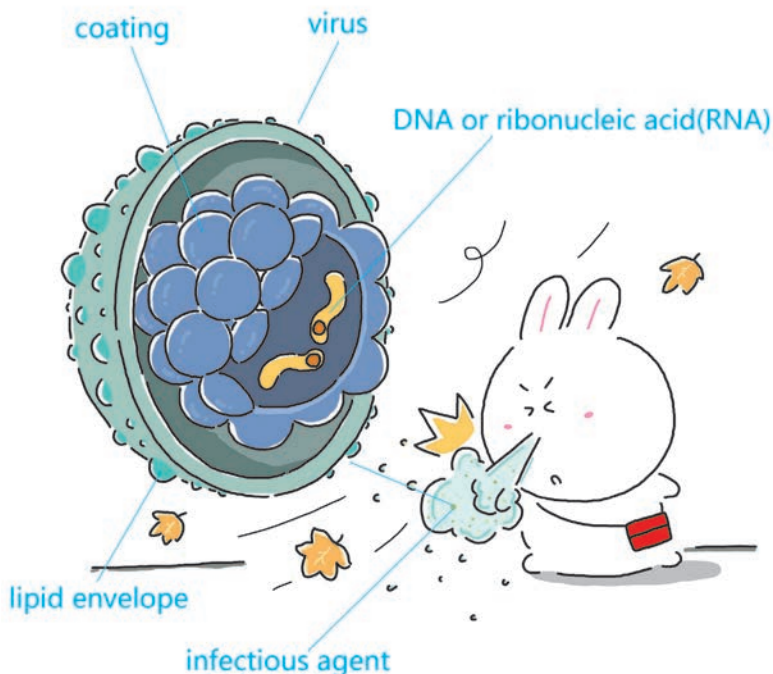
loop [lu:p] n. 环

coil [kɔɪl] n. (一)卷

resilient [ri'zɪliənt] adj. 有弹性的

shell [ʃel] n. 外壳

Virus, infectious agent found in virtually all life forms, including humans, animals, plants, fungi, and bacteria, consist of genetic material — either deoxyribonucleic acid (DNA) or ribonucleic acid (RNA) — surrounded by a protective coating of protein, called a capsid, with or without an outer lipid envelope.



注释

virus ['vaɪərəs] n. 病毒

infectious agent [ɪn'fektʃəs-'eɪdʒənt] n. 传染因子

virtually ['vɜ:tʃʊəli] adv. 实际上

fungi ['fʌndʒaɪ] n. 真菌类 (singular: fungus ['fʌŋɡəs])

consist of [kən'sɪst-əv] v. 由……组成

ribonucleic acid [ˌraɪbənju:'kli:k-'æsɪd] n. 核糖核酸(略作RNA)

coating ['kəʊtɪŋ] n. 外壳

capsid ['kæpsɪd] n. 衣壳 (包裹着一个病毒粒子的蛋白质外壳)

lipid ['lɪpɪd] n. 脂质

envelope ['envəleʊp] n. 包裹性的结构或包皮, 例如膜, 或者病毒的包膜

The most important classes of biomolecules are nucleic acids, proteins, carbohydrates, and lipids.

Nucleic acids are responsible for storing and transferring genetic information. They are enormous molecules made up of long strands of subunits, called bases which are arranged in a precise sequence. These are "read" by other components of the cell and used as a guide in making proteins.

Proteins are large molecules built up of small subunits called amino acids. The proteins of greatest interest to biochemists are the enzymes, which serve as catalysts of chemical reactions.

Carbohydrates are the basic fuel molecules of the cell. They contain carbon, hydrogen, and oxygen in approximately equal amounts. Green plants and some bacteria use a process known as photosynthesis to make simple carbohydrates (sugars) from carbon dioxide, water, and sunlight. Animals, however, obtain their carbohydrates from foods. Once a cell possesses carbohydrates, it may break them down to yield chemical energy or use them as raw material to produce other biomolecules.

Lipids are fatty substances that play a variety of roles in the cell. Some are held in storage for use as high-energy fuel; others serve as essential components of the cell membrane.

注释

class [kla:s] n. 类

biomolecule [baieu'molikju:l] n. 生物分子

nucleic acid [nju:'kli:ik-'æsid] n. 核酸

protein ['prəuti:n] n. 蛋白质

carbohydrate [kɑ:bəu'haidreit] n. 碳水化合物

lipid ['lipid,'laipid] n. 脂质, 油脂

be responsible for [bi:'ri'sponsəbl-fɔ:] n. 为……负责

enormous [i'nɔ:məs] adj. 巨大的, 庞大的

strand [strænd] n. 线、束、串

subunit [sʌb'ju:nit] n. 子(分,亚)组

base [beis] n. 碱基: 嘌呤(腺嘌呤和鸟嘌呤)或嘧啶(胸嘧啶、胸腺嘧啶和尿嘧啶)的一种

precise [pri'sais] adj. 精确的

sequence ['si:kwəns] n. 顺序, 序列

amino acid [æminəu-'æsid] n. 氨基酸

biochemist [baieu'kemist] n. 生物化学家

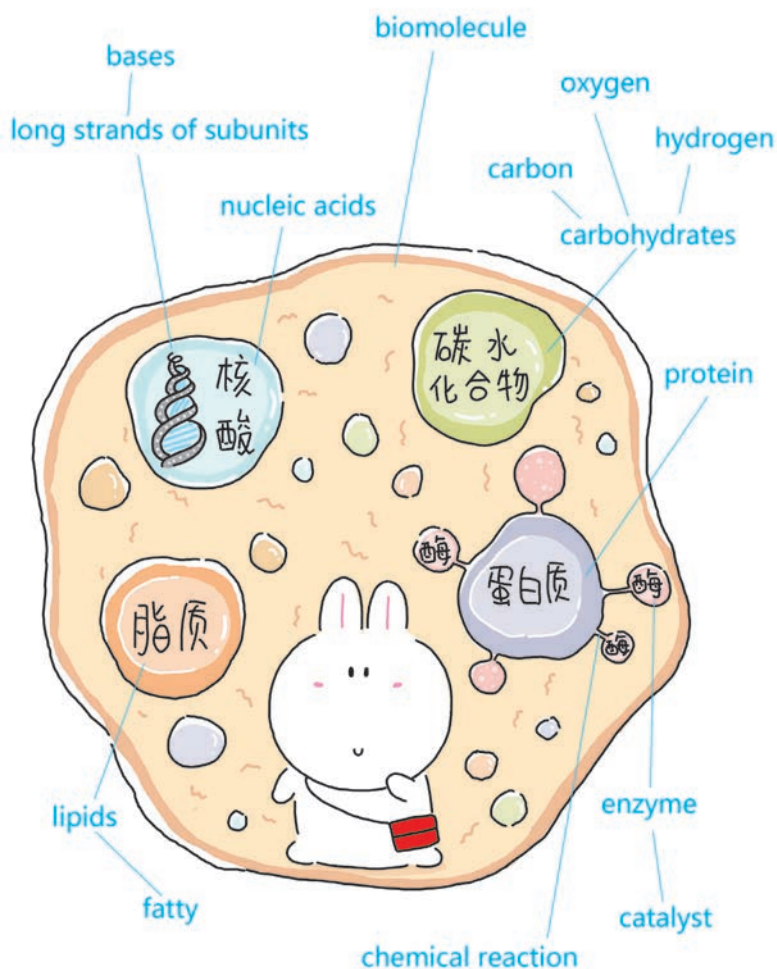
enzyme ['enzaim] n. 酶

catalyst ['kætəlist] n. 催化剂

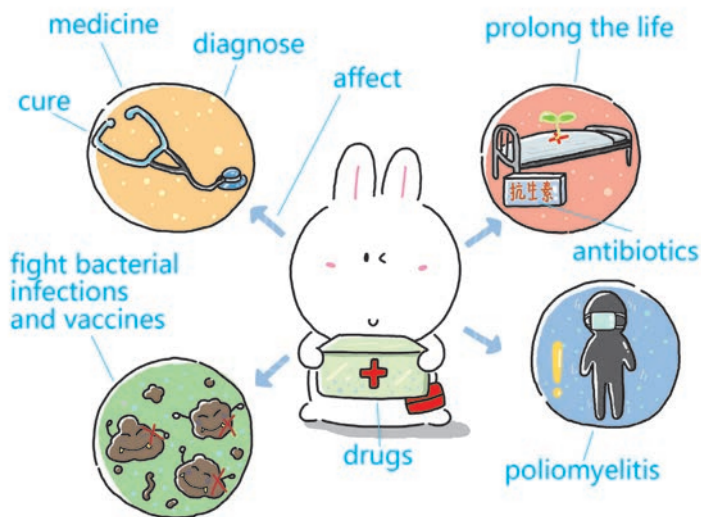
chemical reaction ['kemikəl-ri'ækʃən] n. 化学反应

fuel [fjuəl] n. 燃料, 能量

carbon ['kɑ:bən] n. 碳
 hydrogen ['haɪdrədʒən] n. 氢
 oxygen ['ɒksɪdʒən] n. 氧
 approximately [ə'prɒksɪmɪtli] adv. 大约
 carbon dioxide ['kɑ:bən-daɪ'ɒksaɪd] n. 二氧化碳
 obtain [əb'teɪn] v. 获取
 cell membrane [sel-'mem,breɪn] n. 细胞膜
 yield [ji:ld] v. 产出, 生出
 fatty ['fæti] adj. 含脂肪的, 脂肪状的
 a variety of [ə-və'raɪəti-əv] adj. 多种多样的
 break down ['breɪk-daʊn] v. 分解



The availability of new and more effective drugs — substances that affect the function of living cells used in medicine to diagnose, cure, prevent the occurrence of diseases and disorders, and prolong the life of patients with incurable conditions, such as antibiotics, which fight bacterial infections, and vaccines which prevent diseases caused by bacteria and viruses, helped increase the average American's life span from about 60 years in 1900 to about 78 years in 2005. During the 20th century, drugs enabled the eradication of smallpox, once a widespread and often fatal disease. By the early 21st century, vaccines had led to the near eradication of poliomyelitis, once feared as a cause of paralysis.

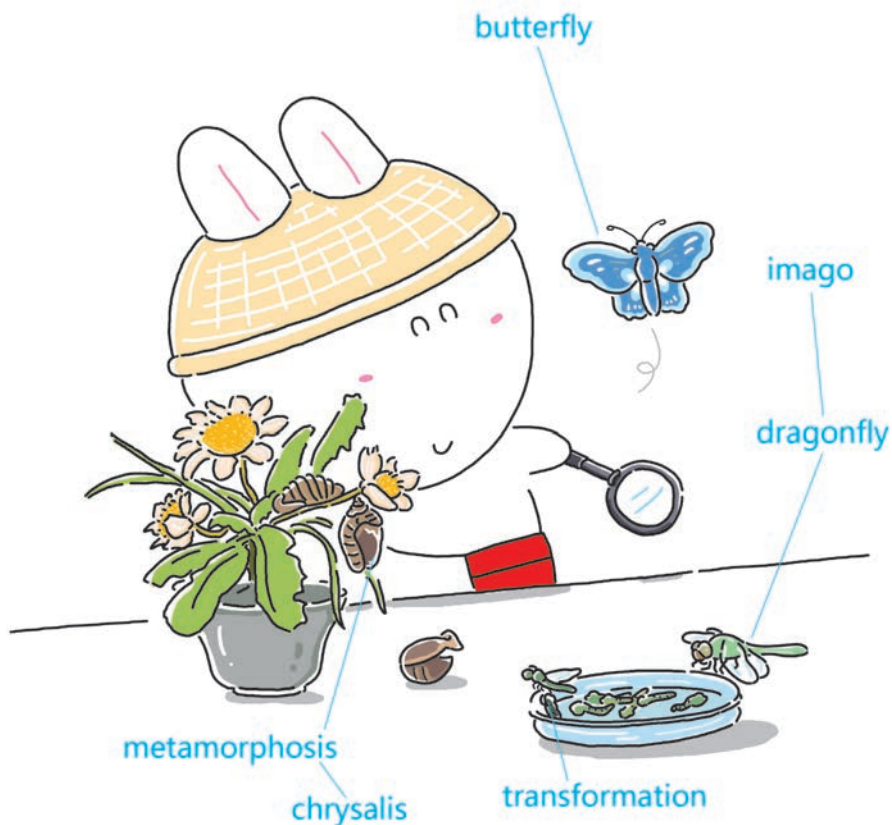


注释

drug [drʌg] n. 药物
 affect [ə'fekt] v. 影响
 medicine ['medisin] n. 医学, 药
 diagnose ['daɪəgnəʊz] v. 诊断
 cure [kjʊə] v. 治疗
 disorder [dis'ɔ:də] n. 失调, 紊乱
 prolong [prə'lon] v. 延长
 condition [kən'dɪʃən] n. 病痛
 antibiotics [ˌæntɪbaɪ'ɒtɪks] n. 抗生素, 抗生学
 infection [ɪn'fekʃən] n. 传染, 感染

vaccine ['væksɪn] n. 疫苗
 life span ['laɪf-spæn] n. 寿命
 eradication [ɪˌrædɪ'keɪʃən] n. 根除
 smallpox ['smɔ:l,pɒks] n. 天花
 widespread ['waɪdspreɪd] adj. 普遍的
 fatal ['feɪtəl] adj. 致命的
 poliomyelitis [ˌpəʊliəmaɪə'laitɪs] n. 小儿麻痹症, 急性骨髓灰白质炎
 paralysis [pə'relɪsɪs] n. 麻痹, 瘫痪

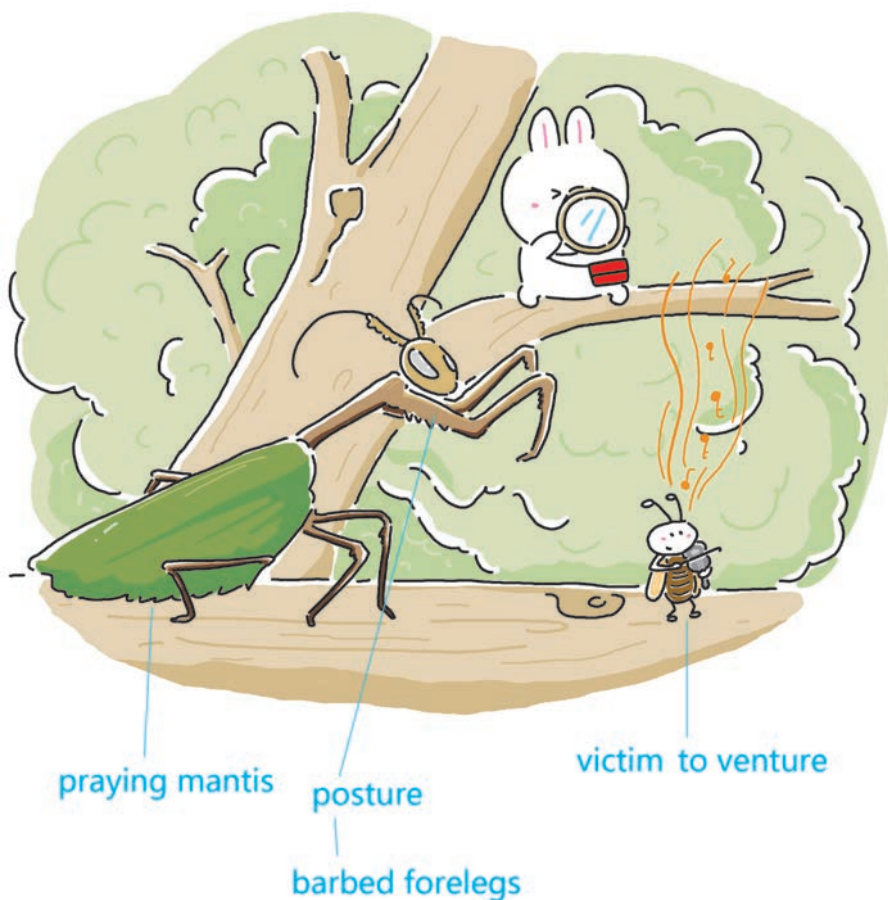
For sheer variety and abundance, insects rank among the most successful animals on Earth. Dragonflies, unlike butterflies, do not create a chrysalis or undergo complete metamorphosis. Instead, they undergo a series of small transformations until they finally develop into the imago — the adult, winged form.



注释

- sheer [ʃiə] adj. 绝对的
 abundance [əˈbʌndəns] n. 丰富
 butterfly ['bʌtəflai] n. 蝴蝶
 dragonfly ['dræɡənflai] n. 蜻蜓
 chrysalis ['krisəlis] n. 蛹
 metamorphosis [ˌmetə'mɔ:fəsis] n. 变态, 蜕变
 transformation [ˌtrænsfə'meɪʃən] n. 转变
 imago ['iːmeigəu] n. 成虫

The praying mantis is a carnivore that feeds on smaller insects, such as flies, crickets, and grasshoppers. Its name reflects the prayer-like posture it assumes, while waiting for its victims to venture within the reach of its long, barbed forelegs.

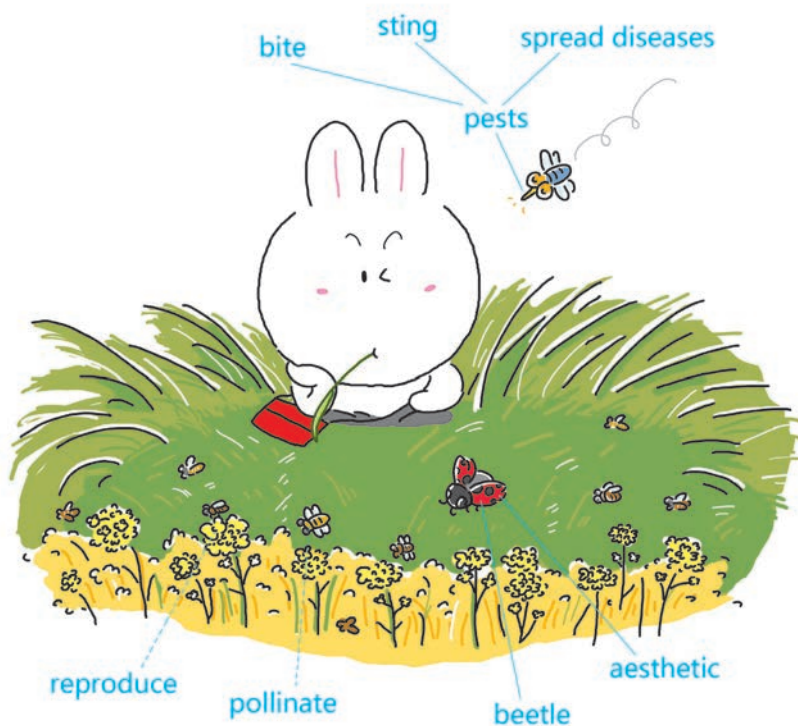


注释

praying mantis ['preiɪŋ-'mæntɪs] n. 螳螂
 carnivore ['kɑ:nɪvɔ:] n. 食肉动物
 fly [flaɪ] n. 苍蝇
 cricket ['krikit] n. 蟋蟀
 grasshopper ['grɑ:s,hɒpə] n. 蝗虫

posture ['pɒstʃə] n. 姿势
 assume [ə'sju:m] v. 采用
 victim ['vɪktɪm] n. 牺牲品
 venture ['ventʃə] v. 冒险
 barbed foreleg [bɑ:bd-'fɔ:leg] n. 有倒钩的前肢

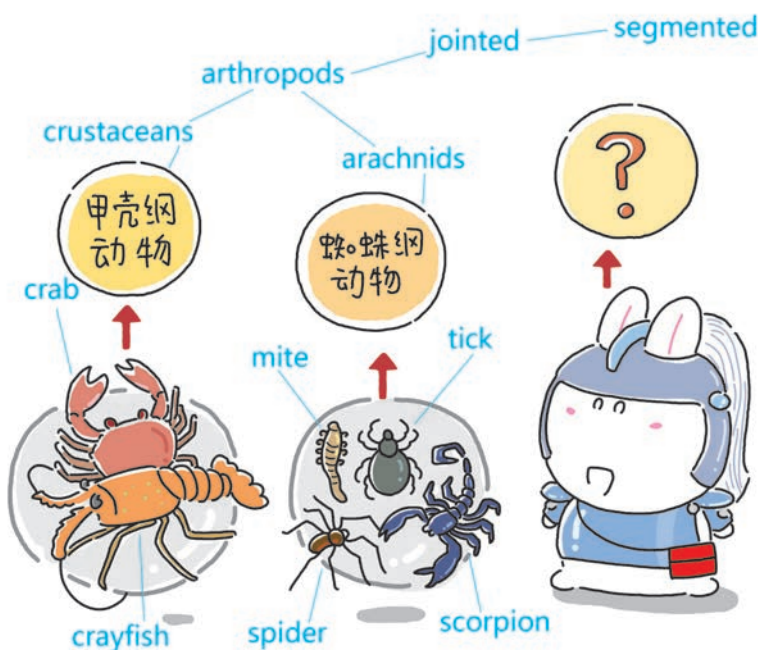
Insects are often regarded as pests because some bite, sting, spread diseases, or compete with humans for crop plants. Nevertheless, without insects to pollinate flowers, the human race would soon run out of food because many of the crop plants that we rely on would not be able to reproduce. In addition, insects are of aesthetic importance — some insects, such as dragonflies, beetles, and butterflies, are widely thought to be among the most beautiful of all animals.



注释

- pest [pest] n. 害虫
 bite [bait] v. 咬
 sting [stin] v. 螫
 spread [spred] v. 扩散, 传播
 disease [di'zi:z] n. 疾病
 pollinate ['polineit] v. 授粉, 传粉
 rely on [ri'lai-ɔn] v. 依赖
 reproduce [ri:prə'dju:s] v. 繁殖
 aesthetic [i:s'θetik] adj. 美学的
 beetle ['bi:tl] n. 甲虫

Insects are invertebrates, animals without backbones. They belong to a category of invertebrates called arthropods, which all have jointed legs, segmented bodies, and a hard outer covering called an exoskeleton. Two other well-known groups of arthropods are crustaceans, which include crayfish and crabs, and arachnids, which include spiders, ticks, mites, and scorpions. Many types of arthropods are commonly called bugs, but not every "bug" is an insect. Spiders, for example, are not insects, because they have eight legs and only two main body segments.

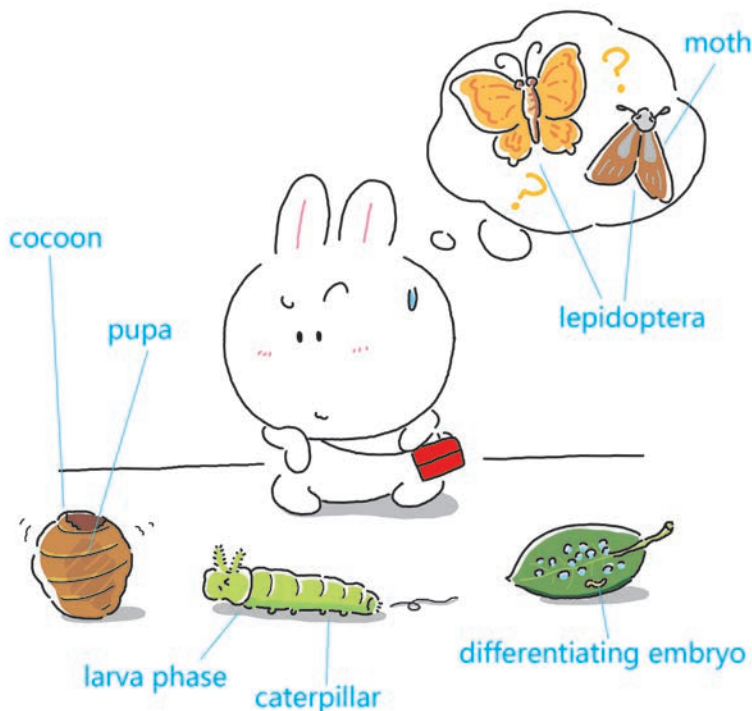


注释

invertebrate [in've:tibrət] n. 无脊椎动物
 arthropod ['ɑ:θrəpəd] n. 节肢动物
 jointed ['dʒɔɪntɪd] adj. 有关节的
 segmented [seg'mentɪd] adj. 分段的
 exoskeleton [eksəu'skelitən] n. 外骨骼
 crustacean [krʌ'steɪʃən] n. 甲壳纲动物
 crayfish ['kreɪfɪʃ] n. 龙虾

crab [kræb] n. 蟹
 arachnid [ə'ræknɪd] n. 蜘蛛纲动物
 spider ['spaɪdə] n. 蜘蛛
 tick [tɪk] n. (寄生于体大动物的吸血小虫) 壁虱
 mite [maɪt] n. 螨类
 scorpion ['skɔ:pɪən] n. 蝎子
 bug [bʌg] n. 虫子

Caterpillar, larval stage of butterflies and moths, members of the order Lepidoptera, is corresponding in this special order to the grub, maggot, or larva phase in the life history of other insects. The caterpillar develops like any other larva from the segmented egg and differentiating embryo, and undergoes several moltings or ecdyses. It later falls into a quiescent pupa stage, and the pupa is usually sheathed in a silken cocoon.

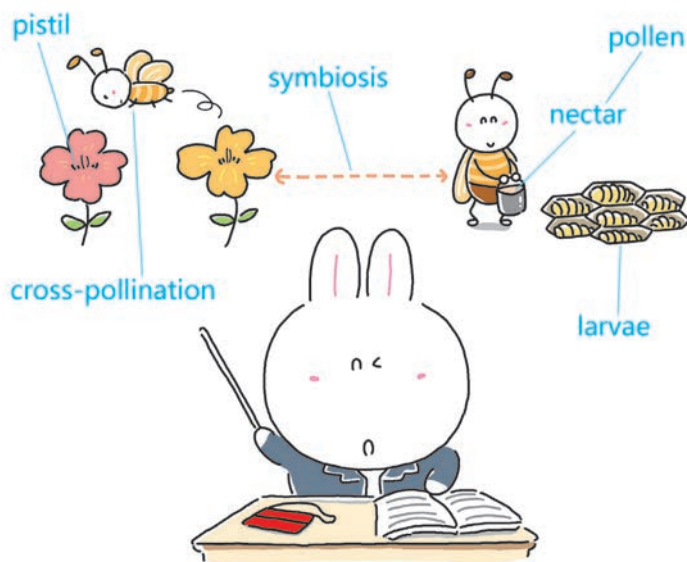


注释

caterpillar ['kætəpɪlə] n. 毛虫
moth [mɒθ] n. 蛾
larval ['lɑ:vəl] adj. 幼虫状态的
Lepidoptera [ˌlepə'dɒptərə] n. 鳞翅目
grub [grʌb] n. 幼虫
maggot ['mæɡət] n. 蛆
larva phase ['lɑ:və-feɪz] n. 幼虫阶段
segmented [seg'mentɪd] adj. 分段的
differentiating [ˌdɪfə'renʃieɪtɪŋ] adj. 分化

embryo ['embriəu] n. 胚胎
molting ['mœlɪŋ] n. 蜕皮
ecdysis ['ekdisɪs] n. 蜕皮
(plural: ecdyses ['ekdisi:z])
quiescent [kwɪ'esənt] adj. 休眠的
pupa ['pjʊ:pə] n. 蛹 (plural: pupae ['pjʊ:pi:])
sheath [ʃi:θ] v. 包裹
silken ['sɪlkən] adj. 柔软光滑的
cocoon [kə'ku:n] n. 茧

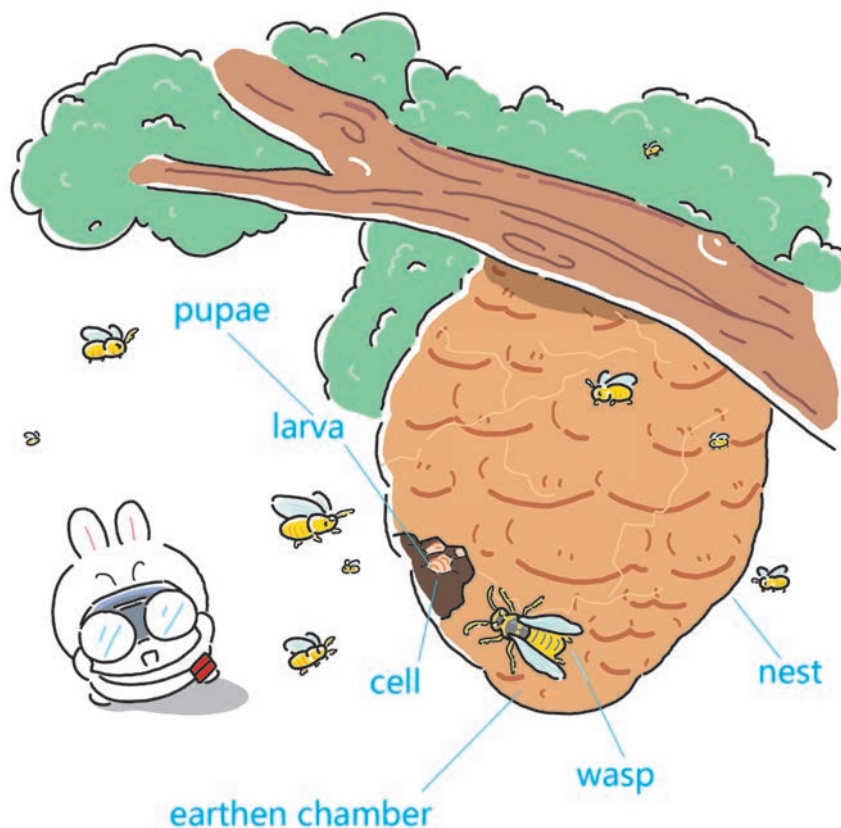
Bees are dependent on pollen as a protein source and on flower nectar or oils as an energy source. Adult females collect pollen primarily to feed their larvae. The pollen they inevitably lose in going from flower to flower is important to the plant because some of it lands on the pistils of other flowers of the same species, achieving cross-pollination. Bees are, in fact, the most important pollinating insects, and their interdependence with plants makes them an excellent example of the type of symbiosis known as mutualism, an association between unlike organisms that is beneficial to both parties.



注释

- pollen ['pɒlɪn] n. 花粉
 nectar ['nektə] n. 花蜜
 larva ['lɑ:və] n. (昆虫的) 幼虫 (plural: larvae ['lɑ:vi:])
 pistil ['pɪstɪl] n. 雌蕊
 cross-pollination [ˌkrɒs-pələ'neɪʃən, ˌkrɒs-] n. 异花传粉
 interdependence [ˌɪntə'dɪpendəns] n. 相互依赖
 symbiosis [ˌsɪmbaɪ'əʊsɪs] n. 共生 (现象)
 mutualism ['mju:tʃuəlɪzəm, 'mju:tʃuəlɪzəm] n. 共生
 beneficial [ˌbenɪ'fɪʃəl] adj. 有益的

The primitive bees, like their relatives the wasps, are solitary. Each female makes her own burrow, in which she constructs earthen chambers to contain her young. Some bees are communal. They are like solitary bees except that several females of the same generation use the same nest, each making her own cells for housing her eggs, larvae, and pupae.



注释

primitive [ˈprɪmɪtɪv] adj. 自然的
wasp [wɒsp] n. 黄蜂
solitary [ˈsɒlɪtəri] adj. 独居的
burrow [ˈbʌrəʊ] n. 洞穴
earthen [ˈɜːθən] adj. 土制的
chamber [ˈtʃeɪmbə] n. 房间

communal [ˈkɒmjʊnəl] adj. 群居的
nest [nest] n. 巢
cell [sel] n. 蜂房
larva [ˈlɑːvə] n. 幼虫
pupa [ˈpjuːpə] n. 蛹

The eusocial, or truly social, bees live in large colonies consisting of females of two overlapping generations: mothers (queens) and daughters (workers). Males, called drones, play no part in the colony's organization and only mate with the queens. Larvae are fed progressively — that is, cells are opened as necessary or are left open so that workers can tend the larvae. Parasitic, or cuckoo, bees are those that do not forage, or make nests themselves, but use the nests and food of other species of bees to provide for their parasitic young.

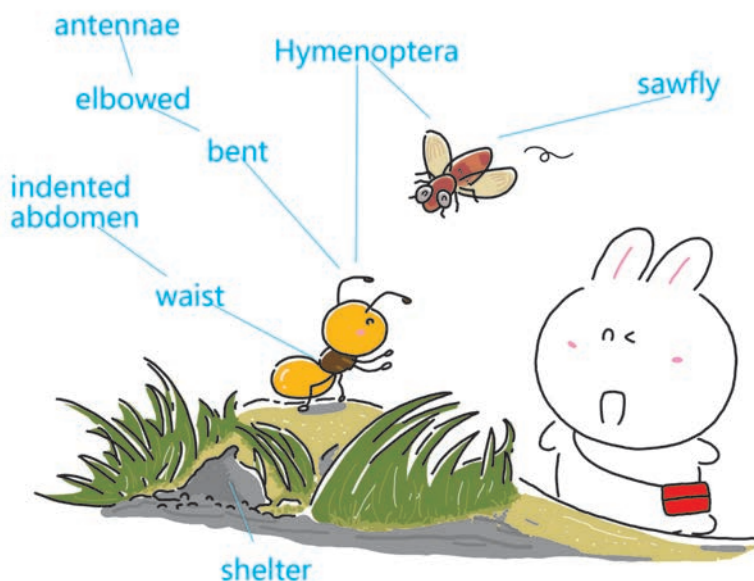


注释

eusocial [ju:'səʊʃəl] adj. 完全群居的
truly social ['tru:li-'səʊʃəl] n. 完全社会性的
colony ['kɒləni] n. 群体
overlapping ['əʊvə'læpiŋ] n. 重叠
drone [draʊn] n. 雄蜂
progressively [prə'ɡresɪvli] adv. 逐步地

tend [tend] v. 照顾
parasitic [ˌpærə'sɪtɪk] adj. 寄生的
cuckoo ['kuku:] n. 杜鹃; adj. 寄生的
cuckoo bee ['kuku:'bi:] n. 寄生蜂
forage ['fɒrɪdʒ] v. 觅食

Ant colonies have elaborate social structures in which the various activities necessary for the feeding, shelter, and reproduction of the colony are divided among specially adapted individuals. Ants belong to an order of insects called the Hymenoptera, a group that also includes bees, wasps, and sawflies. Some species of wasps and bees resemble ants in that they live in colonies and are, therefore, said to be social, but ants are the only hymenopterans in which every species is social. Ants are distinguished from other hymenopterans in that they have bent, or elbowed, antennae and an indented abdomen that forms a narrow waist.



注释

elaborate [i'laebəreɪt] adj. 复杂的

shelter ['feltə] n. 遮蔽处

adapted [ə'dæptɪd] adj. 适合的

Hymenoptera ['haimi,nɒptərə] n. 膜翅目

sawfly ['sɔ:flaɪ] n. 叶蜂

resemble [ri'zeɪbl] v. 类似

in that [ɪn - ðæt] n. 因为

hymenopteran ['haimi'nɒptərən] n. 膜翅目昆虫

distinguish [dis'tɪŋɡwɪʃ] v. 区别

bent [bent] adj. 弯的

elbowed ['elbəʊd] adj. 像肘一样弯曲的

antenna [æn'tenə] n. 触须

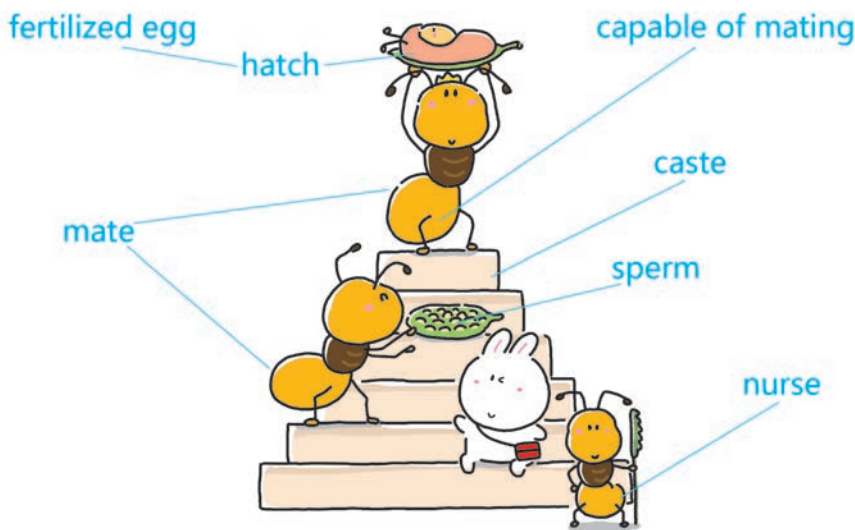
(plural: antennae [æn'teni:])

indented [ɪn'dentɪd] adj. 锯齿状的

abdomen ['æbdəmən] n. 腹部

waist [weɪst] n. 昆虫腹部的较窄部位

Members of an ant colony typically fall into categories known as castes, each with a different role. The majority of colony members are female worker ants that are unable to mate. Worker ants do not have wings and perform most of the work of the colony, including searching for food, nursing young, and defending the colony against ants from other colonies. Queens are larger than worker ants and are the only females of the colony capable of mating. Queens are born with wings, which they break off after mating. They mate with winged male ants, later using the sperm from the mating to produce fertilized eggs, which are hatched to produce more worker ants and a new generation of queens.



注释

caste [kɑ:st] n. 有社会组织的昆虫的阶级

mate [meɪt] v. 交配

perform [pə'fɔ:m] v. 执行

nurse [nɜ:s] v. 护理

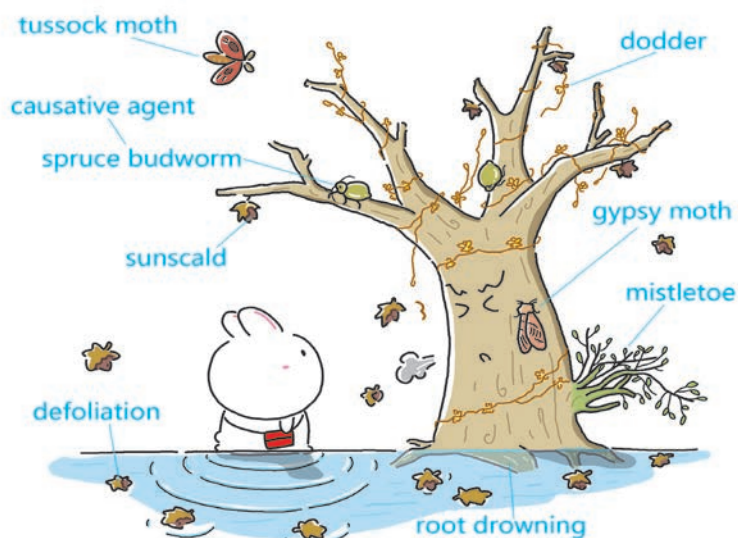
capable of ['keɪpəbl-əv] adj. 有能力的

sperm [spɜ:m] n. 精子

fertilized egg ['fɜ:tilaɪzd-eg] n. 受精卵

hatch [hætʃ] v. 孵化

Insects and diseases are a continuing menace to forests. Various insects, such as the gypsy moth, the tussock moth, and the spruce budworm, devastate extensive areas by defoliation. Other insects serve as carriers for the causative agents of diseases that destroy trees. Parasitic tree diseases may be caused by bacteria, fungi, viruses, and nematodes, or by such parasitic plants as the mistletoe or dodder. Noninfectious diseases of trees include sunscald; drought injury; root drowning, or suffocation; nutritional excesses or deficiencies; winter injury; and injury from smoke, gases, and fumes.

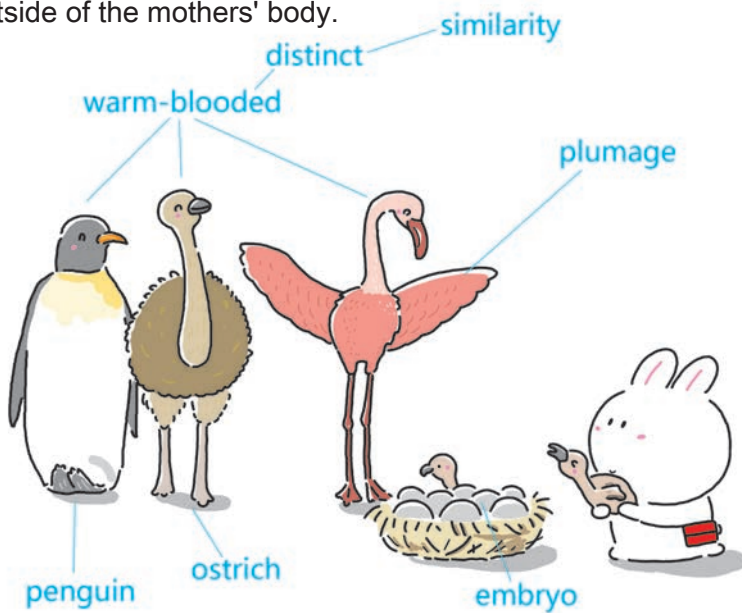


注释

menace ['menəs] n. 威胁
gypsy moth ['dʒɪpsi-moθ] n. 舞毒蛾
tussock moth ['tʌsək-moθ] n. 毒蛾
spruce budworm ['spru:s-'bʌdwə:m] n. 云杉蚜虫
devastate ['devəsteɪt] v. 糟蹋
defoliation [di:'fəʊli'eɪʃən] n. 落叶
causative agent ['kɔ:zətɪv-'eɪdʒənt] n. 病原体
bacteria [bæk'tɪəriə] n. 细菌
fungi ['fʌndʒaɪ] n. 真菌
virus ['vaɪərəs] n. 病毒

nematode ['nemətəʊd] n. 线虫
mistletoe ['mɪsltəʊ] n. 槲寄生
dodder ['dɒdə] n. 菟丝子
sunscald ['sʌnskɔ:lɪd] n. 日灼病
drought injury ['draʊt-'ɪndʒəri] n. 旱害
root drowning ['ru:t-'draʊnɪŋ] n. 根部溺死
suffocation [sʌfə'keɪʃən] n. 窒息
nutritional [nju:'trɪʃənəl] adj. 营养的
deficiency [di'fɪʃənsi] n. 缺乏
fume [fju:m] n. 浓烟

Birds are the only animals with feathers, collectively called plumage, although some other animals, such as insects and bats, also have wings. Nearly all birds can fly, and even flightless birds, such as ostriches and penguins, are evolved from flying ancestors. Many characteristics and behaviors of birds are distinct from all other animals, but there are some similarities. Like mammals, birds have four-chambered hearts and are warm-blooded — having a relatively constant body temperature that enables them to live in a wide variety of environments. Like reptiles, birds develop from embryos in eggs outside of the mothers' body.



注释

plumage ['plu:mɪdʒ] n. 鸟类的羽毛

ostrich ['ɒstri:tʃ] n. 鸵鸟

penguin ['pɛŋɡwɪn] n. 企鹅

ancestor ['ænsɪstə] n. 祖先

distinct [dɪ'stɪŋkt] adj. 截然不同的

similarity [ˌsɪmɪ'lærɪti] n. 相似

chamber ['tʃeɪmbə] n. 腔

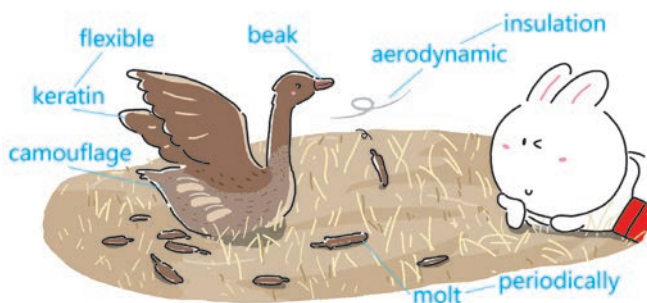
warm-blooded ['wɔ:m-blʌdɪd] adj. (动物) 恒温的

constant ['kɒnstənt] adj. 恒久不变的

reptile ['reptail] n. 爬行动物

embryo ['embriəu] n. 胚胎

The main component of feathers is keratin, a flexible protein that also forms the hair and fingernails of mammals. Feathers provide the strong yet lightweight surface area needed for powered, aerodynamic flight. They also serve as insulation, trapping pockets of air to help birds conserve their body heat. The varied patterns, colors, textures, and shapes of feathers help birds to signal their age, sex, social status, and species identity to one another. Some birds have plumage that blends in with their surroundings to provide camouflage, helping these birds escape notice by their predators. Birds use their beaks to preen their feathers, often making use of oil from a gland at the base of their tails. Preening removes dirt and parasites, and keeps feathers waterproof and supple. Because feathers are nonliving structures that cannot repair themselves when worn or broken, they must be renewed periodically. Most adult birds molt — lose and replace their feathers — at least once a year.

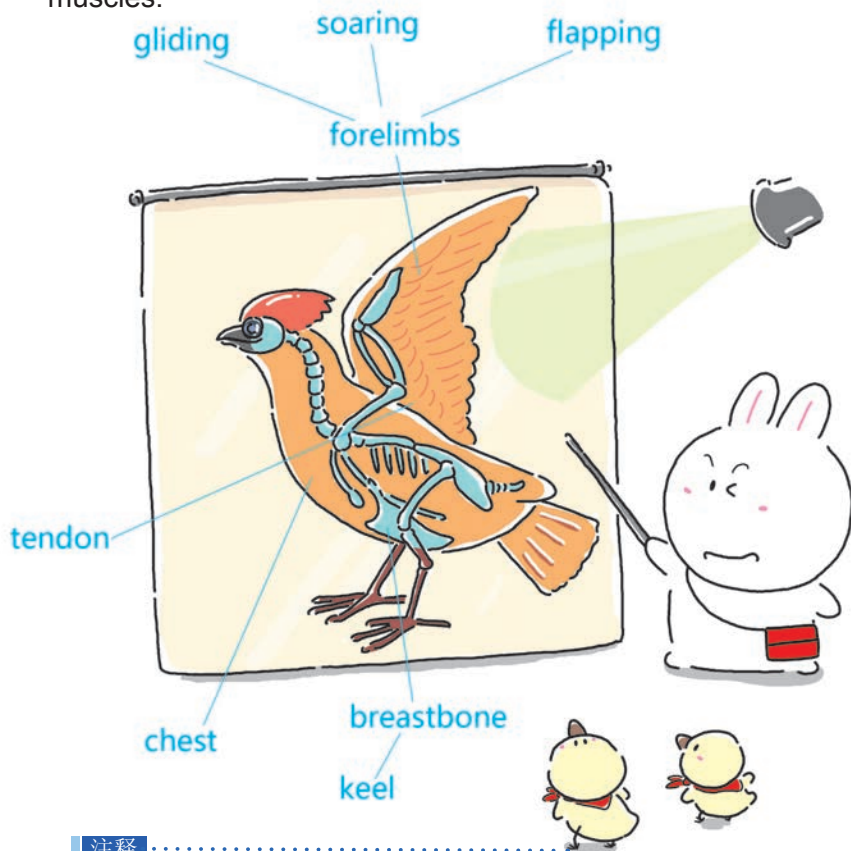


注释

keratin ['kerətin] n. 角质
flexible ['fleksəbl] adj. 柔韧的
mammal ['mæməl] n. 哺乳动物
aerodynamic [ˌɛərəʊdaɪ'næmik] adj. 空气动力学的
insulation [ˌɪnsju'leɪʃən] n. 绝缘, 绝热
trap [træp] v. 捕捉
conserve [kən'sɜ:v] v. 保存
texture ['tekstʃə] n. 质地
social status [ˌsəʊʃəl-'steɪtəs] n. 社会地位
blend in ['blend-in] v. 与(环境等)协调
camouflage ['kæməflɑ:ʒ] n. 伪装

predator ['predətə] n. 捕食者
beak [bi:k] n. 喙
preen [pri:n] v. (鸟)用嘴整理(羽毛)
gland [glænd] n. 腺
tail [teɪl] n. 尾
parasite ['pærəsait] n. 寄生虫
waterproof ['wɔ:təpru:f] adj. 防水的
supple ['sʌpl] adj. 柔软的
worn [wɔ:n] adj. 用旧了的, 磨破了的
periodically [ˌpiəri'ɒdɪkəli] adv. 周期性地
molt [məʊlt] v. 换毛

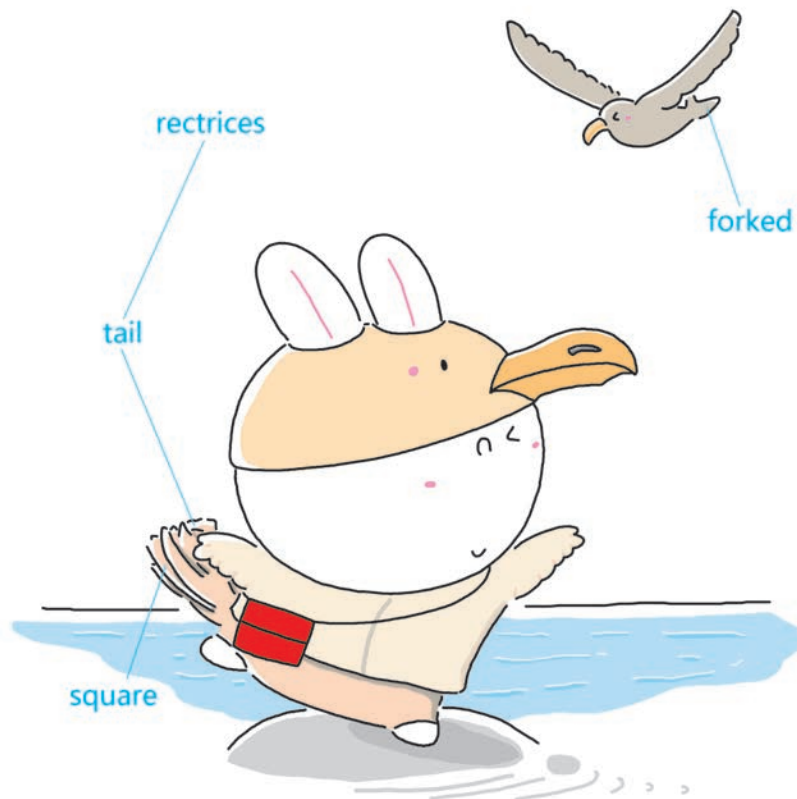
Bird wings are highly modified forelimbs with a skeletal structure resembling that of arms. The shape of a bird's wings influences its style of flight, which may consist of gliding, soaring, or flapping. Flight muscles are located in the chest and are attached to the wings by large tendons. The breastbone, a large bone shaped like the keel of a boat, supports the flight muscles.



注释

- forelimb ['fɔ:lɪm] n. 前肢
 resemble [ri'zembəl] v. 类似
 glide [glaid] v. 滑行
 soar [sɔ:] v. 振翅飞行
 flap [flæp] v. 拍动
 chest [tʃest] n. 胸部
 tendon ['tendən] n. 腱
 breastbone ['brestbəʊn] n. 胸骨
 keel [ki:l] n. (船的)龙骨

Nearly all birds have tails, the paired flight feathers of which, called rectrices, extend from the margins of a bird's tail. Tails may be square, rounded, pointed, or forked, depending on the lengths of the rectrices and the way they terminate.



注释

tail [teɪl] n. 尾

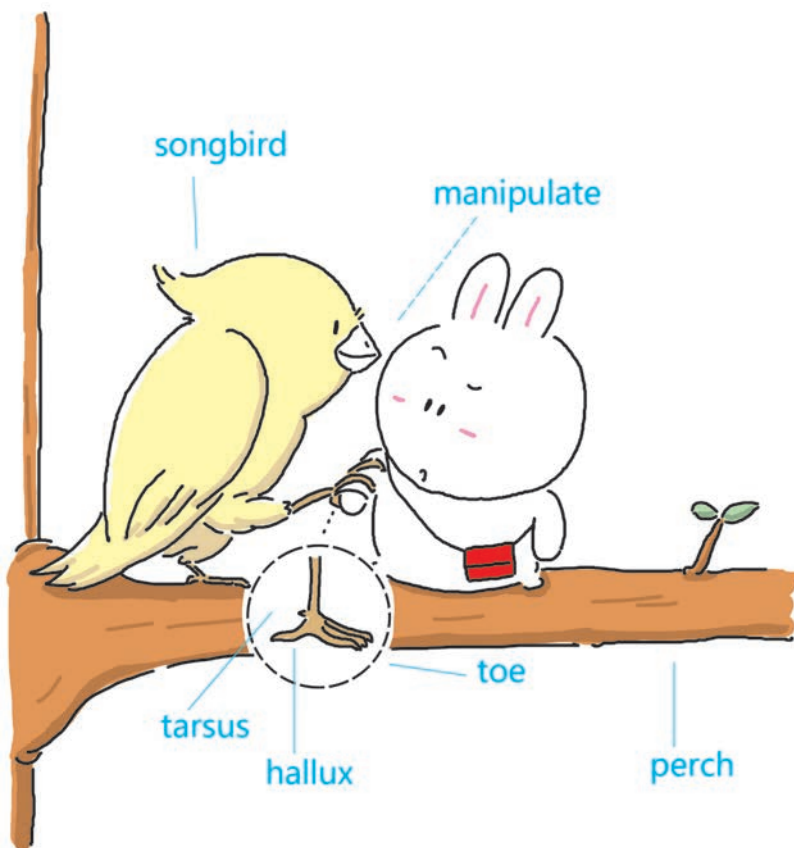
rectrix ['rektriks] n. 舵羽(鸟尾上的硬毛) (plural: rectrices ['rektrɪsɪs])

square [skwɛə] adj. 正方形的

forked [fɔ:kt] adj. 叉状的

terminate ['tɜ:mɪneɪt] v. 结束

Birds have two legs; the lower part of each leg is called the tarsus. Most birds have four toes on each foot, and in many birds, including all songbirds, the first toe, called a hallux, points backwards. Bird toes are adapted in various species for grasping perches, climbing, swimming, capturing prey, and carrying and manipulating food.



注释

tarsus ['tɑ:səs] n. 跖骨

toe [təʊ] n. 脚趾

songbird ['sɒŋbə:d] n. 鸣禽

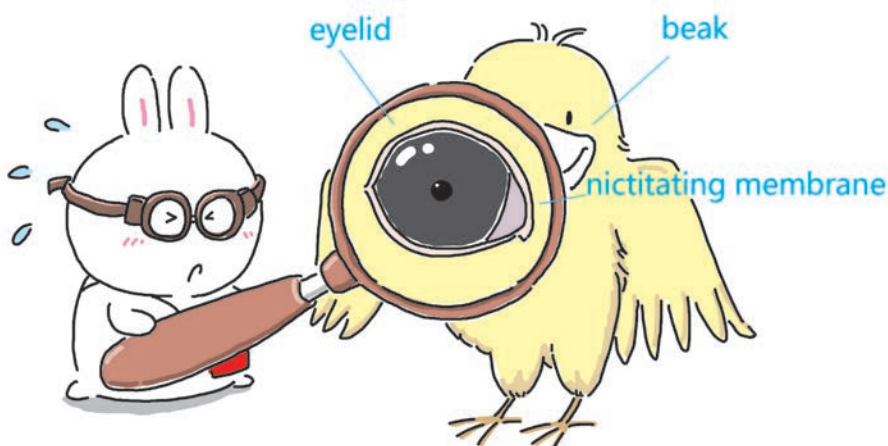
hallux ['hæləks] n. (鸟)后趾, (人)大拇趾

adapted [ə'dæptɪd] adj. 适合的

perch [pɜ:tʃ] n. 栖木: 供鸟栖息的枝条

manipulate [mə'nɪpjuleɪt] v. 处理

Instead of heavy jaws with teeth, modern birds have toothless, lightweight jaws, called beaks or bills. The eyes of birds are protected by three eyelids: an upper lid resembling that of humans, a lower lid that closes when birds sleeps, and a third lid, called a nictitating membrane, that sweeps across the eye sideways, starting from the side near the beak. This lid is a thin, translucent fold of skin that moistens and cleans the eyes and protects it from wind and bright light.



注释

jaw [dʒɔ:] n. 颞

beak [bi:k] n. 喙

bill [bil] n. 喙

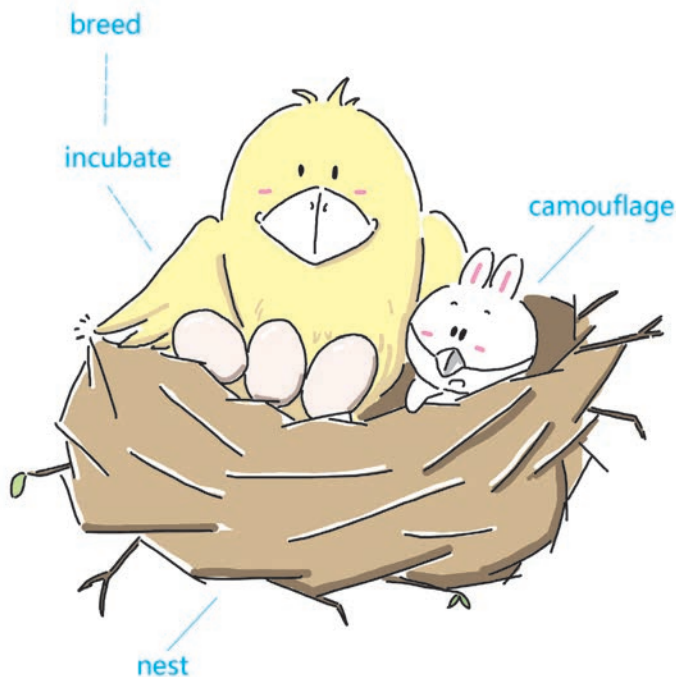
eyelid ['aɪlɪd] n. 眼睑

nictitating membrane ['nɪktɪˌteɪtɪŋ-'membreɪn] n. 瞬膜：鸟类、爬行动物和一些哺乳动物的内生透明眼睑，闭上可保护和滋润眼睛

translucent [trænz'lu:snt] adj. 半透明的

moisten ['moɪsn] v. 使潮湿

For breeding, most birds build nests, which help them to incubate or warm the developing eggs. Nests sometimes offer camouflage from predators and physical protection from the elements. Nests may be elaborate constructions or mere scrapes on the ground.



注释

- breed [bri:d] v. 生育; 繁殖
 nest [nest] n. 巢
 incubate ['inkjə,beit] v. 孵化
 camouflage ['kæməflɑ:ʒ] n. 伪装
 element ['elimənt] n. 环境
 elaborate ['ilæbəreit] adj. 精心制作的
 scrape [skreip] n. 浅坑

The best-known birds are the birds of prey. Some, including hawks, eagles, and falcons, are active during the daytime. Others, notably owls, are nocturnal, or active at night. Birds of prey have hooked beaks, strong talons or claws on their feet, and keen eyesight and hearing. Scavengers that feed on dead animals are also considered birds of prey.



注释

bird of prey ['bɜːd-əv-'preɪ] n. 食肉鸟, 猛禽

hawk [hɔːk] n. 鹰

falcon ['fɔːlkən] n. 猎鹰

owl [aʊl] n. 猫头鹰

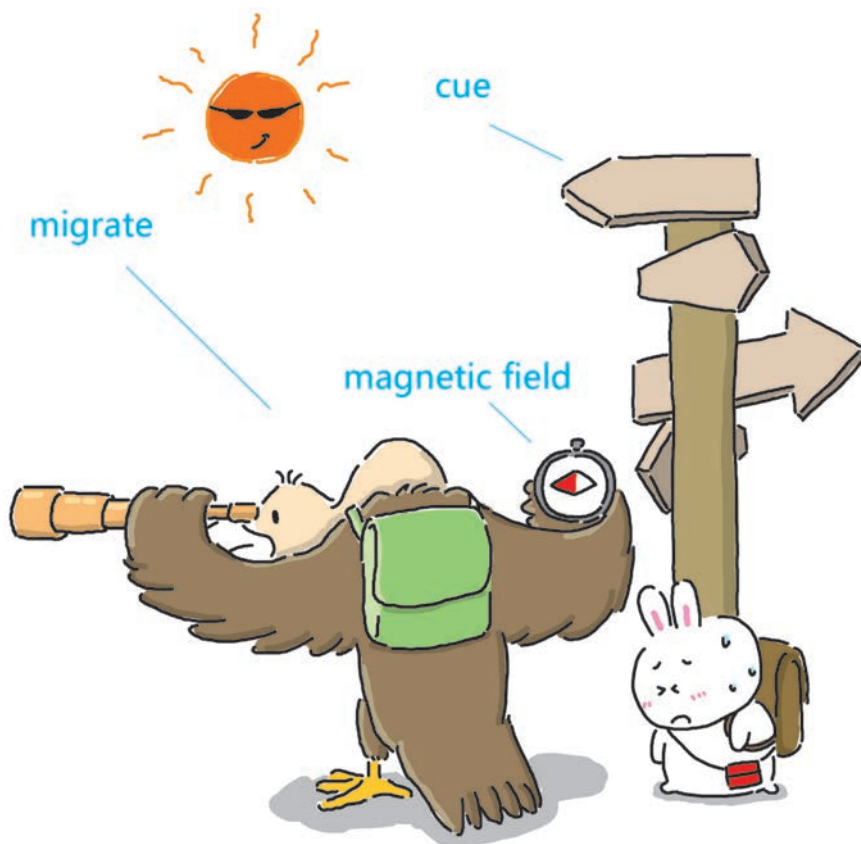
nocturnal [nɒk'tɜːnl] adj. 夜间的

beak [bi:k] n. 喙

talon ['tælən] n. 爪

scavenger ['skævɪndʒə] n. 食腐动物

Migrating birds use a variety of cues to find their ways. These include the positions of the Sun during the day and the stars at night; the earth's magnetic field; and visual, olfactory, and auditory landmarks.



注释

migrate [mai'greit] v. 迁徙

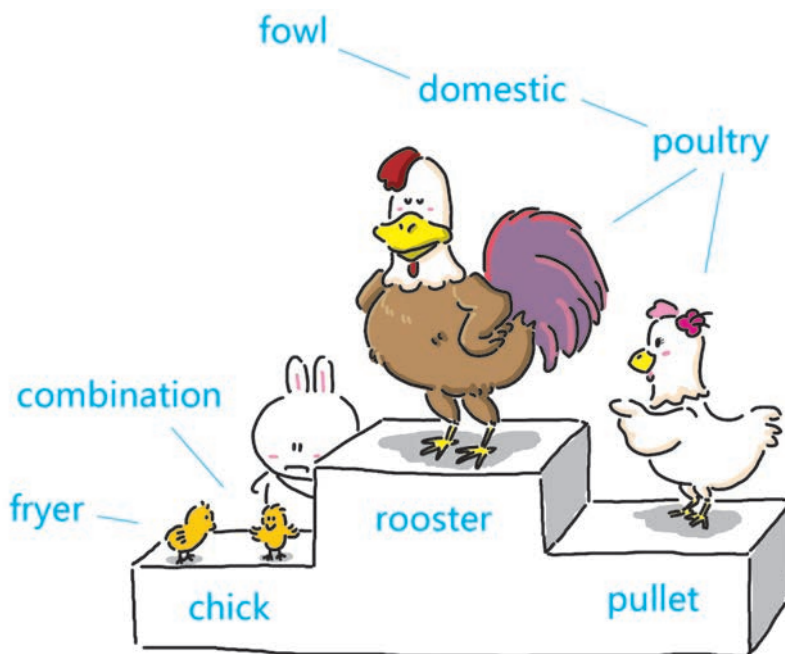
cue [kju:] n. 提示

magnetic field [mæg'netik-fi:ld] n. 磁场

olfactory [ɒ'fæktəri] adj. 嗅觉的

auditory ['ɔ:ditəri] adj. 听觉的

Except in combinations such as waterfowl and wildfowl, in modern usage, the word fowl usually is restricted to the common domestic fowl or chicken. In poultry markets, fowl commonly means a full-grown female bird. Young birds of both sexes, such as broilers and fryers, are called chickens. On poultry farms, male chickens are called roosters or cocks; females, especially those more than a year old, are called hens; females less than a year old are called pullets; very young chickens of either sex are called chicks; and castrated males are called capons.

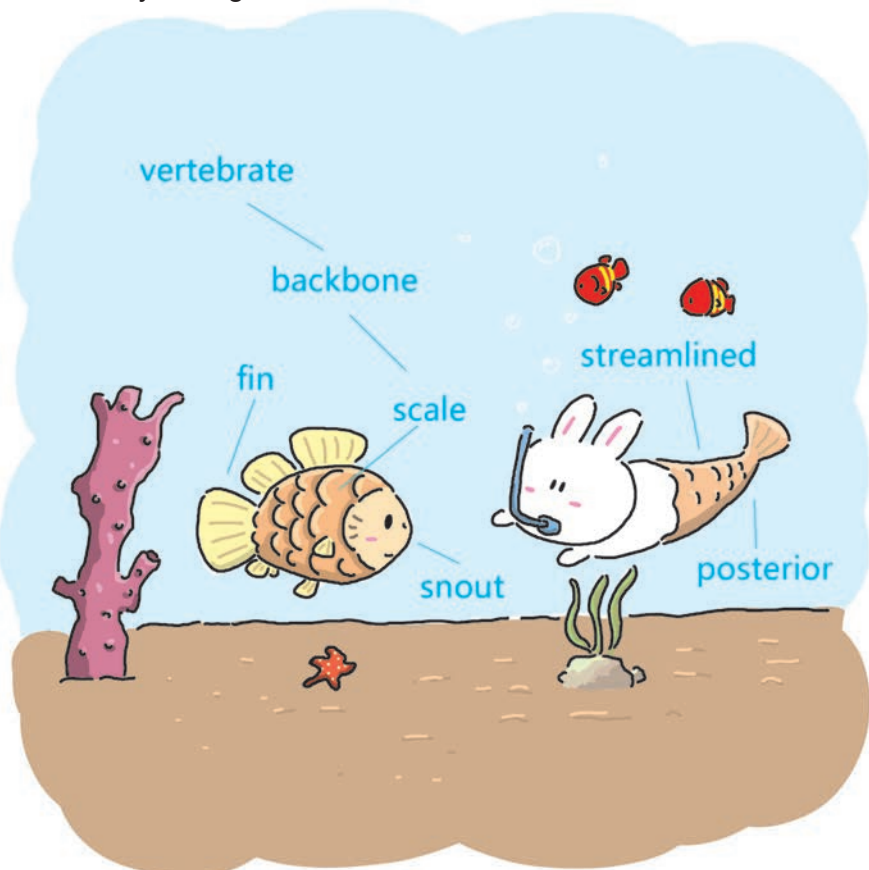


注释

combination [ˌkɒmbɪˈneɪʃən] n. 组合
 waterfowl ['wɔ:tə,fəʊl] n. 水鸟
 wildfowl ['waɪld,fəʊl] n. 野禽
 fowl [faʊl] n. 家禽
 domestic [də'mestɪk] adj. 驯养的
 poultry ['pɔ:ltʃəri] n. 家禽
 broiler ['brɔɪlə] n. 适于烤焙的嫩鸡

fryer [fraɪə] n. 适于油炸的嫩鸡
 rooster ['ru:stə] n. 公鸡
 pullet ['pʊlɪt] n. 小母鸡
 chick [tʃɪk] n. 小鸡
 castrate [kæ'streɪt] v. 阉割
 capon ['keɪpən] n. 阉鸡

All fishes are vertebrates (animals with backbones) with gills for breathing, most of them have fins for swimming, scales for protection, and streamlined bodies generally with pointed snouts and posteriors and broad propulsive tails for moving easily through the water.



注释

vertebrate ['vɜ:tɪbrɪt] n. 脊椎动物

backbone ['bækbəʊn] n. 脊椎

gill [gɪl] n. 腮

fin [fɪn] n. 鱼鳍

scale [skeɪl] n. 鱼鳞

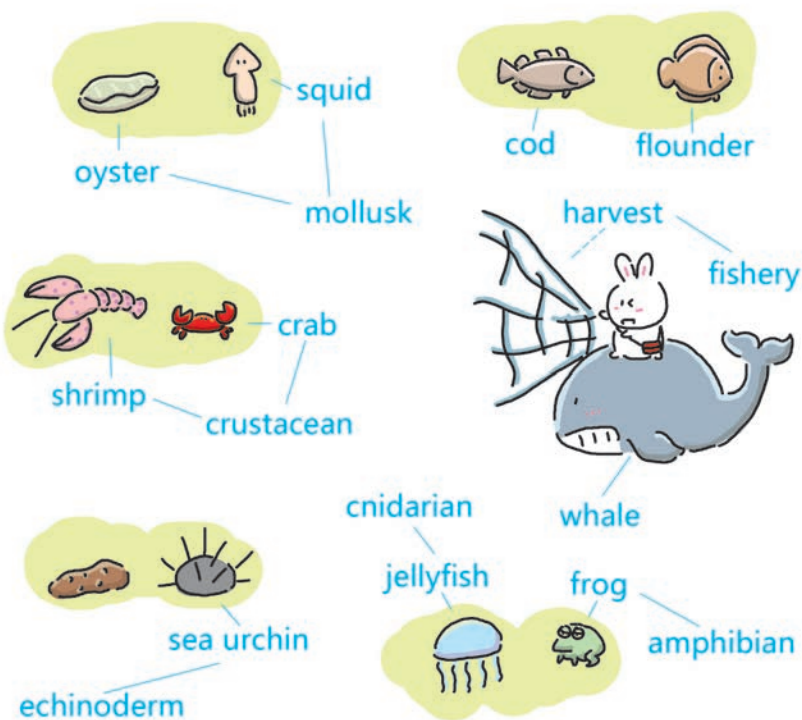
streamlined ['stri:mleɪnd] adj. 流线型的

snout [snaʊt] n. 口鼻部

posterior [pɒ'stɪəriə] n. 身体后部

propulsive [prə'pʌlsɪv] adj. 有推进力的

The term fishery is also used to describe the species of fish being harvested (because most of them are edible), which include familiar finned fish species, like cods and flounders; mollusks, including oysters and squids; and crustaceans, such as shrimps and crabs. Lesser-known fisheries include echinoderms, like sea urchins; some amphibians, including frogs; and cnidarians, such as jellyfish. Even the harvest of whales is usually considered a fishery.



注释

fishery ['fɪʃəri] n. 食用鱼类, 捕鱼业

harvest ['hɑ:vɪst] v. 收获, 捕捞

edible ['edɪbl] adj. 可食用的

cod [kɒd] n. 鳕鱼

flounder ['flaʊndə] n. 比目鱼

mollusk ['mɒləsk] n. 软体动物

oyster ['ɔɪstə] n. 牡蛎

squid [skwɪd] n. 鱿鱼

crustacean [krə'steɪʃən] n. 甲壳纲动物

shrimp [ʃrɪmp] n. 虾

crab [kræb] n. 蟹

echinoderm [i'kainə,də:m] n. 棘皮类动物

sea urchin [si:'ə:tʃɪn] n. 海胆

amphibian [æm'fɪbiən] n. 两栖动物

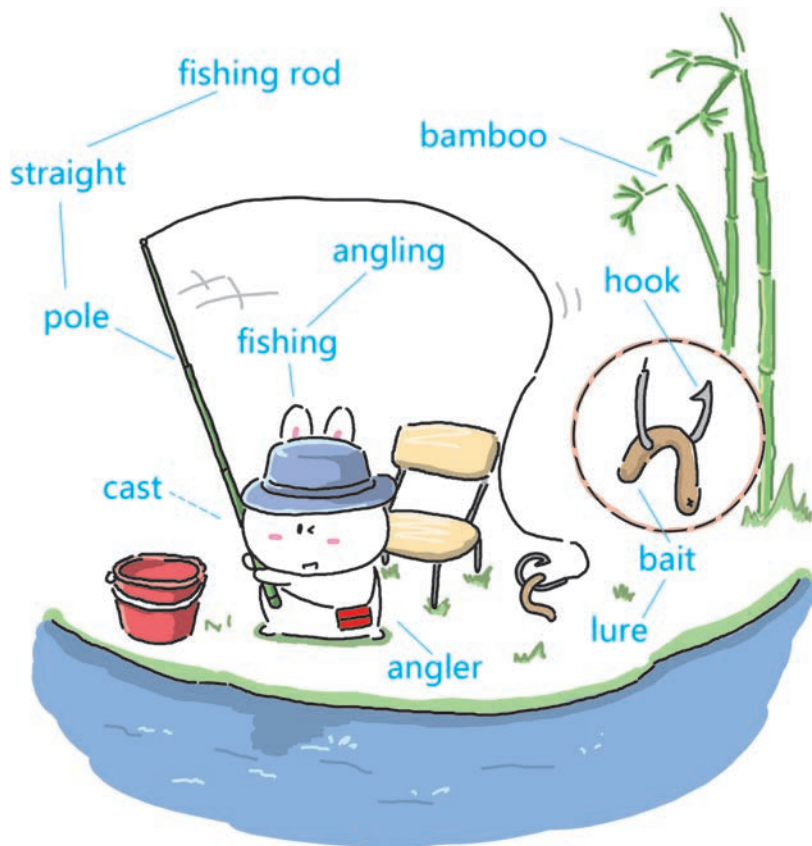
frog [frɒg] n. 青蛙

cnidarian [naɪ'deəriən] n. 刺胞动物

jellyfish ['dʒelɪfɪʃ] n. 水母

whale [hweɪl] n. 鲸

Angling refers to fishing for sport, recreation, and relaxation. A fishing rod is a long, straight, flexible pole made of bamboo, fiberglass, or graphite that an angler uses to cast hook with bait or lures into the water.

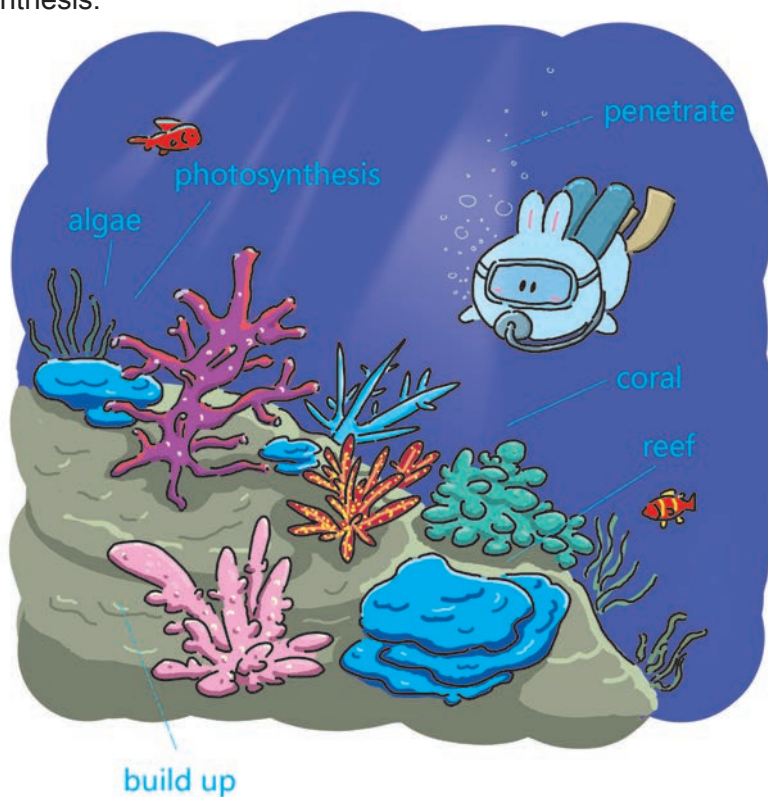


注释

angling ['æŋɡlɪŋ] n. 钓鱼, 钓鱼术
 fishing ['fɪʃɪŋ] n. 钓鱼
 recreation [,rekrɪ'eɪʃn] n. 消遣
 fishing rod ['fɪʃɪŋ-rɒd] n. 钓竿
 straight [streɪt] adj. 直的
 flexible ['fleksəbl] adj. 有弹性的
 pole [pəʊl] n. 杆
 bamboo [bəm'bu:] n. 竹子

fiberglass ['faɪbəɡlɑ:s] n. 玻璃纤维
 graphite ['græfəɪt] n. 石墨
 angler ['æŋɡlə] n. 垂钓者
 cast [kɑ:st] v. 抛
 hook [hʊk] n. 鱼钩
 bait [beɪt] n. 饵
 lure [lʊə] n. 诱饵

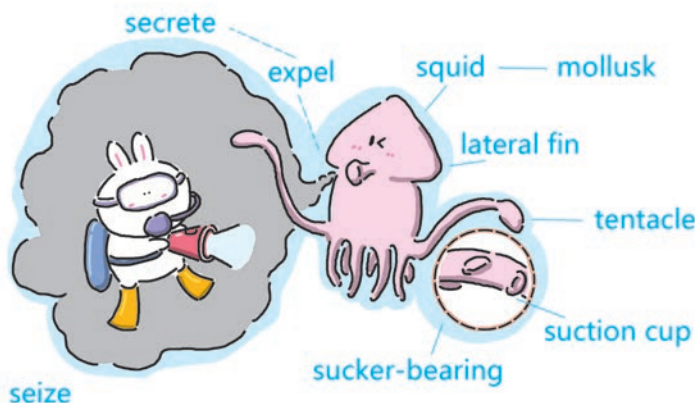
Coral reefs are built up by limestone skeletons of coral, and are found only in tropical saltwater where bright sunlight can penetrate, because corals cannot exist without the symbiotic algae that live in coral tissues and require sunlight for photosynthesis.



注释

- coral ['kɒrəl] n. 珊瑚, 珊瑚虫
 reef [ri:f] n. 礁, 暗礁
 build up ['bɪld-ʌp] v. 堆积
 limestone ['laɪm, stəʊn] n. 石灰岩
 skeleton ['skelɪn] n. 骨骼
 tropical ['trɒpɪkəl] adj. 热带的
 saltwater ['sɔ:lt, wɔ:tə] n. 盐水
 penetrate ['penɪtreɪt] v. 穿透
 symbiotic [ˌsɪmbaɪ'ɒtɪk] adj. 共生的
 algae ['ældʒi:] n. 藻类, 海藻 (singular: alga ['ælgə])
 tissue ['tɪʃu:] n. 组织
 photosynthesis [ˌfəʊtəʊ'sɪnθəsis] n. 光合作用

Squids are carnivorous mollusks belonging to the same class as the cuttlefish and octopus. The body of squids, stiffened by an interior cartilaginous skeleton, is spherical or cigar-shaped, with two lateral fins. Around the mouth are eight sucker-bearing arms and two contractile tentacles with spatulate tips; on the latter are four rows of suction cups encircled by rings of chitinous (horny) hooks. The contractile tentacles, longer than the rest, are used to seize the prey and pass it to the shorter arms, which hold it to be torn by strong jaws shaped like a parrot's beak. Squids can swim faster than any other invertebrate by rapidly expelling water from the mantle cavity through the "funnel". Many deep-sea squids are bioluminescent. They shoot out a cloud of dark ink when pursued; one genus secretes luminescent ink.



注释

carnivorous [kɑ:'nɪvərəs] adj. (动物)食肉的

mollusk ['mɒləsk] n. 软体动物

cuttlefish ['kʌtlɪʃ] n. 乌贼, 墨鱼

octopus ['ɒktəpəs] n. 章鱼

stiffen ['stɪfn] v. (使)变硬

interior [ɪn'tɪəriə] adj. 内部的, 内地的

cartilaginous [kɑ:'tɪlædʒɪnəs] adj. 软骨的, 软骨质的

spherical ['sferɪkəl] adj. 球形的

cigar-shaped [sɪ'gɑ:-'feɪpt] adj. 雪茄形的

lateral fin ['lætərəl-'fɪn] n. 侧鳍

sucker-bearing ['sʌkə-'beərɪŋ] adj. 长着吸盘的

contractile [kən'træktail] adj. 会缩的

tentacle ['tentəkl] n. 触角, 触须

spatulate ['spætjʊlɪt] adj. 竹片状的

row [rəʊ] n. 一排, 一行

suction cup ['sʌkʃən-kʌp] n. 吸盘

chitinous ['kaitɪnəs] adj. 壳质的

horny ['hɔ:ni] adj. 角状的

seize [si:z] v. 抓住, 捉住

prey [prei] n. 被掠食者, 战利品

torn [tɔ:n] v. (tear的过去分词) 撕

jaw [dʒɔ:] n. 颌, 颚

parrot ['pærət] n. 鹦鹉

invertebrate [ɪn'vɜ:tɪbrət] n. 无脊椎动物

expel [ɪks'pel] v. 喷出, 发射

mantle cavity ['mæntl-'kævɪti] n. 外套腔

funnel ['fʌnl] n. 漏斗

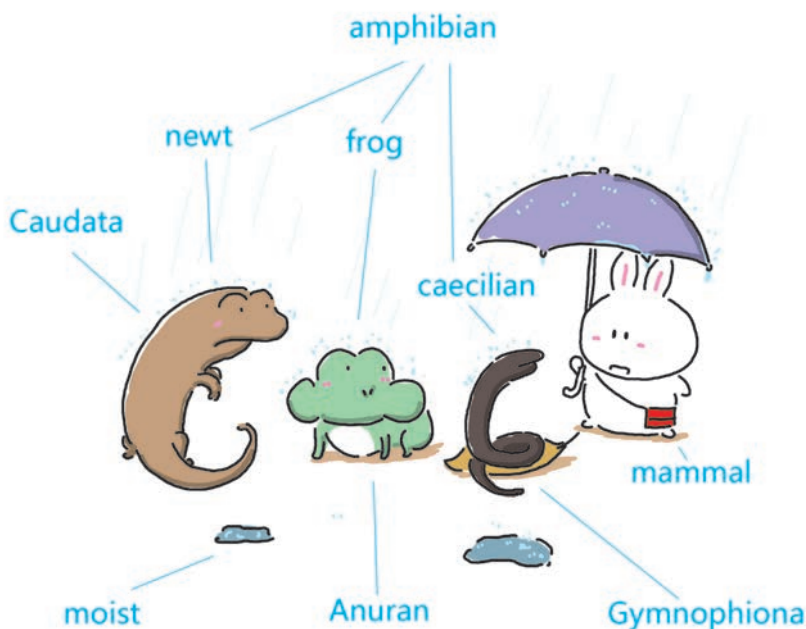
bioluminescent [baɪəʊ,lʊ:'mi:nesnt] adj. 发(冷)光的

pursue [pə'sju:] v. 追捕, 追击

genus ['dʒɪ:nəs] n. 种, 类

secrete [sɪ'kri:t] v. 分泌

Amphibians have moist and hairless skin, through which water can pass in and out, and were the first animals with backbones to adapt to life on land. They are the ancestors of reptiles, which in turn gave rise to mammals and birds. All amphibians belong to one of three main groups: the caudate or tailed amphibian, which includes the sirens, salamanders, and newts; the anuran, or tailless amphibian, which includes frogs and toads; and the gymnophiona, which is made up of the wormlike caecilians.



注释

amphibian [æm'fibiən] n. 两栖动物

moist [moɪst] adj. 潮湿的

adapt [ə'dæpt] v. 适应

ancestor ['ænsəstə] n. 祖先

reptile ['reptail] n. 爬行动物

mammal ['mæməl] n. 哺乳动物

Caudata [kɑ:'deɪtə] n. 有尾目

siren ['saɪərɪn] n. 土鳗

salamander ['sæləmændə] n. 火蜥蜴

newt [nju:t] n. 蝾螈

Anuran [ə'njuərən] n. 无尾目

frog [frɒg] n. 青蛙

toad [təʊd] n. 蟾蜍

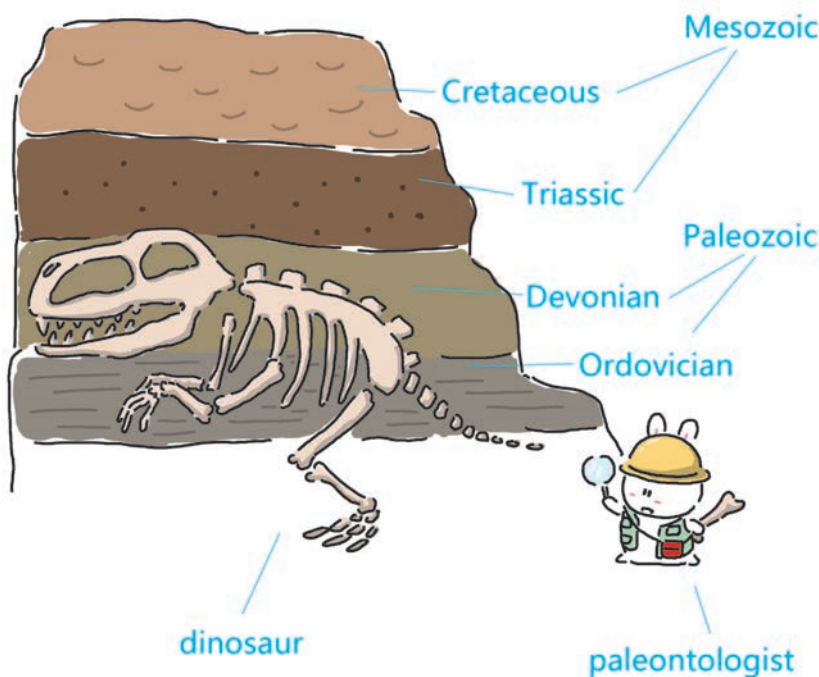
Gymnophiona ['dʒɪmɒfɪnə] n. 无足目

make up of ['meɪk-ʌp-əv] n. 构成, 组成

caecilian [si:'sɪliən] n. 蚓螈(无足目约150种两栖动物的统称)

UNIT090

Paleontologists recognize several mass extinctions in the past 500 million years, which occurred in Ordovician, Devonian, Paleozoic, Mesozoic, Triassic, and Cretaceous Period, respectively. The most recent mass extinction was at the end of Cretaceous Period, and resulted in the loss of 76% of all species, most notably the dinosaurs.



注释

paleontologist [ˌpeɪlɪənˈtɒlədʒɪst] n. 古生物学者

mass extinction [mæs-ɪksˈtɪŋkʃən] n. 大量消亡

Ordovician [ɔːdəˈvɪʃən] n. 奥陶纪

Devonian [deˈvəʊniən] n. 泥盆纪

Paleozoic [pæliəˈzəʊɪk] n. 古生代

Mesozoic [mesəʊˈzəʊɪk] n. 中生代

Triassic [traɪˈæsiːk] n. 三叠纪

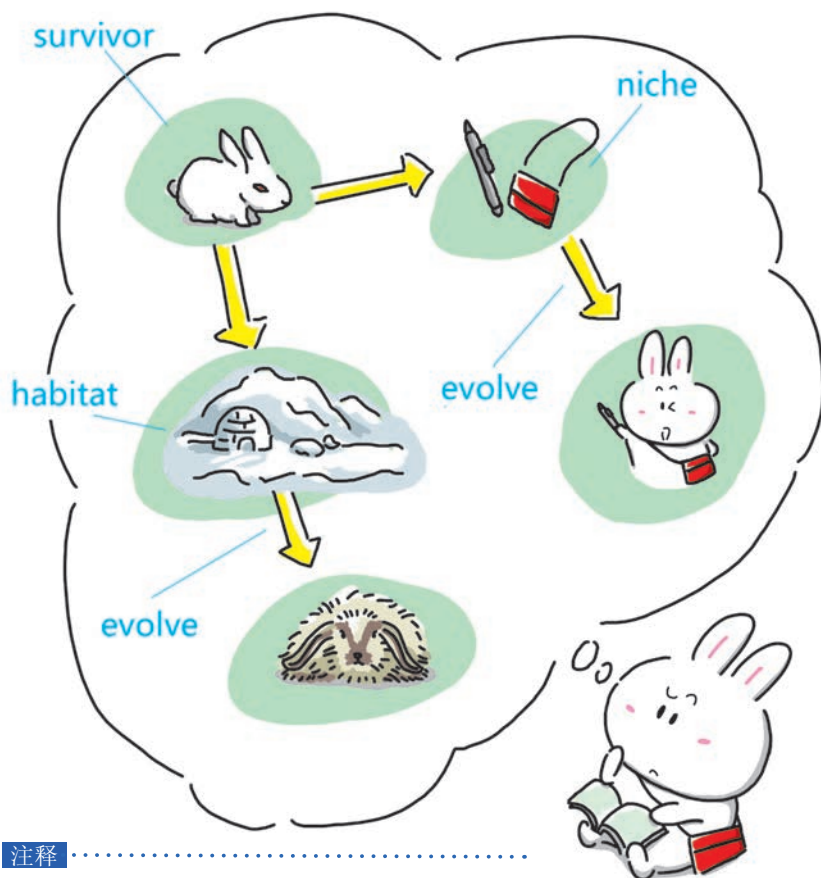
Cretaceous [kriˈteɪsiəs] n. 白垩纪

respectively [rɪsˈpektɪvli] adv. 分别地

species [ˈspiːsiːz] n. 物种

dinosaur [ˈdaɪnəsɔː] n. 恐龙

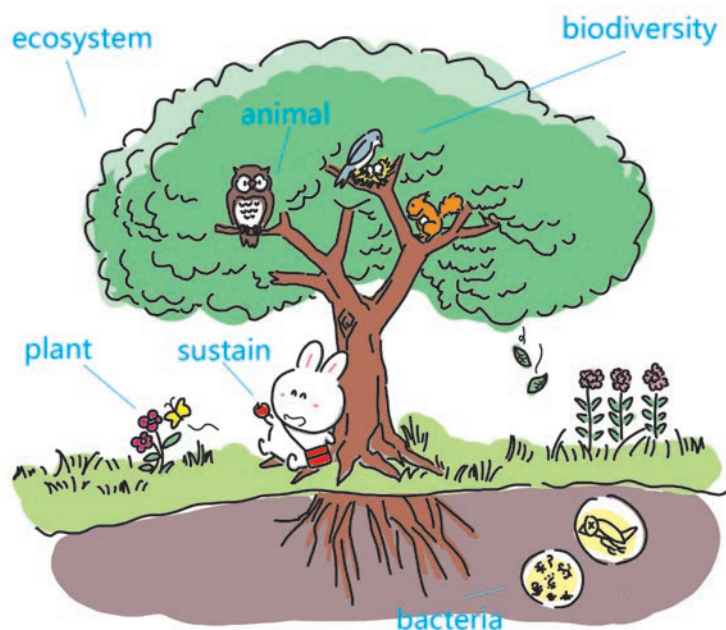
In mass extinctions, entire groups of species — such as families, orders, and classes — die out, creating opportunities for the survivors to exploit new habitats. In their new niches, the survivors evolve new characteristics and habits and, consequently, develop into entirely new species.



注释

- family ['fæmili] n. 科
order ['ɔ:də] n. 目
class [klɑ:s] n. 纲
survivor [sə'vaivə] n. 幸存者
exploit [iks'plɔit] v. 开拓
habitat ['hæbitæt] n. 栖息地
niche [niʃ] n. 合适的环境
evolve [i'vɒlv] v. 进化

As the sum of all species living in the world's ecosystems, known as biodiversity, dwindles, so too go many of the resources on which we depend. Humans use at least 40,000 different plants, animals, fungi, bacteria, and virus species for food, clothing, shelter, and medicines. In addition, the fresh air we breathe, the water we drink, cook and wash with, and the many chemical cycles — including the nitrogen cycle and the carbon cycle, so vital to sustain life — depend on the continued health of ecosystems and the species within them.



注释

ecosystem ['i:kəu,sistem] n. 生态系统

biodiversity [ˌbaɪəʊdaɪ'və:səti] n. 生物多样性

dwindle ['dwindl] v. 减少

plant [plɑ:nt] n. 植物

animal ['æniməl] n. 动物

fungi ['fʌndʒai] n. 真菌类

(singular: fungus ['fʌŋgəs])

bacteria [bæk'tɪəriə] n. 细菌 (singular: bacterium [bæk'tɪəriəm])

virus ['vaɪərəs] n. 病毒

chemical cycle ['kemikəl-'saɪkl] n. 化学循环

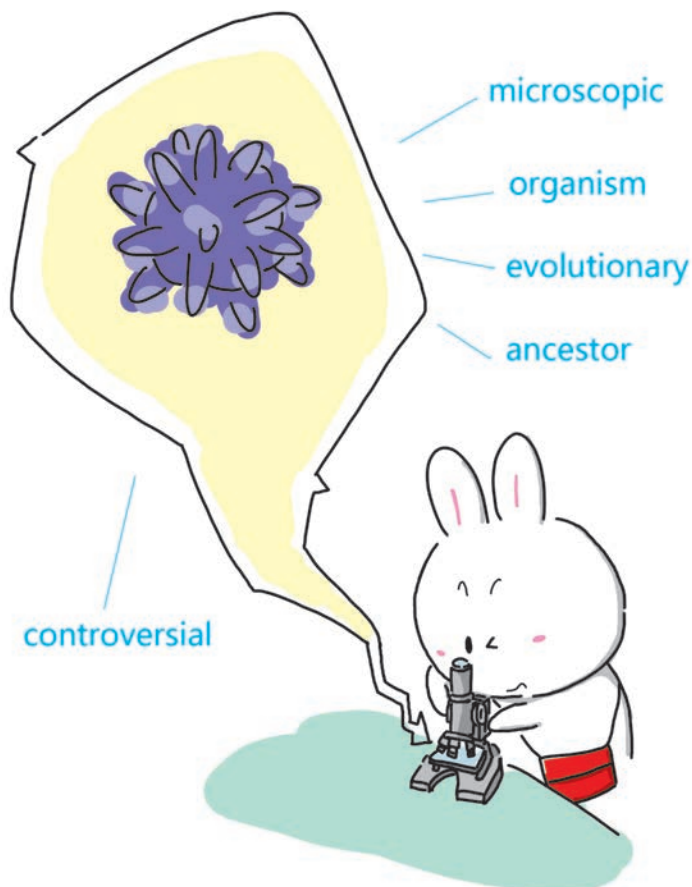
nitrogen cycle ['naɪtrədʒən-'saɪkl] n. 氮循环

carbon cycle ['kɑ:bən-'saɪkl] n. 碳循环

vital ['vaɪtəl] adj. 极重要的

sustain [sə'steɪn] v. 维持

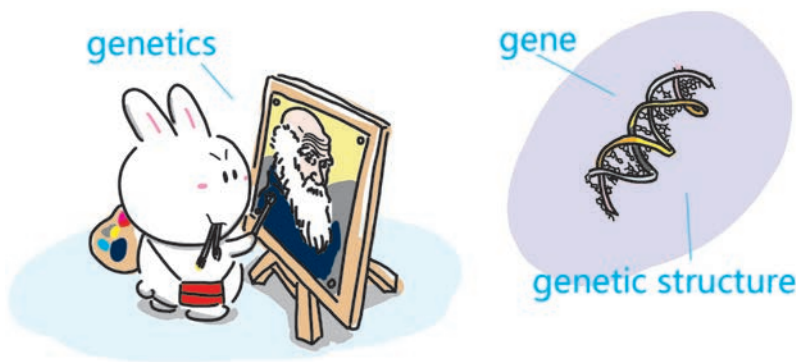
A central and historically controversial component of evolutionary theory is that all living organisms, from microscopic bacteria to plants, insects, birds, and mammals, share a common ancestor.



注释

controversial [ˌkɒntreɪvəːʃəl] adj. 有争议的
 component [kəmˈpəʊnənt] n. 组成部分
 evolutionary [ˌiːvəˈluːʃənəri] adj. 进化的
 organism [ˈɔːɡənizəm] n. 有机物
 microscopic [ˌmaɪkrəˈskɒpɪk] adj. 极微小的
 ancestor [ˈænsɪstə] n. 祖先

Hostility as to evolutionary theory gave way to acclaim as scientists vigorously debated, explored, and built on Darwin's theory of natural selection. As the 20th century unfolded, scientific advances revealed the detailed mechanisms missing from Darwin's theory. Study of the complex chemistry of all organisms unveiled the structure of genes as well as how they were duplicated, altered, and passed from generation to generation. New molecular biology techniques compared the genetic structures of different species, enabling scientists to determine heretofore undetectable evolutionary relationships between species. Today, evolution is recognized as the cornerstone of modern biology. Uniting such diverse scientific fields as cell biology, genetics, paleontology, and even geology and statistics, the study of evolution reveals an exquisitely complex interaction of the forces that act upon every life form on Earth.

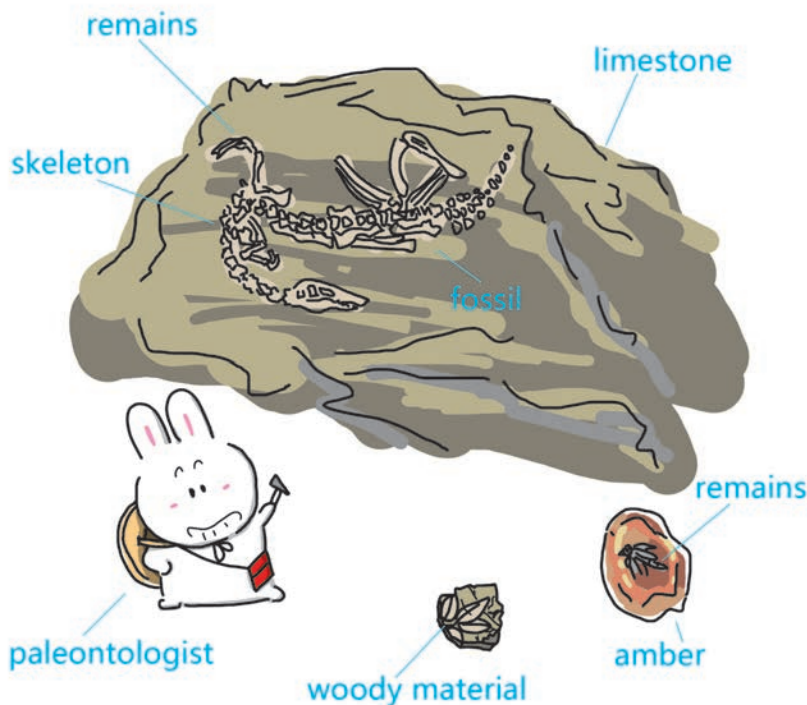


注释

hostility [hɒs'tɪlɪti] n. 反对
 give way to ['gɪv-'wei-tu] v. 退让
 acclaim [ə'kleɪm] n. 称赞
 vigorously ['vɪɡərəsli] adv. 竭力地
 unfold [ʌn'fəʊld] v. 展开
 reveal [rɪ'vi:l] v. 揭示
 unveil [ʌn'veɪl] v. 揭去
 gene [dʒi:n] n. 基因
 duplicate ['dʒu:pɪkɪt] v. 复制
 alter ['ɔ:lteɪ] v. 改变
 genetic structure [dʒi'netɪk-'strʌktʃə] n. 遗传结构

heretofore ['hi:tu'fɔ:] adv. 迄今为止
 undetectable [ʌndɪ'tektəbl] adj. 未被发现的
 cornerstone [ˌkɔ:nə'stəʊn] n. 基础
 cell biology [sel-'baɪ'ɒlədʒi] n. 细胞生物学
 genetics [dʒi'netɪks] n. 遗传学
 paleontology [ˌpæliən'tɒlədʒi] n. 古生物学
 geology [dʒi'ɒlədʒi] n. 地质学
 statistics [stə'tɪstɪks] n. 统计学
 exquisitely ['ekskwɪzɪtli] adv. 非常地

Fossils are most commonly found in limestone, sandstone, and shale (a kind of sedimentary rock). Remains of organisms can also be found trapped in natural asphalt, amber, and ice. The hard indigestible skeletons and shells of animals and the woody material of plants are usually preserved best. Fossils of organisms made of soft tissue that decays readily are rarer. Paleontologists use fossils to learn how life has changed and evolved throughout earth's history.



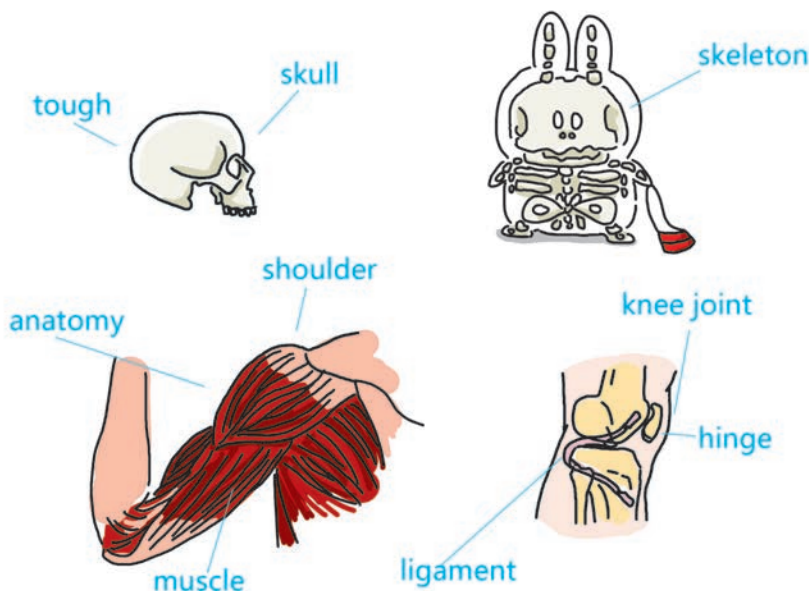
注释

fossil ['fɒsl] n. 化石
limestone ['laɪm.stəʊn] n. 石灰岩
sandstone ['sænd.stəʊn] n. 砂岩
shale [ʃeɪl] n. 页岩
sedimentary rock [sedɪ'mentəri-rɒk] n. 沉积岩
remains ['riːmeɪnz] n. 遗体
organism ['ɔːɡənɪzəm] n. 有机物
trap [træp] v. 困住
asphalt ['æsfælt] n. 沥青
amber ['æmbə] n. 琥珀

indigestible [ɪnɪ'dʒestəbl̩] adj. 难消化的
skeleton ['skelɪtn̩] n. 骨骼
shell [ʃel] n. 外壳
woody material [wudi-mə'tɪəriəl] n. 木质材料
preserve [prɪ'zɜːv] v. 保存
tissue ['tɪʃuː] n. 组织
decay [di'keɪ] v. 腐烂
paleontologist [ˌpeɪlɪən'tɒlədʒɪst] n. 古生物学者
evolve [ɪ'vɒlv] v. 进化

Basic Human Anatomy: Musculoskeletal System

The human skeleton consists of more than 200 bones, bound together by tough and relatively inelastic connective tissues called ligaments. The different parts of the body vary greatly in their degrees of movement, which are effected by contractions of the skeletal muscles, to which the bones are attached by tendons. Thus, the arm at the shoulder is freely movable, whereas the knee joint is definitely limited to a hinge-like action, and the bones composing the skull are immovable.



注释

anatomy [ə'neɪtəmi] n. 人体解剖

musculoskeletal [ˌmʌskjʊləʊ'skelitəl] adj. 肌(与)骨骼的

skeleton ['skelɪtn] n. 骨骼

bind [baɪnd] v. 绑定, 缔结

tough [tʌf] adj. 坚硬的

inelastic [ɪni'leɪstɪk] adj. 无弹性的

tissue ['tɪʃu:] n. 组织

ligament ['lɪgəmənt] n. 韧带

vary ['vɛəri] v. 不同, 变化

contraction [kən'trækʃən] n. 收缩

muscle ['mʌsl] n. 肌肉

skeletal muscle ['skelɪtɪ-'mʌsl] n. 骨骼肌

attach [ə'tætʃ] v. 贴上; 系; 附上

tendon ['tendən] n. 腱

shoulder ['ʃəʊldə] n. 肩

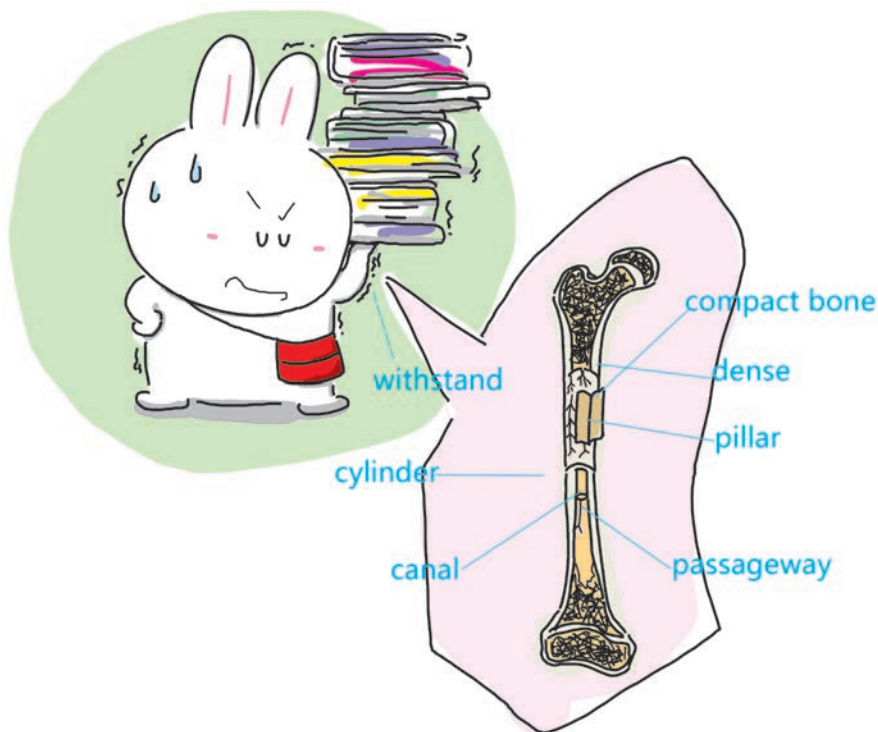
knee joint [ni:-dʒɔɪnt] n. 膝关节

hinge [hɪndʒ] n. 枢纽, 门枢, 节点

compose [kəm'pəʊz] v. 组成

skull [skʌl] n. 颅骨

There are two main types of bone. Compact bone, which makes up most of the bone of arms and legs, is very dense and hard on the outside. The structural units of compact bone are osteons, elongated cylinders that act as weight-bearing pillars, and are able to withstand any mechanical stress placed on the bone. The center of each osteon contains a hollow canal that acts as a central passageway for blood vessels and nerves.



注释

- compact bone [kəm'pækt-bəʊn] n. 密质骨
 make up ['meɪk-ʌp] v. 组成
 dense [dens] adj. 密集的
 osteon ['ɒstɪən] n. 骨单位(密质骨构造的基本单位)
 elongated ['i:lŋ,geɪtɪd] adj. 被拉长的
 cylinder ['sɪlɪndə] n. 圆柱体
 pillar ['pɪlə] n. 柱状物
 withstand [wɪð'stænd] v. 经受, 承受
 canal [ke'næl] n. 导管
 passageway ['pæsɪdʒ,wei] n. 出入口

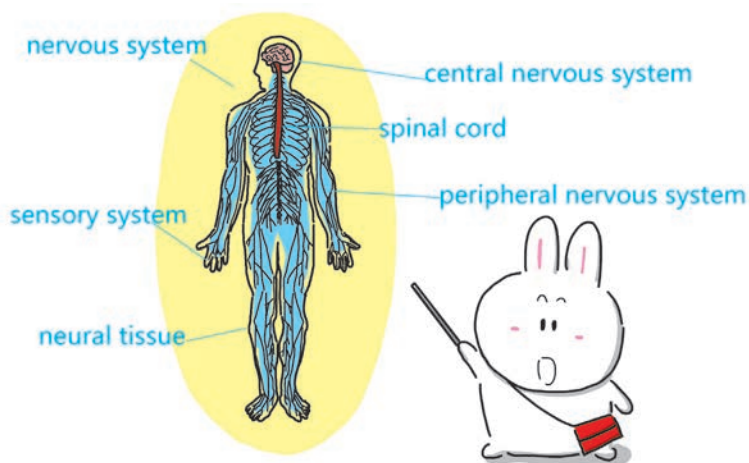
In some bones, internal to the compact bone is spongy bone, also known as cancellous bone, composed of a honeycomb network of bones that act as supporting beams. Spongy bone is designed to bear stress from several directions, such as that exerted on the pelvis in bending or stretching. The spaces between the trabeculae are filled with red bone marrow, containing the blood vessels that nourish spongy bone. Spongy bone is found in bones of the pelvis, ribs, breastbone, vertebrae, skull, and at the ends of the arm and leg bones.



注释

- spongy bone ['spɒndʒi-bəʊn] n. 松质骨
 cancellous ['kænsiləs] adj. 网状骨质的
 cancellous bone ['kænsiləs-bəʊn] n. 松质骨
 honeycomb ['hʌnikəʊm] n. 蜂窝
 beam [bi:m] n. 梁
 pelvis ['pelvis] n. 骨盆
 bend [bend] v. (使)弯曲
 stretch [stretʃ] v. 伸展
 trabecula [trə'bekjule] n. 小梁
 marrow ['mæərəʊ] n. 骨髓
 breastbone ['brestbəʊn] n. 胸骨
 vertebra ['vɜ:tibrə] n. 脊椎骨

The nervous system has two divisions: the central nervous system, including the brain and spinal cord, and the peripheral nervous system, including all neural tissue of motor and sensory systems. Impulses go to the central nervous system through sensory nerves and are carried away from it by motor nerves. The motor system is further divided into the somatic (or skeletal) nervous system and the autonomic nervous system. The somatic motor system allows voluntary control over skeletal muscle with a few exceptions. The autonomic nervous system is largely involuntary, and controls cardiac and smooth muscles and glands.

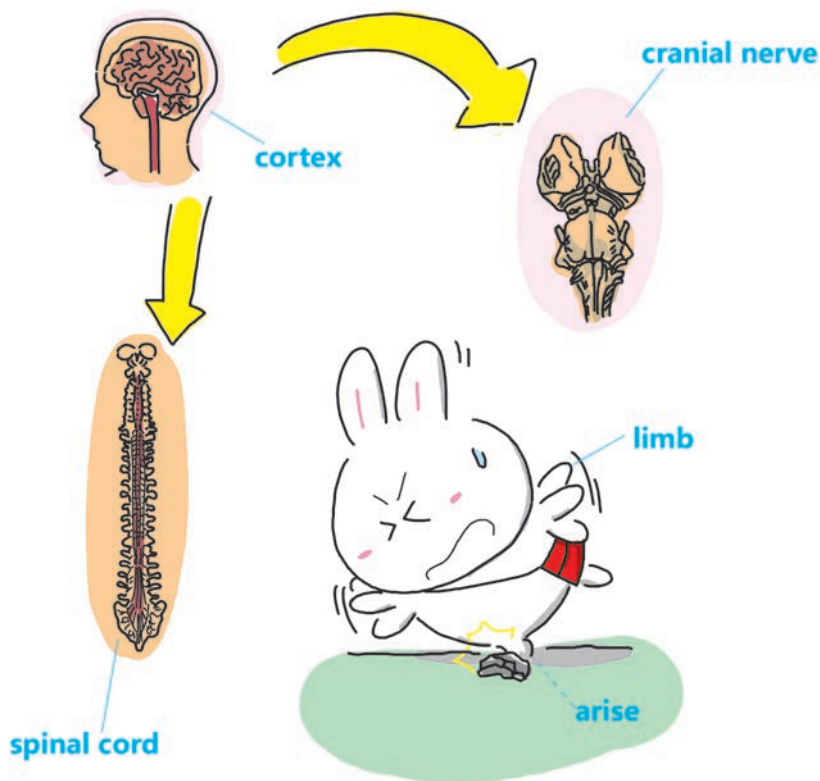


注释

- nervous system ['nə:vəs-'sistəm] n. 神经系统
 division [di'viʒən] n. 部分
 central nervous system ['sentrəl-'nə:vəs-'sistəm] n. 中枢神经系统
 spinal cord ['spainl-kɔ:d] n. 脊髓
 peripheral nervous system [pə'riferəl-'nə:vəs-'sistəm] n. 周围神经系统
 neural tissue ['njuərel-'tisju:] n. 神经组织
 motor ['məutə] n. 运动肌
 sensory system ['sensəri-'sistəm] n. 感觉系统
 impulse ['impʌls] n. 神经脉冲
 somatic nervous system [sə'mætik-'nə:vəs-'sistəm] n. 躯体神经系统
 skeletal nervous system ['skelitl-'nə:vəs-'sistəm] n. 骨骼神经系统
 autonomic nervous system [ɔ:təu'nɒmik-'nə:vəs-'sistəm] n. 自主神经系统
 voluntary ['vɒləntəri] adj. 自主的
 cardiac ['kɑ:diæk] adj. 心脏(病)的
 smooth muscle [smu:ð-'mʌsl] n. 平滑肌
 gland [glænd] n. 腺

UNIT100

Voluntary movement of head, limbs, and body is caused by nerve impulses arising in the motor area of the cortex of the brain, and carried by cranial nerves or by nerves that emerge from the spinal cord to connect with skeletal muscles.



注释

limb [lim] n. 肢

arise [ə'raɪz] v. 起, 发生

cortex ['kɔ:tɛks] n. 大脑皮层

cranial nerve ['kreɪnjəl-nə:v] n. 颅神经

emerge from [i'mə:dʒ-frəm] v. 自……出现

spinal cord ['spainl-kɔ:d] n. 脊髓

Movement may occur also in direct response to an outside stimulus; thus, a tap on the knee causes a jerk, and a light shone into the eye makes the pupil contract. These involuntary responses are called reflexes.



注释

in response to [in-'spons-tu] prep. 相应，回应

stimulus ['stimjʊləs] n. 刺激

tap [tæp] n. 轻打

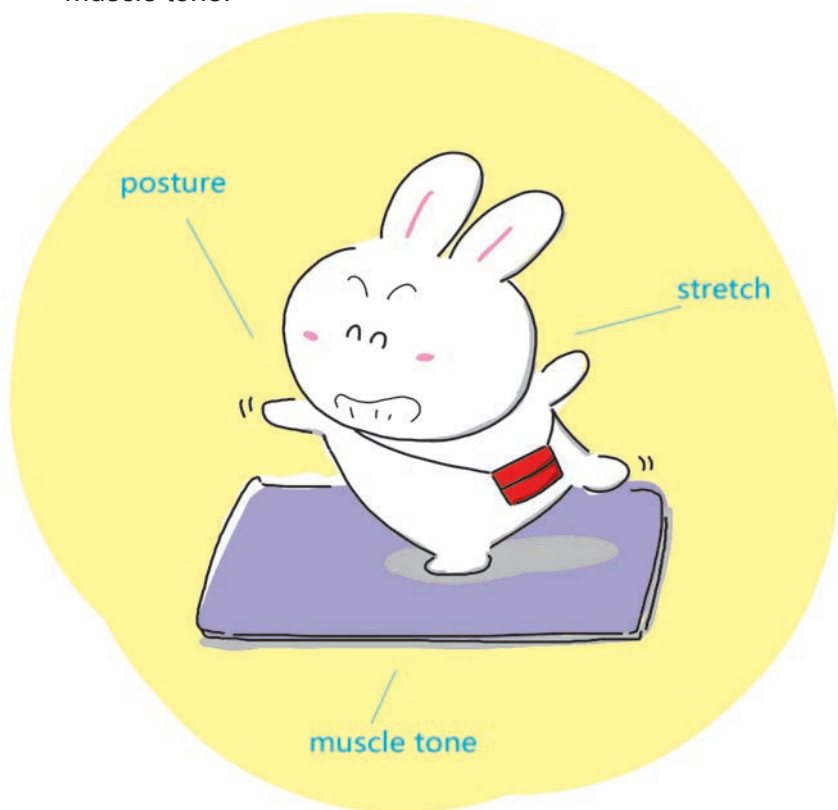
jerk [dʒə:k] n. 抽搐

pupil ['pju:pəl] n. 瞳孔

contract [kən'trækt] v. 收缩

reflex ['ri:fleks] n. 反射

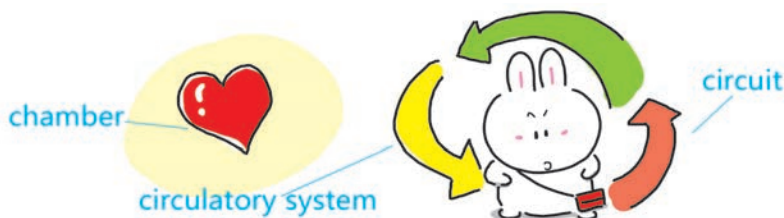
Muscular contractions do not always cause actual movement. A small fraction of the total number of fibers in most muscles are usually contracting. This serves to maintain the posture of a limb and enables the limb to resist passive elongation or stretch. This slight continuous contraction is called muscle tone.



注释

- contraction [kən'trækʃən] n. 收缩
 fraction ['frækʃən] n. 小部分
 fiber ['faɪbə] n. 纤维
 posture ['pɒstʃə] n. 姿势
 resist [rɪ'zɪst] v. 抵抗; 对抗
 passive elongation ['pæsɪv-i:lɒŋ'geɪʃən] n. 被动延伸
 stretch [stretʃ] n. 伸展
 continuous [kən'tɪnjuəs] adj. 连续的
 muscle tone ['mʌsl-təʊn] n. 肌肉弹性, 肌肉张力

In passing through the system, blood pumped by the heart follows a winding course through the right chambers of the heart into the lungs, where it picks up oxygen and back into the left chambers of the heart. From these it is pumped into the main artery, the aorta, which branches into increasingly smaller arteries until it passes through the smallest, known as arterioles. Beyond the arterioles, the blood passes through a vast amount of tiny thin-walled structures called capillaries. Here, the blood gives up its oxygen and its nutrients to the tissues, and absorbs from them carbon dioxide and other waste products of metabolism. The blood completes its circuit by passing through small veins that join to form increasingly larger vessels until it reaches the largest veins, the inferior and superior venae cavae, which return it to the right side of the heart. Blood is propelled mainly by contractions of the heart and arteries; contractions of skeletal muscle also contribute to circulation. Valves in the heart and in the veins ensure its flow in one direction.



注释

circulatory system [sə:kju'leɪtəri - 'sistem]
n. 循环系统

pump [pʌmp] v. 输送

winding ['waɪndɪŋ] adj. 弯曲的

chamber ['tʃeɪmbə] n. 心室

pick up ['pɪk-ʌp] v. 获得

oxygen ['ɒksɪdʒən] n. 氧气

artery ['ɑ:təri] n. 动脉

aorta [ei'ɔ:tə] n. 主动脉

arteriole [ɑ:'tɪəriəl] n. 小动脉

vast [vɑ:st] adj. 大量的

capillary [ke'pɪləri] n. 毛细血管

give up ['gɪv-ʌp] v. 把……输送到……

nutrient ['nju:triənt] n. 营养品

tissue ['tɪʃu:] n. 组织

absorb [əb'sɔ:b] v. 吸收

carbon dioxide ['kɑ:bən-dai'ɒksaɪd] n. 二氧化碳

metabolism [mə'tæbələzəm] n. 新陈代谢

circuit ['sɜ:kit] n. 一圈

vein [veɪn] n. 静脉

vessel ['vesəl] n. 血管

inferior [ɪn'fɪəriə] adj. 处于下方的

superior [sju'piəriə] adj. 处于上方的

venae cavae ['vi:ni:-'kɪvi:] n. 腔静脉

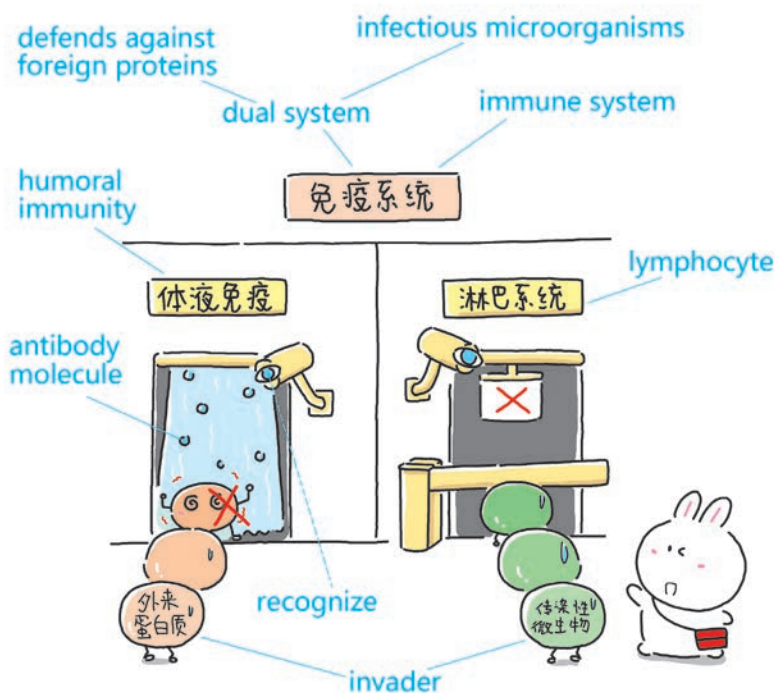
(singular: vena ['vi:nə] cava ['ki:və])

propel [prə'pel] v. 推动

circulation [sə:kju'leɪʃən] n. 循环

valve [vælv] n. 瓣膜

The body defends itself against foreign proteins and infectious microorganisms by means of a complex dual system that depends on recognizing a portion of the surface pattern of the invader. The two parts of the system are termed cellular immunity, in which lymphocytes are the effective agent, and humoral immunity, based on the action of antibody molecules.



注释

immune system ['ɪmjʊ:n-'sɪstəm] n. 免疫系统

defend [dɪ'fend] v. 防卫

foreign ['fɒrɪn] adj. 异质的, 外来的

protein ['prəʊti:n] n. 蛋白质

infectious [ɪn'fekʃəs] adj. 有传染性的

microorganism [ˌmaɪkrəʊ'ɔ:ɡənɪzəm] n. 微生物

dual system ['dju:əl-'sɪstəm] n. 双重系统

recognize ['rekəɡnaɪz] v. 认出

invader [ɪn'veɪdə] n. 侵入物

lymphocyte ['lɪmfəʊsaɪt] n. 淋巴细胞

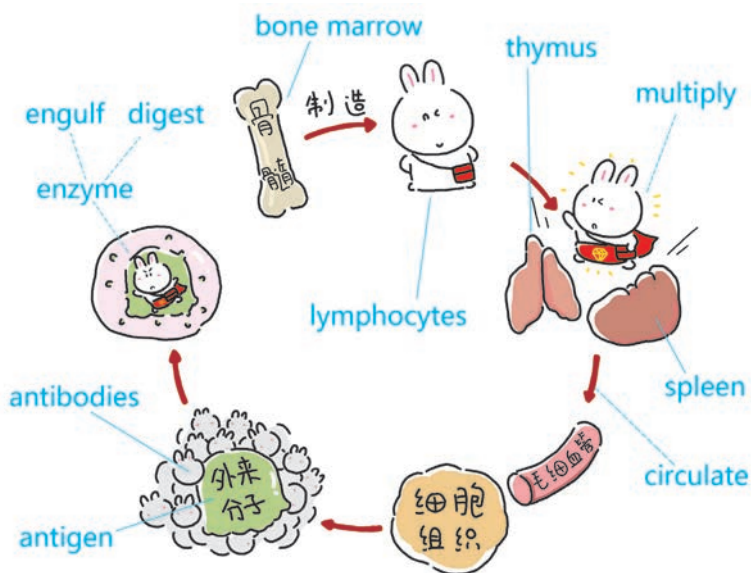
agent ['eɪdʒənt] n. 作用物

cellular immunity ['seljʊlə-'ɪmjʊ:nəti] n. 细胞免疫

humoral immunity ['hju:mərəl-'ɪmjʊ:nəti] n. 体液免疫

antibody molecule [æntɪ'bɒdi-'mɒlɪkjʊ:l] n. 抗体分子

Lymphocytes, which resemble blood plasma in composition, are manufactured in the bone marrow and multiply in the thymus and spleen. They circulate in the bloodstream, penetrating the walls of blood capillaries to reach the cells of tissues. When particular lymphocytes recognize a foreign molecular pattern (termed an antigen), they release antibodies in great numbers. Antibodies attach themselves to the antigen, and in that way mark them for destruction by other substances in the body's defense arsenal. These are primarily complement, a complex of enzymes that makes holes in foreign cells and phagocytes, cells that engulf and digest foreign matter.

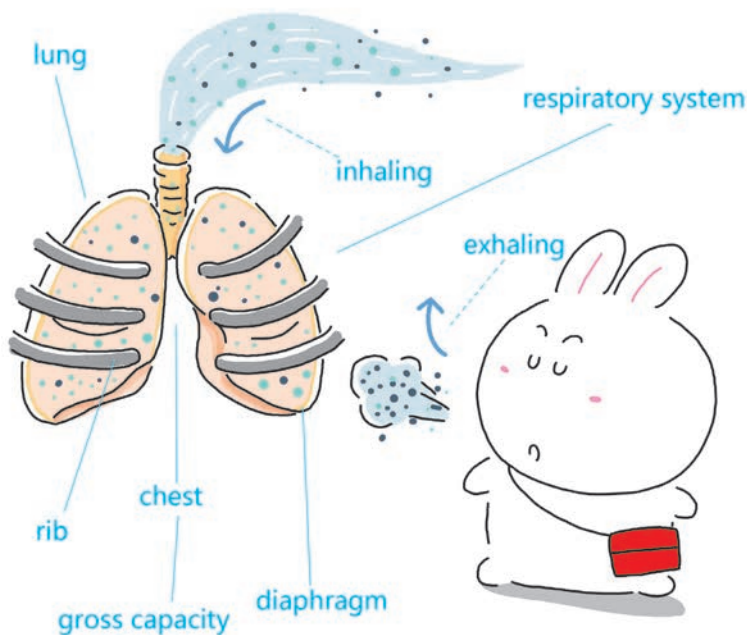


注释

resemble [riˈzeɪbl̩] v. 类似于
 plasma [ˈplæzmə] n. 血浆
 bone marrow [bəʊn-ˈmæərəʊ] n. 骨髓
 multiply [ˈmʌltɪplaɪ] v. 增加
 thymus [ˈθaɪməs] n. 胸腺
 spleen [spliːn] n. 脾
 circulate [ˈsɜːkjuleɪt] v. 循环
 penetrate [ˈpenɪtreɪt] v. 穿透
 antigen [ˈæntɪdʒən] n. 抗原

antibody [ˈæntɪˌbɒdɪ] n. 抗体
 attach to [əˈtætʃ-tuː] v. 把……放在
 arsenal [ˈɑːsənəl] n. 武器库
 complement [ˈkɒmplɪmənt] n. 补体
 enzyme [ˈenzaim] n. 酶
 phagocyte [ˈfægeusaɪt] n. 噬菌细胞
 engulf [enˈɡʌlf] v. 吞没
 digest [daɪˈdʒest] v. 消化

In the lungs, oxygen enters tiny capillaries, where it combines with hemoglobin in the red blood cells, and is carried to the tissues. Inhaling draws into the lungs air that is higher in oxygen and lower in carbon dioxide; exhaling forces from the lungs air that is higher in carbon dioxide and lower in oxygen. Changes in the size and gross capacity of the chest are controlled by the contractions of the diaphragm and of the muscles between the ribs.



注释

respiratory system [ris'piəreteri-'sistem] n. 呼吸系统

lung [lʌŋ] n. 肺

hemoglobin [hi:məu'gleubin] n. 血色素

inhale [in'heil] v. 吸气

exhale [eks'heil] v. 呼气

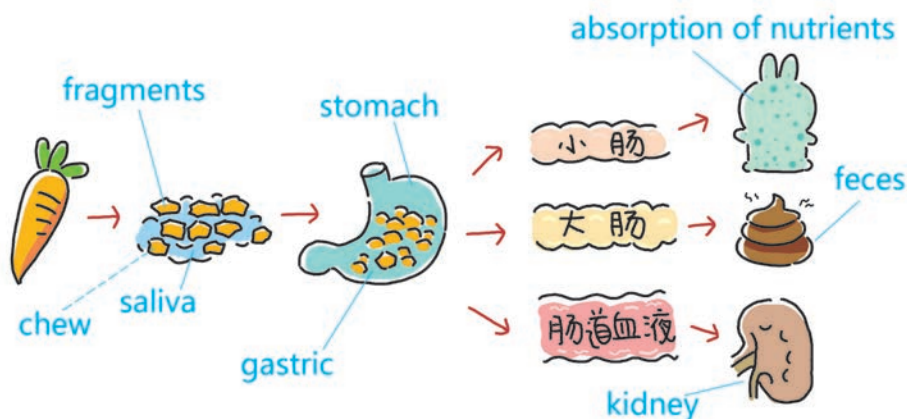
gross capacity [grəus-kə'pæsiti] n. 总容量

chest [tʃest] n. 胸腔

diaphragm ['daɪəfræm] n. 隔膜

rib [rib] n. 肋骨

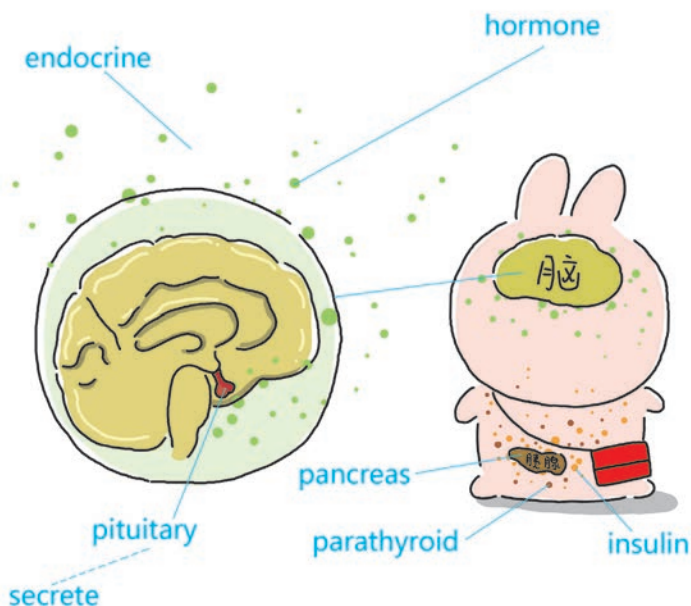
After food is broken into fragments by chewing and mixed with saliva, digestion begins. The food passes down the gullet into the stomach, where the process is continued by the gastric and intestinal juices. Thereafter, the mixture of food and secretions, called chyme, is pushed down the alimentary canal by peristalsis, rhythmic contractions of the smooth muscle of the gastrointestinal system. Absorption of nutrients from chyme occurs mainly in the small intestine; unabsorbed food and secretions and waste substances from the liver pass to the large intestine and are expelled as feces. Water and water-soluble substances travel via the bloodstream from the intestines to the kidneys, which absorb all the constituents of the blood plasma except its proteins. The kidneys return most of the water and salts to the body, while excreting other salts and waste products, along with excess water, as urine.



注释

- excretory ['eksəri:tɔ:ri] adj. 排泄的
 fragment ['frægmənt] n. 碎片
 chew [tʃu:] v. 咀嚼
 saliva [sə'laivə] n. 唾液
 digestion [di'dʒestʃən] n. 消化
 gullet ['gʌlɪt] n. 食管
 stomach ['stʌmək] n. 胃
 gastric ['gæstri:k] adj. 胃的
 intestinal [in'testɪnəl] adj. 肠的
 juice [dʒu:s] n. 体液
 secretion [si'kri:ʃən] n. 分泌物
 chyme [kaim] n. 食糜: 部分消化的食物的稠状半流体物, 从胃进入十二指肠
 alimentary canal [æli'mentəri-kə'næl] n. 消化道
 peristalsis [ˌperi'stælsɪs] n. 蠕动
 gastrointestinal [ˌgæstrəuɪn'testənəl] adj. 肠胃的
 absorption [əb'sɔ:pʃən] n. 吸收
 nutrient ['nju:triənt] n. 营养物质
 small intestine [smɔ:l-in'testɪn] n. 小肠
 waste substance [weɪst-'sʌbstəns] n. 废物
 liver ['lɪvə] n. 肝脏
 large intestine [lɑ:dʒ-in'testɪn] n. 大肠
 expel [ɪks'pel] v. 排出
 feces ['fi:si:z] n. 粪便
 water-soluble ['wɔ:tə-sɒljəbəl] n. 可溶于水的
 via ['vaɪə] prep. 通过
 kidney ['kɪdni] n. 肾
 constituent [kən'stitjuənt] n. 成分
 excrete [eks'kri:t] v. 排泄, 分泌
 urine ['juəri:n] n. 尿

In addition to the integrative action of the nervous system, the control of various body functions is exerted by the endocrine glands. An important part of this system, the pituitary, lies at the base of the brain. This master gland secretes a variety of hormones. The posterior lobe of the pituitary secretes vasopressin, which acts on the kidney to control the volume of urine. Other glands in the endocrine system are the pancreas, which secretes insulin and the parathyroid, which secretes a hormone that regulates the quantity of calcium and phosphorus in the blood.

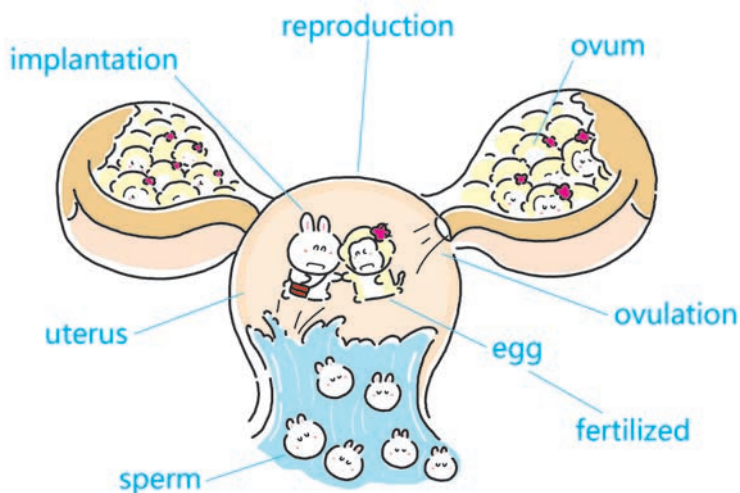


注释

endocrine ['endaukrain] adj. 内分泌的
 exert [ig'zə:t] v. 行使 (职权等), 使受 (影响等)
 gland [glænd] n. 腺
 pituitary [pi'tju:itəri] n. (脑) 垂体
 secrete [si'kri:t] v. 分泌
 hormone ['hɔ:məun] n. 荷尔蒙
 posterior [pɔ'stiəriə] adj. 后面的, 尾部的
 posterior lobe [pɔ'stiəriə-ləub] n. 后叶

vasopressin [væsəu'presin] n. 血管加压素, 加压素
 pancreas ['pæŋkriəs] n. 胰, 胰腺
 insulin ['insjulin] n. 胰岛素
 parathyroid [pære'θairɔid] adj. 副甲状腺的
 regulate ['regjuleit] v. 控制
 calcium ['kælsiəm] n. 钙
 phosphorus ['fɒsfərəs] n. 磷

Reproduction is accomplished by the union of male sperm and the female ovum. In coitus, the male organ ejaculates more than 250 million sperm into the vagina, from which some make their way to the uterus. Ovulation, the release of an egg into the uterus, occurs approximately every 28 days; during the same period, the uterus is prepared for the implantation of a fertilized ovum by the action of estrogens. If a male cell fails to unite with a female cell, other hormones cause the uterine wall to slough off during menstruation. From puberty to menopause, the processes of ovulation, preparation, and menstruation are repeated monthly except for periods of pregnancy.

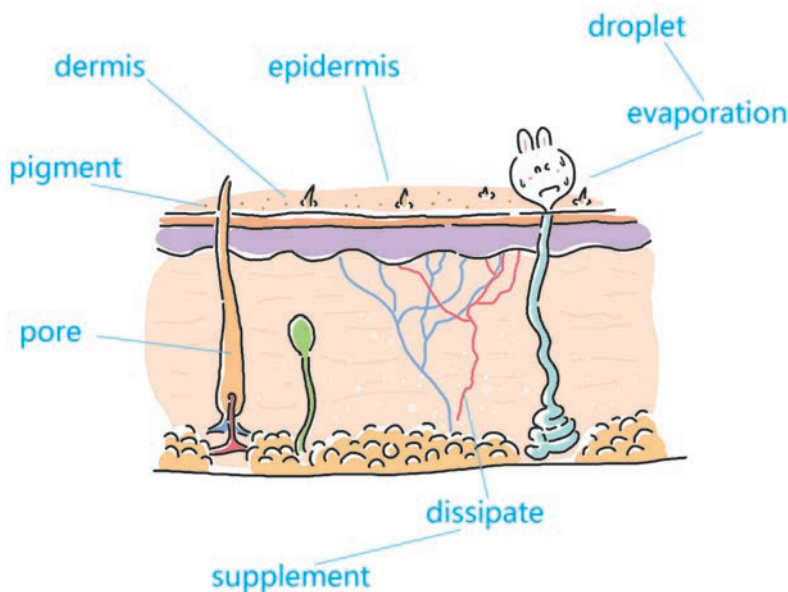


注释

reproduction [ˌriːprəˈdʌkʃən] n. 繁殖, 生殖
 accomplish [əˈkɒmplɪʃ] v. 完成
 sperm [spɜːm] n. 精子
 ovum [ˈəʊvəm] n. 卵子
 coitus [ˈkɔɪtəs] n. 性交
 ejaculate [iˈdʒækjuleɪt] v. 射精
 vagina [vəˈdʒaɪnə] n. 阴道
 uterus [ˈjuːtərəs] n. 子宫
 ovulation [ˌɒvjuˈleɪʃən] n. 排卵

egg [eg] n. 卵
 implantation [ˌɪmplɑːnˈteɪʃən] n. 胚胎植入
 fertilized [ˈfɜːtɪlaɪzɪd] adj. 已受精的
 estrogen [ˈestɹədʒən] n. 雌激素
 slough off [ˈslau-ɔf] v. 脱落
 menstruation [ˌmenstruˈeɪʃən] n. 月经期间
 puberty [ˈpjʊːbɪti] n. 青春期
 menopause [ˈmeneˌpəʊz] n. 绝经期
 pregnancy [ˈpregnənsi] n. 怀孕

The inner layer of the skin, called the dermis, contains sweat glands, blood vessels, nerve endings, and the bases of hair and nails. The outer layer, the epidermis, is only a few cells' thick. It contains pigments, pores, and ducts, and its surface is made of dead cells that it sheds from the body. The sweat glands excrete waste and cool the body through evaporation of fluid droplets; the blood vessels of the dermis supplement temperature regulation by contracting to preserve body heat and expanding to dissipate it. Separate kinds of receptors convey pressure, temperature, and pain. Fat cells in the dermis insulate the body, and oil glands lubricate the epidermis.



注释

dermis ['dɜ:mɪs] n. 真皮

nail [neɪl] n. 指甲

epidermis [ˌepɪ'dɜ:mɪs] n. 表皮

pigment ['pɪgmənt] n. 色素

pore [pɔ:] n. 毛孔

duct [dʌkt] n. 管：管状体内通道，尤指运输腺分泌物的管道

shed [ʃed] v. 脱落，蜕（皮），脱（毛）

excrete [eks'kri:t] v. 分泌，排出

evaporation [i,væpə'reɪʃən] n. 蒸发；蒸腾（作用）

droplet ['drɒplɪt] n. 小滴

supplement ['sʌplɪmənt] v. 补充

dissipate ['dɪsɪpeɪt] v. 使耗散

receptor [rɪ'septə] n. 感受器

convey [kən'veɪ] v. 转达

insulate ['ɪnsjuleɪt] v. 隔热

lubricate ['lu:brikeɪt] v. 使润滑

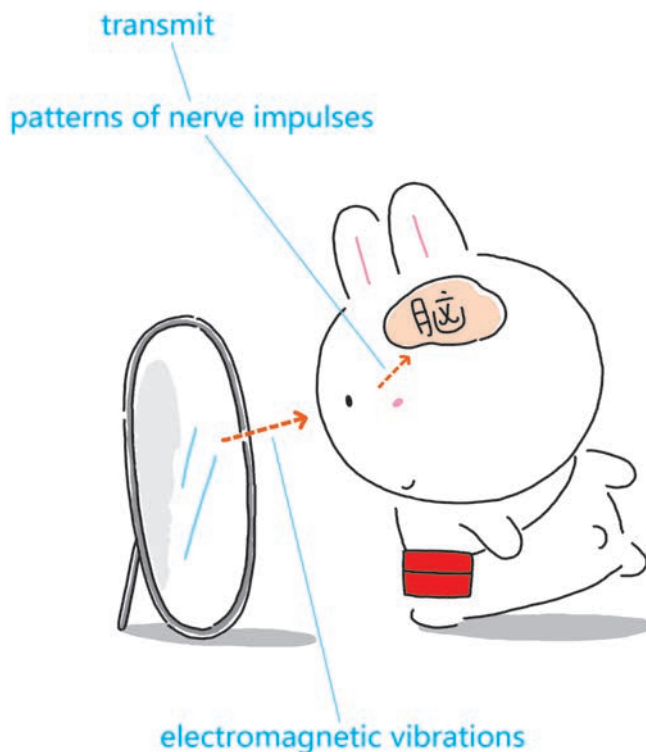
Only vertebrates, or animals with backbones, have ears. Invertebrate animals, such as jellyfish and insects, lack ears, but have other structures or organs that serve similar functions. Human ears are capable of perceiving an extraordinarily wide range of changes in loudness, the tiniest audible sound being about 1 trillion times less intense than a sound loud enough to cause the ear pain. Besides loudness, the human ear can detect a sound's pitch, which is related to a sound's vibration frequency, and of course, the tone. Another sonic phenomenon, known as masking, occurs because lower-pitched sounds tend to deafen the ear to higher-pitched sounds.



注释

- jellyfish ['dʒelɪfɪʃ] n. 水母
 lack [læk] v. 缺乏; 缺少
 perceive [pə'si:v] v. 察觉
 extraordinarily [ik'strɔ:dənərɪli] adv. 格外地
 loudness ['laʊdnɪs] n. 响度
 audible ['ɔ:dəbl] adj. 听得见的
 detect [dɪ'tekt] v. 察觉到
 pitch [pɪtʃ] n. 音高
 vibration consequence [vaɪ'breɪʃən-'kɒnsɪkwəns] n. 振动频率
 sonic ['sɒnɪk] adj. 声音的
 masking ['mɑ:skɪŋ] n. 遮蔽
 deafen ['defn] v. 使聋

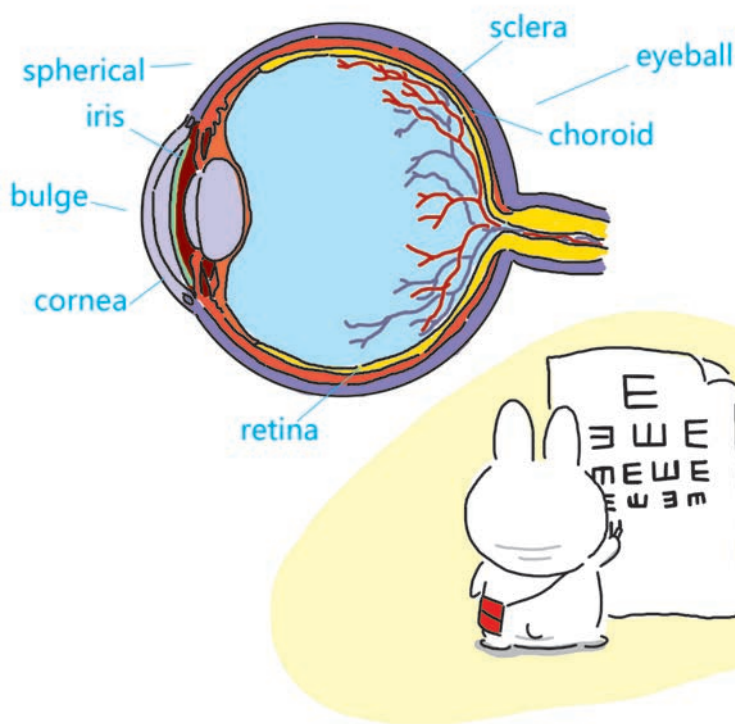
The actual process of seeing is performed by the brain rather than by the eye. The function of eyes is to translate the electromagnetic vibrations of light into patterns of nerve impulses that are transmitted to the brain.



注释

- process ['prəuses] n. 过程
 perform [pə'fɔ:m] v. 执行
 electromagnetic [iˌlektərəmæg'netik] adj. 电磁的
 vibration [vai'breɪʃən] n. 振动
 pattern ['pætən] n. 模式
 transmit [trænz'mɪt] v. 传送, 传导

The human eyeball, is a spherical structure with a pronounced bulge on its forward surface. The outer part of the eye is composed of three layers of tissue. The outside layer is the sclera, a protective coating. At the front of the eyeball, it is continuous with the bulging transparent cornea. The middle layer of the coating of the eye is the choroid, which is continuous with the ciliary body, and with the iris, which lies at the front of the eye. The innermost layer is the light-sensitive retina.

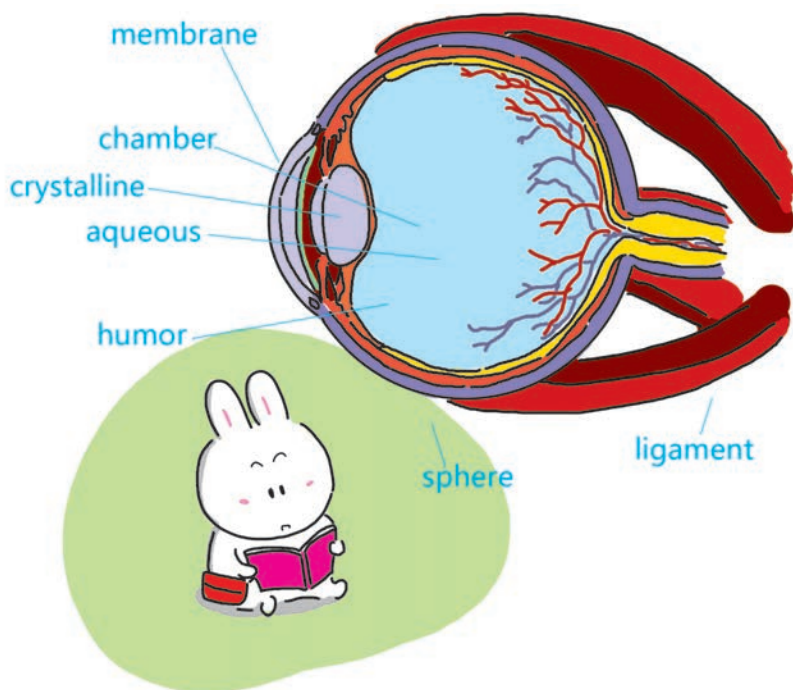


注释

eyeball ['aɪbɔ:l] n. 眼球
 spherical ['sferɪkəl] adj. 球形的
 pronounced [prə'naʊnst] adj. 显著的
 bulge [bʌldʒ] n. 凸出部分
 sclera ['skliərə] n. 巩膜
 coating ['kəʊtɪŋ] n. 覆盖层

cornea ['kɔ:niə] n. 角膜
 choroid ['kɔ:ɹɔɪd] n. 脉络膜
 ciliary body ['sɪliəri-'bɒdɪ] n. 睫状体
 iris ['aɪərɪs] n. 虹膜
 innermost ['ɪnə,məʊst] adj. 最里面的
 retina ['retnə] n. 视网膜

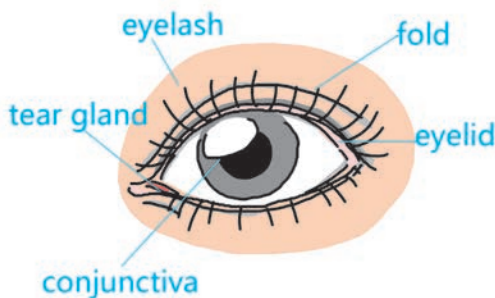
The cornea is a tough five-layered membrane through which light is admitted to the interior of the eye. Behind the cornea is a chamber filled with clear watery fluid, the aqueous humor, which separates the cornea from the crystalline lens. The lens itself is a flattened sphere constructed of a large number of transparent fibers arranged in layers. It is connected by ligaments to a ringlike muscle. This muscle, by flattening the lens or making it more nearly spherical, changes its focal length.



注释

- membrane ['membrein] n. 膜
 admit [əd'mit] v. 许可进入
 chamber ['tʃeɪmbə] n. 有机物体内部封闭的腔；室
 aqueous ['eɪkwɪəs] adj. 水的
 humor ['hju:mə] n. 体液
 crystalline ['krɪstəlɪn] adj. 水晶的
 lens [lenz] n. 透镜，镜头
 flatten ['flætən] v. 使变平
 sphere [sfɪə] n. 球(体)
 ligament ['lɪɡəmənt] n. 系带，韧带
 focal ['fəʊkəl] adj. 焦点的

Several structures, not parts of the eyeball, contribute to the protection of the eye. The most important of these are the eyelids, two folds of skin and tissue, upper and lower, that can be closed by means of muscles to form a protective covering over the eyeball against excessive light and mechanical injury. The eyelashes, a fringe of short hairs growing on the edge of either eyelid, act as a screen to keep dust particles and insects out of the eyes when the eyelids are partly closed. Inside the eyelids is a thin protective membrane, the conjunctiva, which doubles over to cover the visible sclera. Each eye also has a tear gland, situated at the outside corner of the eye. The salty secretion of these glands lubricates the forward part of the eyeballs when the eyelids are closed, and flushes away any small dust particles or other foreign matter on the surface of the eyes.



注释

eyelid ['aɪlɪd] n. 眼睑

fold [fəʊld] n. 层

excessive [ɪk'sesɪv] adj. 过度的, 过分的

injury ['ɪndʒəri] n. 伤害

eyelash ['aɪləʃ] n. 睫毛

fringe [frɪndʒ] n. (地毯等的) 穗, 须边, 流苏

conjunctiva [kɒndʒʌŋk'taɪvə] n. (眼球) 结膜

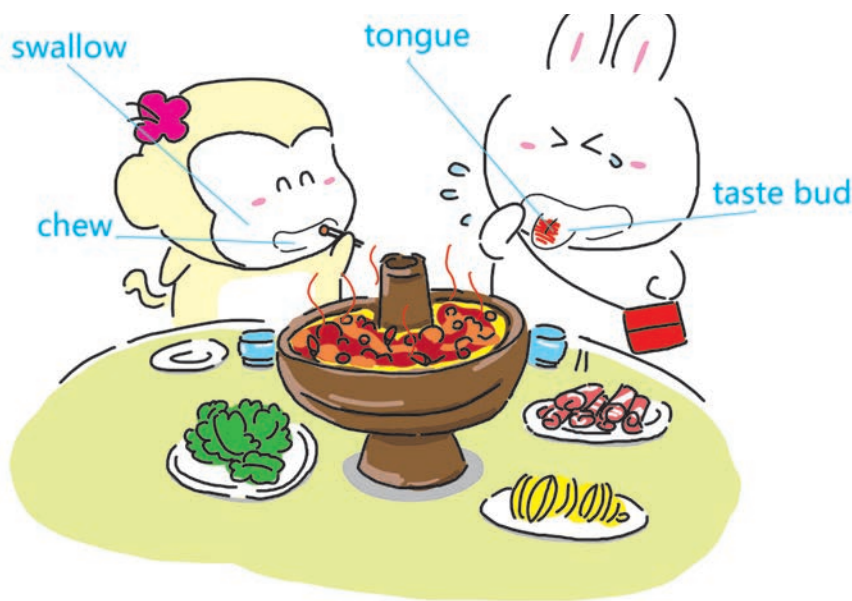
tear gland ['tiə-'glænd] n. 泪腺

lubricate ['lu:brikeit] v. 加油润滑

flush [flʌʃ] v. 冲掉

particle ['pɑ:tɪkl] n. 微粒

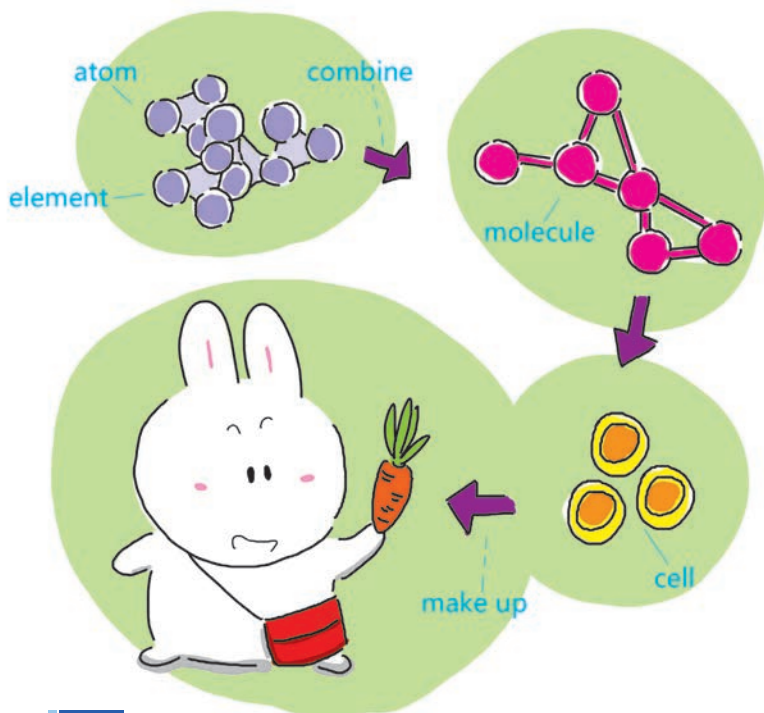
The tongue serves as an organ of taste, with taste buds scattered over its surface and concentrated toward the back of the tongue. In chewing, the tongue holds the food against the teeth; in swallowing, it moves the food back into the pharynx, and then into the esophagus when the pressure of the tongue closes the opening of the trachea or windpipe. It also acts, together with the lips, teeth, and hard palate, to form word sounds.



注释

- tongue [tʌŋ] n. 舌头
 taste bud ['teɪst-'bʌd] n. 味蕾
 scatter over ['skæ-tə-'əʊvə] v. 分布在……
 chew [tʃu:] v. 咀嚼
 swallow ['swɒləʊ] v. 吞咽
 pharynx ['færɪŋks] n. 咽
 esophagus [i(:)'sɒfəgəs] n. 食道
 trachea [trə'ki:ə] n. 气管
 windpipe ['wɪndpaɪp] n. 气管
 hard palate [hɑ:d-'pæli:t] n. 硬腭

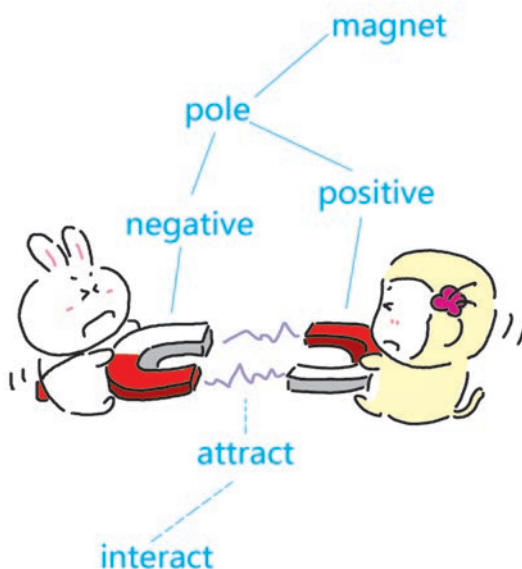
Understanding atoms is key to understanding the physical world. More than 100 different elements exist in nature, each with its own unique atomic makeup. The atoms of these elements react with one another, and combine in different ways to form a virtually unlimited number of chemical compounds. When two or more atoms combine, they form a molecule. Millions of molecules join together to make up the cells in humans or in any other plants, or animals.



注释

- atom ['ætəm] n. 原子
 element ['eliment] n. 元素
 unique [ju:'ni:k] adj. 独特的
 makeup ['meikʌp] n. 结构
 react [ri'ækt] v. 起反应
 combine [kəm'beɪn] v. 结合
 virtually ['vɜ:tʃuəli] adv. 实际上
 compound ['kɒpaund] n. 化合物
 molecule ['mɒlikju:l] n. 分子
 cell [sel] n. 细胞
 make up ['meɪk-ʌp] v. 构成, 组成

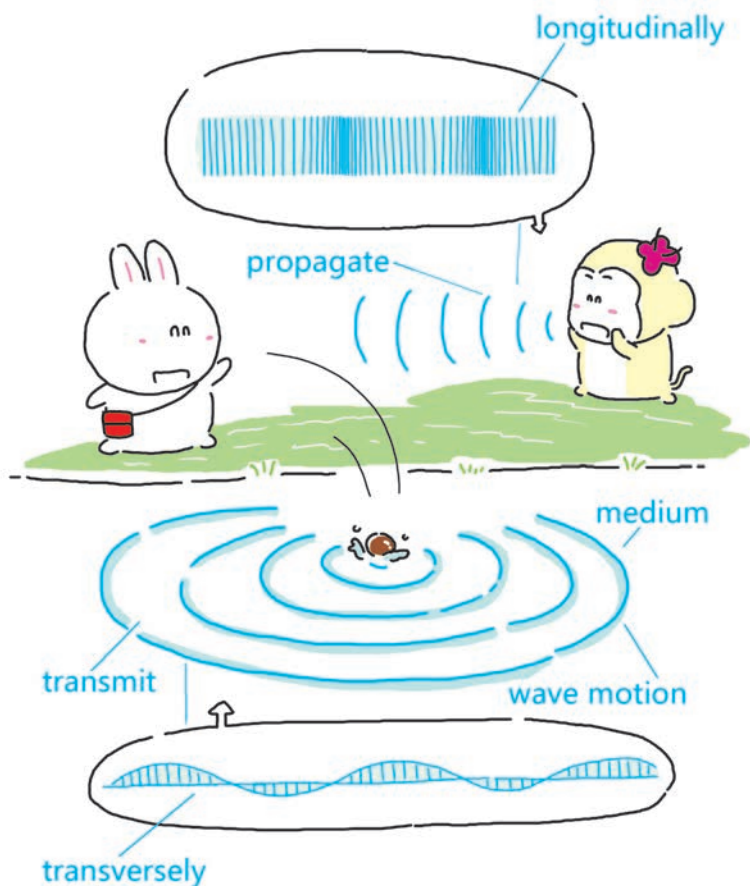
Atoms are made of smaller particles. An atom consists of a cloud of electrons, surrounding a small dense nucleus of protons and neutrons. Electrons and protons have a property called electric charge, which affects the way they interact with each other and with other electrically charged particles. Electrons carry a negative electric charge, while protons have a positive electric charge. The negative charge is the opposite of the positive charge, and, like the opposite poles of a magnet, these opposite electric charges attract one another. Conversely, like charges (negative and negative, or positive and positive) repel one another. The attraction between an atom's electrons and its protons holds the atom together. The nucleus contains nearly all of the mass of the atom, but it occupies only a tiny fraction of the space inside the atom. If an atom was magnified until it was as large as a football stadium, the nucleus would be about the size of a grape. Atoms of the same element that differ in mass number are called isotopes.



注释

particle ['pɑ:tɪkl] n. 微粒
 electron [i'lektrɒn] n. 电子
 dense [dens] adj. 密集的
 nucleus ['nju:kliəs] n. 原子核
 proton ['prəʊtɒn] n. 质子
 neutron ['nju:trɒn] n. 中子
 electric charge [i'lektrik-tʃɑ:dʒ] n. 电荷
 interact [ˌɪntər'ækt] v. 相互作用
 negative ['negətɪv] adj. 负的
 positive [pə'zɪtɪv] adj. 正的
 opposite [ˈɒpəzɪt] n. 对立物
 pole [pəʊl] n. 磁极, 电极
 magnet ['mæɡnɪt] n. 磁铁
 conversely [ˈkɒnvə:sli] adv. 相反地
 like charges ['laɪk-tʃɑ:dʒɪz] n. 同性电荷
 attract [ə'trækt] v. 吸引
 repel [ri'pel] v. 排斥
 attraction [ə'trækʃən] n. 引力
 mass [mæs] n. 质量
 occupy [ˈɒkjʊpaɪ] v. 占用
 fraction ['frækʃən] n. 小部分
 magnify ['mæɡnɪfaɪ] v. 放大
 stadium ['steɪdɪəm] n. 体育场
 property [ˈprɒpəti] n. 性质
 grape [greɪp] n. 葡萄
 mass number ['mæs-'nʌmbə] n. (原子)质量数
 isotope ['aɪsəʊtəʊp] n. 同位素

In general, waves can be propagated or transmitted, transversely or longitudinally. In both cases, only the energy of wave motion is propagated through the medium; no portion of the medium itself actually moves very far.



注释

- propagate ['prɒpeɪt] v. 传播
 transmit ['trænz'mɪt] v. 传导
 transversely ['trænzvə'sli] adv. 横向地
 longitudinally [lɒndʒɪ'tjuːdɪnli] adv. 纵向地
 wave motion ['weɪv-'məʊʃən] n. 波动
 medium ['miːdɪəm] n. 介质

As the energy of wave motion is propagated outward from the center of disturbance, the individual air molecules that carry the sound move back and forth, parallel to the direction of wave motion. Thus, a sound wave is a series of alternate increases and decreases of air pressure. Each individual molecule passes the energy on to neighboring molecules, but after the sound wave has passed, each molecule remains in about the same location.



注释

energy ['enədʒi] n. 能量

disturbance [dis'tɜ:bəns] n. 扰动

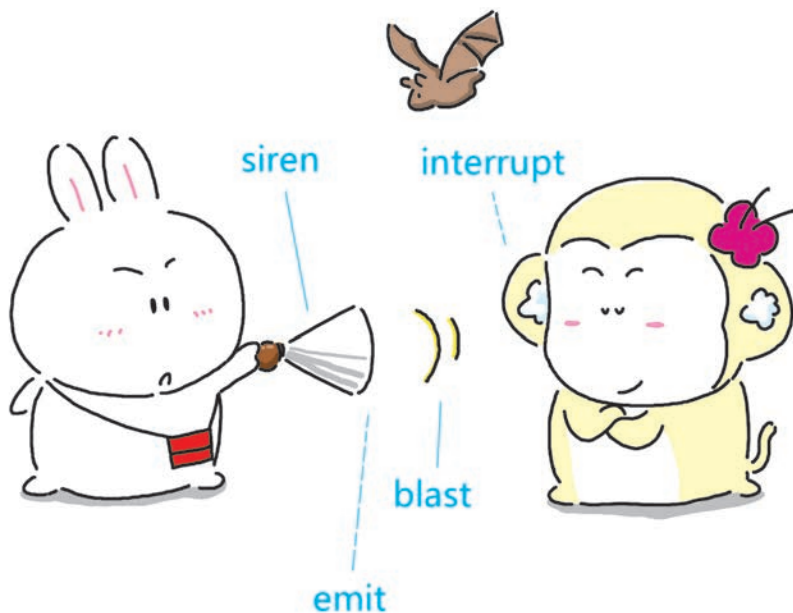
back and forth ['bæk-ænd-'fɔ:θ] adv. 来回地

alternate [ɔ:l'tə:nit] adj. 交替的

air pressure [εə-'preʃə] n. 气压

neighboring ['neibəriŋ] adj. 邻近的

Sounds can be produced at a desired frequency by different methods. Sirens emit sound by means of an air blast interrupted by a toothed wheel with 44 teeth.



注释

frequency ['fri:kwənsi] n. 频率

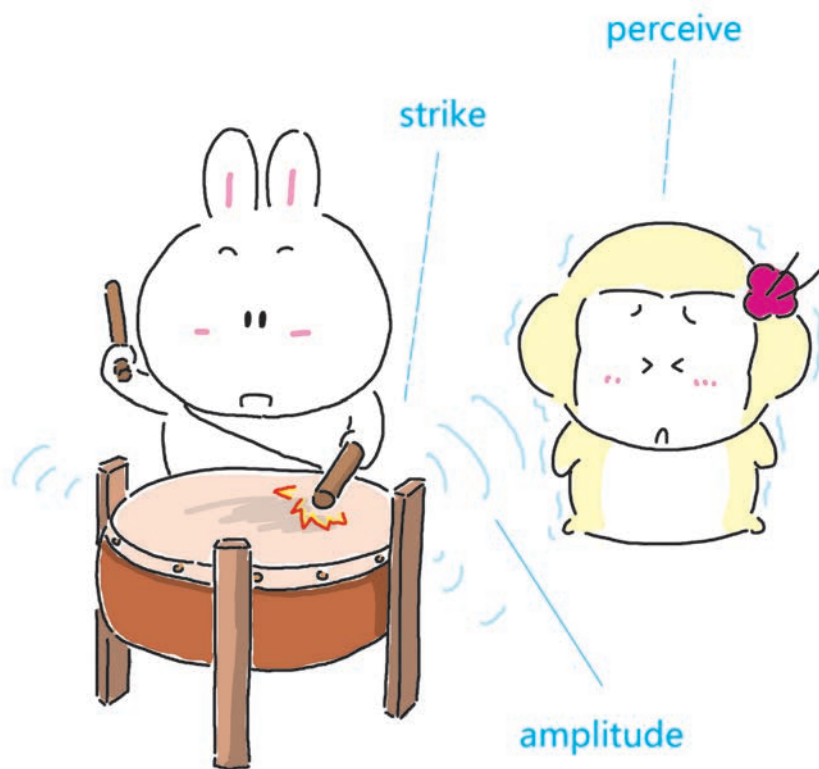
siren ['saɪərən] n. 汽笛

emit ['i:mɪt] v. 发出

blast [blɑ:st] n. 一股 (强而突然的气流)

interrupt [ˌɪntə'rapt] v. 中断

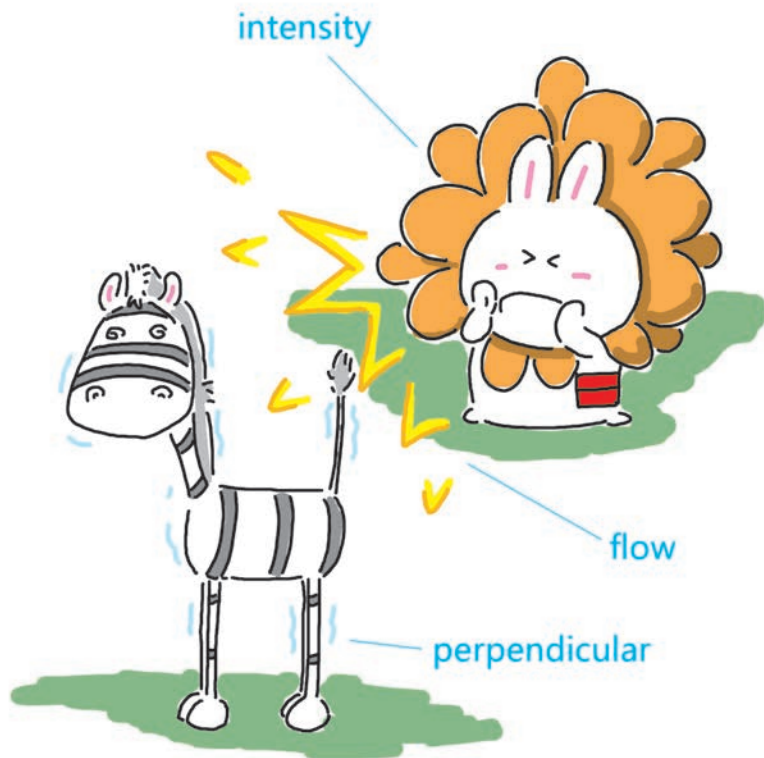
The amplitude of a sound wave is the degree of motion of air molecules within the wave, which corresponds to the changes in air pressure that accompany the wave. The greater the amplitude of the wave, the harder the molecules strike the eardrum, and the louder the sound that is perceived.



注释

amplitude [ˈæmplɪtjuːd] n. 振幅
 accompany [əˈkʌmpəni] v. 伴随
 strike [straɪk] v. 击打
 eardrum [ˈiədɹʌm] n. 耳膜
 perceive [pəˈsiːv] v. 感到

The distance at which a sound can be heard depends on its intensity. Intensity is the average rate of flow of energy per unit area perpendicular to the direction of propagation, similar to the rate at which a river flows through a gate in a dam.



注释

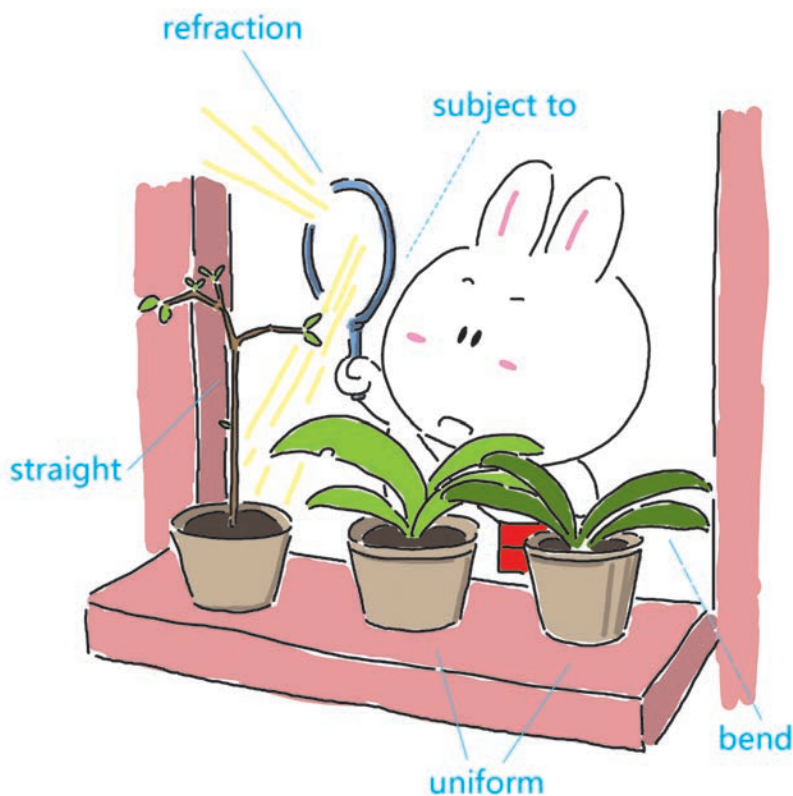
intensity [in'tensiti] n. 强度

flow [fləu] n. 传播

perpendicular [ˌpə:pən'dikjulə] adj. 垂直的

dam [dæm] n. 坝

Sound moves forward in a straight line when traveling through a medium having uniform density. Like light, however, sound is subject to refraction, which bends sound waves from their original path.



注释

straight [streit] adj. 笔直的

uniform ['ju:nifo:m] adj. 相同的

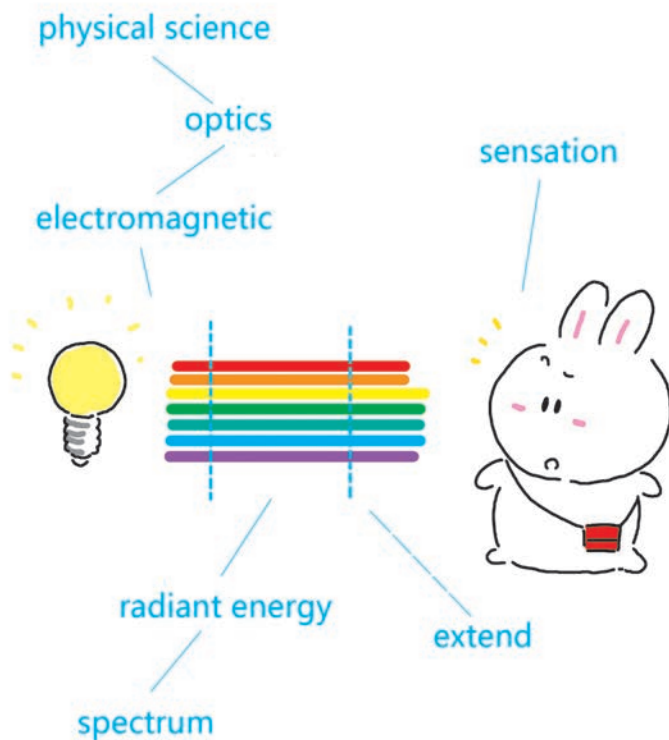
density ['densiti] n. 密度

subject to ['sʌbdʒikt-tu:] v. 受……支配

refraction [ri'frækʃən] n. 折射

bend [bend] v. 使弯曲

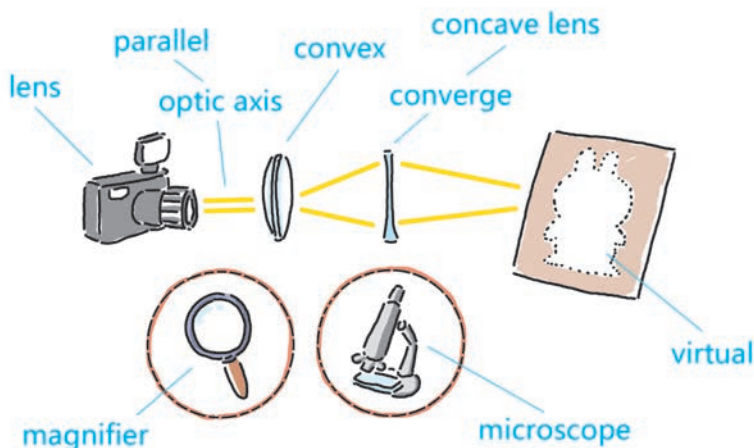
Optics is a branch of physical science dealing with the propagation and behavior of light. In a general sense, light is that part of the electromagnetic spectrum that extends from X rays to microwaves, and includes the radiant energy that produces the sensation of vision.



注释

- optics ['ɒptiks] n. 光学
 physical science ['fɪzɪkəl-'saɪəns] n. 物理科学
 propagation [,prɒpə'geɪʃən] n. 传播
 electromagnetic [i,lekt'rəʊmæg'netɪk] adj. 电磁的
 spectrum ['spektrəm] n. 光谱
 extend [ɪks'tend] v. 延伸
 X ray ['eks-reɪ] n. X光
 microwave ['maɪkrəweɪv] n. 微波
 radiant energy ['reɪdiənt-'enədʒi] n. 放射性能量
 sensation [sen'seɪʃən] n. 感觉

Lenses made with surfaces of small radius have shorter focal lengths. A lens with two convex surfaces will always refract rays parallel to the optic axis, so that they converge to a focus on the side of the lens opposite to the object. Concave lenses form only virtual, erect, and diminished images. If the object distance is smaller than the focal length of the lens, the image is virtual, erect, and larger than the object. Observer is then using the lens as a magnifier or simple a microscope. The angle subtended at the eye by this virtual enlarged image is greater than would be the angle subtended by the object if it is at the normal viewing distance. The ratio of these two angles is the magnifying power of the lens.

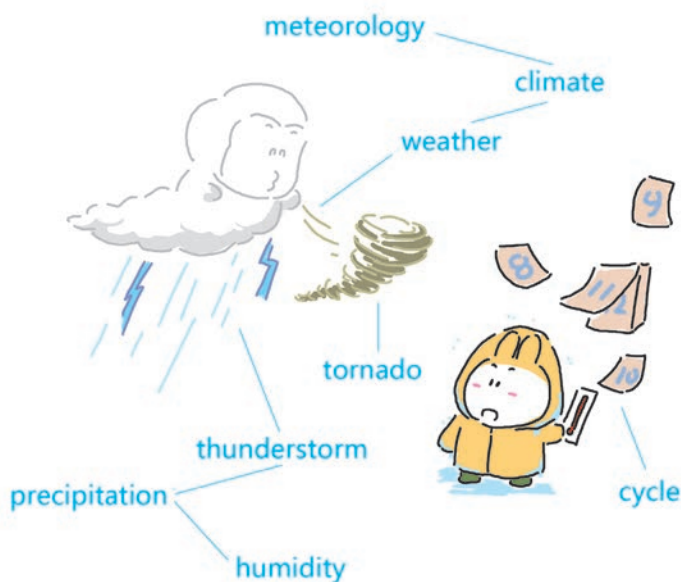


注释

lens [lenz] n. 镜头
 radius ['reɪdʒəs] n. 半径(plural: radii ['reɪdiəi])
 focal length ['fəʊkəl-lenθ] n. 焦距
 convex ['kɒn'veks] adj. 凸的, 凸面的
 refract [ri'frækt] v. 使折射
 parallel ['pærəlel] adj. 平行的
 optic axis ['ɒptɪk-'æksɪs] n. 光轴
 converge [kən've:dʒ] v. 会聚
 concave lens [kɒn'keɪv - lenz] n. 凹透镜

erect [i'rekt] adj. 竖立的
 diminish [dɪ'mɪnɪʃ] v. 减少
 virtual ['vɜ:tʃuəl] adj. 虚的
 magnifier ['mæɡnɪfaɪə] n. 放大镜
 microscope ['maɪkrəskəʊp] n. 显微镜
 subtend [səb'tend] v. 对向
 ratio ['reɪʃiəʊ] n. 比, 比率
 angle ['æŋɡl] n. 角, 角度

The elements of weather include temperature, humidity, cloudiness, precipitation, wind, and pressure. These elements are organized into various weather systems, such as monsoons, areas of high and low pressure, thunderstorms, and tornadoes. All weather systems have well-defined cycles and structural features, and are governed by the laws of heat and motion. These conditions are studied in meteorology, the science of weather and weather forecasting. Weather differs from climate, which is the weather that a particular region experiences over a long period of time.

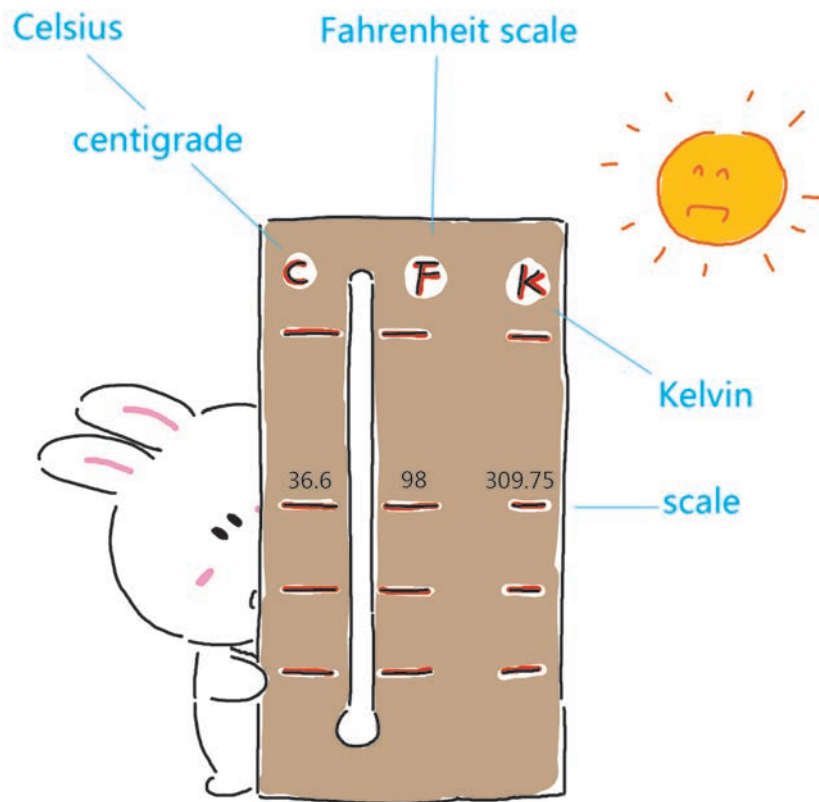


注释

weather ['weðə] n. 天气
 humidity ['hju:'mɪdɪtɪ] n. 湿度
 precipitation [pri:si'pi:teɪʃən] n. 降水
 pressure ['preʃə] n. (气) 压
 various ['vɛəriəs] adj. 不同的, 好几个
 monsoon [mɒn'su:n] n. 季风
 thunderstorm ['θʌndəstɔ:m] n. 雷雨
 tornado [tɔ:'neɪdəu] n. 龙卷风
 well-defined [,wel-dɪ'faɪnd] adj. 明确的

cycle ['saɪkl] n. 循环, 周期
 feature ['fi:tʃə] n. 特征
 govern ['gʌvən] v. 支配
 motion ['məʊʃən] n. 运动
 meteorology [mi:tiə'ɒlədʒi] n. 气象学
 weather forecast ['weðə-'fɔ:kə:st] n. 天气预报
 region ['ri:dʒən] n. 地区
 climate ['klaɪmɪt] n. 气候

Scientists use the Kelvin, or absolute, scale and the Celsius, or centigrade, scale to measure temperature. Most nations use the Celsius scale, although the United States continues to use the Fahrenheit scale.



注释

Kelvin ['kelvin] n. 开氏温标

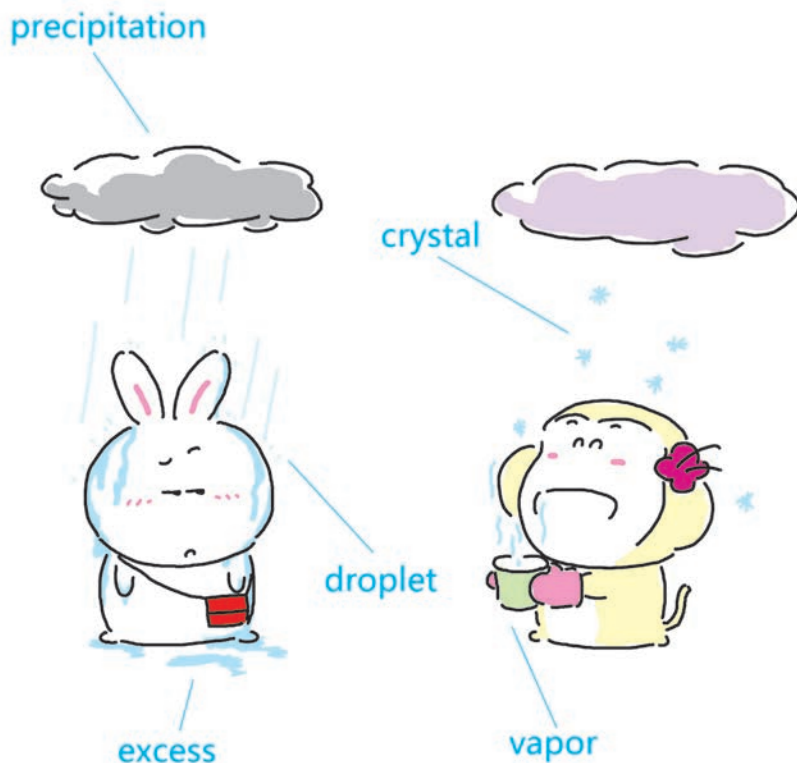
scale [skeil] n. 刻度

Celsius ['selsjəs] adj. 摄氏的

centigrade ['sentigreid] adj. 摄氏温度的

Fahrenheit scale ['færən, hait-skeil] n. 华氏温标

Most clouds and almost all precipitation are produced by the cooling of air as it rises. When air temperature is reduced, excess water vapor in the air condenses into liquid droplets or ice crystals to form clouds or fog.



注释

precipitation [priˌsɪpiˈteɪʃən] n. 降水

reduce [riˈdjuːs] v. 减少

excess [ˈekses] adj. 过量的

vapor [ˈveɪpə] n. 蒸气

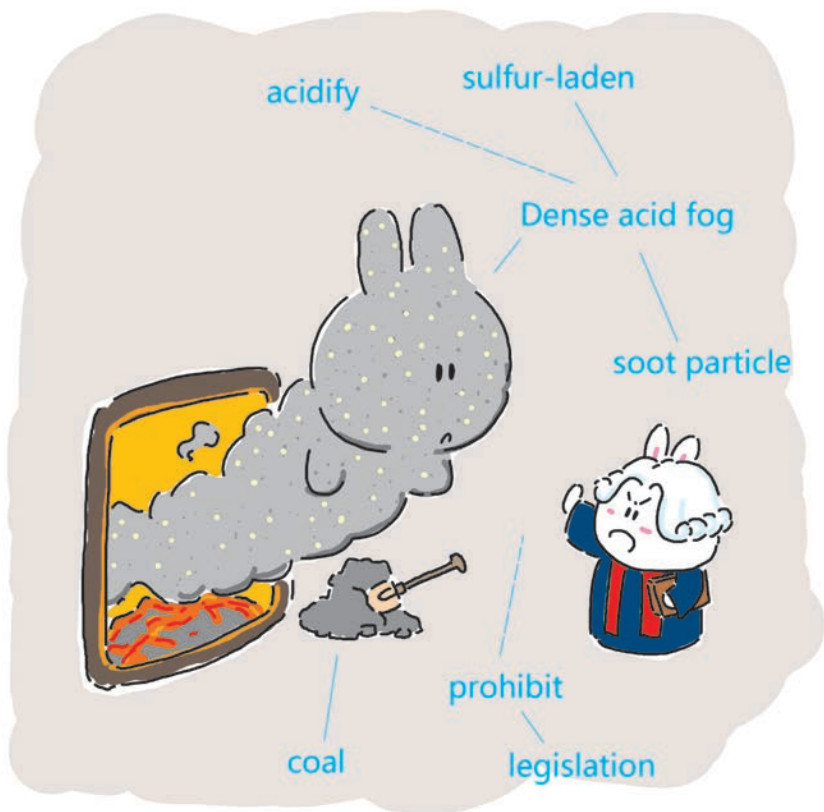
condense [kənˈdens] v. 浓缩, 凝结

droplet [ˈdrɒplɪt] n. 小滴

crystal [ˈkrɪstəl] n. 晶体

fog [fɒɡ] n. 雾

Fog is thickened and acidified when the air is filled with sulfur-laden soot particles produced by the burning of coal. Dense acid fogs that killed thousands of people in London up to 1956 led to legislation that prohibited coal burning in cities.



注释

thicken ['θɪkən] v. 使变厚, 使变浓

acidify [ə'sɪdaɪ] v. 酸化

sulfur-laden ['sʌlfər-'leɪdn] adj. 含有硫黄的

soot particle ['sʊt-'pɑ:tɪkl] n. 煤烟颗粒

coal [kəʊl] n. 煤

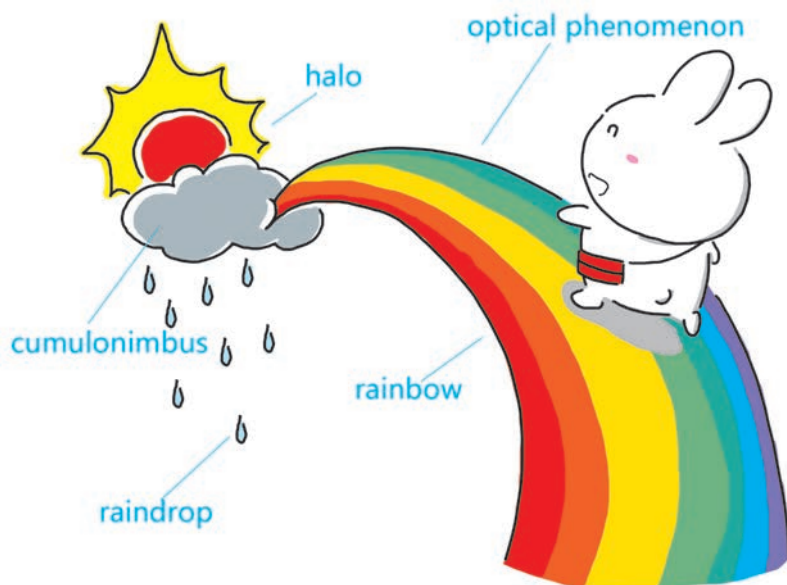
dense [dens] adj. (烟、雾等) 浓密的

acid fog ['æsɪd-fɒg] n. 酸雾

legislation [ˌledʒɪs'leɪʃən] n. 法规、法案, 立法

prohibit [prə'hɪbɪt] v. 禁止

Optical phenomena, such as rainbows and halos, occur when light shines through cloud particles. Rainbows are seen when sunlight from behind the observer strikes the raindrops falling from cumulonimbus clouds. The raindrops act as tiny prisms, bending and reflecting the different colors of light back to the observer's eyes at different angles and creating bands of color. Halos are seen when sunlight or moonlight in front of the observer strikes ice crystals, and then passes through high thin cirrostratus clouds.



注释

optical phenomena ['optikəl-fi'nominə] n. 光学现象

rainbow ['reinbəu] n. 彩虹

halo ['heilu] n. 日晕

strike [straik] v. 照在……上

raindrop ['reindrɒp] n. 雨滴

cumulonimbus ['kju:mjələu'nimbəs] n. 积雨云

prism ['prizəm] n. 棱镜

bend [bend] v. 使弯曲

reflect [ri'flekt] v. 反射

angle ['æŋgl] n. 角, 角度

band [bənd] n. (光、频)带, (波)段

cirrostratus ['sɪrəu'stra:təs] n. 卷层云

Precipitation takes a variety of forms, including rain, drizzle, freezing rain, snow, hail, and ice pellets or sleet. Snowflakes are either single ice crystal or clusters of ice crystals. Hailstones are balls of ice which consist of clusters of raindrops that have collided and frozen together. Large hailstones only occur in violent thunderstorms, in which strong updrafts keep the hailstones suspended in the atmosphere long enough to grow large.



注释

a variety of [ə-və'raɪəti-əv] adj. 多种多样的

drizzle ['drɪzl] n. 细雨

freezing rain ['fri:zɪŋ-reɪn] n. 冻雨

hail [heɪl] n. 冰雹

ice pellet [aɪs-'pelɪt] n. 结冰的小球

sleet [sli:t] n. 雨夹雪，雨淞

snowflake ['snəʊfleɪk] n. 雪花

cluster ['klʌstə] n. 串

hailstone ['heɪl,steɪn] n. (一粒) 冰雹

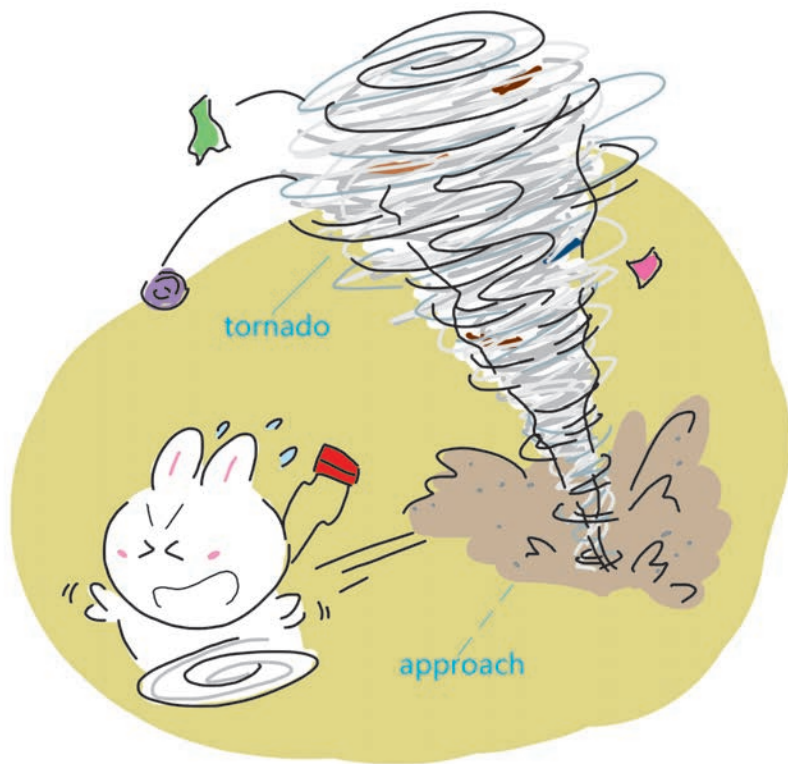
collide [kə'laɪd] v. 碰撞

violent ['vaɪələnt] adj. 猛烈的

updraft ['ʌpdra:ft] n. 上升气流

suspend [sə'spend] v. 悬浮 (大气、液体中)

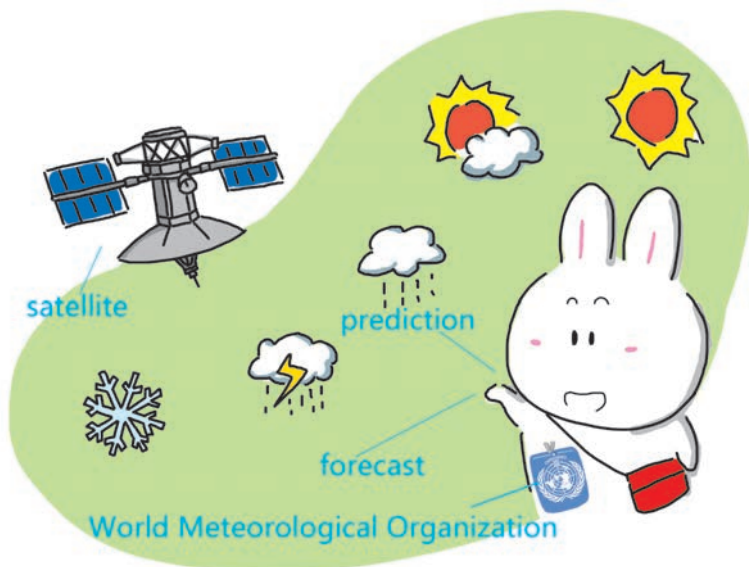
Wind is the horizontal movement of air. Wind speeds in hurricanes and typhoons exceed 120 km/h near the storm's center and may approach 320 km/h. The highest wind speeds at the surface of the Earth — as high as 480 km/h — occur in tornadoes. Except for these storms, wind speed usually increases with height to the top of the troposphere.



注释

- horizontal [ˌhɒrɪˈzɒntəl] adj. 水平的
hurricane [ˈhʌrɪkən] n. 飓风
typhoon [taɪˈfuːn] n. 台风
exceed [ɪkˈsiːd] v. 超过 (限度、范围)
approach [əˈprəʊtʃ] v. 接近，逼近
tornado [tɒˈneɪdəʊ] n. 龙卷风
height [haɪt] n. 海拔
troposphere [ˈtrɒpəʊsfiə] n. 对流层

Since the early 20th century, great strides have been made in weather prediction, largely as a result of computer development, but also because of instrumentation such as satellites and radars. Weather data from around the world are collected by the World Meteorological Organization, the National Weather Service, and other agencies and entered into computer models that apply the laws of motion and of the conservation of energy and mass to produce forecasts. In some cases, these forecasts have provided warning of major storms as much as a week in advance. However, because the behavior of weather systems is chaotic, it is impossible to forecast the details of weather more than about two weeks in advance.



注释

stride [straɪd] n. 进步

prediction [pri'dɪkʃən] n. 预报

satellite ['sætəlaɪt] n. 人造卫星

World Meteorological Organization [wɜːldˌmi:tɪə'lɒdʒɪkəlˌɔːɡənai'zeɪʃən] n. 世界气象组织

National Weather Service ['næʃənəlˌweðə'se:vɪs] n. 国家天气服务局

agency ['eɪdʒənsi] n. 机构, 机关

apply [ə'plaɪ] v. 应用

law of motion ['ləʊ-əv-'məʊʃən] n. 运动定律

laws conservation of energy/mass ['ləʊ-zˌkɒnsə'veɪʃən-əv-'enədʒi/mæss] n. 能量/质量守恒定律

forecast ['fɔːkɑːst] n. 预报

in advance [ɪn-əd'vɑːns] adv. 预先

chaotic [keɪ'ɒtɪk] adj. 无秩序的

The three major subdivisions of the world ocean are the Atlantic Ocean, the Pacific Ocean, and the Indian Ocean, which are bounded by the continental masses. The two minor subdivisions of the world ocean are the Southern Ocean, bounded by the Antarctic Circumpolar Current to the north and the Antarctica to the south, and the Arctic Ocean, almost landlocked except between Greenland and Europe. From the shorelines of the continents, a submerged part of the continental mass, called the continental shelf, extends seaward an average distance of 75 km; it varies in width from nearly zero to 1,500 km. The shelf gives way abruptly at a depth of about 200 m to a steeper zone known as the continental slope, which descends about 3,500 m. The continental rise, a gradually sloping zone of sediment that is considered part of the ocean bottom, extends about 600 km from the base of the continental slope to the flat abyssal plains of the deep-ocean floor. In the central parts of the oceans are the midocean ridges, which are extensive mountain chains with inner troughs that are heavily intersected by cracks, called fracture zones. The ridge system seems to merge into the continents in several areas, such as the Red Sea and the Gulf of California, and such areas are regions of great geologic activity, characterized by volcanoes, or earthquakes and faults. The midocean ridges play a key role in plate tectonics (movements in the Earth's crust), for it is from the inner troughs of these ridges that molten rock upwells from the Earth's mantle and spreads laterally on both sides, adding new material to the Earth's rigid crustal plates.

注释

oceanography [ˌəʊjəˈnɒɡrəfi] n. 海洋学

subdivision [ˈsʌbdɪˌvɪʒən] n. 部分

bound [baʊnd] v. 标明界限

continental mass [ˌkɒntɪˈnental-mæs] n. 大陆块体

Antarctic Circumpolar Current [ænˈtɑːktɪk-səˈkəmˈpəʊlə-ˈkʌrənt] n. 南极绕极流

Antarctica [ænˈtɑːktɪkə] n. 南极洲

Arctic Ocean [ˈɑːktɪk-ˈəʊjən] n. 北冰洋

landlocked [ˈlændlɒkt] adj. 被陆地包围的

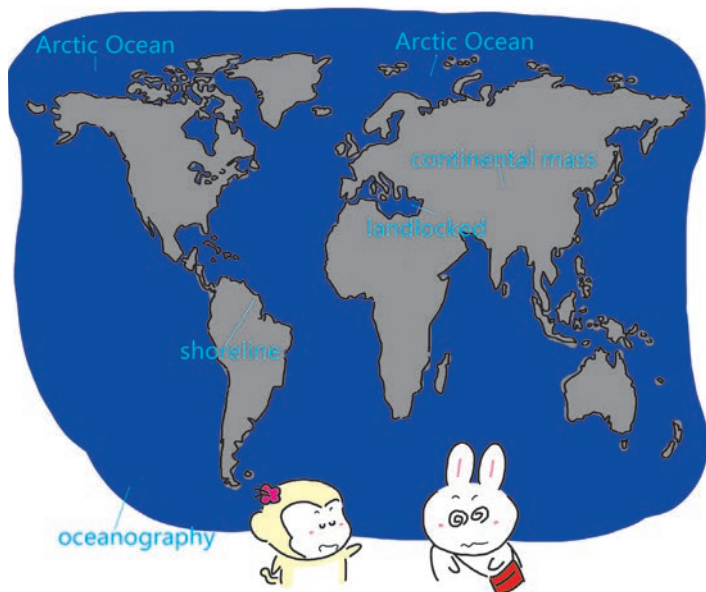
shoreline [ˈʃɔːlaɪn] n. 海岸线

submerged [səbˈmɜːdʒd] adj. 水下的

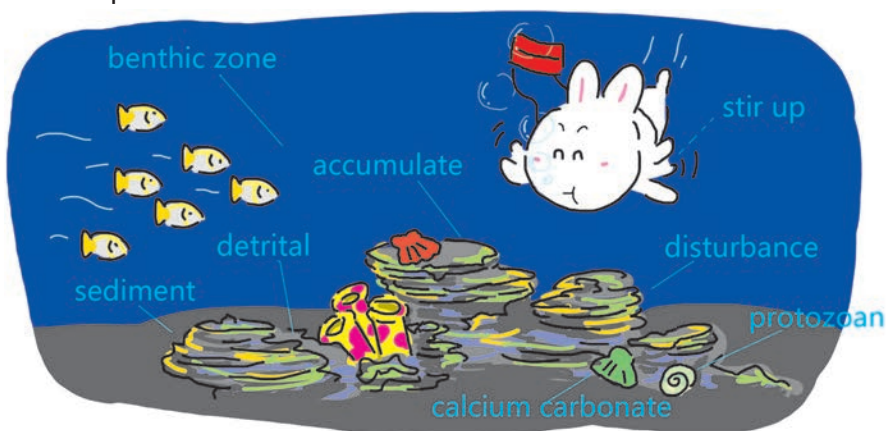
continental shelf [ˌkɒntɪˈnental-ʃelf] n. 大陆架

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shelf [ʃelf] n. 架
 abruptly [ə'brʌptli] adv. 突然地
 steep [sti:p] adj. 陡峭的
 zone [zəʊn] n. 地带
 continental slope [ˌkɒntɪ'nentəl-sleɪp] n. 大陆斜坡
 descend [di'send] v. 下降
 sloping ['sleɪpɪŋ] adj. 倾斜的
 sediment ['sedɪmənt] n. 沉积物
 abyssal [ə'bisəl] adj. 深海的
 ridge [rɪdʒ] n. 海脊
 trough [trɔ:f] n. 深海槽
 intersect [ˌɪntə'sekt] v. 交叉
 crack [kræk] n. 裂缝
 fracture zone ['fræktʃə-zəʊn] n. 断层带
 merge [mə:dʒ] v. 并入
 geologic activity [dʒiə'lɒdʒɪk-æk'tɪvɪtɪ] n. 地质活跃带
 volcano [vɒl'keɪnəʊ] n. 火山
 earthquake ['ə:θkweɪk] n. 地震
 fault [fɔ:lt] n. 断层
 plate tectonics ['pleɪt-tek'tɒnɪks] n. 板块构造论
 crust [krʌst] n. 地壳
 molten ['mɒltən] adj. 熔化的
 upwell [ʌp'wel] v. 涌起
 mantle ['mæntl] n. 地幔
 laterally ['lætərəli] adv. 侧面地
 crustal plate ['krʌstəl-'pleɪt] n. 地壳板块



The ocean floor is covered by an average of 0.5 km of sediment, which consists of rock particles and organic remains. Clay minerals, which are formed by the weathering of continental rocks and carried out to sea by rivers and wind, are usually abundant in the deep sea. Thick deposits of such detrital material are often found near mouths of rivers and on continental shelves; fine particles of clay are spread through the ocean and accumulate slowly on the deep-ocean floor. These sediments are stirred up and periodically redistributed by fierce current-generated disturbances that are called benthic storms, because they occur in the sparsely populated deep-sea habitat known as the benthic zone. Also accumulating as sediment in the benthic zone is the calcium carbonate shells of small organisms, such as foraminifera and the siliceous shells of marine protozoans.



注释

sediment ['sedɪmənt] n. 沉积物

organic remains [ɔ:'gænik - ri'meɪns] n. 有机残留物

clay mineral ['kleɪ-'mɪnərəl] n. 粘土矿物

weathering ['weðərɪŋ] n. 侵蚀

carry out ['kæri-'aʊt] v. 传送

abundant [ə'bʌndənt] adj. 丰富的

deposit [di'pɒzɪt] n. 沉淀物

detrital [di'traɪtəl] adj. 由岩屑形成的

mouth of river ['maʊθ-əv-'rɪvə] n. 入海口

accumulate [ə'kju:mjuleɪt] v. 堆积

stir up ['stɜ:-ʌp] v. 搅动

fierce [fɪəs] adj. 猛烈的

disturbance [dɪs'tɜ:bəns] n. 局部运动

benthic storm ['benθɪk-stɔ:m] n. 深海风暴

sparsely [spɑ:sli] adv. 稀少地

habitat ['hæbɪtæt] n. 栖息地

benthic zone ['benθɪk-zəʊn] n. 深海区域

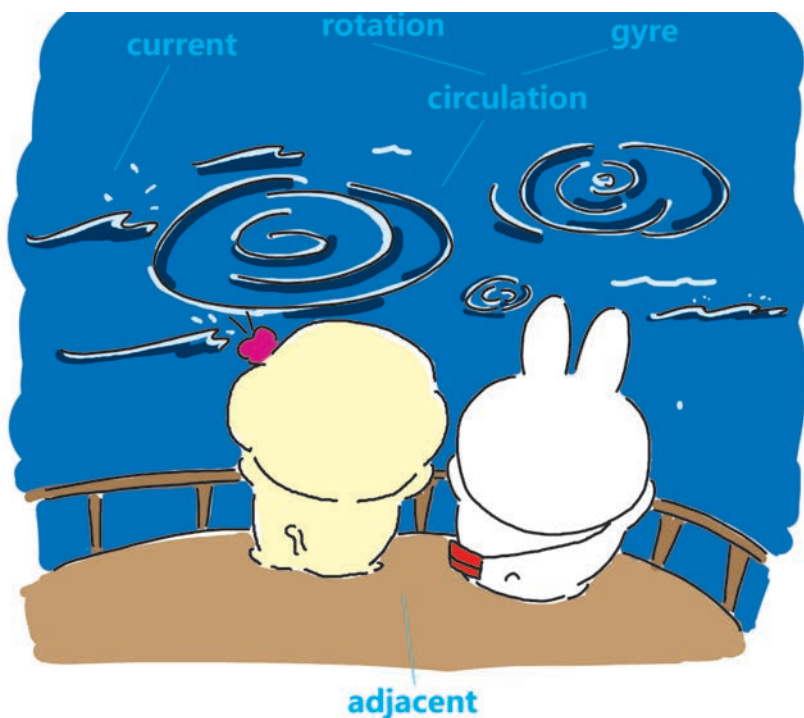
calcium carbonate ['kælsiəm-'kɑ:bəneɪt] n. 碳酸钙

foraminifera [fə'reɪmə'nɪfərə] n. 有孔虫类

siliceous [sɪ'lɪʃəs] adj. 由硅石组成的

protozoan [prəuteu'zəʊən] n. 原生动物

The surface currents of the ocean are characterized by large gyres, or currents that are kept in motion by prevailing winds, but the direction of which is altered by the rotation of the Earth. Although the surface circulation of the ocean is a function of winds and the rotation of the earth, the deeper circulation in the oceans is a function of density differences between adjacent water masses.



注释

current ['kʌrənt] n. 水流

gyre ['dʒaɪə] n. 漩涡

motion ['məʊʃən] n. 运动

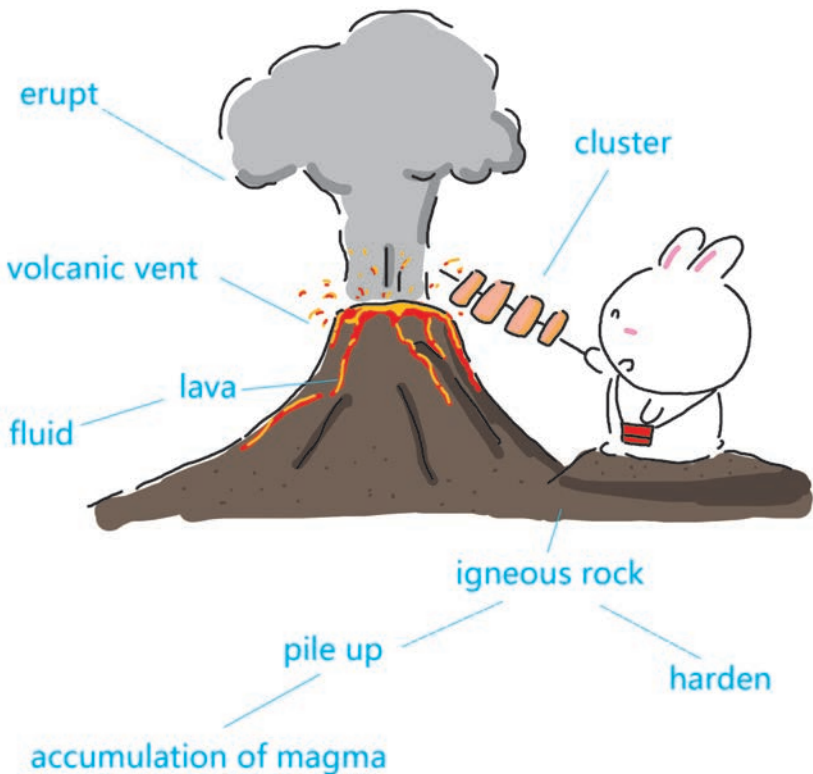
rotation [rəʊ'teɪʃn] n. 自转

circulation [ˌsə:kju'leɪʃən] n. 循环

density ['densɪti] n. 比重

adjacent [ə'dʒeɪsənt] adj. 紧挨着的

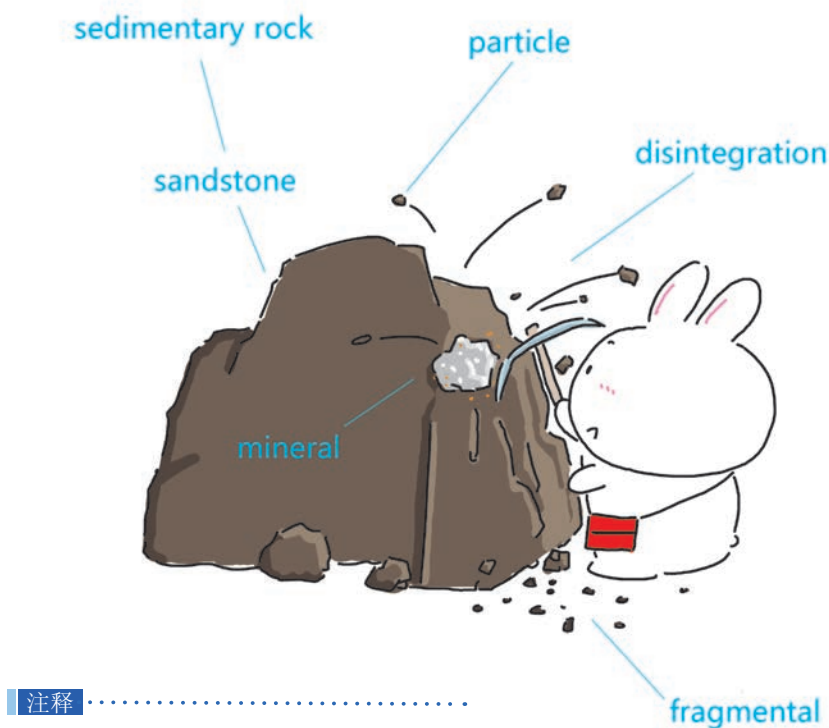
All volcanoes are formed by the accumulation of magma (molten rock that forms below the Earth's surface). Magma can erupt through one or more volcanic vents, which can be a single opening, a cluster of openings, or a long crack called a fissure vent. It forms deep within the Earth, generally within the upper part of the mantle, or less commonly, within the base of the Earth's crust. Some lava (molten rock that erupts at the Earth's surface) can be as thick and viscous as toothpaste, while other lava can be as thin and fluid as warm syrup and flows rapidly down the sides of a volcano. Once lava hardens, it forms igneous rock. After many eruptions, the volcanic materials pile up around the vent or vents. These piles form a topographic feature, such as a hill, mountain, plateau, or crater that we recognize as a volcano.



注释

volcano [vɒl'keɪnəʊ] n. 火山
 accumulation [ækju:mju'leɪʃ(ə)n] n. 堆积物
 magma ['mægmə] n. 岩浆
 molten ['məʊltən] adj. 熔化的
 erupt [ɪ'rʌpt] v. 喷出
 volcanic vent [vɒl'kænik-vent] n. 火山口
 vent [vent] n. 孔, 口, 裂口
 cluster ['klʌstə] n. 串
 a cluster of [ə-'klʌstə-əv] adj. 成群的, 成串的
 crack [kræk] n. 断裂处
 fissure vent ['fɪʃə-vent] n. 断裂口
 fissure ['fɪʃə] n. 裂缝, 裂隙
 mantle ['mæntl] n. 地幔
 crust [krʌst] n. 地壳
 lava ['lɑ:və] n. 熔岩
 thick [θɪk] adj. 厚的
 viscous ['vɪskəs] adj. 粘性的
 toothpaste ['tu:θpeɪst] n. 牙膏
 thin [θɪn] adj. 薄的
 fluid ['flu:ɪd] adj. 易流动的
 syrup ['sɪrəp] n. 糖浆
 harden ['hɑ:dn] v. 变硬, 凝固
 igneous rock ['ɪgniəs-rɒk] n. 火成岩
 pile up ['pail-ʌp] v. 堆积
 pile [pail] n. 堆积, 堆
 topographic [tɒpə'græfɪk] adj. 地形学上的
 plateau ['plætəʊ] n. 高地
 crater ['kreɪtə] n. 坑, 火山[喷火]口; (月球上的) 环形山
 recognize ['rekəɡnaɪz] v. 识别

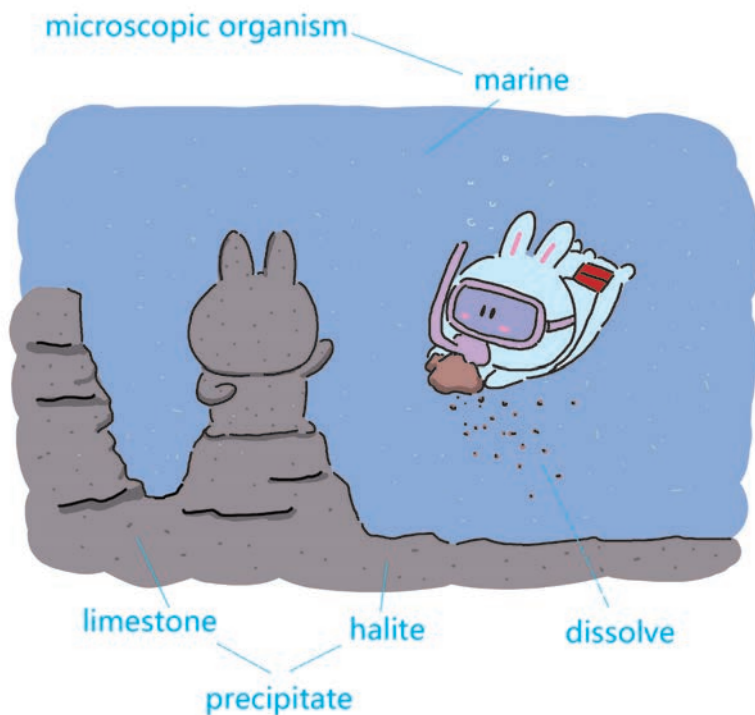
Sedimentary rocks are classified, according to their manners of origin, into mechanical or chemical sedimentary rocks. Mechanical rocks, or fragmental rocks, are composed of mineral particles produced by the mechanical disintegration of other rocks and transported, without chemical deterioration, by flowing water. They are carried into larger bodies of water, where they are deposited in layers. Shale and sandstone are common sedimentary rocks of mechanical origin.



注释

- sedimentary rock [sedi'mentəri-rɒk] n. 沉积岩
 manner ['mænə] n. 样式
 fragmental [fræg'mentəl] adj. 碎屑状的
 mineral ['minərəl] n. 矿物, 矿石
 particle ['pɑ:tɪkl] n. 颗粒
 disintegration [dis,ɪntɪ'greɪʃən] n. 分裂, 崩解
 deterioration [di'tiəriə'reiʃən] n. 磨损, 腐化, 变质
 deposit [di'pɒzɪt] v. 沉积, 沉淀
 layer ['leɪə] n. 地层
 shale [ʃeɪl] n. 页岩
 sandstone ['sændstəʊn] n. 砂岩

The materials making up chemical sedimentary rocks may consist of the remains of microscopic marine organisms precipitated on the ocean floor, as in the case of limestone. They may also have been dissolved in water circulating through the parent rock formation, and then deposited in a sea or lake by precipitation from the solution. Halite, gypsum is formed by the evaporation of salt solutions and the consequent precipitation of the salts.



注释

make up ['meik-ʌp] v. 组成

consist of [kən'sist-əv] v. 由……组成

microscopic organism

[maɪkrə'skɒpɪk-'ɔ:ɡənɪzəm] n. 微生物

marine [mə'ri:n] adj. 海洋的

precipitate [pri'sɪpɪteɪt] v. 沉淀

in the case of [ɪn-ðə-'keɪs-əv] adv. 在……的情况下

limestone ['laɪm,steɪn] n. 石灰岩

dissolve [dɪ'zɒlv] v. 溶解

circulate ['sə:kjuleɪt] v. 循环

solution [sə'lu:ʃən] n. 溶液

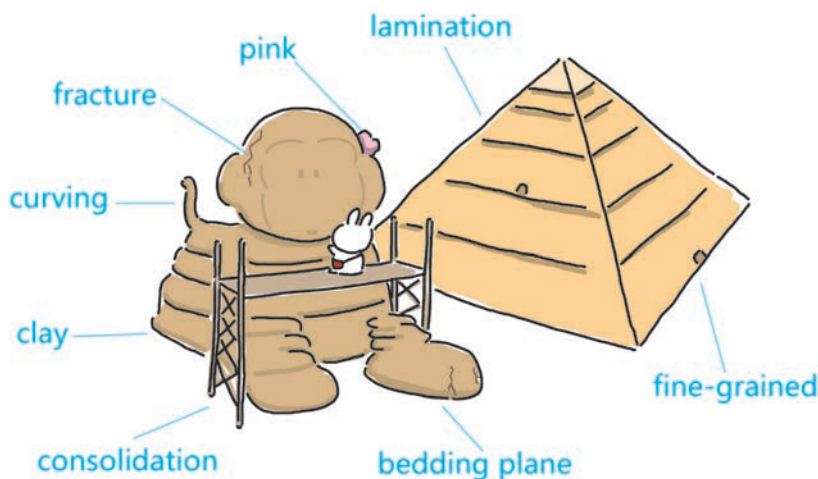
halite ['hælaɪt] n. 岩盐

gypsum ['dʒɪpsəm] n. 石膏

evaporation [ɪ.væpə'reɪʃən] n. 蒸发(作用)

salt solution ['sɔ:lt-sə'lu:ʃən] n. 盐溶液

Most shales (common name applied to fine-grained varieties of sedimentary rocks formed by the consolidation of beds of clay or mud) exhibit fine laminations that are parallel to the bedding planes and along which the rock breaks in an irregular curving fracture. Most varieties of shales are colored in various shades of gray, but other colors, such as red, pink, green, brown, and black, are often present. Shales are soft enough to be scratched with a knife, and feel smooth and almost greasy to the touch. Many shales yield oil when distilled by heat, and the sedimentary rocks containing larger quantities of oil are called oil shales. Widely distributed throughout the world, oil shales are a source of oil for countries lacking petroleum.



注释

shale [ʃeɪl] n. 页岩

fine-grained ['faɪn-'greɪnd] adj. 纹理细腻的, 平整的

consolidation [kən'sɒlɪ'deɪʃən] n. 加固

clay [kleɪ] n. 黏土

exhibit [ɪg'zɪbɪt] v. 显示, 展出

lamine [læ'mɪneɪt] n. 层叠结构

lamine to [læ'mɪneɪt-tu] adj. 与……平行/相似的

bedding plane ['bedɪŋ-'pleɪn] n. (岩层)层平面, 底层

curving ['kɜ:vɪŋ] adj. 弯曲的

fracture ['fræktʃə] n. 岩石的裂缝或断层

shades of gray [ʃeɪdz-əv-'greɪ] n. 灰度梯度
(深浅不同的各种灰色)

pink [pɪŋk] n. 粉红色

scratch [skrætʃ] v. 刮擦

greasy ['greɪzi] adj. 油腻的

yield [ji:lɪd] v. 生出, 出产; 产生

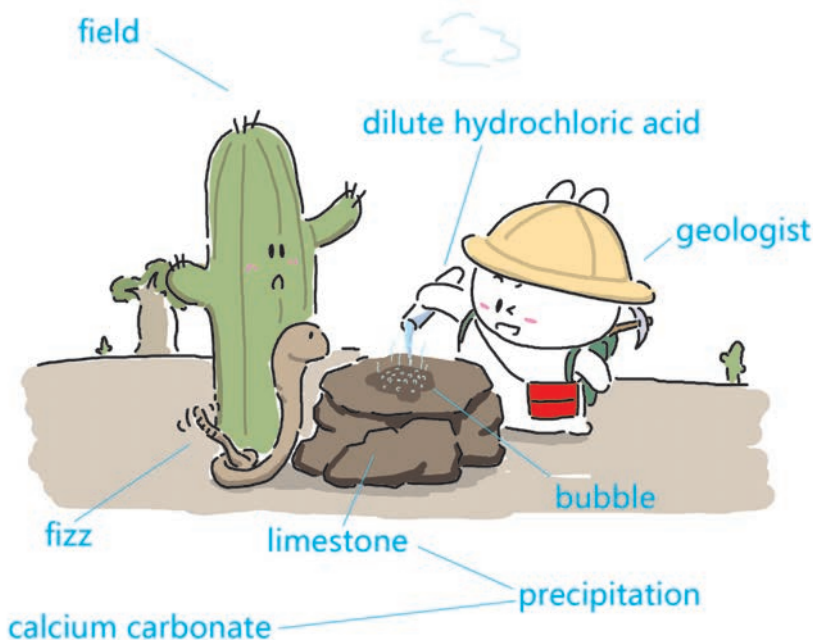
distill [dɪ'stɪl] v. 蒸馏

oil shale [ɔɪl-'ʃeɪl] n. 油母页岩

distribute [dɪ'strɪbjʊ:t] v. 分布, 散布 (over)

petroleum [pɪ'treʊliəm] n. 石油

All limestone forms from the precipitation of calcium carbonate from water. Calcium carbonate leaves solutions in many ways and each way produces a different kind of limestone. When a drop of dilute hydrochloric acid is placed on a piece of limestone, the acid reacts with the calcite and forms bubbles of carbon dioxide. This "fizz" reaction is so characteristic of limestone that many geologists carry a small bottle of dilute hydrochloric acid into the field for a rapid and easy identification of limestone.



注释

limestone ['laɪm,steɪn] n. 石灰岩

precipitation [prɪ'sɪpɪ'teɪʃən] n. 沉淀

calcium carbonate ['kælsiəm-'kɑ:bəneɪt] n. 碳酸钙

solution [sə'lju:ʃən] n. 溶液

dilute [dai'lju:t] adj. 稀释的

hydrochloric acid [ˌhaɪdrə'klo:ɹɪk-'æsɪd] n. 盐酸

acid ['æsɪd] n. 酸

calcite ['kælsait] n. 方解石

bubble ['bʌbl] n. 气泡

carbon dioxide ['kɑ:bən-dai'ɒksaɪd] n. 二氧化碳

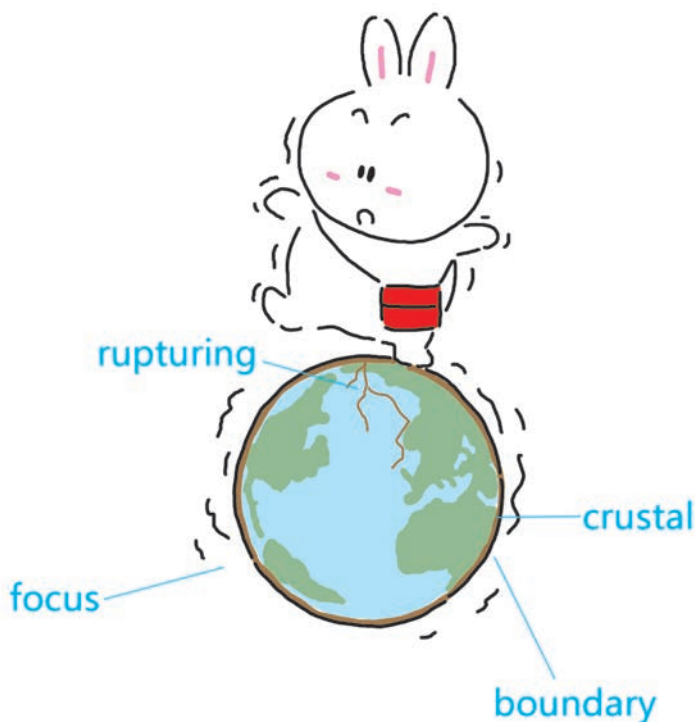
fizz [fɪz] n. 嘶嘶声

geologist [dʒɪ'ɒlədʒɪst] n. 地质学家

field [fi:ld] n. 野外

identification [aɪ,deɪntɪfɪ'keɪʃən] n. 鉴定

The point within the Earth along the rupturing geological fault where an earthquake originates is called the focus, or hypocenter. Seismologists know from observations that most earthquakes originate as shallow-focus earthquakes, and most of them occur near plate boundaries — areas where the Earth's crustal plates move against each other.



注释

rupture ['rʌptʃə] adj. 断裂的

fault [fɔ:lt] n. 断层

focus ['fəukəs] n. 震源 (“震中”是epicenter ['epi,sentə])

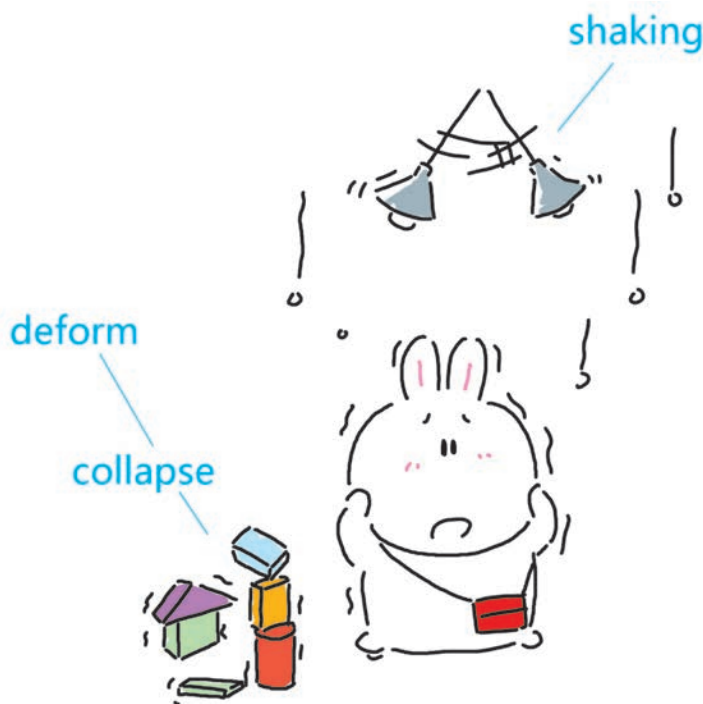
hypocenter ['haipəu,sentə] n. 震源

seismologist [saiz'mələdʒist] n. 地震学家

boundary ['baundəri] n. 边界

crustal ['krʌstəl] adj. 地壳的

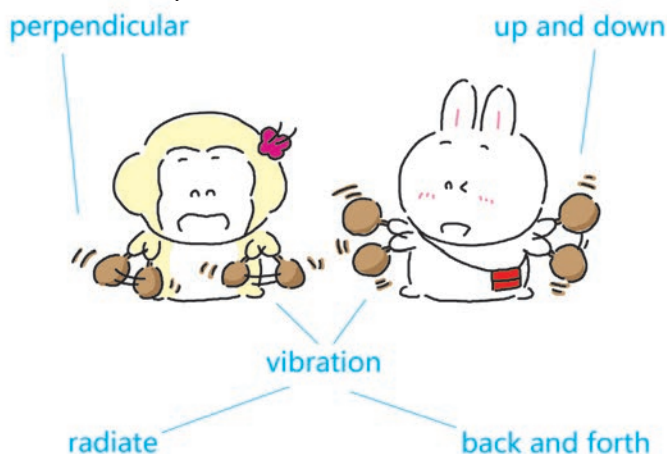
The destruction an earthquake causes depends on its magnitude and duration, or the amount of shaking that occurs. A structure's design and the materials used in its construction also affect the amount of damage the structure incurs. Earthquakes vary from small, imperceptible shaking to large shocks felt over thousands of kilometers. Earthquakes can deform the ground, make buildings and other structures collapse, and create tsunamis.



注释

- magnitude ['mægnitju:d] n. 震级
 duration [djʊə'reiʃən] n. 持续时间
 shaking ['ʃeɪkɪŋ] n. 摇晃
 incur [in'kə:] v. 招致
 vary ['veəri] v. 不同，变化
 imperceptible [ˌɪmpə'septɪbl] adj. 极细微的
 deform [di'fɔ:m] v. (使) 变形
 collapse [kə'læps] n. 倒塌
 tsunami [tsju:'nɑ:mɪ] n. 海啸

The sudden movement of rocks along a fault causes vibrations that transmit energy through the Earth in the form of waves, called body waves. There are two types of body waves: primary, or P, waves, and secondary, or S, waves. Body waves, or P and S waves, radiate out from the rupturing fault starting at the focus of the earthquake. P waves are compression waves, because the rocky material in their path moves back and forth in the same direction as the wave travels alternately, compressing and expanding the rock. P waves are the fastest seismic waves; they travel in strong rock at about 6 to 7 km per second. P waves are followed by S waves, which shear or twist, rather than compress the rock they travel through. S waves travel at about 3.5 km per second. S waves cause rocky material to move either side to side or up and down perpendicular to the direction the waves are traveling, thus shearing the rocks. Both P and S waves help seismologists to locate the focus and epicenter of an earthquake.

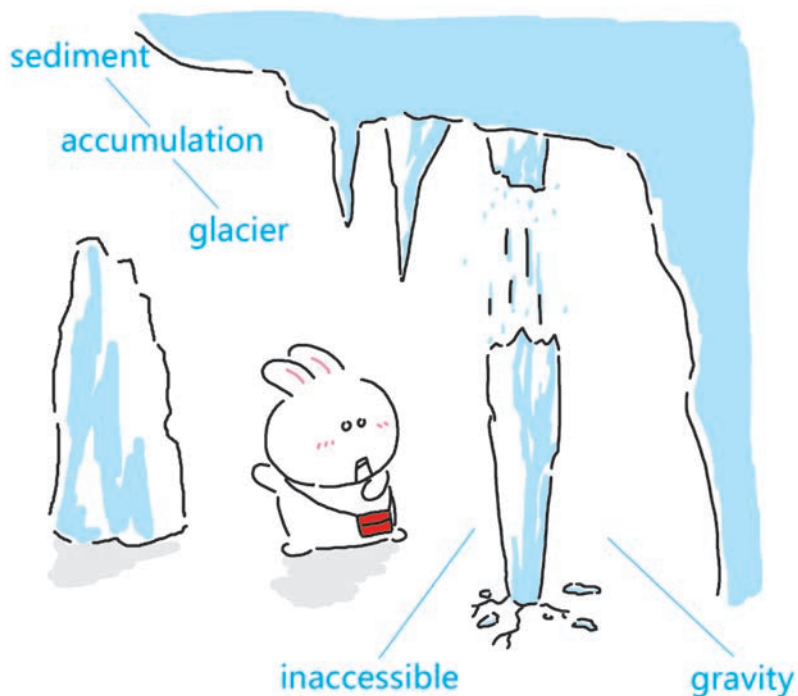


注释

vibration [ˈvaɪˈbreɪʃən] n. 振动
 transmit [ˈtrænzmit] v. 传送
 primary [ˈpraɪməri] adj. 初级的
 secondary [ˈsekəndəri] adj. 第二等级的
 primary wave [ˈpraɪməri-weɪv] n. 纵波
 secondary wave [ˈsekəndəri-weɪv] n. 横波
 radiate [ˈreɪdiət] v. 散发
 compression [kəmˈpreʃən] n. 压缩
 back and forth [ˈbæk-ænd-ˈfoːθ] adv. 来回地

alternate [ɔːlˈtəːnɪt] adj. 交替的
 seismic wave [ˈsaɪzmik-weɪv] n. 地震波
 shear [ʃeə] v. 剪切
 twist [twɪst] n. 扭曲
 side to side [ˈsaɪ-tu-ˈsaɪd] adv. 从一边到另一边
 up and down [ʌp-ænd-ˈdaʊn] adv. 上上下下
 perpendicular [ˌpɜːpənˈdɪkjələ] adj. 垂直的
 epicenter [ˈepɪsəntə] n. 震中

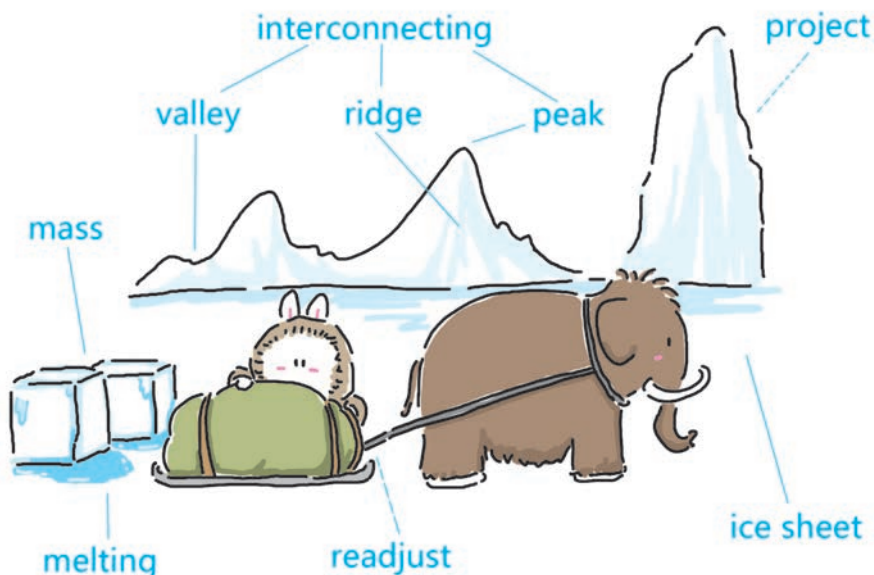
Glaciers, an enduring accumulation of ice, snow, water, rocks, and sediments that moves under the influence of gravity, is an intriguing part of the Earth's natural environment, and its majestic beauty in wild and inaccessible mountain settings is unparalleled.



注释

- glacier ['glæsjə] n. 冰川
 accumulation [ækju:mju'lei(ə)n] n. 堆积物
 sediment ['sediment] n. 沉积物
 gravity ['græviti] n. 重力
 intriguing [in'tri:ɡin] adj. 引人入胜
 majestic [mə'dʒestik] adj. 雄伟的
 inaccessible [inæk'sesəbl] adj. 难以接近
 unparalleled [ʌn'pærəleɪd] adj. 无可比拟的

Ice sheets are the largest ice masses found on Earth, covering huge land areas. The ice sheet in Antarctica covers 13 million sq km. It is over 4 km thick and its weight has depressed the continent below sea level in many places. If this weight was removed, the continent would slowly rise and readjust itself, as Europe still does after the melting of the ice sheet that covered that continent during the last Ice Age. Ice caps are smaller than ice sheets. They form when snow and ice fill a basin or cover a plateau to a considerable depth. Ice fields develop where large interconnecting valley glaciers are separated by mountain peaks and ridges that project through the ice.

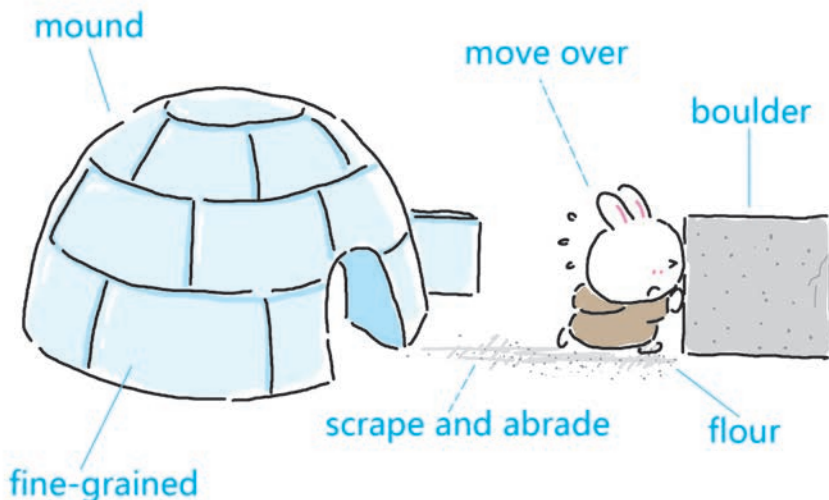


注释

ice sheet [aɪs-ʃi:t] n. 冰原
 mass [mæs] n. 块
 depress [dɪ'pres] v. 使降低
 readjust [ri:'dʒʌst] v. 重新调整
 melting ['meltn] n. 融化
 Ice Age [aɪs-eɪdʒ] n. 冰河时期
 ice cap [aɪs-kæp] n. 冰冠
 basin ['beɪsn] n. 盆地
 plateau ['plætəu] n. 高地

considerable [kən'sɪdərəbl] adj. 相当大的
 ice field [aɪs-fi:ld] n. 冰原
 interconnecting ['ɪntəkə'nektɪŋ] adj. 互相连接的
 valley ['væli] n. (山) 谷
 peak [pi:k] n. 山顶
 ridge [rɪdʒ] n. 山脊
 project [prə'dʒekt] v. 凸出

As glaciers move over bedrock, they scrape and abrade its surface, producing fine-grained rock flour. Glaciers can also pluck away rocks up to boulder size, and transport and deposit them along the margins of the glacier down in the valleys. The glaciers deposit these materials as till, a sediment consisting of mud, sand, gravel, and boulders. Much of this material is deposited in long mounds called moraines. Lateral moraines are formed on each side of a valley glacier where abraded sediment and plucked rocks are deposited.



注释

move over ['mu:v-əuvə] v. 挪开

bedrock ['bedrɒk] n. 岩床

scrape [skreɪp] v. 刮擦

abrade [ə'breɪd] v. 磨损

flour ['flaʊə] n. 细软的粉末

pluck away ['plʌk-ə'weɪ] v. 冰川冲走 (岩石)

boulder ['bəʊldə] n. 巨砾

deposit [di'pɒzɪt] v. 使沉积

margin ['mɑ:dʒɪn] n. 边缘

till [tɪl] n. 冰渍

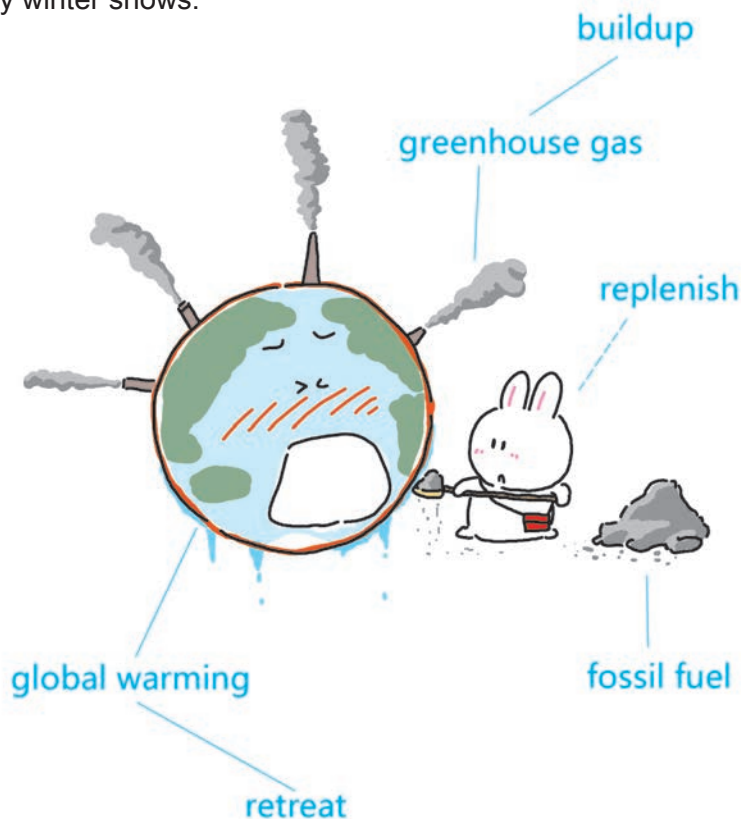
gravel ['grævəl] n. 砂砾

mound [maʊnd] n. 堆

moraine [mɔ'reɪn] n. 冰碛

lateral moraine ['læterəl-mɔ'reɪn] n. 冰川侧碛

Many glaciologists believe that the current worldwide retreat of glaciers is influenced by global warming. They believe global warming is caused by the build-up of greenhouse gases in the atmosphere, since humans began using fossil fuels during the Industrial Revolution. Higher average temperatures are causing glaciers to melt faster than they can be replenished by winter snows.



注释

glaciologist [ˈgleɪʃɪˈɒlədʒɪst] n. 冰河学家

retreat [riˈtri:t] n. 冰川后退

global warming [ˈɡleɪbəl-ˈwɔːmɪŋ] n. 全球变暖

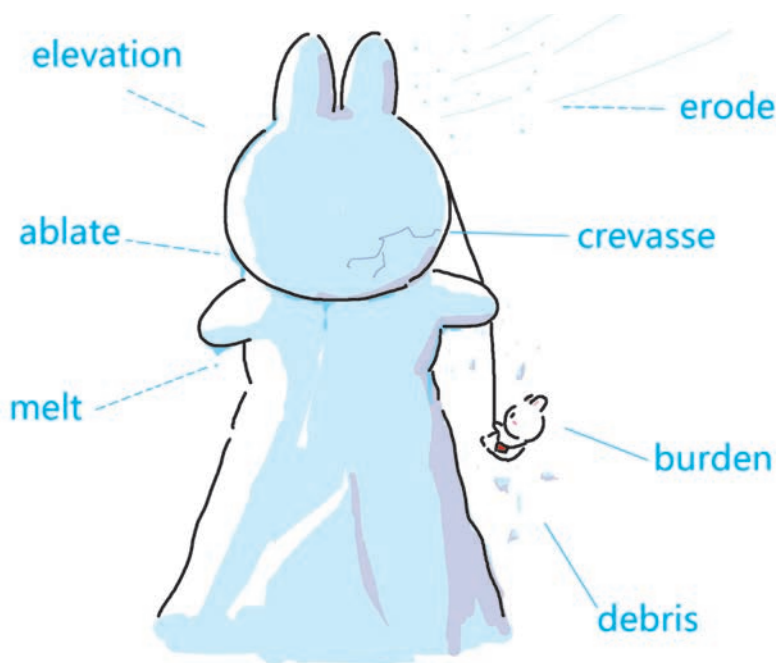
buildup [ˈbɪld-ʌp] n. 累积

greenhouse gas [ˈɡriːnhaʊs-ɡæs] n. 温室气体

fossil fuel [ˈfɒsl-fjuəl] n. (煤、石油、天然气等) 矿物燃料

replenish [riˈplenɪʃ] v. 补充

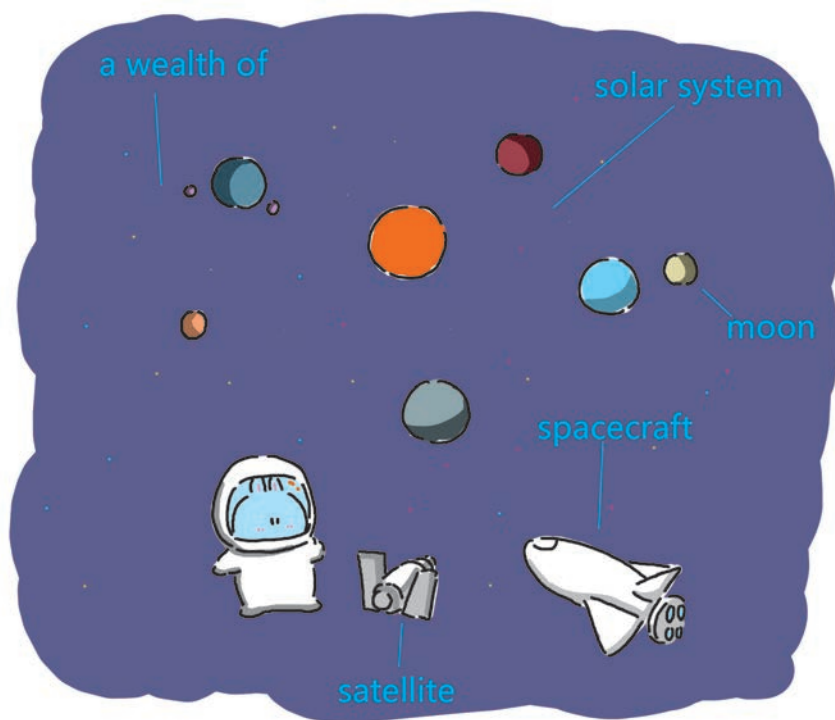
The farther a glacier moves from its source of accumulation into warmer-lower-elevation regions, the more likely it is to ablate or melt. As the ice ablates, it begins to deposit the material it has eroded in the mountains, as well as any rock debris that has fallen into its crevasses. Just as streams of liquid water create landforms when they deposit their loads of material, streams of frozen water — glaciers — drop their burden and shape the landscape as they recede.



注释

- elevation [ˈeliːveɪʃən] n. 海拔
 ablate [æbˈleɪt] v. 融化
 melt [melt] v. 融化
 erode [iˈrəʊd] v. 侵蚀
 debris [ˈdeɪəri:] n. 碎片
 crevasse [kriˈvæs] n. 裂缝
 burden [ˈbɜːdən] n. 负载
 landscape [ˈlændskeɪp] n. 地形
 recede [riˈsiːd] v. 后退

The Moon is the only natural satellite of the Earth. The natural satellites of the other planets in the solar system are also sometimes referred to as moons. Telescopes have revealed a wealth of lunar details since their invention in the 17th century, and spacecraft have contributed further knowledge since the 1950s.



注释

moon [mu:n] n. 月球

satellite ['sætəlaɪt] n. 人造卫星

solar system ['səʊlə-'sɪstəm] n. 太阳系

refer to sb / sth as ... [rɪ'fe:-tu:-'sʌmbədi/'sʌmθɪŋəz] v. 称某人/某物为……

reveal [rɪ'vi:l] v. 揭示, 显示

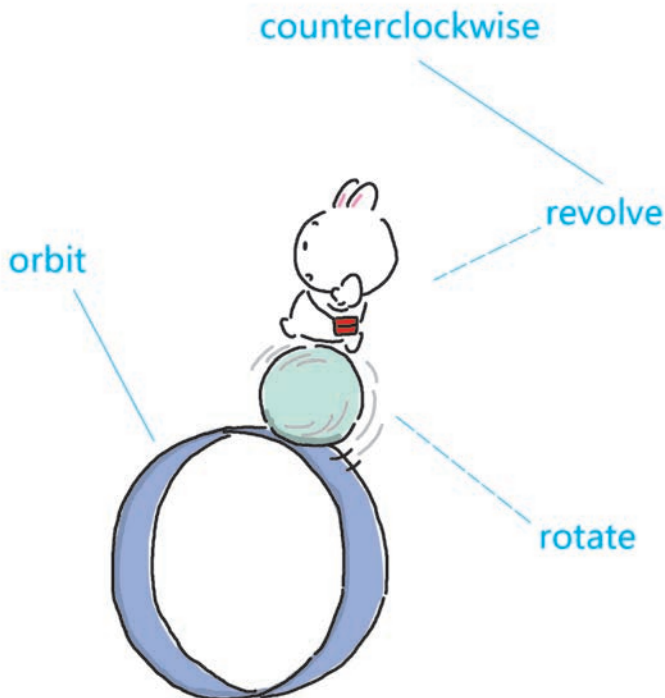
a wealth of [ə-'welθ-əv] adj. 很多的

lunar ['lu:nə] adj. 月球的

spacecraft ['speɪskrɑ:ft] n. 宇宙飞船

contribute [kən'trɪbjut] v. 贡献

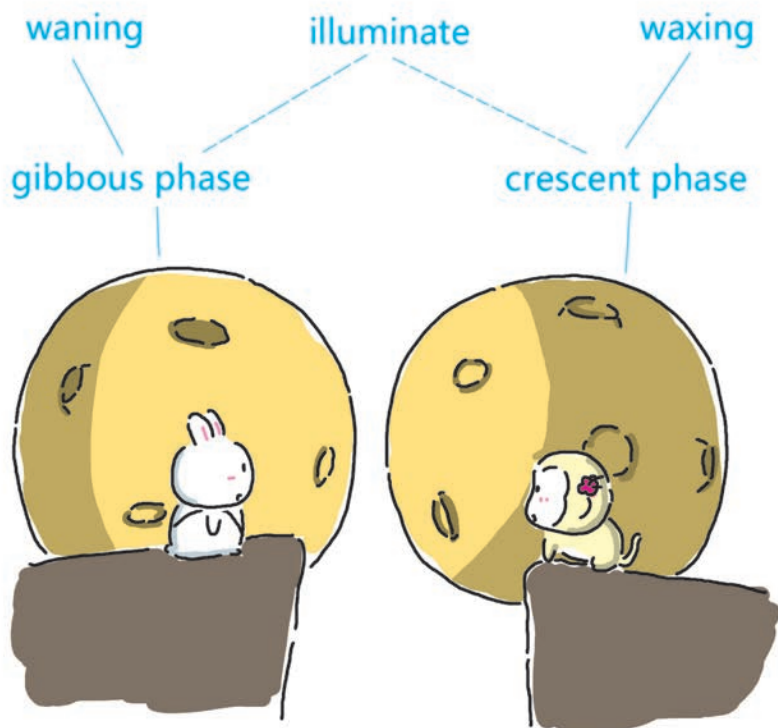
As the Moon orbits Earth in a counterclockwise direction, the Earth itself rotates counterclockwise (from west to east) on its axis, and revolves around the Sun in a counterclockwise orbit.



注释

orbit ['ɔ:bit] v. 绕 (吸引中心 运转 n. 轨道
 counterclockwise [ˌkaʊntə'klokwaɪz] adj. 逆时针方向的
 rotate [rəu'teɪt] v. 自转
 axis [æksɪs] 轴 (plural: axes [æksɪ:z])
 revolve [rɪ'vɒlv] v. 公转

The Moon is full when it is farther away from the Sun than the Earth; it is new when it is closer. When it is more than half illuminated, it is said to be in gibbous phase. When it is less than half illuminated, it is said to be in crescent phase. The Moon is said to be waning as it progresses from full to new, and to be waxing as it proceeds from new to full.



注释

full [ful] adj. 满月

new [nju:] adj. 新月 (的)

illuminate ['ilju:mineit] v. 照亮, 使明亮

gibbous ['gibəs] adj. 光亮部大于半圆的

phase [feiz] n. 位相

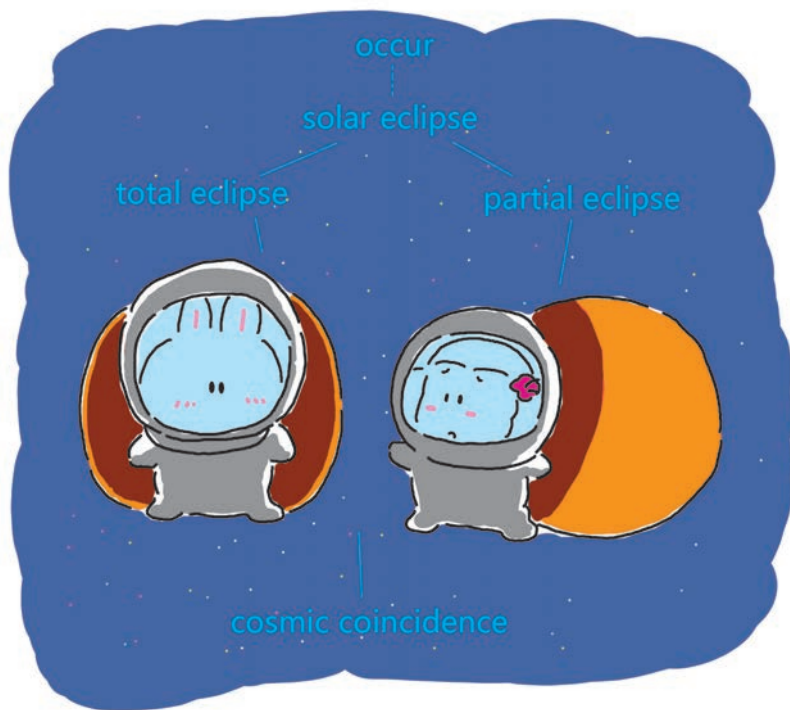
crescent ['kresnt] adj. 新月形的

crescent phase ['kresnt-feiz] n. 新月位相

waning ['weinin] adj. (月亮) 渐亏的

waxing ['wæksin] adj. (月亮) 渐盈的

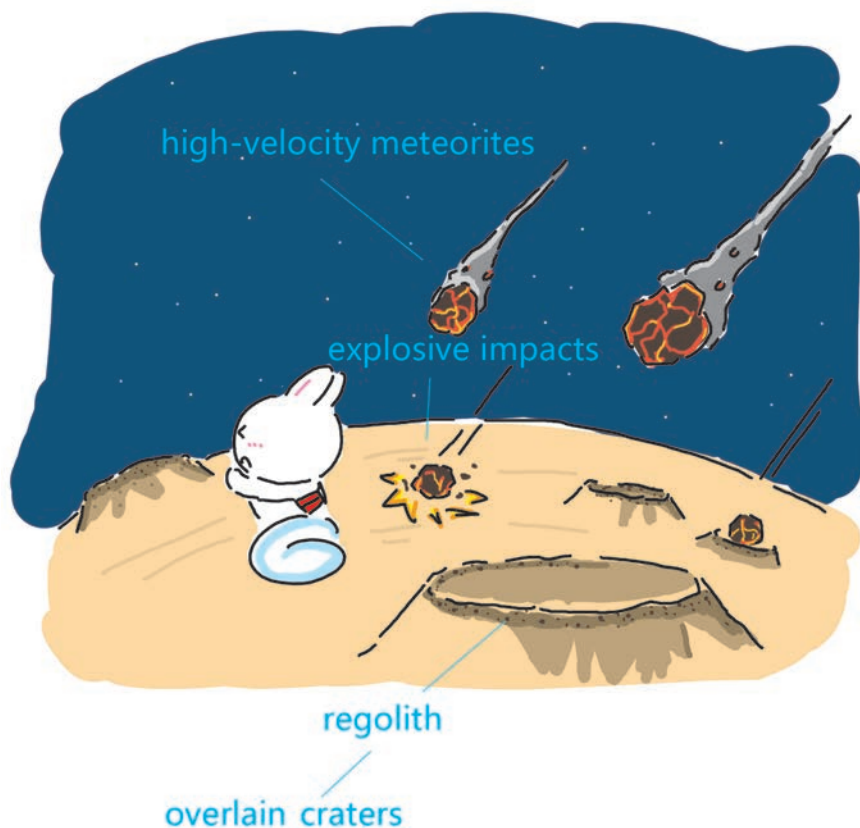
By a cosmic coincidence, the apparent sizes of the disk of the Moon and the disk of the Sun are approximately the same when seen from the Earth. If the Moon's orbit lays exactly in the plane of the Earth's orbit around the Sun, a solar eclipse would occur somewhere on the Earth every month at new moon. However, solar eclipses occur only about 2 to 5 times a year. Partial eclipses, when the Moon only partially covers the disk of the Sun, happen more often than total eclipses.



注释

- cosmic ['kɒzmɪk] adj. 宇宙的
 coincidence [kəʊ'ɪnsɪdəns] n. 巧合
 approximately [ə'prɒksɪmɪtli] adv. 大约
 lay [leɪ] v. 位于
 plane [pleɪn] n. 平面
 eclipse ['ɪ'kɪps] n. 食
 occur [ə'kɜː] v. 出现, 发生
 solar eclipse ['səʊlə-'ɪ'kɪps] n. 日食
 partial eclipse ['pɑːʃəl-'ɪ'kɪps] n. 偏食
 total eclipse ['təʊtəl-'ɪ'kɪps] n. 日全食

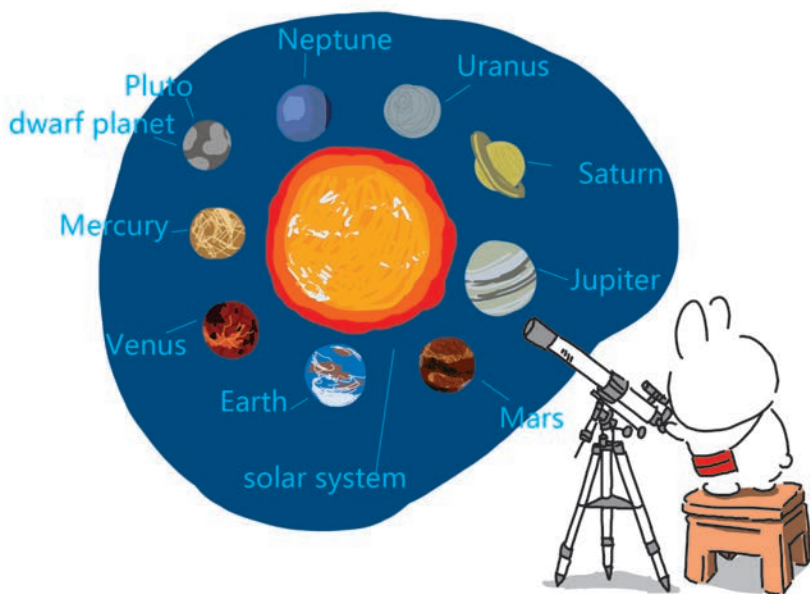
The Moon's surface is covered with craters overlain by a layer of soil called regolith. Nearly all the craters are formed by explosive impacts of high-velocity meteorites.



注释

- crater ['kreɪtə] n. 环形山
 overlie ['əʊvə'laɪ] v. 在……上面
 regolith ['regəlɪθ] n. 风化层
 explosive ['eks'pləʊsɪv] adj. 爆炸 (性) 的
 impact ['ɪmpækt] n. 冲击
 velocity ['vi'lɒsɪtɪ] n. 速率
 meteorite ['mi:tjəraɪt] n. 陨星

Eight major or classical planets are currently recognized by the International Astronomical Union (IAU), the body that gives official names to objects in the solar system. The planets are commonly divided into two groups: the inner planets (Mercury, Venus, Earth, and Mars) and the outer planets (Jupiter, Saturn, Uranus, and Neptune). The inner planets are small and are composed primarily of rock and iron. The outer planets are much larger and consist mainly of hydrogen, helium, and ice. Pluto, historically counted as the ninth planet, does not belong to either group, but was reclassified as a dwarf planet by the IAU in 2006.

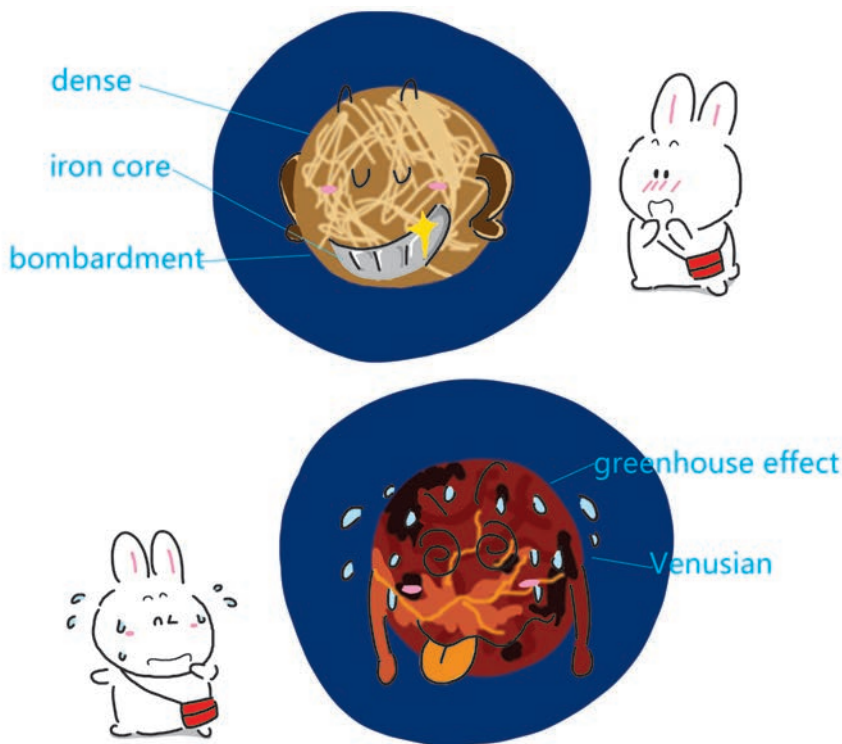


注释

major ['meɪdʒə] adj. 主要的
 classical ['klæsɪkəl] adj. 传统的, 古典的, 经典的
 recognized ['rekəɡnaɪzd] adj. 公认的, 经过验证的
 solar system ['səʊlə-'sɪstəm] n. 太阳系
 divide [dɪ'vaɪd] v. 划分
 Mercury ['mæ:kjʊəri] n. 水星
 Venus ['vi:nəs] n. 金星
 Earth [ə:θ] n. 地球
 Mars [mɑ:z] n. 火星
 Jupiter ['dʒu:pɪtə] n. 木星
 Saturn ['sætən] n. 土星
 Uranus ['ju:ərənəs] n. 天王星

Neptune ['neptju:n] n. 海王星
 compose [kəm'pəʊz] v. 组成
 (be composed of: 由……组成)
 primarily ['praɪməri] adv. 主要地
 consist of [kən'sɪst-əv] v. 由……组成
 mainly ['meɪnli] adv. 主要地
 hydrogen ['haɪdrədʒən] n. 氢
 helium ['hi:lɪəm] n. 氦
 Pluto ['plu:təu] n. 冥王星
 dwarf planet [dwo:f-'plæniit] n. 矮星

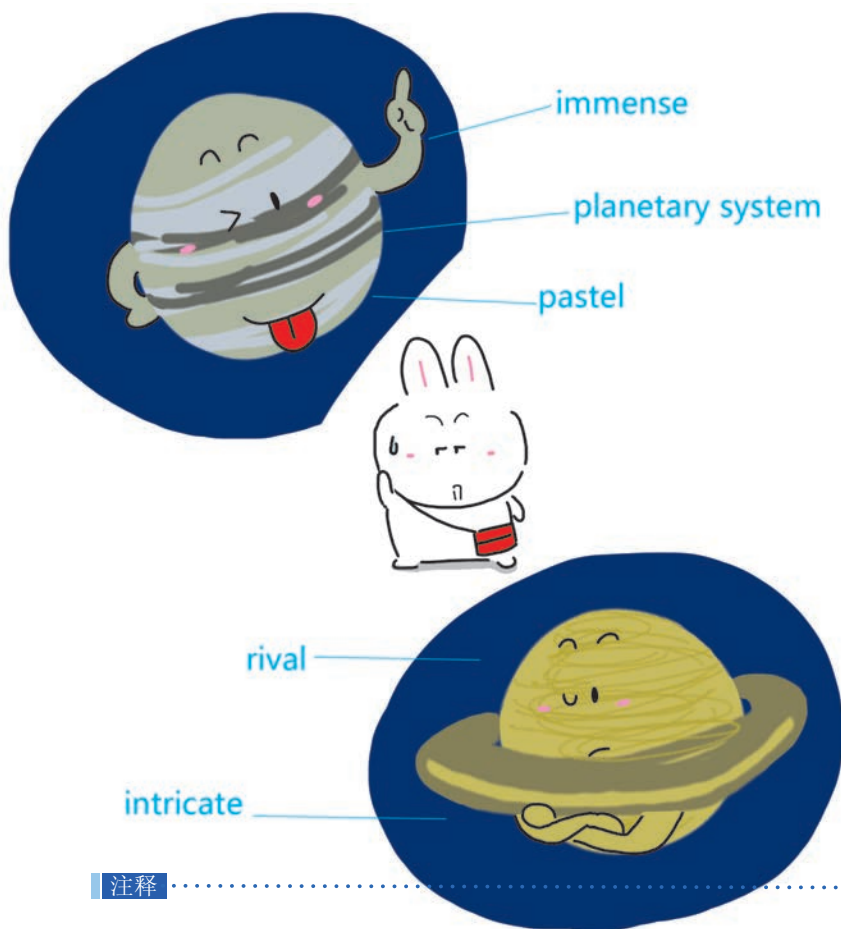
Mercury is surprisingly dense, apparently because it has an unusually large iron core. With only a transient atmosphere, Mercury has a surface that still bears the record of bombardment by asteroidal bodies early in its history. Venus has a carbon dioxide atmosphere 90 times thicker than that of Earth, causing an efficient greenhouse effect by which the Venusian atmosphere is heated.



注释

- dense [dens] adj. 密度大的
 iron core ['aiən-kɔ:] n. 由铁组成的核
 transient ['trænzɪənt] adj. 瞬间变化的
 bombardment [bɒm'bɔ:dmənt] n. 撞击
 asteroidal body ['æstərɔɪd-l-'bɒdɪ] n. 行星体
 carbon dioxide ['kɑ:bən-dai'ɒksaɪd] n. 二氧化碳
 greenhouse effect ['ɡri:nhaʊs-i'fekt] n. 温室效应
 Venusian [vi:n'ju:siən] adj. 金星的

Jupiter is the largest of the planets. Its hydrogen and helium atmosphere contains pastel-colored clouds, and its immense magnetosphere, rings, and satellites make it a planetary system unto itself. Saturn rivals Jupiter, with a much more intricate ring structure and a similar number of satellites.



注释

hydrogen ['haɪdrədʒən] n. 氢

helium ['hi:ljəm] n. 氦

pastel [pæ'stel] adj. 色彩浅而柔和的

immense [ɪ'mens] adj. 极大的

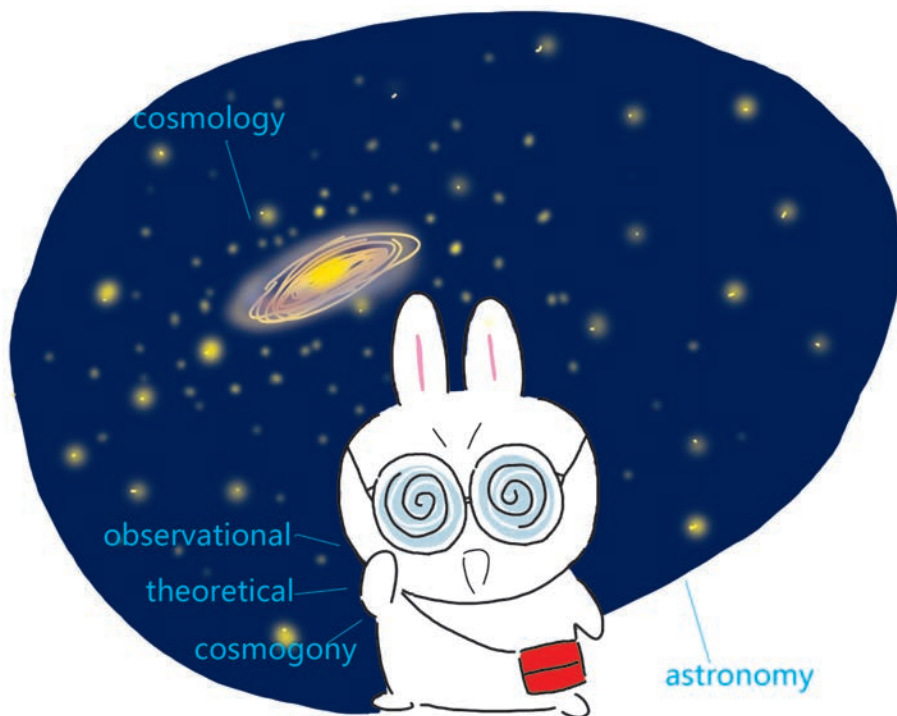
magnetosphere [mæg'ni:təʊsfɪə] n. 磁层

planetary system [plænɪtəri-'sɪstəm] n. 行星系

rival ['raɪvəl] v. 相匹敌

intricate ['ɪntrɪkɪt] adj. 复杂的

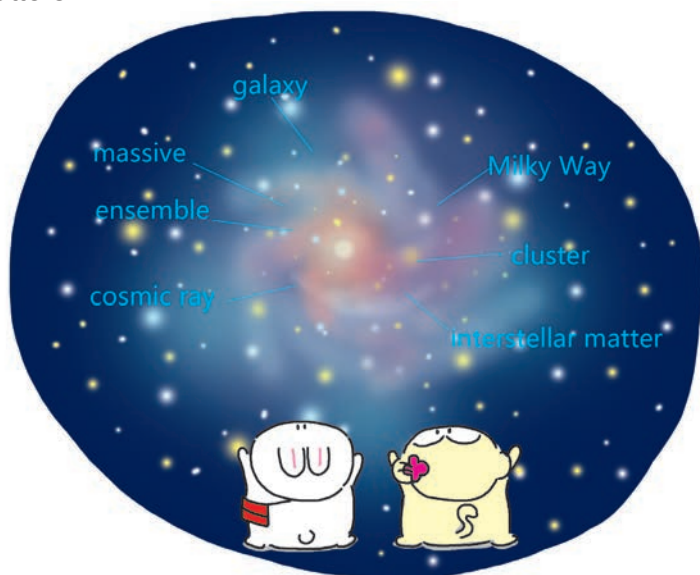
Cosmology is a branch of astronomy, but the observational and theoretical techniques used by cosmologists involve a wide range of other sciences, such as physics and chemistry. Cosmology is distinguished from cosmogony, which used to mean the study of the origin of the universe but now usually refers only to the study of the origin of the solar system.



注释

- cosmogony [kɒz'mɒɡəni] n. 宇宙起源论
 cosmology [kɒz'mɒlədʒi] n. 宇宙天体学
 astronomy [æs'trɒnəmi] n. 天文学
 observational [ˌɒbzə'veɪʃənəl] adj. 观测的
 theoretical [θiə'retikəl] adj. 理论的
 a wide range of [ə-'waɪd-'reɪndʒ-əv] adj. 广泛的
 distinguish [dɪ'stɪŋɡwɪʃ] v. 区别
 origin ['ɒrɪdʒɪn] n. 起源

A galaxy is a massive ensemble of hundreds of millions of stars, all gravitationally interacting, and orbiting about a common center. Astronomers estimate that there are about 125 billion galaxies in the universe. All the stars visible to the unaided eyes from the Earth belong to the Earth's galaxy, the Milky Way. The Sun, with its associated planets, is just one star in this galaxy. Besides stars and planets, galaxies contain clusters of stars; atomic hydrogen gas; molecular hydrogen; complex molecules composed of hydrogen, nitrogen, carbon, and silicon, among others; and cosmic rays, which are collectively called interstellar matters.

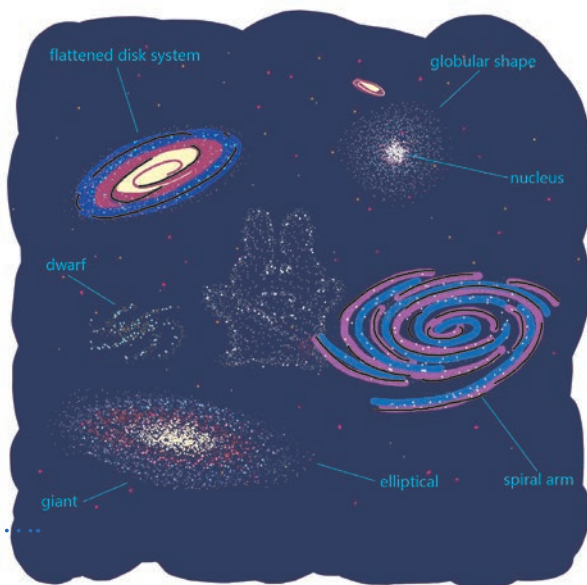


注释

galaxy ['gæləksi] n. 星系
 massive ['mæsɪv] adj. 大量的
 ensemble [ɒn'sɒmbl] n. 整体
 gravitationally [ˌgrævɪ'teɪʃnəli] adv. 引力地
 interact [ˌɪntər'ækt] v. 相互作用
 orbit ['ɔ:bit] v. 绕 (吸引中心) 运转
 unaided [ˌʌn'eɪdɪd] adj. 独立的
 unaided eye [ˌʌn'eɪdɪd-aɪ] n. 肉眼
 Milky Way ['mɪlki:-wei] n. 银河
 associated [ə'səʊʃeɪtɪd] adj. 关联的
 cluster ['klʌstə] n. 星团
 atomic hydrogen gas [ə'tɒmɪk-'haɪdrədʒən-gæs]
 n. 原子氢气

molecular hydrogen [mə'lekjələ-'haɪdrədʒən] n. 分子氢
 complex molecule ['kɒmpleks-'mɒlɪkjʊ:l] n. 复杂分子
 nitrogen ['naɪtrədʒən] n. 氮
 silicon ['sɪlɪkən] n. 硅
 among others [ə'mʌŋ-'ʌðəz] adv. 及其他
 cosmic ray ['kɒzmɪk-reɪ] n. 宇宙射线
 interstellar matter [ˌɪntə(:)'stɛlə-'mæteɪ] n. 星际物质

When viewed or photographed with a large telescope, only the nearest galaxies exhibit individual stars. For most galaxies, only the combined light of all the stars is detected. Galaxies exhibit a variety of forms. Some have an overall globular shape, with a bright nucleus. Such galaxies, called ellipticals, contain a population of old stars, usually with little apparent gas or dust, and few newly formed stars. Elliptical galaxies come in a vast range of sizes, from giant to dwarf. In contrast, spiral galaxies are flattened disk systems containing not only some old stars but also large populations of young stars, much gas and dust, and molecular clouds that are the birthplace of stars. Often the regions containing bright young stars and gas clouds are arranged in long spiral arms that can be observed to wind around the galaxy. Generally a halo of faint older stars surrounds the disk; a smaller nuclear bulge often exists, emitting two jets of energetic matter in opposite directions.



注释

exhibit [ig'zibit] v. 呈现

detect [di'tekt] v. 探测

globular shape ['glɒbjʊlə-feɪp] n. 球形

nucleus ['nju:kliəs] n. 核子

elliptical [i'liptikəl] n. 椭圆星系

giant ['dʒaɪənt] adj. 巨大的

dwarf [dwo:f] adj. 矮小的

flattened disk system ['flætn-disk-'sistəm]
n. 平底圆盘状系统

population [ˌpɒpjʊ'leɪʃən] n. 个数

molecular cloud [məʊ'lekjʊlə-klaʊd] n. 分子云

spiral arm ['spaiərel-ɑ:m] n. 旋臂

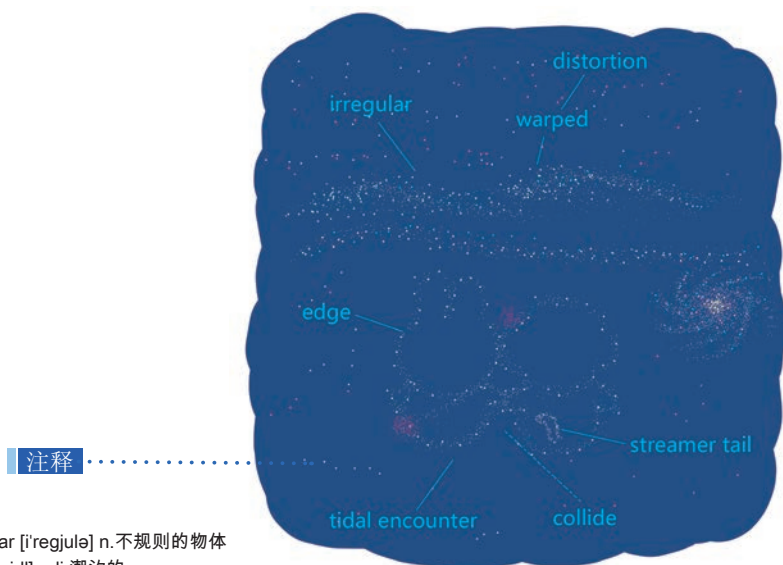
halo ['heɪləʊ] n. 光环

bulge [bʌldʒ] n. 凸出部分

emit [i'mit] v. 喷射

jet [dʒet] n. 喷射物

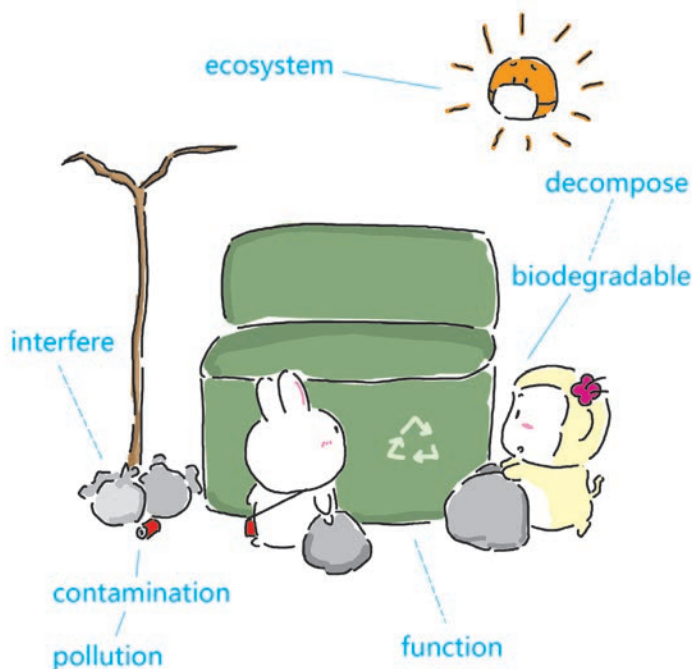
Other disk-like galaxies, with no overall spiral form, are classified as irregular. These galaxies also have large amounts of gas, dust, and young stars, but no arrangement of a spiral form. They are usually located near larger galaxies, and their appearance is probably the result of a tidal encounter with the more massive galaxy. Some extremely peculiar galaxies are located in close groups of two or three, and their tidal interactions have caused distortions of spiral arms, producing warped disks and long streamer tails. Ring galaxies, for example, form when a small galaxy collides with the center of a spiral galaxy. An intense ring of stars forms at the outer edges of the new, combined galaxy. Quasars are objects that appear stellar or almost stellar, but their enormous redshifts identify them as objects at very large distances. They are probably closely related to radio galaxies.



注释

- irregular [i'regjulə] n. 不规则的物体
 tidal ['taɪdl] adj. 潮汐的
 encounter [in'kauntə] n. 相遇
 peculiar [pi'kju:ljə] adj. 罕见的
 interaction [ˌɪntər'ækʃən] n. 相互作用
 distortion [dis'tɔ:ʃən] n. 扭曲变形
 warped [wɔ:pɪd] adj. 弯曲的
 streamer tail ['stri:mə-teɪl] n. 带状的尾巴
 collide [kə'laid] v. 碰撞
 edge [edʒ] n. 边缘
 quasar ['kweɪzɑ:] n. 类星体
 stellar ['steɪlə] adj. 恒星的
 redshift ['red-ʃɪft] n. 红移

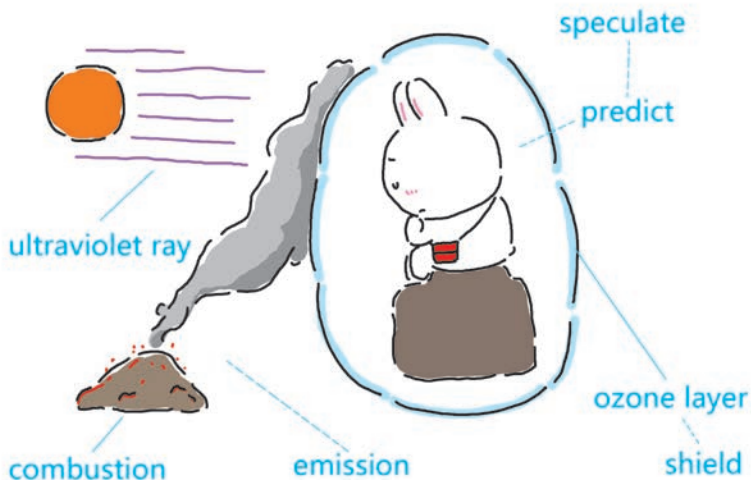
Pollution refers to contamination of the Earth's environment with materials that interfere with human health, the quality of life, or the natural functioning of ecosystems. There are two main categories of pollutants. The biodegradable are materials, such as sewage, that rapidly decompose by natural processes. The nondegradable are materials that either do not decompose or decompose slowly in the natural environment.



注释

- pollution [pə'luʃən] n. 污染 (作用)
 contamination [kən'tæmɪ'neɪʃən] n. 污染, 污物
 interfere [ɪntə'fɪə] v. 有害于 (with)
 function ['fʌŋkʃən] v. (机器、机制等) 运行, 发挥作用
 ecosystem ['i:kəsɪstəm] n. 生态系统
 pollutant [pə'lu:tənt] n. 污染物
 biodegradable [baɪəʊdɪ'greɪdəbl] adj. 可以进行分解和还原处理的
 sewage ['sjuɪdʒ] n. 阴沟污物; 污水
 decompose [di:kəm'pəuz] v. 分解, 分析
 nondegradable ['nɒndɪ'greɪdəbl] adj. 不可以进行分解和还原处理的

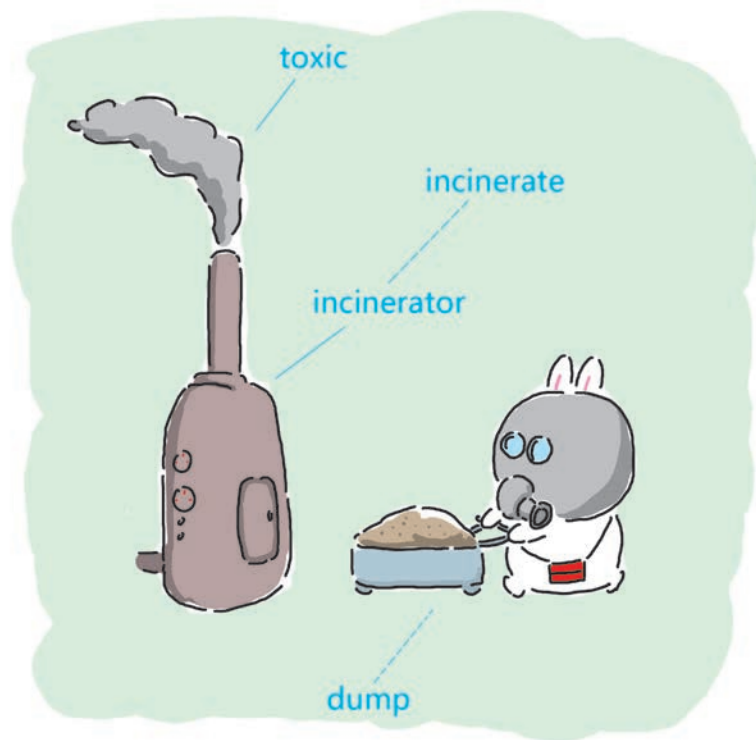
Because of the complex relationships among the many types of organisms and ecosystems, environmental contamination may have far-reaching consequences that are not immediately obvious or that are difficult to predict. For instance, scientists can only speculate on some of the potential impacts of the depletion of the ozone layer, the protective layer in the atmosphere that shields the Earth from the Sun's harmful ultraviolet rays. Another major effect of pollution is the tremendous cost of pollution clean-up and prevention. The global effort to control emissions of carbon dioxide, a gas produced from the combustion of fossil fuels such as coal or oil, or of other organic materials like wood, is one such example.



注释

- far-reaching ['fɑ:-'ri:tʃɪŋ] adj. 深远的, 广泛的
 consequence ['kɒnsɪkwəns] n. 后果, 影响
 immediately [i'mi:diətli] adv. 直接地, 立即
 predict ['pri:dɪkt] v. 预测
 speculate ['spekjuleɪt] v. 思索, 推测 (about, on, upon)
 impact ['ɪmpækt] n. 影响, 效力
 ozone layer ['əʊzəʊn-'leɪə] n. 臭氧层
 shield [ʃi:ld] v. 防御, 保护
 ultraviolet ray [ˌʌltrə'vaɪələt-reɪ] n. 紫外线
 tremendous [tri'mendəs] adj. 可怕的, 惊人的
 emission [i'mɪʃən] n. 发出 (物), 放射 (物)
 combustion [kəm'bʌstʃən] n. 燃烧
 fossil fuel ['fɒsl-'fjuəl] n. (煤、石油、天然气等) 矿物燃料

Research has shown that low-income populations and minorities do not receive the same protection from environmental contamination as do higher-income communities. Toxic waste incinerators, chemical plants, and solid waste dumps are often located in low-income communities because of a lack of organized, informed community involvement in municipal decision-making processes.



注释

minority [mai'nɒrɪti] n. 少数

toxic ['tɒksɪk] adj. 有毒的

incinerate [ɪn'sɪnəreɪt] v. (把……) 烧成灰, 烧掉, 焚化

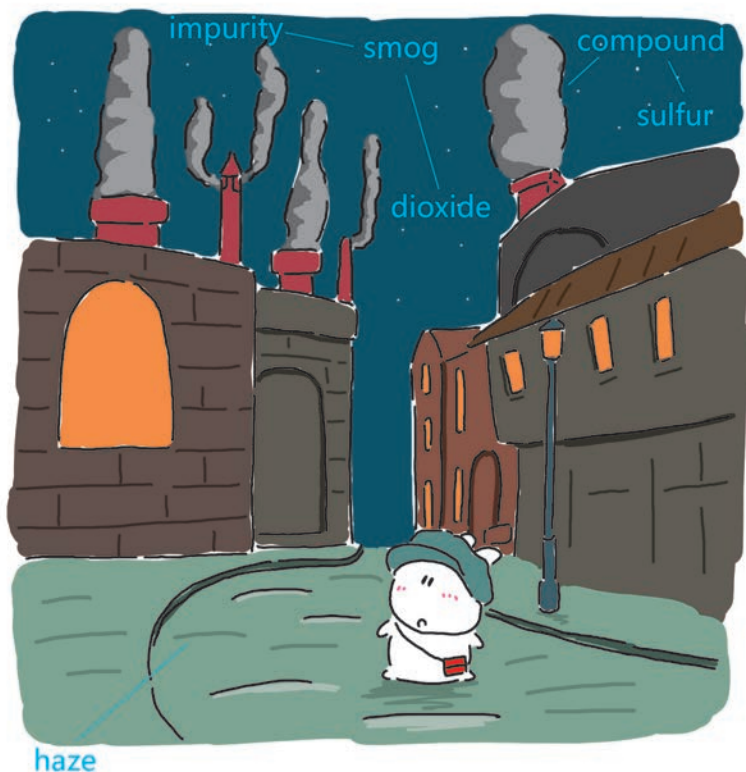
incinerator [ɪn'sɪnəreɪtə] n. (垃圾的) 焚化炉

waste dumps ['weɪst-'dʌmps] v. 废料堆

informed [ɪn'fɔ:md] adj. 见闻广的, 有情报根据的

municipal [mju'nɪsɪpəl] adj. 市政的

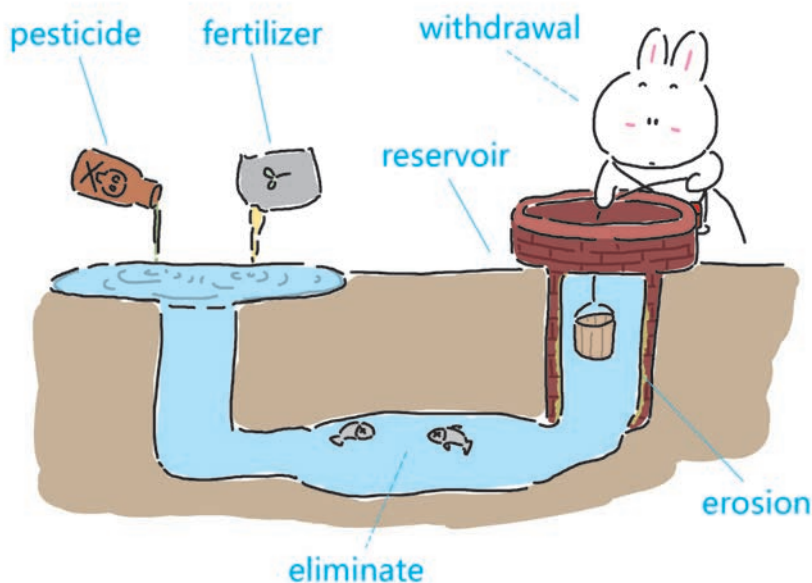
Urban air pollution is commonly known as smog. The dark London smog that Evelyn wrote of is generally a smoky mixture of carbon monoxide and organic compounds from incomplete combustion of fossil fuels such as coal, and sulfur dioxide from impurities in the fuels. As the smog ages and reacts with oxygen, organic and sulfuric acids condense as droplets, increasing the haze.



注释

- smog [smɒɡ] n. 烟雾
 compound ['kɒmpaʊnd] n. 混合物
 sulfur ['sʌlfə] n. 硫 (碱)
 dioxide [daɪ'ɒksaɪd] n. 二氧化物
 impurity [ɪm'pjʊərɪti] n. 不洁, 杂质
 age [eɪdʒ] v. 老化, 陈化
 condense [kən'dens] v. 浓缩, 凝结
 droplet ['drɒplɪt] n. 小滴
 haze [heɪz] n. 霾, 烟雾, 雾

The demand for fresh water rises continuously as the world's population grows. From 1940 to 1990, withdrawals of fresh water from rivers, lakes, reservoirs, and other sources increased fourfold. Sewage, industrial wastes, and agricultural chemicals such as fertilizers and pesticides are the main causes of water pollution. Erosion, the wearing away of topsoil by wind and rain, also contributes to water pollution. Soil and silt washed from logged hillsides, plowed fields, or construction sites, can clog waterways and kill aquatic vegetation. Even small amounts of silt can eliminate desirable fish species.

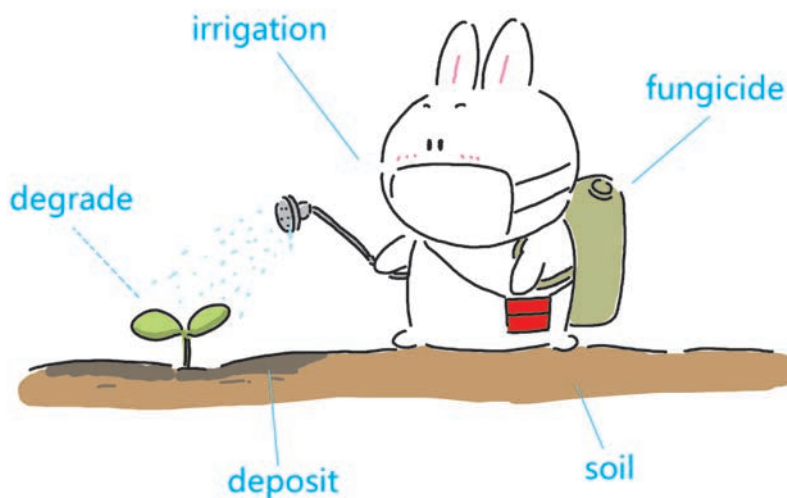


注释

withdrawal [wið'drɔ:əl] n. & v. 提取, 收回
 reservoir ['rezəvwa:] n. 贮水池, 水库
 fourfold ['fɔ:fəʊld] n. 四倍
 fertilizer ['fɜ:tilaɪzə] n. 肥料
 pesticide ['pestisaɪd] n. 杀虫剂
 erosion [i'rəʊʒən] n. 腐蚀, 侵蚀 (作用)
 silt [sɪlt] n. 沙, 淤泥

logged [lɒgd] adj. 低湿的, 浸透的 (土地)
 plow [plau] v. 犁; 耕
 construction site [kən'strʌkʃən-saɪt] n. 建筑工地
 clog [klɒg] v. 阻塞, 塞满
 aquatic [ə'kwætɪk] adj. 水生的, 水栖的
 eliminate [ɪ'limineɪt] v. 除去, 消灭

Unhealthy soil management methods have seriously degraded soil quality, caused soil pollution, and enhanced erosion. Treating the soil with chemical fertilizers, pesticides, and fungicides interferes with the natural processes occurring within the soil and destroys useful organisms such as bacteria, fungi, and other microorganisms. Improper irrigation practices in areas with poorly drained soil may result in salt deposits that inhibit plant growth and may lead to crop failure.



注释

- soil [sɔɪl] n. 土地, 土壤
 degrade [di'greɪd] v. 使剥蚀
 fungicide ['fʌndʒɪsaɪd] n. 杀真菌剂
 interfere [ɪntə'fɪə] v. 有害于
 microorganism [ˌmaɪkrəʊ'ɔ:ɡənɪzəm] n. 微生物
 improper [ɪm'prɒpə] adj. 不适当的, 错误的
 irrigation [ˌɪrɪ'ɡeɪʃən] n. 灌溉
 practice ['præktɪs] n. 做法, 技术
 drain [dreɪn] v. (土地) 排水, (资源等) 逐渐枯竭
 deposit [dɪ'pɒzɪt] n. 淤积, 沉淀
 inhibit [ɪn'hɪbɪt] v. 抑制

Solid wastes are unwanted solid materials such as garbage, paper, plastics and other synthetic materials, metals, and wood. Billions of tons of solid wastes are thrown out annually. Areas where wastes are buried, called landfills, are the cheapest and most common disposal method for solid wastes worldwide. But landfills quickly become overfilled and may contaminate air, soil, and water. Incineration, or burning, of waste reduces the volume of solid wastes, but produces dense ashen wastes (some of which become airborne) that often contain dangerous concentrations of hazardous materials, such as heavy metals and toxic compounds.



注释

garbage [ˈgɑːbɪdʒ] n. 垃圾, 废料

plastic [ˈplæːstɪk] n. 塑料

synthetic [sɪnˈθetɪk] adj. 合成的

landfill [ˈlændfɪl] n. 垃圾填埋洼地

disposal method [dɪsˈpəʊzəl-ˈmeθəd] n. 处置方法

overfill [ˈəʊvəˈfɪl] v. 把……装得溢出

incinerate [ɪnˈsɪnəreɪt] n. 焚化

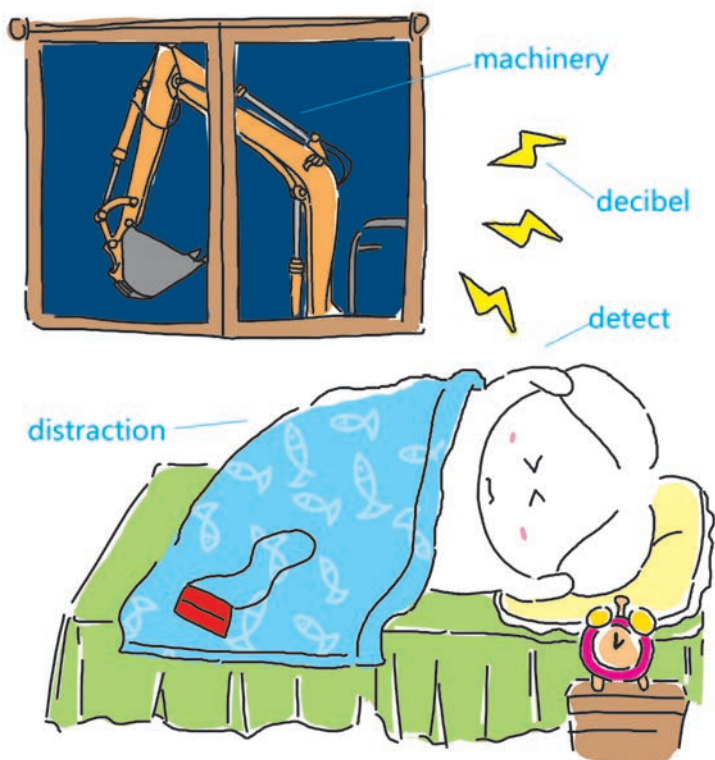
dense [dens] adj. (烟、雾等) 浓密的, 浓厚的

ashen [ˈæʃ(ə)n] adj. 灰的, 灰色的

airborne [ˈeəboːn] adj. (可以通过) 空气传播的

hazardous [ˈhæzədəs] adj. 危险的

Noise, such as that produced by airplanes, traffic, or industrial machinery, is considered a form of pollution. Noise pollution is at its worst in densely populated areas. It can cause hearing loss, stress, high blood pressure, sleep loss, distraction, and productivity loss. Measured in decibels, noise intensity can range from zero, the quietest sound the human ear can detect, to over 160 decibels.



注释

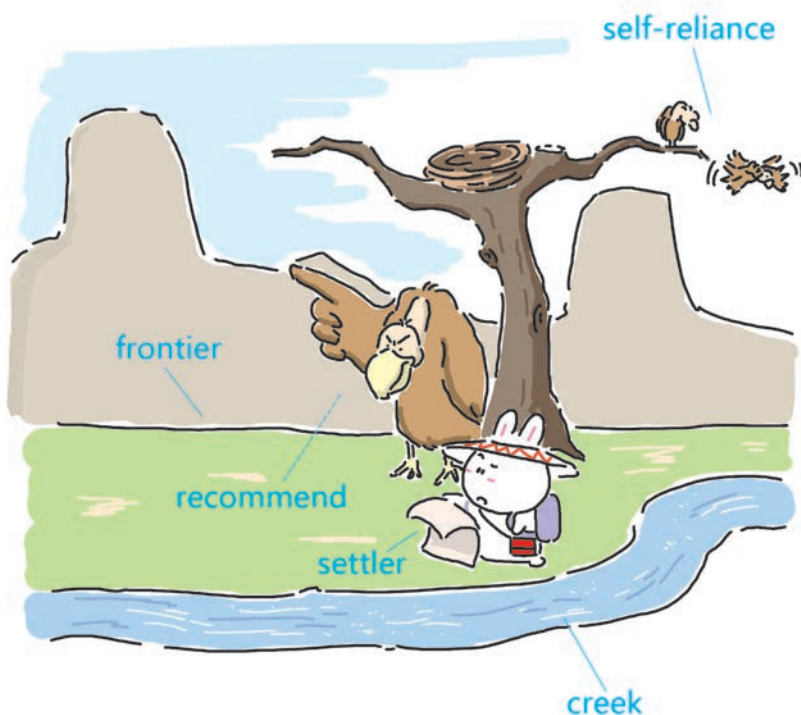
machinery [mə'ʃi:nəri] n. 机械设备

distraction [dis'trækʃən] n. 使人分心的事, 心乱, 心烦

decibel ['desibel] n. 分贝

detect [di'tekt] v. 察觉到

Settlers on the frontier lived on lands that were sparsely populated. Although frontier life demanded self-reliance, settlers often wanted government help for improving the new region by building roads, recommending crops, running surveys, certifying claims, and dredging creeks. They also wanted the federal government to provide protection for them from Native Americans who were occasionally hostile to settlers moving onto their lands.

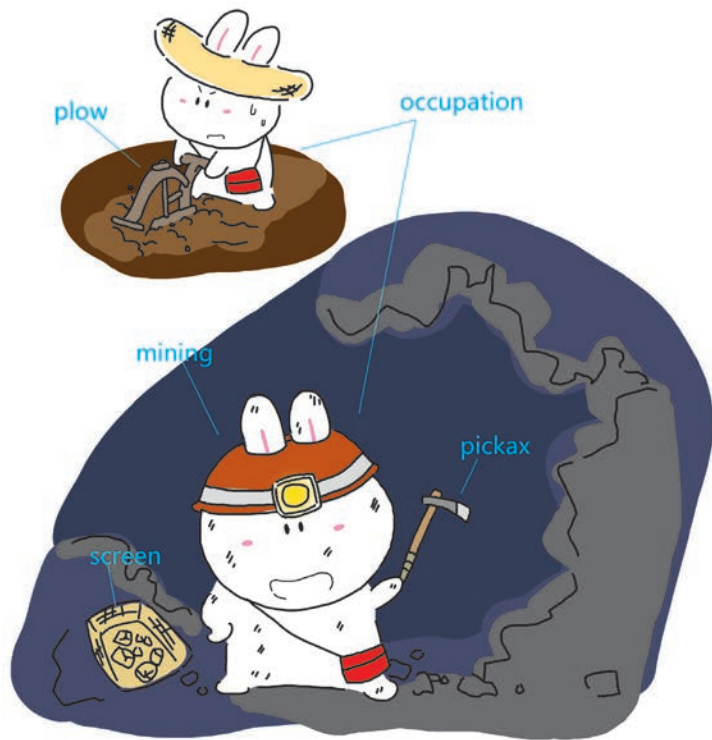


注释

settler ['setl̩ə] n. 移民者
 frontier ['frʌntiə] n. 边境
 sparsely [spɔːsli] adv. 稀少地
 self-reliance ['self-rɪ'leɪns] n. 依靠自己
 recommend [ˌrekə'mend] v. 推荐
 running survey [ˈrʌnɪŋ-'səːveɪ] n. 勘测

certify ['səːtɪfaɪ] v. 证明
 dredge [dredʒ] v. 用挖掘机清除、加深或加宽
 creek [kri:k] n. 小溪
 federal ['fedərəl] adj. 联邦的
 hostile ['hɒstail] adj. 敌对的
 (be hostile to : 对……有敌意)

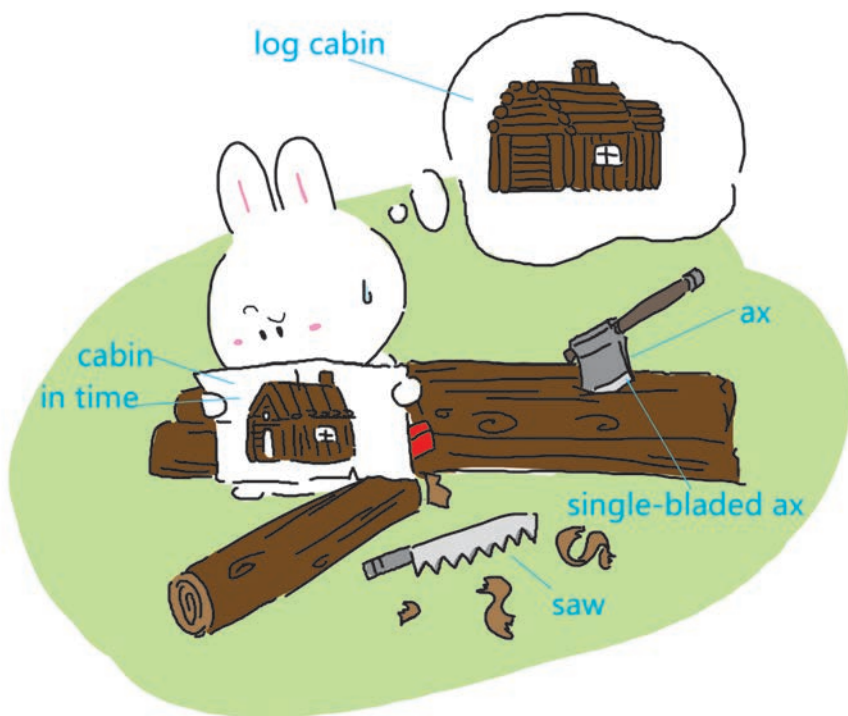
On the frontier, tools varied with time, place, and occupation. In agriculture, of course, settlers depended on plow. If they turned to ranching, they needed a lariat (or rope), a branding iron, and a pair of pliers. If they took up mining, they carried a pickax, a screen, and a skillet.



注释

- occupation [ˌɒkjʊˈpeɪʃən] n. 职业
 plow [plau] n. 犁
 ranch [rɑːntʃ] v. 经营牧场
 lariat ['læriət] n. 套索
 rope [rəʊp] n. 绳索
 branding iron ['brændɪŋ-'aɪən] n. 烙铁
 pliers ['plaɪəz] n. 老虎钳
 mining ['maɪnɪŋ] n. 采矿
 pickax ['pɪ,kæks] n. 镐
 screen [skri:n] n. 筛子
 skillet ['skɪlɪt] n. 煎锅 (= frying pan)

Until the people on the frontier moved beyond the forests, in the second half of the 19th century, they had to cut down trees to clear space in which to erect a log cabin, and, in time, to grow crops. So long as they remained in the forests, the most important and most widely used tool was the single-bladed ax which could be used to cut down trees, make firewood, and provide the wood for a log cabin. Some saws were used, although they remained primitive through the first 200 years of frontier America.



注释

erect [i'rekt] v. 搭建

cabin ['kæbin] n. 小木屋

in time ['in-taim] adv. 最终

ax [æks] n. 斧头 (也做ax, plural: axes ['æksi:z])

single-bladed ax ['singl-bleidid-æks] n. 单刃斧

log cabin [log-'kæben] n. 小木屋

saw [sɔ:] n. 锯

primitive ['primitiv] adj. 原始的

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Frontier houses were built in all sizes and shapes and to all degrees of comfort. Logs for a log cabin were hewn from whatever was available; oak and pine were the favorite woods. Settlers might add split logs for floors; cover the roof with bark, thatch, or shingles; or build a log fireplace and chimney daubed with mud. As the mud filling the holes on the outside of the cabin dried, the settlers might smooth and whitewash it.



注释

hew [hju:] v. 砍 [past: hewn]

oak [əuk] n. 橡木

pine [pain] n. 松树

split [split] adj. 劈开的

bark [bɑ:k] n. 树皮

thatch [θætʃ] n. 茅草

shingle ['ʃɪŋɡl] n. 木瓦

fireplace ['faɪəpleɪs] n. 壁炉

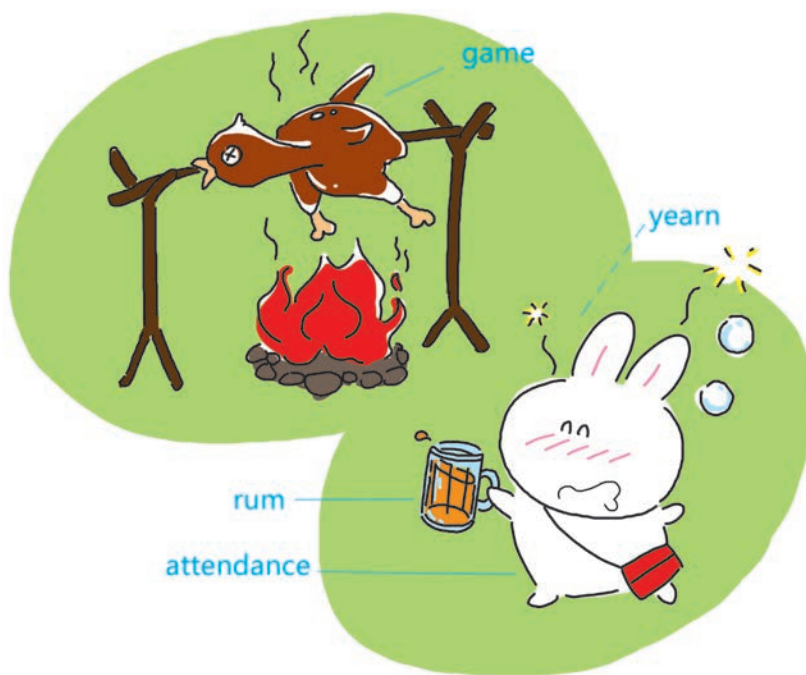
chimney ['tʃɪmni] n. 烟囱

daub [dɔ:b] v. 涂抹

smooth [smu:ð] v. 使光滑

whitewash ['(h)waitwɔʃ] v. 用石灰水粉刷

People on the frontier amused themselves primarily by visiting neighbors. A man's visit to a tavern or the family's attendance at a church revival provided events to be recalled in detail through the long cold evenings of the winter. The frontier family yearned for other faces. If people lived close enough, the raising of a log cabin could be a great social event. The women would bake for days, game would be killed and salted down, and, unless the religion of the frontier community forbade it, rum and whiskey would be collected for a work-and-play party that might last as long as several days.



注释

tavern ['tævə:n] n. 酒馆

attendance [ə'tendəns] n. 出席

revival [ri'vaɪvəl] n. 复兴大会 (尤指旨在促进基督教信仰的布道会)

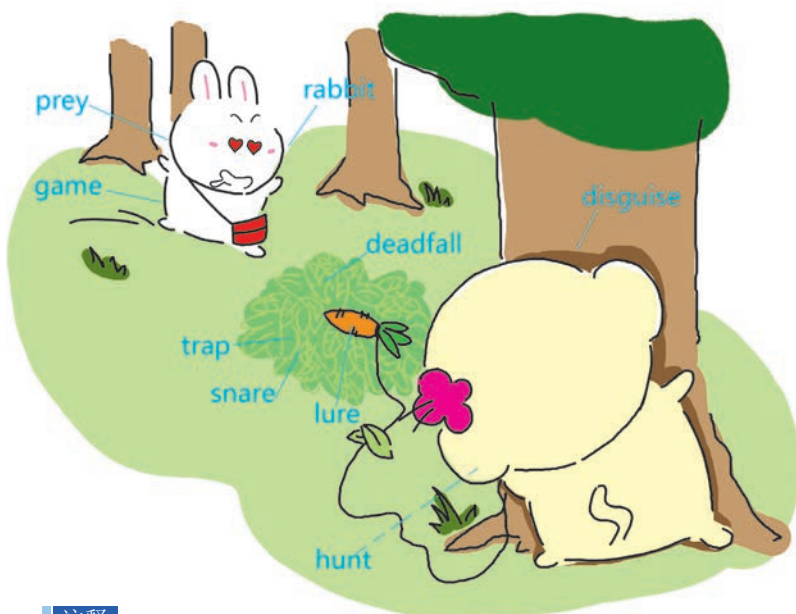
yearn [jɜ:n] v. 向往

game [geɪm] n. 猎物

forbid [fə'bid] v. 禁止

rum [rʌm] n. 朗姆酒

Northeast Indians hunted a variety of game, large and small: deer, rabbit, squirrel, beaver, and various birds, such as turkey, partridge, duck, and goose. They also hunted moose, elk, and bear. Some peoples living near the prairies of the Mississippi River Valley hunted the North American bison, or buffalo. In addition to hunting with spears, bows and arrows, and clubs, they used traps, snares, and deadfalls (traps designed to cause heavy objects, such as logs, to fall, disabling or killing prey). They used disguises to get close to animals, lured prey with animal calls, and set fires to drive animals toward the hunters or traps.



注释

hunt [hʌnt] v. 狩猎
game [geɪm] n. 猎物
rabbit ['ræbɪt] n. 野兔
squirrel ['skwɪrəl] n. 松鼠
beaver ['bi:və] n. 海狸
turkey ['tɜ:kɪ] n. 火鸡
partridge ['pɑ:trɪdʒ] n. 鹌鹑
moose [mu:s] n. 驼鹿
elk [elk] n. 麋鹿
prairie ['præəri] n. 大草原
bison ['baɪsn] n. 野牛
buffalo ['bʌfələʊ] n. 北美野牛

spear [spiə] n. 矛
bow [bəʊ] n. 弓
arrow ['ærəʊ] n. 箭
club [klʌb] n. 棍棒
trap [træp] n. 陷阱
snare [sneə] n. 陷阱
deadfall ['dedfɔ:l] n. 陷阱
disable [dis'eɪbl] v. 使残废
disguise [dis'gaɪz] n. 伪装
lure [luə] v. 引诱
prey [preɪ] n. 猎物

Northeast Indians also fished rivers, streams, lakes, and ponds. They caught fish with harpoons, hooks, nets, and traps. Peoples living along the Atlantic Coast depended on shellfish for part of their diet. Wild plant foods were also an important food source, including berries, nuts, roots, stalks, and leaves. Some tribes along the western Great Lakes collected a tall grass with an edible grain, referred to as a wild rice. Peoples living in maple country collected sap from the trees in early spring, and boiled it down into maple syrup and sugar.



注释

stream [stri:m] n. 河流

pond [pɒnd] n. 池塘

harpoon [hɑ:'pu:n] n. 鱼叉

hook [huk] n. 鱼钩

net [net] n. 网

shellfish ['ʃel,fɪʃ] n. 贝类

berry ['beri] n. 浆果

nut [nʌt] n. 坚果

root [ru:t] n. 根

stalk [sto:k] n. 茎

tribe [traɪb] n. 部落

edible grain ['edibl-'greɪn] n. 食用谷物

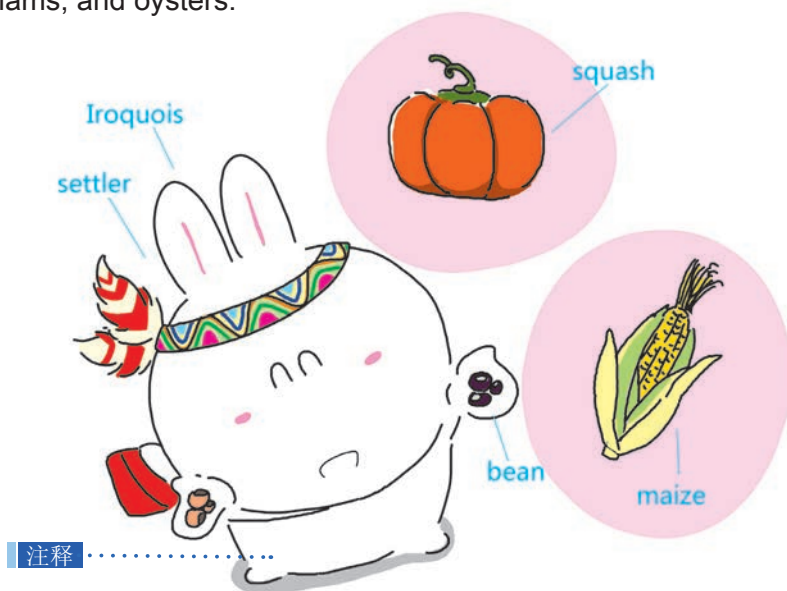
maple ['meɪpl] n. 枫树

sap [sæp] n. 树液

boil [boɪl] n. 煮

maple syrup ['meɪpl-'sɪrəp] n. 枫糖浆: 由糖枫汁液做成的甜糖浆

Most Northeast peoples supplemented a hunting-gathering diet with farming. The Iroquois thought of their three most important crops — maize (corn), beans, and squash. Corn stalks supported the vines of bean plants, while the large-leaved squash plants served to block weed growth. Algonquian peoples introduced the Pilgrims and other early settlers in their homelands to these cultivated crops in addition to many wild foods, including maple sugar, cranberries, blueberries, lobsters, clams, and oysters.



supplement ['sʌplɪmənt] v. 补充

Iroquois ['iɾəkwɔɪ, 'iɾəkwɔɪz] n. 易洛魁族人 (北美印第安人的一个民族)

maize [meɪz] n. 玉米

bean [bi:n] n. 豆

squash [skwɒʃ] n. 南瓜

corn stalk [kɔ:n-stɔ:k] n. 玉米杆

vine [vaɪn] n. 藤蔓

weed [wi:d] n. 杂草

Algonquian [æl'gɔŋkwɪ:ən, -ki:ən] n. 阿尔贡金族人 (北美印第安人的一个民族)

pilgrim ['pɪlgrɪm] n. 移民者: 1620年, 在新英格兰建立普利茅斯殖民地的英国主张脱离国教者

settler ['setlɜ] n. 移民者

cultivate ['kʌltɪveɪt] v. 耕种

in addition to [ɪn-ə'dɪʃən-tu:] prep. 除……之外还

cranberry ['krænbəri] n. 蔓越橘

blueberry ['blu:bəri] n. 蓝莓

lobster ['lɒbstə] n. 龙虾

clam [klæm] n. 蛤

oyster ['ɔɪstə] n. 牡蛎

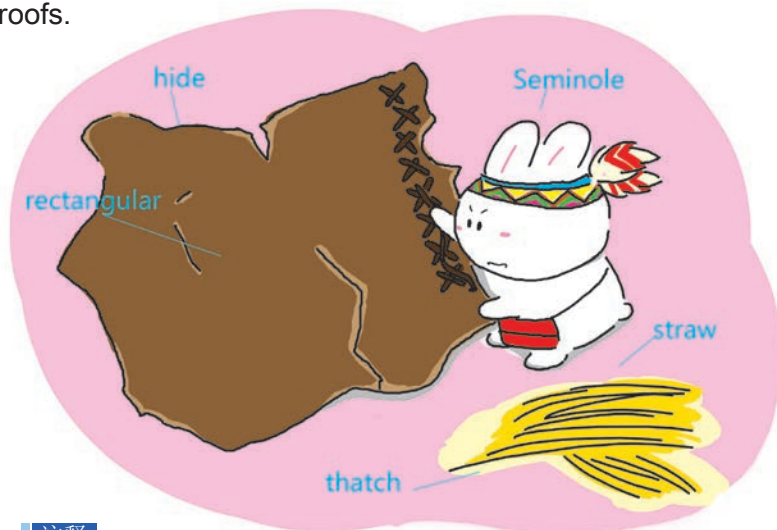
Family played an important role in Northeast Indian society. Most tribes were further organized into clans — clusters of related families who claimed a common ancestor. The Iroquois was a matrilineal society, with descent and property passing through the female line. Each clan was headed by an elder woman, known as the clan mother, who owned the crops and the communal dwellings and held great political power. They elected tribal chiefs, who were generally male, retained the right to veto actions they opposed, and had to approve declarations of war. Unlike the Iroquois, the Algonquian was a patrilineal society, with descent and property traced through the male line.



注释

- tribe [traɪb] n. 部落
 clan [klæn] n. 部族
 ancestor ['ænsɪstə] n. 祖先
 matrilineal [ˌmætrɪ'liːniəl] adj. 母系的
 descent [dɪ'sent] n. 血统
 clan mother [klæn-'mʌðə] n. 族长
 dwelling ['dwelɪŋ] n. 住处, 居住空间
 chief [tʃi:f] n. 首领
 retain [ri'teɪn] v. 保留
 veto ['vi:təu] v. 否决
 declaration [ˌdekle'reɪʃən] n. 宣言
 patrilineal [ˌpætrɪ'liːniəl] adj. 父系的
 trace [treɪs] v. 顺着; 追溯

Most Southeast peoples located their villages along river valleys and planted their crops in nearby fields. Homes and public buildings were typically rectangular or, less frequently, circular. Most structures were constructed of wattle and daub, a type of architecture in which branches and vines were tied over pole frameworks and covered with a mixture of mud or clay. Sometimes structures were covered with plant materials, including thatch — made from straw, reeds, rushes, and grasses — as well as woven mats, bark, bamboo stalks, and palm fronds. Animal hides were also used as coverings. For swampy areas the Seminole people built chickees, distinctive open-sided houses on stilts with wooden platforms and thatched roofs.



注释

locate [ləu'keit] v. 设置, 确定……的位置

valley ['væli] n. 流域, 河谷

rectangular [rek'tæŋgjulə] adj. 矩形的

circular ['sɜ:kjulə] adj. 圆形的

wattle ['wɒtl] n. 柳栅: 用嫩枝、芦苇或枝条编成的构架, 用以筑墙、篱笆和屋顶

daub [do:b] n. 涂料: 柔软粘稠的涂抹原料, 如泥灰、润滑脂或泥浆

branch [brɑ:nʃ] n. 树枝

vine [vain] n. 藤蔓

framework ['freimwɜ:k] n. (框架) 结构

mud [mʌd] n. 泥

clay [klei] n. 黏土

thatch [θætʃ] n. 茅草

straw [stro:] n. 麦秆

rush [rʌʃ] n. 灯心草

mat [mæt] n. 草垫

bark [bɑ:k] n. 树皮

palm frond ['pɑ:m - frɒnd] n. 棕榈叶

hide [haɪd] n. 兽皮

swampy ['swɒmpi] adj. 沼泽的

Seminole ['semineʊl] n. 塞米诺族印第安人

chickee ['tʃiki:] n. (西米诺尔印第安人的) 棕榈叶顶棚屋

distinctive [dis'tɪŋktɪv] adj. 独特的, 有特色的

stilt [stɪlt] n. 支撑物

In warm weather, Southeast Indian men typically wore only breechcloths, usually of deerskin. Women typically wore wraparound plant-fiber skirts and shell necklaces. In cold weather, men wore deerskin shirts, leggings, and moccasins; women wore deerskin capes and moccasins. For ceremonial purposes, tribal leaders and priests wore capes of feathers. Among some Southeast tribes, men plucked out their hair with shell tweezers, and tattooed themselves with designs representing exploits in war and with totems (symbols that serve as an emblem of a family or clan). Elaborate tattoos also adorned some Southeast women.

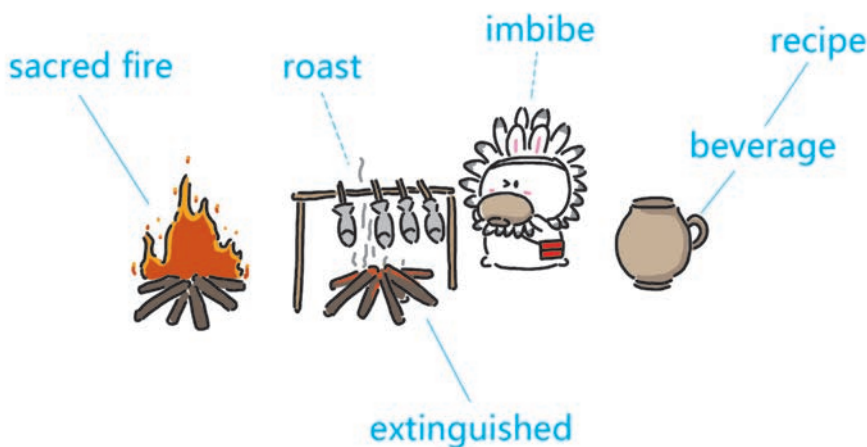


注释

breechcloth ['bri:tʃkləθ] n. 短裤
 deerskin ['di:skɪn] n. 鹿皮
 wraparound ['ræpə,raʊnd] adj. 裹身的
 necklace ['nekli:s] n. 项链
 legging ['legɪn] n. 紧身裤
 cape [keɪp] n. 披肩, 短斗篷
 moccasin ['mɒkəsɪn] n. 鹿皮靴
 ceremonial [ˌseri'məʊnjəl] adj. 仪式场合的
 priest [pri:st] n. 祭司
 capes of feathers [keɪps-əv-'feðəz] n. 扎满羽毛的披肩
 pluck out ['plʌk-aʊt] v. 拔掉

tweezers ['twi:zəz] n. 镊子
 tattoo [tə'tu:] v. 刺花纹
 exploit [ɪks'plɔɪt] n. 功绩[勋], 英勇的行为
 totem ['təʊtəm] n. 图腾
 emblem ['embləm] n. 象征
 elaborate [ɪ'læbəreɪt] adj. 精巧的, 精细的
 adorn [ə'dɔ:n] v. 装饰

The Green Corn Ceremony was an annual renewal and thanks-giving festival performed by the Cherokee, Creek, and other Southeast tribes. It was held in mid or late summer, when the corn was ready for roasting. The ceremony lasted from four to eight days and included ritual fasting, dancing, and feasting. Old fires were extinguished, and a new sacred fire was lit from which every household obtained fire. New tools, weapons, and clothing were made. Wrongdoers were forgiven for most crimes except murder. A beverage known as the Black Drink — so named by English traders because of its dark color — was believed to purify spiritually all those who imbibed it. Different tribes had different recipes for this ritual tea, made from varying species of holly, tobacco, and other plants.

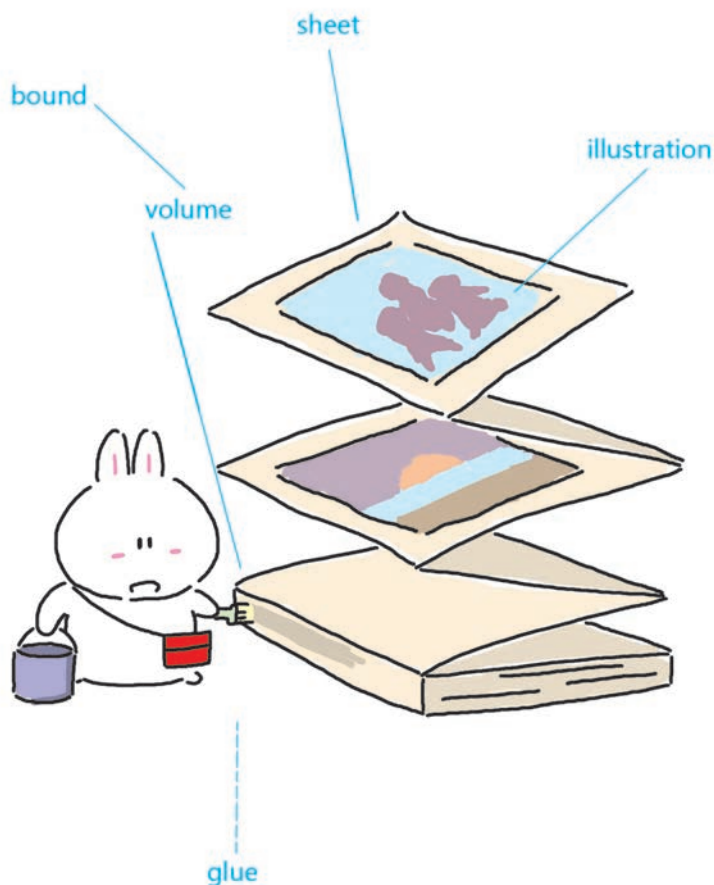


注释

renewal [ri'njuəl] n. 更新
 festival ['festəvəl] adj. 节日的
 Cherokee ['tʃərə'ki:] n. 切罗基族
 (北美易洛魁人的一支)人
 Creek [kri:k] n. 美国克里克族印第安人
 roast [rəʊst] v. 烘烤
 ceremony ['seriməni] n. 仪式
 ritual ['ritʃuəl] adj. (宗教) 仪式的
 feast [fi:t] n. 宴会

extinguish [iks'tɪŋɡwɪʃ] v. 熄(灯), 灭(火), 熄灭
 sacred fire ['seɪkɪd-'faɪə] n. 祭祀的火焰
 wrongdoer ['rɔ:ŋ,du:ə,'rɔ:ŋ-] n. 做坏事的人
 beverage ['bevərɪdʒ] n. 饮料
 purify ['pjʊərɪfaɪ] v. 净化
 imbibe [im'baɪb] v. 喝
 recipe ['resəpi] n. 秘方
 holly ['hɒli] n. 冬青树

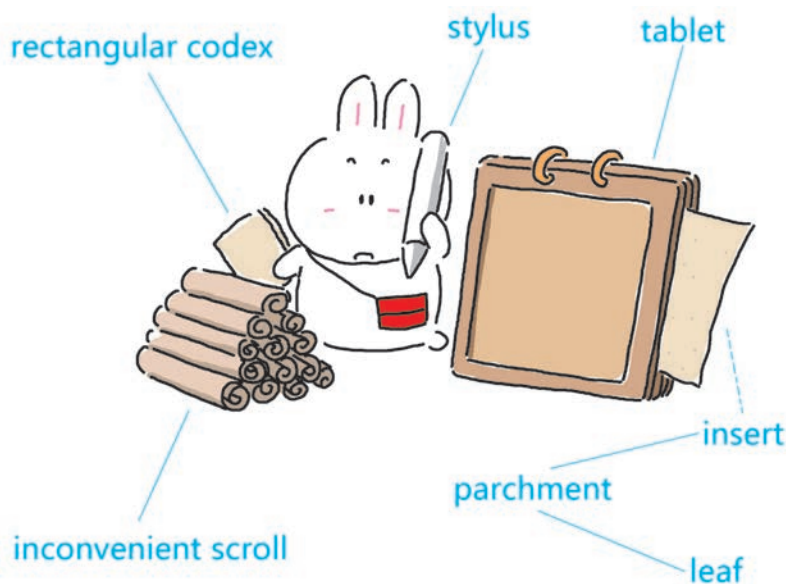
Book refers to a volume of many sheets of paper bound together, containing text, illustrations, music, photographs, or other kinds of information. The pages are sewn or glued together on one side and bound between hard or soft paper covers.



注释

- volume ['vɒljʊ:m] n. 卷
 sheet [ʃi:t] n. (一) 张
 bound [baʊnd] adj. 连在一起的
 illustration [ˌɪlə'streɪʃən] n. 插图
 sew [sew] v. 缝合
 glue [glu:] v. 粘合

The 4th century also marked the culmination of a gradual process, began about the 1st century, in which the inconvenient scroll was replaced by the rectangular codex (Latin for book), the direct ancestor of the modern book. The codex, as first used by the Greeks and Romans for business accounts or school work, was a small ringed notebook consisting of two or more wooden tablets covered with wax, which could be marked with a stylus, smoothed over, and reused many times. Additional leaves, made of parchment, were sometimes inserted between the tablets.

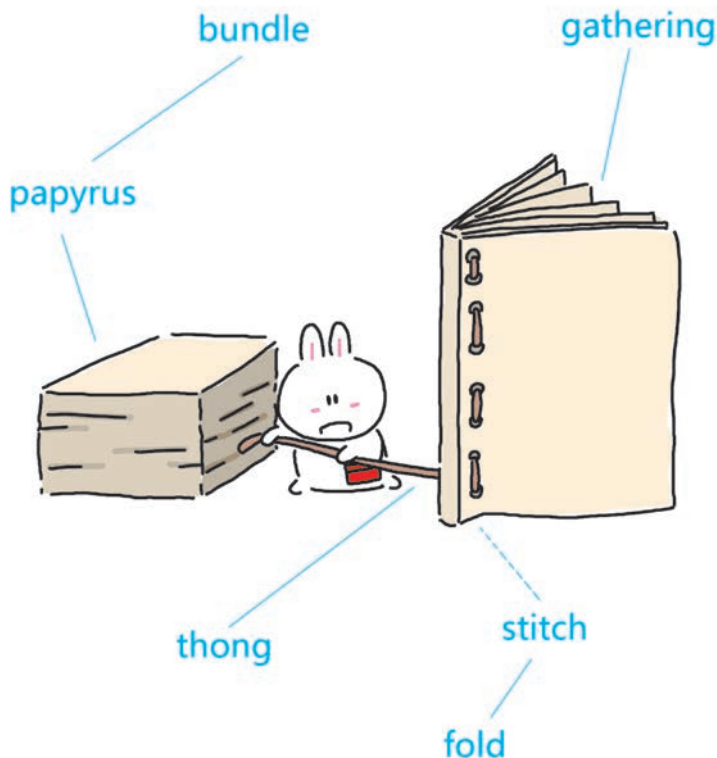


注释

culmination [kʌlmiˈneɪʃ(ə)n] n. 顶点
 gradual [ˈɡrædʒuəl] adj. 逐渐的
 inconvenient [ˌɪnkənˈviːnjənt] adj. 不方便的
 scroll [skrɔːl] n. 书卷
 rectangular [ˌrekˈtæŋɡjələ] adj. 矩形的
 codex [ˈkəudeks] n. 抄本
 (plural: codices [ˈkəudisiːz])
 ancestor [ˈænsɪstə] n. 祖先

account [əˈkaʊnt] n. 账目
 tablet [ˈtæblɪt] n. 写字板
 wax [wæks] n. 蜡
 stylus [ˈstɑɪləs] n. (古人刻写蜡板的) 尖笔
 smooth over [smuːð-ˈəʊvə] v. 擦除
 leaf [liːf] n. 书页
 parchment [ˈpɑːtʃmənt] n. 羊皮纸
 insert [ˈɪnˈsɜːt] v. 插入, 夹进

In time, the codex came to consist of many sheets of papyrus or, later, parchment, gathered in small bundles folded in the middle. These gatherings were laid one upon the other, stitched together through the folds, and attached to wooden boards by thongs.



注释

in time [in-taim] adv. 最终

consist of [kən'sist-əv] v. 由……组成

papyrus [pə'paiərəs] n. (古代) 纸 (莎) 草纸 (plural: papyri [pə'paiəri])

bundle ['bʌndl] n. 捆

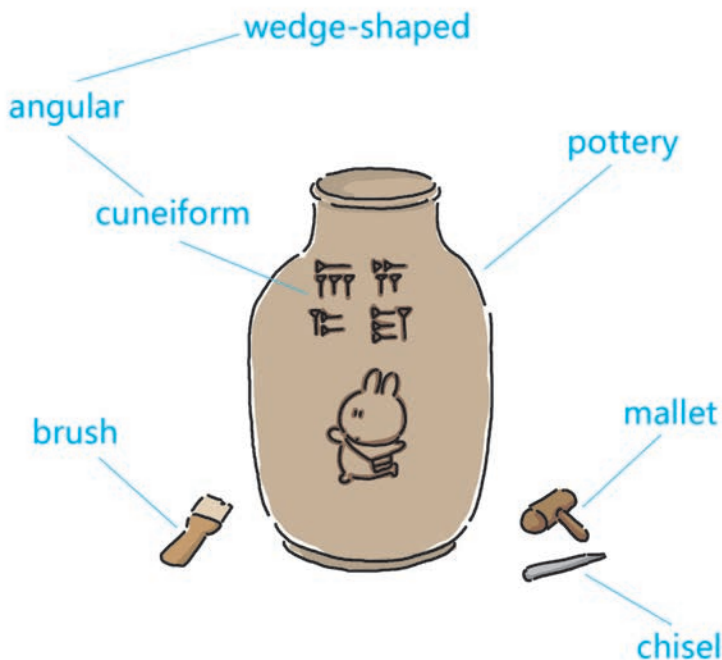
gathering [ˈgæðərɪŋ] n. 订成书贴的书页

stitch [stɪtʃ] v. 缝合

fold [fəʊld] n. 折页

thong [θɒŋ] n. 皮带

The earliest form of Western writing was cuneiform, made by pressing an angular stick of three or four sides into soft clay that was then baked, making these wedge-shaped marks permanent. The next major developments in writing tools were the use of the brush and of the mallet and chisel by the Greeks. Writing found on ancient Greek pottery was done with a small round brush, and early Greek letters were incised on stone with a metal chisel driven by a mallet. Romans, using broad-edged tools, introduced variations in the width of alphabetic marks.

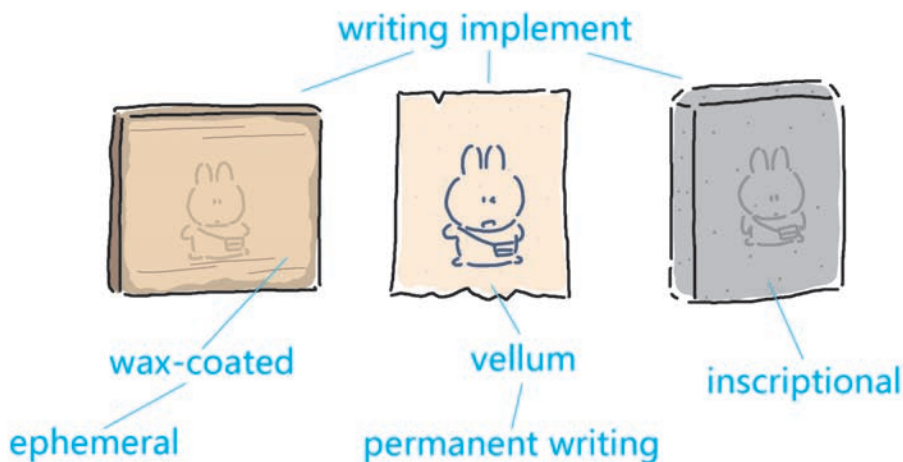


注释

cuneiform ['kju:ni:fɔ:m] n. 楔形文字
 press [pres] v. 压
 angular ['æŋgjʊlə] adj. 有角的
 clay [klei] n. 黏土
 bake [beik] v. 烧硬
 wedge-shaped ['wedʒeɪpt] adj. 楔形的
 permanent ['pɜ:mənənt] adj. 持久的
 brush [brʌʃ] n. 刷子

mallet ['mælit] n. 木槌
 chisel ['tʃizl] n. 凿子
 pottery ['pɒtəri] n. 陶器
 incise [in'saiz] v. 刻
 variation [vɛəri'eɪʃən] n. 变体
 alphabetic [ælfə'betik] adj. 照字母次序的

By the beginning of the 1st century AD, Roman writing implements varied according to both the purpose of the writing and the surface used. Ephemeral writing and school exercises were often done with pointed styluses, made of metal or bone on small wax-coated wooden tablets. Letters were scratched on the waxed surface with the pointed end of the stylus and erasures were made with the other, blunt end of the same tool. Permanent writing was done on papyrus with a reed cut to a point and dipped in ink. Flat brushes and reeds cut with a broad edge were used on smooth surfaces, such as specially prepared animal skins (vellum or parchment) and plaster or stone walls. Inscriptional writing was done with mallet and chisel, but the style of these inscribed letters, with their variations from thick to thin strokes.



注释

writing implement ['raɪtɪŋ-'ɪmplɪmənt] n. 书写用具

ephemeral [ɪ'femərəl] adj. 短暂的

wax-coated [wæks-'kəʊtɪd] adj. 涂蜡的

scratch [skrætʃ] v. 刻划

blunt [blʌnt] adj. (笔尖等) 无尖锋的, 钝的

permanent writing ['pɜ:mənənt-'raɪtɪŋ] n. 能够长时间保留的书写

reed [ri:d] n. 芦苇

dip [dɪp] v. 浸蘸

vellum ['veləm] n. 牛皮纸

plaster [plɑ:stə] n. 胶泥, 灰泥, 涂墙泥

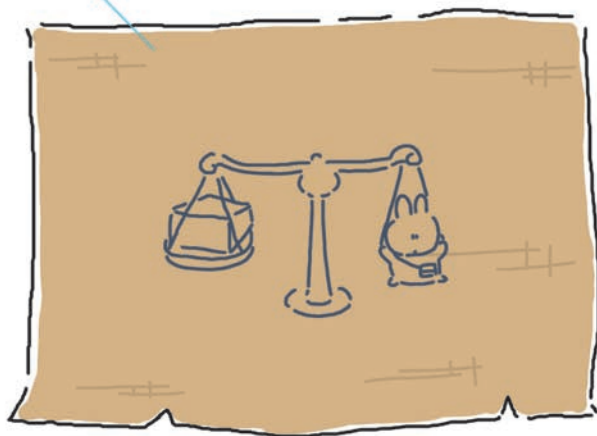
inscriptional [ɪn'skrɪpʃənəl] adj. 碑文的

stroke [strəʊk] n. 笔划

UNIT188

As the size of writing became smaller, both writing tools and surfaces changed. Vellum or parchment books replaced the papyrus roll, and the quill replaced the reed pen.

papyrus



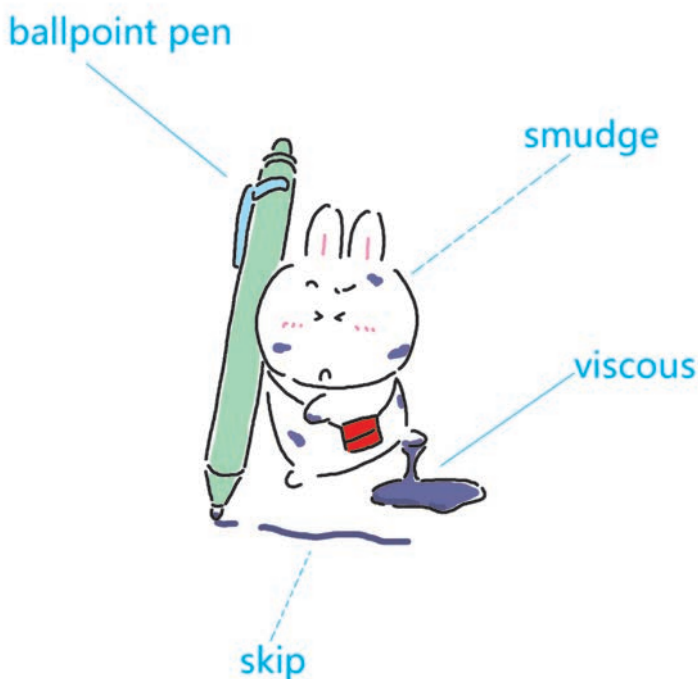
quill

注释

papyrus [pə'paɪərəs] n. 莎草纸

quill [kwɪl] n. 羽毛笔

The success of ball pen was based upon a viscous, oil-based ink. Early ballpoint pens did not write well; they tended to skip, and the slow-drying oil-based ink smudged easily.



注释

ball pen [bɔ:l-pen] n. 圆珠笔

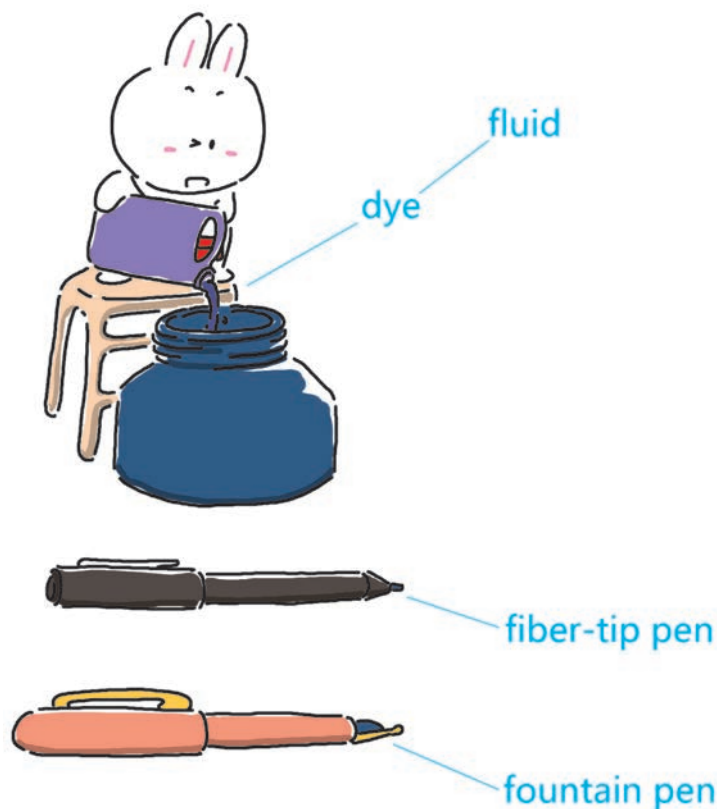
viscous ['vɪskəs] adj. 黏性的

skip [skip] v. 滑, 笔触中途不出油

smudge [smʌdʒ] v. 弄脏, 涂去, 使模糊

UNIT190

Unlike its predecessors, the fiber-tip pen uses dye as a writing fluid. As a result, the fiber-tip pen can produce a wide range of colors unavailable in ballpoint and fountain pen inks.



注释

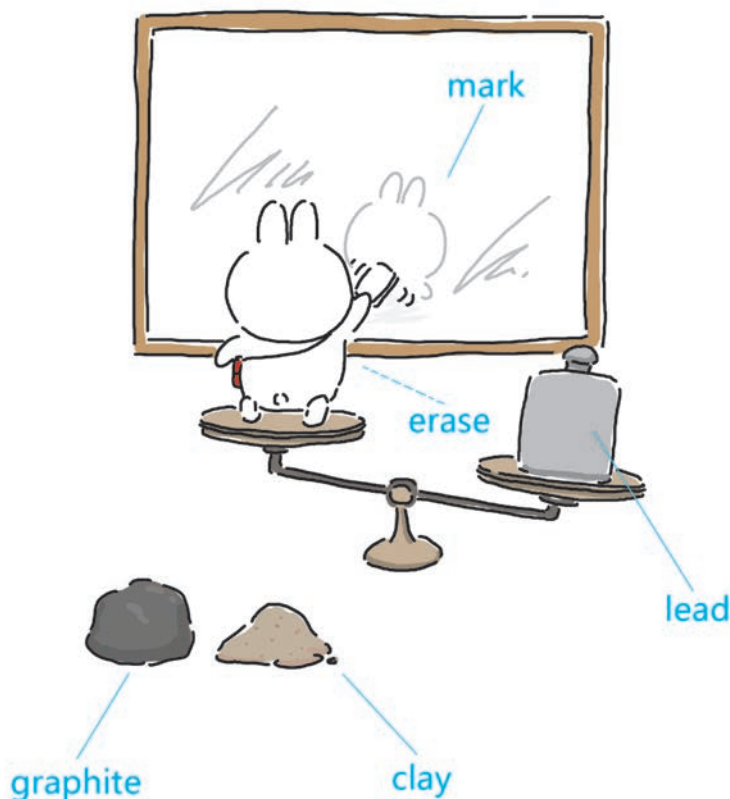
fiber-tip pen ['faɪbə-tɪp-'pen] n. 纤维笔

dye [daɪ] n. 染料

fluid ['flu:ɪd] n. 液

fountain pen ['faʊntɪn-'pen] n. 自来水钢笔

One of the most popular tools for ephemeral writing is the pencil. Pencil marks, unlike those made by writing implements using fluids, can be easily erased. Although commonly called lead pencils, they do not contain any of that metal but are composed of a mixture of graphite (a form of carbon) and clay.



注释

mark [mɑ:k] n. (笔) 迹, 痕迹

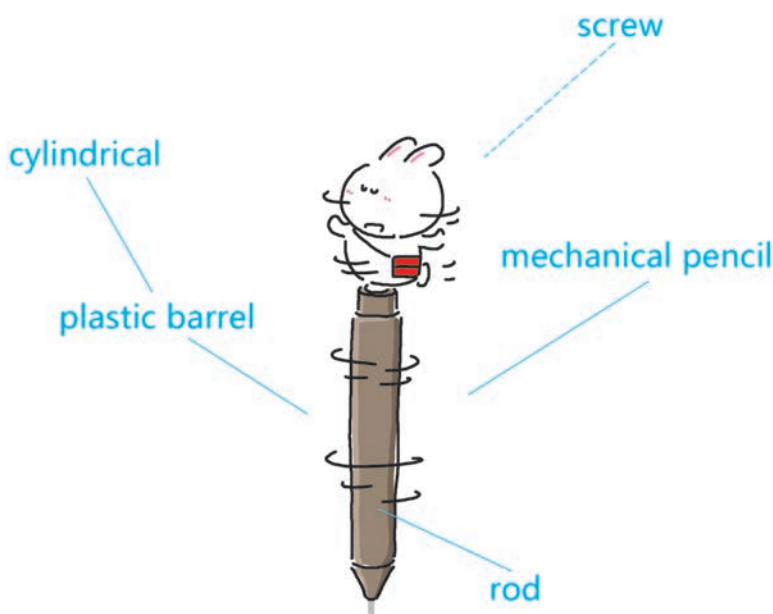
erase ['i:reiz] v. 擦掉

lead [led] n. 铅

graphite ['græfait] n. 石墨

clay [klei] n. 黏土

The mechanical pencil, patented in 1877, consists of a cylindrical piece of pencil lead inserted into a metal or plastic barrel against a movable rod that can be adjusted by a screwing motion to expose part of the lead.



注释

mechanical pencil [mi'kænikəl-'pensl] n. 自动铅笔

patent ['peɪtənt] v. 获得专利

cylindrical [sɪ'lɪndrɪk(ə)] adj. 圆柱的

plastic barrel ['plɑ:stɪk-'bærəl] n. 塑料管

rod [rɒd] n. 杆

screw [skru:] v. 拧, 转

expose [ɪks'pəʊz] v. 使暴露, 使露出

Word List

-A-

- o a cluster of [ə-'klʌst-əv] adj.成群的, 成串的
- o a variety of [ə-və'raɪəti-əv] adj.多种多样的
- o a wealth of [ə-welθ-əv] adj.很多的
- o a wide range of [ə-waɪd-reɪndʒ-əv] adj.广泛的
- o abdomen ['æbdəmən] n.腹部
- o ablate [æb'leɪt] v.融化
- o abrade [ə'breɪd] v.磨损
- o abruptly [ə'brʌptli] adv.突然地
- o absorb [əb'sɔ:b] v.吸收
- o absorption [əb'sɔ:pʃən] n.吸收
- o abundance [ə'bʌndəns] n.丰富
- o abundant [ə'bʌndənt] adj.丰富的
- o abyssal [ə'bisəl] adj.深海的
- o acclaim [ə'kleɪm] n.称赞
- o accompany [ə'kʌmpəni] v.伴随
- o accomplish [ə'kɒmplɪʃ] v.完成
- o account [ə'kaʊnt] n.账目
- o account for [ə'kaʊnt-fɔ:] v.占据
- o accumulate [ə'kju:mjuleɪt] v.堆积
- o accumulation [əkju:mju'leɪʃ(ə)n] n.堆积物
- o acid ['æsid] n.酸
- o acid fog ['æsid-fɒg] n.酸雾
- o acidify [ə'sɪdɪfaɪ] v.酸化
- o acrobat ['ækroʊbæt] n.杂技演员
- o action ['ækʃən] n.机械装置
- o actually ['æktʃuəli] adv.实际上
- o adapt [ə'dæpt] v.适应
- o adapted [ə'dæptɪd] adj.适合的
- o adjacent [ə'dʒeɪsənt] adj.紧挨着的
- o administrative [əd'mɪnɪstrətɪv] adj.管理的
- o admit [əd'mɪt] v.许可进入
- o adorn [ə'dɔ:n] v.装饰
- o aerodynamic [ˌeərəʊdaɪ'næmɪk] adj.空气动力学的

- o aerophone ['eərə,fəun] n. 管乐器
- o aesthetic [i:s'θetik] adj. 美学的
- o affect [ə'fekt] v. 影响
- o age [eidʒ] v. 老化, 陈化
- o agency ['eidʒənsi] n. 机构, 机关
- o agent ['eidʒənt] n. 媒介; 作用物
- o air pressure [εə-'preʃə] n. 气压
- o airborne ['εəbɔ:n] adj. (可以通过) 空气传播的
- o algae ['ældʒi:] n. 藻类, 海藻 (singular: alga ['ælgə])
- o Algonquian [æl'gɔŋkwi:ən,-ki:ən] n. 阿尔贡金族人 (北美印第安人的一个民族)
- o align [ə'lain] v. 排列
- o alignment [ə'lainmənt] n. 直线排列
- o alimentary canal [æli'mentəri-kənæl] n. 消化道
- o alphabetic [ælfə'betik] adj. 照字母次序的
- o alpine region ['ælpain-'ri:dʒən] n. 高山区
- o alter ['ɔ:lteɪ] v. 改变
- o alternate [ɔ:l'tə:nit] adj. 交替的
- o aluminum [æljʊ:'minjəm] n. 铝
- o amateur ['æmətə] n. 业余艺术家
- o amateur hobbyist ['æmətə-'hɔbiist] n. 业余爱好者
- o amber ['æmbə] n. 琥珀
- o amino acid ['æminəu-'æsid] n. 氨基酸
- o among others [ə'mʌŋ-'lðəz] adv. 及其他
- o amphibian [æm'fibiən] n. 两栖动物
- o amplitude ['æmplitju:d] n. 振幅
- o anatomy [ə'nætəmi] n. 人体解剖
- o ancestor ['ænsistə] n. 祖先
- o angiosperm ['ændʒiəspə:m] n. 被子植物
- o angle ['æŋgl] n. 角, 角度
- o angler ['æŋglə] n. 垂钓者
- o angling ['æŋglin] n. 钓鱼, 钓鱼术
- o angular ['æŋgjulə] adj. 有角的
- o animal ['æniməl] n. 动物
- o anodized coating ['ænəudaizd-'kəutiŋ] n. 电镀层, 电镀膜
[anodize v. 阳极氧化: 用电解方式给 (金属面) 镀上一层保护性或装饰性的氧化物]
- o Antarctic Circumpolar Current [æn'ta:ktik,-sə:kəm'pəulə-'kʌrənt]
n. 南极绕极流
- o Antarctica [æn'ta:ktikə] n. 南极洲
- o antenna [æn'tenə] n. 触须
- o anthropologist [ænθrə'pɒlədʒist] n. 人类学家
- o antibiotics [æntibai'ɒtiks] n. 抗生素, 抗生学

- o antibody ['æntɪˌbɒdɪ] n. 抗体
- o antibody molecule ['æntɪˌbɒdi-'mɒlɪkjʊ:l] n. 抗体分子
- o antigen ['æntɪdʒən] n. 抗原
- o anuran [ə'njuərən] n. 无尾目
- o aorta [ei'ɔ:tə] n. 主动脉
- o appeal [ə'pi:l] v. 对……有吸引力
- o applause [ə'plɔ:z] n. 喝彩
- o apply [ə'plai] v. 涂, 搽, 撒; 应用
- o apprentice [ə'prentɪs] n. 学徒
- o approach [ə'prəʊtʃ] v. 接近, 逼近
- o approximately [əprɒksɪ'metli] adv. 大约
- o aquatic [ə'kwætɪk] adj. 水生的, 水栖的
- o aqueous ['eɪkwɪəs] adj. 水的
- o arachnid [ə'ræknaɪd] n. 蛛纲动物
- o arch [ɑ:tʃ] n. 拱 (连柱的)
- o architect ['ɑ:kɪtekt] n. 建筑师
- o architecture ['ɑ:kɪtektʃə] n. 建筑, 建筑学
- o Arctic Ocean ['ɑ:ktɪk-'əʊjən] n. 北冰洋
- o arise [ə'raɪz] v. 起, 发生
- o arrow ['ærəʊ] n. 箭
- o arsenal ['ɑ:sənəl] n. 武器库
- o arteriole [ɑ:'tɪriəʊl] n. 小动脉
- o artery ['ɑ:təri] n. 动脉
- o arthropod ['ɑ:θrəpɒd] n. 节肢动物
- o artisan [ɑ:'ti:zən] n. 工匠
- o ash wood ['æʃ-wu:d] adj. 栉木的
- o ashen ['æʃ(ə)n] adj. 灰的, 灰色的
- o asphalt ['æsfælt] n. 沥青
- o aspiring [əs'paɪərɪŋ] adj. 有志向的
- o associated [ə'səʊfɪeɪtɪd] adj. 关联的
- o assume [ə'sju:m] v. 承担; 采用
- o asteroidal body ['æstərɔɪd'l'bɒdɪ] n. 行星体
- o astronomy [əs'trɒnəmi] n. 天文学
- o atom ['ætəm] n. 原子
- o atomic hydrogen gas [ə'tɒmɪk-'haɪdrədʒən-gæs] n. 原子氢气
- o attach [ə'tætʃ] v. 附加; 贴上; 系; 附上
- o attach to [ə'tætʃ-tu:] v. 把……放在
- o attendance [ə'tendəns] n. 出席
- o attract [ə'trækt] v. 吸引
- o attraction [ə'trækʃən] n. 引力
- o audible ['ɔ:dəbl] adj. 听得见的
- o audience ['ɔ:dʒəns] n. 观众

- o audition [ɔ:'dijən] v. 在试演中评估某一人物
- o auditorium [ɔ:'di:tɔ:riəm] n. 观众席
- o auditory [ɔ:'ditəri] adj. 听觉的
- o augment [ɔ:'g'ment] v. 增补, 扩充
- o autonomic nervous system [ɔ:təu'nɒmik-'nə:vəs-'sistəm] n. 自主神经系统
- o ax [æks] n. 斧头 (也做axe, plural: axes['æksi:z])
- o axis ['æksis] 轴 (plural: axes['æksi:z])

-B-

- o back and forth [bæk-ænd-fɔ:θ] adv. 来回地
- o backbone ['bækbəun] n. 脊椎
- o bacteria [bæk'tiəriə] n. 细菌 (singular: bacterium [bæk'teriəm])
- o bait [beit] n. 饵
- o bake [beik] v. 烘焙; 烧硬
- o ball pen [bɔ:l-pen] n. 圆珠笔
- o ballast ['bæləst] n. 道碴
- o bamboo [bæm'bu:] n. 竹子
- o band [bænd] n. (光、频) 带, (波) 段
- o barbed foreleg [ba:bd-'fɔ:leg] n. 有倒钩的前肢
- o bark [bɑ:k] n. 树皮
- o base [beis] n. 碱基: 嘌呤 (腺嘌呤和鸟嘌呤) 或嘧啶 (胸嘧啶、胸腺嘧啶和尿嘧啶) 的一种
- o basin ['beisn] n. 盆地
- o basketry ['bɑ:skitri] n. 编篮工艺
- o bass [beis] n. 低音部分
- o bass bar [beis-bɑ:r] n. 低音梁
- o bassoon [bə'su:n] n. 低音管, 巴松
- o bazaar [bə'zɑ:] n. 集市
- o be associated with [bi:-ə'səʊʃieitid-wið] v. 与……相关
- o be composed of [bi:-kəm'pəuzd-əv] n. 由……组成
- o be responsible for [bi:-ri'sponsəbl-fɔ:] n. 为……负责
- o beak [bi:k] n. 喙
- o beam [bi:m] n. 梁
- o bean [bi:n] n. 豆
- o beaver ['bi:və] n. 海狸
- o bedding plane ['bediŋ-plein] n. (岩层)层平面, 底层
- o bedrock ['bedrɒk] n. 岩床
- o beech [bi:tʃ] n. 山毛榉
- o beetle ['bi:tl] n. 甲虫
- o bell [bel] n. 铃、钟、或钟琴

- o belly ['beli] n. (弦乐器的) 面板
- o bend [bend] v.使弯曲
- o beneficial [ˌbenɪ'fɪjəl] adj.有益的
- o Bengal [ben'ɡo:l] n.孟加拉
- o bent [bent] adj.弯的
- o benthic storm ['benθɪk-stɔ:m] n.深海风暴
- o benthic zone ['benθɪk-zəʊn] n.深海区域
- o berry ['beri] n.浆果
- o beverage ['bevərɪdʒ] n.饮料
- o bill [bil] n.啄
- o bind [baɪnd] v.绑定, 缔结
- o binder ['baɪndə] n.粘合剂
- o biochemist [ˌbaɪəʊ'kɛmɪst] n.生物化学家
- o biodegradable [ˌbaɪəʊdɪ'greɪdəbl] adj.可以进行分解和还原处理的
- o biodiversity [ˌbaɪəʊdaɪ'vɜ:səti] n.生物多样性
- o bioluminescent [ˌbaɪəʊˌlu:mi'nesnt] adj.发 (冷) 光的
- o biomolecule [ˌbaɪəʊ'mɒlɪkjʊ:l] n.生物分子
- o bird of prey [bə:d-əv-preɪ] n.食肉鸟,猛禽
- o bison ['baɪsn] n.野牛
- o bite [baɪt] v.咬
- o black walnut [blæk-'wɔ:lnt] n.黑胡桃
- o blast [blɑ:st] n.一股 (强而突然的气流)
- o bleach [bli:tʃ] n.漂白剂
- o blend in [blend-in] v.与 (环境等) 协调
- o blender ['blendə] n.搅拌机
- o blind [blaɪnd] n.窗帘 (尤指固定在卷轴上能够拉下来的布质窗帘)
- o blossom ['blɒsəm] n.花簇
- o blueberry ['blu:bəri] n.蓝莓
- o blunt [blʌnt] adj. (笔尖等) 无尖锋的, 钝的
- o boil [boɪl] v&n.煮
- o bombardment [bɒm'bɑ:dmənt] n.撞击
- o bone marrow [bəʊn-'mæərəʊ] n.骨髓
- o boring ['bɔ:rɪŋ] n.钻 (孔)
- o botanist ['bɒtənɪst] n.植物学家
- o boulder ['bəʊldə] n.巨砾
- o bound [baʊnd] v.标明界限; adj. 连在一起的
- o boundary ['baʊndəri] n.边界
- o bow [bəʊ] n.琴弓; 弓
- o bowed string [baʊd-strɪŋ] adj.演奏弦乐的
- o brace [breɪs] n.手摇曲柄钻
- o braise [breɪz] v.炖: 先用油炒, 然后在有盖容器中炖
- o branch [brɑ:ntʃ] n.树枝

- o branding iron ['brændɪŋ-'aɪən] n.烙铁
- o brass [brɑ:s] n.铜管乐器
- o break down [breɪk-daʊn] v.分解
- o breastbone ['brestbəʊn] n.胸骨
- o breechcloth ['bri:tʃklɒθ] n.短裤
- o breed [bri:d] v.生育; 繁殖
- o bring out [brɪŋ-aʊt] v.显示出
- o broil [brɔɪl] v.(用火)烤
- o broiler ['brɔɪlə] n.适于烤焙的嫩鸡
- o brush [brʌʃ] n.刷子
- o bubble ['bʌbl] n.气泡
- o buffalo ['bʌfələʊ] n.北美野牛
- o bug [bʌg] n.虫子
- o build up [bɪld-ʌp] v.堆积
- o building block ['bɪldɪŋ-blɒk] n.结构单元
- o buildup [bɪldʌp] n.累积
- o bulge [bʌldʒ] n.凸出部分
- o bundle ['bʌndl] n.捆
- o burden ['bɜ:dən] n.负载
- o burrow ['bʌrəʊ] n.洞穴
- o butterfly ['bʌtəflaɪ] n.蝴蝶
- o by means of [baɪ-mi:nz-əv] prep. 通过……方式

-C-

- o cabin ['kæbɪn] n.小木屋
- o caecilian [si:'sɪliən] n.蚓螈(无足目约150种两栖动物的统称)
- o calcite ['kælsait] n.方解石
- o calcium ['kælsiəm] n.钙
- o calcium carbonate ['kælsiəm-'kɑ:bəneɪt] n.碳酸钙
- o cameo ['kæmiəʊ] n.浮雕
- o camera ['kæməɹə] n.照相机
- o camera obscura ['kæməɹə-ɒb'skjuərə] n.暗箱
- o camouflage ['kæməflɑ:ʒ] n.伪装
- o campo ['kæmpəʊ] n.南美草原
- o canal [kə'næl] n.运河; 导管
- o cancellous ['kænsɪləs] adj.网状骨质的
- o cancellous bone ['kænsɪləs-bəʊn] n.松质骨
- o cane [keɪn] n.藤条
- o canopy ['kænəpi] n.天篷: 树林中最上面一层
- o capable of ['keɪpəbl-əv] adj.有能力的

- o cape [keɪp] n.披肩, 短斗篷
- o capes of feathers [keɪps-əv-'feðəz] n.扎满羽毛的披肩
- o capillary [kə'pɪləri] n.毛细血管
- o capital-intensive [ˌkæpɪtəl-'ɪntensɪv] adj.资本密集型的
- o capon ['keɪpən] n.阉鸡
- o capsid ['kæpsɪd] n.衣壳 (包裹着一个病毒粒子的蛋白质外壳)
- o carbohydrate [ˌkɑ:bəu'haidreɪt] n.碳水化合物
- o carbon ['kɑ:bən] n.碳
- o carbon cycle ['kɑ:bən-'saɪkl] n.碳循环
- o carbon dioxide ['kɑ:bən-daɪ'ɒksaɪd] n.二氧化碳
- o cardiac ['kɑ:diæk] adj.心脏(病)的
- o carnivore ['kɑ:nɪvɔ:] n.食肉动物
- o carnivorous [kɑ:'nɪvərəs] adj.(动物)食肉的
- o carpenter ['kɑ:pɪntə] n.木匠
- o carry out ['kæri-aʊt] v.传送
- o cartilaginous [ˌkɑ:tɪ'lædʒɪnəs] adj.软骨的, 软骨质的
- o carve [kɑ:v] v.切割
- o cast [kɑ:st] v.投射; 指定角色; 抛
- o caste [kɑ:st] n.有社会组织的昆虫的阶级
- o casting ['kɑ:stɪŋ] n.铸造
- o casting director ['kɑ:stɪŋ-dɪ'rektə] n.负责挑选演员的人, 星探
- o castrate [kæ'streɪt] v.阉割
- o catalyst ['kætəlist] n.催化剂
- o caterpillar ['kætəpɪlə] n.毛虫
- o caudata [kɑ:deɪtə] n.有尾目
- o causative agent ['kɔ:zətɪv-'eɪdʒənt] n.病原体
- o C-clamp [si:-klæmp] n.C型夹
- o ceiling ['si:lɪŋ] n.天花板
- o celebrated ['selɪbreɪtɪd] adj.著名的
- o cell [sel] n.细胞; 蜂房
- o cell biology [sel-baɪ'ɒlədʒɪ] n.细胞生物学
- o cell membrane [sel-'membreɪn] n.细胞膜
- o cell wall [sel-wɔ:l] n.细胞壁
- o cellular immunity ['seljʊlə-'ɪmjʊ:nəti] n.细胞免疫
- o Celsius ['selsjəs] adj.摄氏的
- o cement [si'ment] n.水泥
- o cement concrete [si'ment-'kɒnkri:t] n.水泥混凝土
- o centigrade ['sentɪɡreɪd] adj.摄氏温度的
- o central nervous system ['sentrəl-'nɜ:vəs-'sɪstəm] n.中枢神经系统
- o ceramic [si'ræmɪk] adj.陶器的
- o ceremonial [ˌserɪ'məʊnjəl] adj.仪式场合的
- o ceremony ['serɪməni] n.仪式

- o certify ['sə:tɪfaɪ] v. 证明
- o chamber ['tʃeɪmbə] n. 房间; 腔; 心室; 有机物体内封闭的腔; 室
- o chaotic [keɪ'ɒtɪk] adj. 无秩序的
- o characteristic [ˌkærɪktə'rɪstɪk] n. 特性
- o chemical cycle ['kemɪkəl-'saɪkl] n. 化学循环
- o chemical reaction ['kemɪkəl-ri:'ækʃən] n. 化学反应
- o Cherokee [ˌtʃerə'ki:] n. 切罗基族(北美易洛魁人的一支)人
- o chest [tʃest] n. 胸部; 胸腔
- o chew [tʃu:] v. 咀嚼
- o chick [tʃɪk] n. 小鸡
- o chickee ['tʃɪki:] n. (西米诺尔印第安人的)棕榈叶顶棚屋
- o chief [tʃi:f] n. 首领
- o chimney ['tʃɪmni] n. 烟囱
- o chisel ['tʃɪzl] n. 凿子
- o chitinous ['kəɪtɪnəs] adj. 壳质的
- o chloroplast ['klɒ:rəʊplæst] n. 叶绿体
- o chord [kɔ:d] n. 和弦
- o chordophone ['kɔ:dəʊfəʊn] n. 弦乐器
- o choreographer [ˌkɔ:ri'ɒgrəfə] n. 舞蹈指导
- o choroid ['kɔ:roɪd] n. 脉络膜
- o chromatic [krəʊ'mætɪk] adj. 半音(阶)的, 含半音的
- o chromatically [krəʊ'mætɪkli] adv. 半音地
- o chromosome ['krəʊməsəʊm] n. 染色体
- o chrysalis ['krɪsəɪsɪs] n. 蛹
- o chyme [kaɪm] n. 食糜: 部分消化的食物的稠状半流体物, 从胃进入十二指肠
- o cigar-shaped [sɪ'gɑ:-'feɪpt] adj. 雪茄形的
- o ciliary body ['sɪliəri-'bɒdi] n. 睫状体
- o circuit ['sə:kɪt] n. 一圈
- o circular ['sə:kjələ] adj. 圆形的
- o circulate ['sə:kjuleɪt] v. 循环
- o circulation [ˌsə:kju'leɪʃən] n. 循环
- o circulatory system [sə:kju'leɪtəri-'sɪstəm] n. 循环系统
- o cirrostratus ['sɪrəʊ'strə:təs] n. 卷层云
- o cladding ['klædɪŋ] n. 覆层: 在高温高压下结合到另一金属上的金属外层
- o clam [klæm] n. 蛤
- o clan [klæn] n. 部族
- o clan mother [klæn-'mʌðə] n. 族长
- o clarinet [ˌklærɪ'net] n. 单簧管, 黑管
- o class [klɑ:s] n. 类; 纲
- o classical ['klæsɪkəl] adj. 传统的, 古典的, 经典的
- o clay [kleɪ] n. 黏土, 泥土

- o clay mineral [klei-'minərəl] n.粘土矿物
- o clef [klef] n.谱号
- o clientele [ˌklaɪən'tel] n.顾客
- o climate ['klaɪmɪt] n.气候
- o clog [klog] v.阻塞, 塞满
- o clown [klaun] n.小丑
- o club [klʌb] n.棍棒
- o cluster ['klʌstə] n.串; 星团
- o cnidarian [nai'deəri:ən] n.刺胞动物
- o coal [kəʊl] n.煤
- o coast [kəʊst] n.海岸
- o coating ['kəʊtɪŋ] n.涂层; 外壳; 覆盖层
- o cocoon [kə'ku:n] n.茧
- o cod [kɒd] n.鳕鱼
- o code [kəʊd] n.法规
- o codex ['kəudeks] n.抄本 (plural: codices ['kəudisi:z])
- o cohere [kəu'hiə] v.粘合
- o coil [kɔɪl] v.盘绕; n.(一)卷
- o coincidence [kəu'ɪnsɪdəns] n.巧合
- o coitus ['kɔɪtəs] n.性交
- o collaborative [kə'ləbəreɪtɪv] adj.协作的
- o collapse [kə'læps] n.倒塌
- o collide [kə'laid] v.碰撞
- o colony ['kɒləni] n.群体
- o column ['kɒləm] n.柱 (a colum of air 气柱)
- o combination [ˌkɒmbɪ'neɪʃən] n.组合
- o combine [kəm'beɪn] v.结合
- o combustion [kəm'bʌstʃən] n.燃烧
- o comedy ['kɒmɪdi] n.喜剧
- o commercial [kə'mɜ:ʃəl] adj.商业的
- o communal ['kɒmjʊnəl] adj.群居的
- o commuter [kə'mju:tə] n.通勤者 (定期从一地到另一地旅行的人, 如从郊区到城市并回来)
- o compact bone [kəm'pækt-bəʊn] n.密质骨
- o competitive [kəm'petɪtɪv] adj.竞争的
- o complement ['kɒmplɪmənt] n.补体
- o complex molecule ['kɒmpleks-'mɒlɪkjʊ:l] n.复杂分子
- o component [kəm'pəʊnənt] n.零件, 构件; 组成部分
- o compose [kəm'pəʊz] v.组成 (be composed of: 由……组成)
- o composer [kəm'pəʊzə] n.作曲家
- o composition [ˌkɒmpə'zɪʃən] n.乐曲
- o compound ['kɒmpaʊnd] n.化合物; 混合物

- o compression [kəm'preʃən] n.压缩
- o concave lens [kən'keiv-lenz] n.凹透镜
- o concentric [kən'sentrik] adj.同一中心的
- o concrete ['kɒŋkri:t] n.混凝土; v.浓缩, 凝结
- o condition [kən'diʃən] n.病痛
- o cone [kəʊn] n.球果
- o conform [kən'fɔ:m] v. (to, with) 遵照, 依照
- o conjunctiva [ˌkɒndʒʌŋk'taɪvə] n.(眼球)结膜
- o consequence ['kɒnsɪkwəns] n.后果, 影响
- o conserve [kən'sə:v] v.保存
- o considerable [kən'sɪdərəbl] adj.相当大的
- o consist of [kən'sɪst-əf] v.由……组成
- o consolidation [kən'sɒlɪ'deɪʃən] n.加固
- o constant ['kɒnstənt] adj.恒久不变的
- o constituent [kən'stɪtjuənt] n.成分
- o construction site [kən'strʌkʃən-saɪt] n.建筑工地
- o contact ['kɒntækt] n.关联
- o contamination [kən'tæmɪ'neɪʃən] n.污染, 污物
- o continental mass [ˌkɒntɪ'nentəl-mæs] n.大陆块体
- o continental shelf [ˌkɒntɪ'nentəl-ʃelf] n.大陆架
- o continental slope [ˌkɒntɪ'nentəl-sləʊp] n.大陆斜坡
- o continuous [kən'tɪnjuəs] adj.连续的
- o contour ['kɒntuə] n. (音调或声调的) 升降曲线; 变化方式
- o contract [kən'trækt] v.收缩
- o contractile [kən'træktail] adj.会缩的
- o contraction [kən'trækʃən] n.收缩
- o contribute [kən'trɪbjʊt] v.贡献
- o controversial [ˌkɒntrə'vɜ:ʃəl] adj.有争议的
- o conurbation [ˌkɒnə'beɪʃən] n. (连带卫星城镇和市郊的) 大都市
- o converge [kən'vɜ:dʒ] v.聚合于, 集中于一点, 会聚
- o conversely ['kɒnvɜ:sli] adv.相反地
- o convex ['kɒnveks] adj.凸的, 凸面的
- o convey [kən'veɪ] v.转达
- o conveyance [kən'veɪəns] n.运送
- o cookery ['kʊkəri] n.烹饪
- o coordinate [kəʊ'ɔ:dɪneɪt] v.调整
- o coral ['kɒrəl] n.珊瑚
- o core [kɔ:] n.中心
- o corn stalk ['kɔ:n-stɔ:k] n.玉米杆
- o cornea ['kɔ:niə] n.角膜
- o cornerstone [ˌkɔ:nə'stəʊn] n.基础
- o corresponding to [ˌkɒrəs'pɒndɪŋ - tu:] adj.与……相应的

- o corridor ['kɒrɪdɔ:] n.走廊
- o cortex ['kɔ:teks] n.大脑皮层
- o cosmic ['kɔ:zmɪk] adj.宇宙的
- o cosmic ray ['kɔ:zmɪk-reɪ] n.宇宙射线
- o cosmology [kɔ:z'mɒlədʒi] n.宇宙天体学
- o costume ['kɔ:stju:m] n.装束
- o cotton ['kɒtən] n.棉布
- o counterclockwise [ˌkauntə'klɒkwaɪz] adj.逆时针方向的
- o crab [kræb] n.蟹
- o crack [kræk] n.裂缝; 断裂处
- o craft [krɑ:ft] n.工艺
- o craftspeople [krɑ:ftspi:pl] n.道具师
- o cranberry ['krænbəri] n.蔓越橘
- o cranial nerve ['kreɪnjəl-nə:v] n.颅神经
- o crater ['kreɪtə] n.坑, 火山[喷火]口; (月球上的) 环形山
- o crayfish ['kreɪfɪʃ] n.龙虾
- o creek [kri:k] n.小溪
- o Creek [kri:k] n.美国克里克族印第安人
- o crescent ['kresnt] adj.新月形的
- o crescent phase ['kresnt-feɪz] n.新月位相
- o Cretaceous [kri'teɪsiəs] n.白垩纪
- o crevasse [kri'væs] n.裂缝
- o crew [kru:] n.全体人员
- o cricket ['krikit] n.蟋蟀
- o criteria [krai'tiəriə] n.标准
- o crossbeam ['krɒsbi:m] n.横梁
- o cross-pollination [ˌkrɒs,pɒlə'neɪʃən] n.异花传粉
- o crowd ... to ... [kraʊd-tu:] v.把……挤进……; 把……塞进……
- o crown [kraʊn] n.树冠
- o crust [krʌst] n.地壳
- o crustacean [krʌ'steɪʃən] n.甲壳纲动物
- o crustal ['krʌstəl] adj.地壳的
- o crustal plate ['krʌstəl-pleɪt] n.地壳板块
- o crystal ['krɪstəl] n.晶体
- o crystalline ['krɪstəlɪn] adj.水晶的
- o cube [kju:b] n.立方体
- o cuckoo ['kuku:] n. 杜鹃; adj.寄生的
- o cuckoo bee ['kuku:-'bi:] n.寄生蜂
- o cue [kju:] n.提示
- o cuisine [kwi'zi:n] n.烹饪法
- o culmination [ˌkʌlmi'neɪʃ(ə)n] n.顶点
- o cultivate ['kʌltɪveɪt] v.耕种

- o cumulonimbus ['kju:mjələu'nimbəs] n. 积雨云
- o cuneiform ['kju:ni:fɔ:m] n. 楔形文字
- o cure [kjue] v. 治疗
- o current ['kərənt] n. 水流
- o curtain ['kə:tn] n. 窗帘
- o curving ['kə:viŋ] adj. 弯曲的
- o cuttlefish ['kʌtlfiʃ] n. 乌贼, 墨鱼
- o cycle ['saikl] n. 循环, 周期
- o cylinder ['silində] n. 圆柱体
- o cylindrical [si'lindrik(ə)] adj. 圆柱的
- o cytoplasm ['saitəplæzəm] n. 细胞质

-D-

- o dam [dæm] n. 坝
- o damper ['dæmpə] n. 制音器
- o daub [dɔ:b] v. 涂抹; n. 涂料: 柔软粘稠的涂抹原料, 如泥灰、润滑脂或泥浆
- o deadfall ['dedfɔ:l] n. 陷阱
- o deafen ['defn] v. 使聋
- o debris ['debri:] n. 碎片 (plural: debris ['debri:z])
- o decay [di'kei] v. 腐烂
- o decibel ['desibel] n. 分贝
- o deciduous forest [di'sidzjuəs-'fɔrist] n. 落叶林
- o deciduous monsoon forest [di'sidzjuəs-mɒn'su:n-'fɔrist] n. 落叶季雨林
- o declaration [dek'lə'reiʃən] n. 宣言
- o decompose [di:kəm'pəuz] v. 分解, 分析
- o decorative ['dekorətiv] adj. 装饰性的
- o deem [di:m] v. 视为
- o deerskin ['diəskin] n. 鹿皮
- o defend [di'fend] v. 防卫
- o deficiency [di'fijənsi] n. 缺乏
- o defoliation [di:fəuli'eɪʃən] n. 落叶
- o deform [di'fɔ:m] v. (使) 变形
- o degrade [di'greid] v. 使剥蚀
- o dense [dens] adj. 密集的
- o density ['densiti] n. 密度; 比重
- o deoxyribonucleic acid [di:,ɒksi:,raibənu:'kli:ik-'æsid] n. 脱氧核糖核酸
- o deposit [di'pɔzit] n. 沉淀物; 淤积, 沉淀; v. 沉积, 沉淀; 使沉积
- o depress [di'pres] v. 使降低
- o deregulation [di:regju'leɪʃən] n. 缩小国家对经济干预的范围

- o dermis ['dɜ:mɪs] n. 真皮
- o descend [dɪ'send] v. 下降
- o descent [dɪ'sent] n. 血统
- o detect [dɪ'tekt] v. 察觉到; 探测
- o deterioration [dɪ,tɪərɪə'reɪʃən] n. 磨损, 腐化, 变质
- o detrital [dɪ'traɪtəl] adj. 由岩屑形成的
- o devastate ['devəsteɪt] v. 糟蹋
- o Devonian [de'vəʊniən] n. 泥盆纪
- o diagnose ['daɪəgnəʊz] v. 诊断
- o dialogue ['daɪəlɒɡ] n. 对白
- o diameter [daɪ'æmɪtə] n. 直径
- o diaphragm ['daɪəfræm] n. 隔膜
- o differentiating [ˌdɪfə'renʃeɪtɪŋ] adj. 分化
- o digest [daɪ'dʒest] v. 消化
- o digestion [dɪ'dʒestʃən] n. 消化
- o dilute [daɪ'lju:t] adj. 稀释的
- o diminish [dɪ'mɪnɪʃ] v. 减少
- o dinosaur ['daɪnəsɔ:] n. 恐龙
- o dioxide [daɪ'ɒksaɪd] n. 氧化物
- o dip [dɪp] v. 浸蘸
- o director [dɪ'rektə] n. 导演
- o disable [dɪs'eɪbl] v. 使残废
- o disease [dɪ'zi:z] n. 疾病
- o disguise [dɪs'gaɪz] n. 伪装
- o disintegration [ˌdɪs,ɪntɪ'greɪʃən] n. 分裂, 崩解
- o disorder [dɪs'ɔ:də] n. 失调, 紊乱
- o disposal method [dɪs'pəʊzəl-'meθəd] n. 处置方法
- o dissipate ['dɪsɪpeɪt] v. 使耗散
- o dissolve [dɪ'zɒlv] v. 溶解
- o distasteful [dɪs'teɪstfʊl] adj. (令人) 不愉快的
- o distill [dɪ'stɪl] v. 蒸馏
- o distinct [dɪs'tɪŋkt] adj. 截然不同的
- o distinction [dɪs'tɪŋkʃən] n. 区别
- o distinctive [dɪs'tɪŋktɪv] adj. 独特的, 有特色的
- o distinguish [dɪs'tɪŋɡwɪʃ] v. 区别
- o distortion [dɪs'tɔ:ʃən] n. 扭曲变形
- o distraction [dɪs'trækʃən] n. 使人分心的事, 心乱, 心烦
- o distribute [dɪs'trɪbjʊ:t] v. 分配; 分布, 散布 (over)
- o disturbance [dɪs'tɜ:bəns] n. 扰动; 局部运动
- o diverse [daɪ'vəs] adj. 不同的
- o divide [dɪ'vaɪd] v. 划分
- o division [dɪ'vɪʒən] n. 部分

- o dodder ['dɒdə] n. 菟丝子
- o dome [dəʊm] n. 穹隆 (结构): 类似于穹隆 (弓形、有拱的) 或圆屋顶的结构
- o domestic [də'mestɪk] adj. 家庭的; 驯养的
- o doorway ['dɔ:weɪ] n. 门口
- o dowel ['dəʊəl] n. 暗榫
- o dragonfly ['dræɡənflaɪ] n. 蜻蜓
- o drain [dreɪn] v. (土地) 排水, (资源等) 逐渐枯竭
- o drainer ['dreɪnə] n. 下水道装置, 滤水器
- o drama ['drɑ:mə] n. 戏剧
- o dramaturge ['dræmə,tʃ:dʒ] n. 剧作家
- o draw [drɔ:] v. 拉
- o dredge [dredʒ] v. 用挖掘机清除、加深或加宽
- o drill [drɪl] n. 钻孔机
- o drizzle ['drɪzl] n. 细雨
- o drone [drəʊn] n. 雄蜂
- o droplet ['drɒplɪt] n. 小滴
- o drought injury [draʊt-'ɪndʒəri] n. 旱害
- o drug [drʌɡ] n. 药物
- o drum [drʌm] n. 鼓
- o dry heat [draɪ-'hi:t] n. 干热
- o dual system ['dju:əl-'sɪstəm] n. 双重系统
- o duct [dʌkt] n. 管: 管状体内通道, 尤指运输腺分泌物的管道
- o dulcimer ['dʌlsɪmə] n. 洋琴
- o duplicate ['dju:pəlɪkɪt] v. 复制
- o duration [dʒuə'reɪʃən] n. 持续时间
- o dwarf [dwɔ:f] adj. 矮小的
- o dwarf planet [dwɔ:f-'plænɪt] n. 矮星
- o dwelling ['dwelɪŋ] n. 住处, 居住空间
- o dwindle ['dwɪndl] v. 减少
- o dye [daɪ] n. 染料

-E-

- o eardrum ['iədʌm] n. 耳膜
- o Earth [ə:θ] n. 地球
- o earthen ['ɛ:θən] adj. 土制的
- o earthquake ['ɛ:θkweɪk] n. 地震
- o ebony ['ebəni] n. 乌木
- o ecdysis ['ekdɪsɪs] n. 蜕皮 (plural: ecdyses ['ekdɪsɪ:z])
- o echinoderm ['ɪ'kaɪnə,də:m] n. 棘皮类动物
- o eclipse ['ɪ'klɪps] n. 食

- o ecosystem ['i:kəsistəm] n.生态系统
- o edge [edʒ] n.边缘
- o edible ['edibl] adj.可食用的
- o edible grain ['edibl-grein] n.食用谷物
- o egg [eg] n.卵
- o ejaculate [i'dʒækjuleit] v.射精; adj.复杂的; 精心制作的; 精巧的, 精细的
- o elbowd ['elbəud] adj.像肘一样弯曲的
- o electric charge [i'lektrik-tʃɑ:dʒ] n.电荷
- o electric elevator [i'lektrik-eliveitə] n.电梯
- o electromagnetic [i'lektromæg'netik] adj.电磁的
- o electron [i'lektrɒn] n.电子
- o element ['elimənt] n.环境; 元素
- o elevation [eli'veiʃən] n.海拔
- o eliminate [i'limineit] v.除去, 消灭
- o elk [elk] n.麋鹿
- o elliptical [i'liptikəl] n.椭圆星系
- o elongated ['i:lɒŋgeitid] adj.被拉长的
- o embed [im'bed] v.镶嵌
- o emblem ['embləm] n.象征
- o embryo ['embriəu] n.胚胎
- o emerge from [i'mə:dʒ-frəm] v.自……出现
- o emission [i'miʃən] n.发出(物), 放射(物)
- o emit [i'mit] v.发出; 喷射
- o emulsion [i'mʌlʃən] n.感光乳剂
- o enamel [i'næməl] n.瓷漆: 一种玻璃似的, 烧制在金属、玻璃或瓷器上通常不透明的、保护性或装饰性的覆盖层
- o encaustic [in'kɔ:stik] n.蜡画
- o encounter [in'kauntə] n.相遇
- o end [end] n.目的
- o endocrine ['endəukrain] adj.内分泌的
- o energy ['enədʒi] n.能量
- o energy-efficient ['enədʒi-i'fiʃənt] adj. 能源节约的
- o engulf [in'gʌlf] v.吞没
- o enormous [i'nɔ:məs] adj.巨大的, 庞大的
- o ensemble [ɒn'səmbəl] n.合奏; 整体
- o entertain [entə'tein] v.娱乐
- o envelope ['envələup] n.包裹性的结构或包皮, 例如膜, 或者病毒的包膜
- o enzyme ['enzaim] n.酶
- o ephemeral [i'femərəl] adj. 短暂的
- o epicenter ['episentə] n.震中
- o epidermis [epi'də:mis] n.表皮
- o equator [i'kweitə] n.赤道

- o equivalent interval [i'kwivələnt-'intəvəl] n.等间隔的 音程
- o eradication [i,rædi'keiʃən] n.根除
- o erase [i'reiz] v.擦掉
- o erect [i'rekt] v.竖立, 建立; 搭建; adj.竖立的
- o erode [i'rəud] v.侵蚀
- o erosion [i'rəʊʒən] n.腐蚀, 侵蚀 (作用)
- o erupt [i'rʌpt] v.喷出
- o esophagus [i(:)'sɒfəgəs] n.食道
- o estrogen ['estrədʒən] n.雌激素
- o eukaryotic cell [ju'kæriəutik-sel] n.真核细胞
- o eusocial [ju:'səʊʃəl] adj.完全群居的
- o evaporation [i,væpə'reiʃən] n.蒸发; 蒸腾 (作用)
- o evolutionary [i:və'lu:fənəri] adj.进化的
- o evolve [i'vɒlv] v.进化
- o exceed [ik'si:d] v. 超过 (限度、范围)
- o excess ['ekses] adj.过量的
- o excessive [ik'sesiv] adj.过度的, 过分的
- o excrete [eks'kri:t] v.排泄,分泌; 排出
- o excretory [eks'kri:təri] adj.排泄的
- o exert [ig'zə:t] v.行使 (职权等), 使受 (影响等)
- o exhale [eks'heil] v.呼气
- o exhibit [ig'zibit] v.显示, 显出; 呈现
- o exoskeleton [eksəu'skelitən] n.外骨骼
- o expel [iks'pel] v.喷出, 发射; 排出
- o exploit [iks'plɔit] v.开拓 n. 功绩[勋], 英勇的行为
- o explosive [iks'pləʊsiv] adj.爆炸 (性) 的
- o expose [iks'pəʊz] v.曝光, 暴露; 使暴露, 使露出
- o exposure [iks'pəʊʒə] n.曝光
- o exquisitely ['ekskwizitli] adv.非常地
- o extend [iks'tend] v.延伸
- o extinguish [iks'tingwiʃ] adj.熄 (灯), 灭 (火), 熄灭
- o extraordinarily [ik'strɔ:dənərili] adv.格外地
- o eyeball ['aibɔ:l] n.眼球
- o eye-catching ['ai-keɪtʃɪŋ] adj.引人注目的
- o eyelash ['aɪlæʃ] n.睫毛
- o eyelid ['aɪlɪd] n.眼睑

-F-

- o fabric weaving ['fæbrik-'wi:vɪŋ] n.纺织编织
- o facade [fə'sɑ:d] n. (建筑物的) 外立面, 尤指正面

- o facilitate [fə'siliteit] v. 提供便利
- o Fahrenheit scale ['færənhait-skeil] n. 华氏温标
- o fair [fɛə] n. 集市
- o falcon ['fɔ:lkn] n. 猎鹰
- o family ['fæmili] n. 科
- o far-reaching ['fɑ:-'ri:tʃɪŋ] adj. 深远的，广泛的
- o fashion ['fæʃən] v. 制造
- o fasten ['fɑ:sən] v. 使固定；加固
- o fat [fæt] n. 油脂
- o fatal ['feitəl] adj. 致命的
- o fatty ['fæti] adj. 含脂肪的，脂肪状的
- o fault [fɔ:lt] n. 断层
- o feast [fi:st] n. 宴会
- o feature ['fi:tʃə] n. 特征
- o feces ['fi:si:z] n. 粪便
- o federal ['fedərəl] adj. 联邦的
- o feedback ['fi:dbæk] n. 反馈
- o felt [felt] n. 毛毡
- o fertility rite [fə:'tiliti-rait] n. 丰收仪式
- o fertilized ['fə:tilaizid] adj. 已受精的
- o fertilized egg ['fə:tilaizd-eg] n. 受精卵
- o fertilizer ['fə:tilaizə] n. 肥料
- o festival ['festəvəl] adj. 节日的
- o fiber ['faibə] n. 纤维
- o fiberglass ['faibəglɑ:s] n. 玻璃纤维
- o fiber-tip pen ['faibə-tip-pen] n. 纤维笔
- o field [fi:ld] n. 野外
- o fierce [fiəs] adj. 猛烈的
- o figuration [,figjə'reiʃən] n. 用装饰音装饰
- o file [fail] n. 锉刀
- o filler ['filə] n. 填充料
- o film [film] n. 胶片
- o fin [fin] n. 鱼鳍
- o fine [fain] adj. (区别) 微小的；细微的
- o fine-grained ['fain-'greind] adj. 纹理细腻的，平整的
- o finishing ['finiʃɪŋ] n. 抛光
- o fir [fɛ:] n. 枞树，冷杉
- o fireplace ['faɪəpleis] n. 壁炉
- o fishery ['fiʃəri] n. 食用鱼类，捕鱼业
- o fishing ['fiʃɪŋ] n. 钓鱼
- o fishing rod ['fiʃɪŋ-rɒd] n. 钓竿
- o fissure ['fiʃə] n. 裂缝，裂隙

- o fissure vent ['fɪʃə-vent] n. 断裂口
- o fizz [fɪz] n. 嘶嘶声
- o flank [flæŋk] v. 位于……的两侧
- o flat [flæt] adj. 扁平的
- o flatten ['flætən] v. 使变平
- o flattened disk system ['flætn-disk-'sistəm] n. 平底圆盘状系统
- o flexible ['fleksəbl] adj. 柔韧的; 有弹性的
- o float [fləʊt] v. 漂浮
- o flounder ['flaʊndə] n. 比目鱼
- o flour ['flaʊə] n. 细软的粉末
- o flow [fləʊ] n. 传播
- o fluid ['flu:ɪd] adj. 易流动的; n. 液
- o flush [flʌʃ] v. 冲掉
- o flute [flu:t] n. 竖笛, 横笛
- o fly [flaɪ] n. 苍蝇
- o focal ['fəʊkəl] adj. 焦点的
- o focal length ['fəʊkəl-lenθ] n. 焦距
- o focus ['fəʊkəs] n. 震源 (“震中”是epicenter)
- o fog [fɒg] n. 雾
- o fold [fəʊld] n. 层; 折页
- o folk [fəʊk] adj. 民间的
- o forage ['fɒrɪdʒ] v. 觅食
- o foraminifera [fə'reɪmə'nɪfərə] n. 有孔虫类
- o forbid [fə'bid] v. 禁止
- o forecast ['fɔ:kɑ:st] n. 预报
- o foreign ['fɒrɪn] adj. 异质的, 外来的
- o forelimb ['fɔ:lim] n. 前肢
- o forest ['fɒrɪst] n. 森林
- o forked [fɔ:kt] adj. 叉状的
- o fossil ['fɒsl] n. 化石
- o fossil fuel ['fɒsl-fjuəl] n. (煤、石油、天然气等) 矿物燃料
- o fountain ['faʊntɪn] n. 喷泉
- o fountain pen ['faʊntɪn-pen] n. 自来水钢笔
- o fourfold [fɔ:fəʊld] n. 四倍
- o fowl [faʊl] n. 家禽
- o fraction ['frækʃən] n. 小部分
- o fracture ['fræktʃə] n. 岩石的裂缝或断层
- o fracture zone ['fræktʃə-zəʊn] n. 断层带
- o fragment ['frægmənt] n. 碎片
- o fragmental [fræg'mentəl] adj. 碎屑状的
- o fragrance ['freɪgrəns] n. 香气
- o frame [freɪm] n. 骨架

- o framework ['freimwə:k] n. (框架) 结构
- o freestanding ['fri:stændɪŋ] adj. (雕刻、建筑物等) 独立的 , 自力撑持的
- o freezing rain ['fri:ziŋ-reɪn] n. 冻雨
- o freight [freɪt] n. 货物
- o frequency ['fri:kwənsi] n. 频率
- o fringe [frɪndʒ] n. (地毯等的) 穗 , 须边 , 流苏
- o frog [frɒg] n. 青蛙
- o frontier ['frʌntiə] n. 边境
- o fry [fraɪ] v. 油炸
- o fryer [fraɪə] n. 适于油炸的嫩鸡
- o fuel [fjuəl] n. 燃料, 能量
- o full [ful] n. 满月
- o fume [fju:m] n. 浓烟
- o function ['fʌŋkʃən] v. (机器、机制等) 运行 , 发挥作用
- o fungi ['fʌndʒaɪ] n. 真菌类; 真菌类 (singular: fungus ['fʌŋɡəs])
- o fungicide ['fʌndʒisaɪd] n. 杀真菌剂
- o funnel ['fʌnl] n. 漏斗

-G-

- o galaxy ['gæləksi] n. 星系
- o game [geɪm] n. 猎物
- o garbage ['gɑ:bɪdʒ] n. 垃圾 , 废料
- o gasoline ['gæseslɪ:n] n. 汽油
- o gastric ['gæstri:k] adj. 胃的
- o gastrointestinal [ˌgæstrəʊɪn'testənəl] adj. 肠胃的
- o gathering ['gæðərɪŋ] n. 订成书贴的书页
- o gauge [geɪdʒ] n. 轨距
- o gear [giə] n. 设备; v. 使适合于
- o gelatin ['dʒelətɪn, 'dʒelə'ti:n] n. 凝胶
- o gene [dʒi:n] n. 基因
- o genetic material [dʒi'netɪk-mə'tiəriəl] n. 遗传物质
- o genetic structure [dʒi'netɪk-'strʌktʃə] n. 遗传结构
- o genetics [dʒi'netɪks] n. 遗传学
- o genus ['dʒi:nəs] n. 种, 类
- o geologic activity [dʒiə'lɒdʒɪk-æk'tɪvɪti] n. 地质活跃带
- o geologist [dʒi'ɒlədʒɪst] n. 地质学家
- o geology [dʒi'ɒlədʒi] n. 地质学
- o gesture ['dʒestʃə] n. 造型
- o giant ['dʒaɪənt] adj. 巨大的
- o gibbous ['gɪbəʊs] adj. 光亮部大于半圆的

- o gill [dʒɪl] n. 腮
- o give up [gɪv-ʌp] v. 把……输送到……
- o give way to [gɪv-wei-tu] v. 退让
- o glacier ['glæsjə] n. 冰川
- o glaciologist [ˌgleɪʃɪ'ɒlədʒɪst] n. 冰河学家
- o gland [glænd] n. 腺
- o glaze [gleɪz] n. 釉料
- o glide [glɑɪd] v. 滑行
- o global warming ['gləʊbəl-'wɔːmɪŋ] n. 全球变暖
- o globular shape ['glɒbjʊlə-feɪp] n. 球形
- o glucose ['gluːkəʊs] n. 葡萄糖
- o glue [gluː] v. 粘合
- o gong [gɒŋ] n. 锣
- o Gothic ['gɒθɪk] adj. 哥特式的
- o gouge [gaʊdʒ] n. 半圆凿
- o govern ['gʌvən] v. 支配
- o gradual ['grædʒuəl] adj. 逐渐的
- o grain [greɪn] n. 纹理图案
- o grand [grænd] adj. 大的
- o grant [grɑːnt] v. 授予 (权利); 让与 (财产)
- o grape [greɪp] n. 葡萄
- o graphite ['græfəɪt] n. 石墨
- o grasshopper ['grɑːs,hɒpə] n. 蝗虫
- o grassland ['grɑːslænd] n. 草原
- o grate [greɪt] n. 壁炉
- o gravel ['grævəl] n. 砂砾
- o gravitationally [ˌgrævɪ'teɪʃnəli] adv. 引力地
- o gravity ['grævɪti] n. 重力
- o greasy ['griːsi] adj. 油腻的
- o greenhouse effect ['griːnhaʊs-i'fekt] n. 温室效应
- o greenhouse gas ['griːnhaʊs-gæs] n. 温室气体
- o greywater ['greɪwɔːtə] n. 生活污水
- o gross capacity [grəʊs-kə'pæsɪti] n. 总容量
- o grub [grʌb] n. 幼虫
- o guild [gɪld] n. 协会
- o guitar [gɪ'tɑː] n. 吉他
- o gullet ['gʌlɪt] n. 食管
- o gymnophiona ['dʒɪmɒfɪnə] n. 无足目
- o gymnosperm ['dʒɪmnəʊ'spɜːm] n. 裸子植物
- o gypsum ['dʒɪpsəm] n. 石膏
- o gypsy moth ['dʒɪpsi-mɒθ] n. 舞毒蛾
- o gyre ['dʒaɪə] n. 漩涡

-H-

- o habitat ['hæbitæt] n. 栖息地
- o hail [heil] n. 冰雹
- o hailstone ['heilstəʊn] n. (一粒) 冰雹
- o halide ['hælaɪd] adj. 卤化物(的)
- o halite ['hælaɪt] n. 岩盐
- o hallux ['hæləks] n. (鸟)后趾, (人)大拇趾
- o halo ['heɪləʊ] n. 日晕; 光环
- o hammer ['hæmə] v. 锤打; n. 音锤
- o hard palate [hɑ:d-'pælit] n. 硬腭
- o harden ['hɑ:dn] v. 变硬, 凝固
- o hardwood ['hɑ:dwud] n. 阔叶树
- o harmony ['hɑ:məni] n. 和声
- o harp [hɑ:p] n. 竖琴
- o harpoon [hɑ:'pu:n] n. 鱼叉
- o harvest ['hɑ:vɪst] v. 收获, 捕获
- o hatch [hætʃ] v. 孵化
- o hawk [hɔ:k] n. 鹰
- o hazardous ['hæzədəs] adj. 危险的
- o haze [heɪz] n. 霾, 烟雾, 雾
- o height [haɪt] n. 海拔
- o helium ['hi:lɪəm] n. 氦
- o hemisphere ['hemi,sfiə] n. 半球
- o hemoglobin [,hi:məu'gləubɪn] n. 血色素
- o hemp [hemp] n. 大麻纤维
- o heretofore ['hiətu'fɔ:] adv. 迄今为止
- o hew [hju:] v. 砍
- o hide [haɪd] n. 兽皮
- o hinge [hɪndʒ] n. 枢纽, 门枢, 节点
- o hollow ['hɒləʊ] adj. 中空的
- o hollowed log ['hɒləʊd-lɒg] n. 中空的原木
- o holly ['hɒli] n. 冬青树
- o honeycomb ['hʌnikəʊm] n. 蜂窝
- o hook [huk] n. 鱼钩
- o horizontal [ˌhɒrɪ'zɒntəl] adj. 水平的
- o hormone ['hɔ:məʊn] n. 荷尔蒙
- o horny ['hɔ:ni] adj. 角状的
- o hostile ['hɒstail] adj. 敌对的 (be hostile to : 对……有敌意)
- o hostility [hɒs'tɪlɪti] n. 反对

- o house [haus] v. 包含
- o humidity [hju:'miditi] n. 湿度
- o humor ['hju:mə] n. 体液
- o humoral immunity ['hju:mərəl-i'mju:nəti] n. 体液免疫
- o hunt [hʌnt] v. 狩猎
- o hurricane ['hʌrikən] n. 飓风
- o hydrochloric acid ['haɪdrə'klɔ:rik-'æsid] n. 盐酸
- o hydrogen ['haɪdrədʒən] n. 氢
- o hygienic [hai'dʒi:nik] adj. 卫生学的 (hygiene ['haɪdʒi:n] 卫生学)
- o hymenoptera ['haimi,nɒptərə] n. 膜翅目
- o hypocenter ['haɪpəu,sentə] n. 震源

-I-

- o ice age [ais-eɪdʒ] n. 冰河时期
- o ice cap [ais-kæp] n. 冰冠
- o ice field [ais-fi:ld] n. 冰原
- o ice pellet [ais-'pelɪt] n. 结冰的小球
- o ice sheet [ais-ʃi:t] n. 冰原
- o identification [aɪ,ɪdenti'keɪʃən] n. 鉴定
- o idiophone ['ɪdiəfəʊn] n. 体鸣乐器
- o igneous rock ['ɪgniəs-rɒk] n. 火成岩
- o illuminate ['ɪljʊ:mineɪt] v. 照亮, 使明亮
- o illustration [ɪlə'streɪʃən] n. 插图
- o image ['ɪmɪdʒ] n. 影像
- o imago ['i:meɪgəʊ] n. 成虫
- o imbibe [ɪm'baɪb] v. 喝
- o imitation [ɪmi'teɪʃən] n. 道具
- o immediate [ɪ'mi:diət] adj. 立即的
- o immediately [ɪ'mi:diətli] adv. 直接地, 立即
- o immense [ɪ'mens] adj. 极大的; v. 淹没
- o immune system [ɪ'mju:n-'sɪstəm] n. 免疫系统
- o impact ['ɪmpækt] n. 冲击; 影响, 效力
- o imperceptible [ɪmpə'septɪbl] adj. 极细微的
- o implantation [ɪmplɑ:n'teɪʃən] n. 胚胎植入
- o impression [ɪm'preʃən] n. 印痕
- o improper [ɪm'prɒpə] adj. 不适当的, 错误的
- o improvised text ['ɪmprəvaɪzd-tekst] n. 同步字幕
- o impulse ['ɪmpʌls] n. 神经脉冲
- o impurity [ɪm'pjʊərɪti] n. 不洁, 杂质
- o in addition to [ɪn-ə'dɪʃən-tu:] prep. 除……之外又

- o in advance [in-əd'vɑ:ns] adv.预先
- o in concert [in-'kɒnsət] adv.同时
- o in place [in-'pleis] adv.在适当的位置
- o in response to [in-'ri'spɒns-tu:] prep.相应, 回应
- o in tandem with [in-'tændəm-wið] adv.同……合作
- o in that [in-ðæt] n.因为
- o in the case of [in-ðə-'keis-əv] adv.在……的情况下
- o in the round [in-ðə-raʊnd] adj.雕刻成立体(的), 栩栩如生的, 表现无余的
- o in time [in-taim] adv.最终
- o inaccessible [inæk'sesəbl] adj.难以接近
- o incinerate [in'sinəreit] v. (把……) 烧成灰, 烧掉, 焚化; n. 焚化
- o incinerator [in'sinəreitə] n. (垃圾的) 焚化炉
- o incise [in'saiz] v.刻
- o incomprehensible [in,kɒmpri'hensəbl] adj.不能理解的
- o inconspicuous [inkən'spikjuəs] adj.不显眼的
- o inconvenient [inkən'vi:njənt] adj.不方便的
- o incubate ['inkjubeɪt] v.孵化
- o incur [in'kə:] v.招致
- o indented [in'dentɪd] adj.锯齿状的
- o indigestible [indi'dʒestɪbl] adj.难消化的
- o inelastic [ini'læstɪk] adj.无弹性的
- o infection [in'fekʃən] n.传染, 感染
- o infectious [in'fekʃəs] adj.有传染性的
- o infectious agent [in'fekʃəs-'eidʒənt] n.传染因子
- o inferior [in'fɪəriə] adj.处于下方的
- o informed [in'fɔ:md] adj.见闻广的, 有情报根据的
- o infrastructure ['ɪnfə'strʌktʃə] n.基础设施
- o inhale [in'heil] v.吸气
- o inhibit [in'hibɪt] v.抑制
- o injury ['ɪndʒəri] n.伤害
- o innermost ['ɪnəməʊst] adj.最里面的
- o inscription [in'skripʃən] adj. 碑文的
- o insert [in'sɜ:t] v.插入, 夹进
- o instruct [in'strʌkt] v.教导
- o instrument ['ɪnstrʌmənt] n.乐器
- o insulate ['ɪnsjuleɪt] v. 隔热
- o insulation [ɪn'sju'leɪʃən] n.绝缘, 绝热
- o insulin ['ɪnsjulin] n.胰岛素
- o intensity [ɪn'tensɪti] n.强度
- o intention [ɪn'tenʃən] n.意图, 目的
- o interact [ɪntər'ækt] v.相互作用
- o interaction [ɪntər'ækʃən] n. 相互作用

- o interconnecting ['intəkə'nektɪŋ] adj. 互相连接的
- o interdependence [ˌɪntədɪ'pendəns] n. 相互依赖
- o interfere [ˌɪntə'fiə] v. 有害于 (with)
- o interior [ˌɪn'tɪəriə] adj. 内部的, 内地的
- o interior design [ˌɪn'tɪəriə-dɪ'zain] n. 室内设计
- o interlace [ˌɪntə(:)'leɪs] v. (使) 交织
- o interpretation [ˌɪntə'pri:teɪʃən] n. 演绎
- o interrupt [ˌɪntə'rʌpt] v. 中断
- o intersect [ˌɪntə'sekt] v. 交叉
- o interstellar matter [ˌɪntə(:)'stelə-'mæteɪ] n. 星际物质
- o intestinal [ˌɪn'testɪnəl] adj. 肠的
- o intricate ['ɪntrɪkɪt] adj. 复杂的
- o intriguing [ˌɪn'tri:ɡɪŋ] adj. 引人入胜
- o invader [ˌɪn'veɪdə] n. 侵入物
- o invertebrate [ˌɪn'vɜ:tɪbrət] n. 无脊椎动物
- o involve [ˌɪn'vɒlv] v. 包括
- o iris ['aɪərɪs] n. 虹膜
- o iron core ['aɪən-kɔ:] n. 由铁组成的核
- o Iroquois [ˌɪrəkwɔɪ, ˌɪrəkwɔɪz] n. 易洛魁族人 (北美印第安人的一个民族)
- o irregular [ˌɪ'reɡjələ] n. 不规则的物体
- o irrigation [ˌɪrɪ'geɪʃən] n. 灌溉
- o isotope ['aɪsəutəʊp] n. 同位素
- o ivory ['aɪvəri] n. 象牙

-J-

- o jaw [dʒɔ:] n. 颌, 颚
- o jellyfish ['dʒelɪfɪʃ] n. 水母
- o jerk [dʒɜ:k] n. 抽搐
- o jet [dʒet] n. 喷射物
- o joining ['dʒɔɪnɪŋ] n. 拼接
- o jointed ['dʒɔɪntɪd] adj. 有关节的
- o journeyman ['dʒɜ:nɪmən] n. 熟练工人
- o juggler ['dʒʌɡlə] n. 魔术师
- o juice [dʒu:s] n. 体液
- o Jupiter ['dʒu:pɪtə] n. 木星
- o jute cord [dʒu:t-kɔ:d] n. 黄麻绳

-K-

- o keel [ki:l] n.(船的)龙骨
- o Kelvin ['kelvin] n.开氏温标
- o keratin ['kerətin] n.角质
- o kidney ['kidni] n.肾
- o kiln [kiln] n.窑
- o kitchen ['kitʃin] n.厨房
- o knee joint [ni:-dʒɔint] n.膝关节
- o knife [naif] n.刀
- o knotting ['nɒtɪŋ] n.结形花边 (流苏)

-L-

- o lace [leɪs] v.穿线
- o lack [læk] v.缺乏; 缺少
- o lacquer ['lækə] n.天然漆: 一种用作表面涂层的光滑的树脂般的物质
- o laminate ['læmineɪt] n. 层叠结构
- o laminate to ['læmineɪt-tu] adj. 与……平行/相似的
- o landfill ['lændfɪl] n.垃圾填埋洼地
- o landlocked ['lændləkt] adj.被陆地包围的
- o landlord ['lændlə:d] n.房东
- o landscape ['lændskeɪp] n. 地形
- o large intestine [lɑ:dʒ-in'testɪn] n.大肠
- o lariat ['læriət] n.套索
- o larva ['lɑ:və] n.(昆虫的)幼虫 (plural: larvae [lɑ:vi:])
- o larva phase ['lɑ:və-feɪz] n.幼虫阶段
- o larval ['lɑ:vəl] adj.幼虫状态的
- o lateral fin ['lætərəl-fin] n.侧鳍
- o lateral moraine ['lætərəl-mɔ'reɪn] n.冰川侧碛
- o laterally ['lætərəlli] adv. 侧面地
- o latrine [lə'tri:n] n.公共厕所
- o lava ['lɑ:və] n.熔岩
- o law [lɔ:] n.法律
- o law of motion [lɔ:-əv-'məʊʃən] n.运动定律
- o laws conservation of energy/mass [lɔ:-z-,kɒnsə'veɪʃən-əv-'enədʒi/mæs]
n.能量/质量守恒定律
- o lay [leɪ] v.位于
- o layer ['leɪə] n.地层

- o lead [led] n. 铅
- o leaf [li:f] n. 书页
- o lease [li:s] n. 租约
- o leather ['leðə] n. 皮革
- o ledge [ledʒ] n. 壁架
- o legging ['legɪŋ] n. 紧身裤
- o legislation [,ledʒɪs'leɪʃən] n. 法规、法案，立法
- o lens [lenz] n. 透镜，镜头
- o Lepidoptera [,lepə'dɒptərə] n. 鳞翅目
- o lessee [le'si:] n. 承租人
- o lessor [le'sɔ:] n. 出租人
- o lever ['li:və] n. 控制杆
- o liberalization [,libərəlaɪ'zeɪʃən] n. 自由化
- o life span [laɪf-spæn] n. 寿命
- o ligament ['lɪgəmənt] n. 系带，韧带
- o lightproof ['laɪtpru:f] adj. 不透光
- o like charges [laɪk-tʃɑ:dʒɪz] n. 同性电荷
- o limb [lɪm] n. 肢
- o lime [laɪm] n. 石灰
- o limestone ['laɪm,stəʊn] n. 石灰岩
- o linen ['lɪnɪn] n. 亚麻布
- o liner ['laɪnə] n. 衬里
- o linseed oil ['lɪnsi:d-oɪl] n. 亚麻油
- o lipid ['lɪpɪd,'laɪpɪd] n. 脂质，油脂
- o liver ['lɪvə] n. 肝脏
- o loblolly pine ['lɒblɒli-paɪn] n. 火炬松
- o lobster ['lɒbstə] n. 龙虾
- o locate [ləu'keɪt] v. 设置，确定……的位置
- o log cabin [lɒg-'kæbən] n. 小木屋
- o logged [lɒgd] adj. 低湿的，浸透的（土地）
- o logistics [lə'dʒɪstɪks] n. 物流
- o longitudinally [lɒndʒɪ'tju:dɪnli] adv. 纵向地
- o longleaf pine ['lɒŋ,li:f-paɪn] n. 长叶松
- o loom [lu:m] n. 织布机
- o loop [lu:p] n. 环
- o loudness ['laʊdnɪs] n. 响度
- o lubricate ['lu:brikeɪt] v. 使润滑；加油润滑
- o lumber ['lʌmbə] n. 木材
- o lunar ['lju:nə] adj. 月球的
- o lung [lʌŋ] n. 肺
- o lure [luə] n. 诱饵；v. 引诱
- o lute [lu:t] n. 鲁特琴

- o lymphocyte ['limfəusait] n. 淋巴细胞
- o lyre ['laɪə] n. 里拉, 七弦竖琴 (古希腊的一种弦乐器)
- o lyrical ['lɪrɪkəl] adj. 抒情的

-M-

- o machinery [mə'ʃɪ:nəri] n. 机械设备
- o maggot ['mæɡət] n. 蛆
- o magma ['mægmə] n. 岩浆
- o magnet ['mæɡnɪt] n. 磁铁
- o magnetic field [mæɡ'netɪk-fi:ld] n. 磁场
- o magnetosphere [mæɡ'ni:təʊsfɪə] n. 磁层
- o magnifier ['mæɡnɪfaɪə] n. 放大镜
- o magnify ['mæɡnɪfaɪ] v. 放大
- o magnitude ['mæɡnɪtju:d] n. 量值; 震级
- o mainly ['meɪnli] adv. 主要地
- o maize [meɪz] n. 玉米
- o majestic [mə'dʒestɪk] adj. 雄伟的
- o major ['meɪdʒə] adj. 主要的
- o make up [meɪk-ʌp] v. 构成, 组成
- o make up of [meɪk-ʌp-əv] n. 构成, 组成
- o makeup ['meɪkʌp] n. 化妆品; 结构
- o mallet ['mælit] n. 木槌
- o mammal ['mæməl] n. 哺乳动物
- o manipulate [mə'nɪpjuleɪt] v. (熟练地) 操作、使用 (机器等); 处理
- o manner ['mænə] n. 样式
- o mantle ['mæntl] n. 地幔
- o mantle cavity ['mæntl-'kævɪti] n. 外套腔
- o maple ['meɪpl] n. 枫木; 枫树
- o maple syrup ['meɪpl-'sɪrəp] n. 枫糖浆: 由糖枫汁液做成的甜糖浆
- o marble ['mɑ:bl] n. 大理石
- o marble slab ['mɑ:bl-slæb] n. 大理石板
- o margin ['mɑ:dʒɪn] n. 边缘
- o marine [mə'ri:n] adj. 海洋的
- o maritime ['mæɪtaɪm] adj. 海洋性的
- o mark [mɑ:k] n. (笔) 迹, 痕迹
- o marrow ['mærəʊ] n. 骨髓
- o Mars [mɑ:z] n. 火星
- o masking ['mɑ:skɪn] n. 遮蔽
- o masonry wall ['meɪsənri-wo:l] n. 砖墙, 砖瓦墙
- o mass [mæs] n. 质量; 块

- o mass extinction [mæs-iks'tɪŋkʃən] n.大量消亡
- o mass number [mæs-'nʌmbə] n.(原子)质量数
- o massive ['mæsɪv] adj.大量的
- o master ['mɑ:stə] n.大师
- o masterpiece ['mɑ:stəpi:s] n.杰作
- o mat [mæt] n.垫子; 草垫
- o mate [meɪt] v.交配
- o matrilineal [ˌmætri'li:niəl] adj.母系的
- o mechanical pencil [mi'kænikəl-'pensl] n.自动铅笔
- o mechanism ['mekənɪzəm] n.机械装置
- o medicine ['medɪsɪn] n.医学, 药
- o medium ['mi:djəm] n.介质
- o megalopolis [ˌmegə'ləpəlɪs] n.特大城市
- o melody ['melədi] n.曲调
- o melt [melt] v.融化
- o melting ['meltɪŋ] n.融化
- o membrane ['membrein] n.(动物或植物体内的)薄膜, 隔膜; 膜
- o membranophone [mem'breɪnəfəʊn] n.膜鸣乐器
- o menace ['menəs] n.威胁
- o menopause ['menəpəʊz] n.经绝期
- o menstruation [ˌmenstru'eɪʃən] n.月经期间
- o Mercury ['mæ:kjuri] n.水星
- o merge [mɜ:dʒ] v.并入
- o Mesozoic [ˌmesəʊ'zəʊɪk] n.中生代
- o metabolism [mə'tæbəlaɪzəm] n.新陈代谢
- o metal oxide ['metl-'ɒksaɪd] n.金属氧化物
- o metamorphosis [ˌmetə'mɔ:fəʊsɪs] n.变态, 蜕变
- o meteorite ['mi:tjəraɪt] n.陨星
- o meteorology [ˌmi:tjə'rɒlədʒi] n.气象学
- o metropolis [mi'trɒpəlɪs] n.大城市
- o metropolitan [ˌmetrə'pɒlɪtən] adj.大城市的
- o microbiological [ˌmaɪkrəubaɪə'lɒdʒɪkəl] adj.微生物学的
- o microorganism [ˌmaɪkrəʊ'ɔ:gənɪzəm] n.微生物
- o microscope ['maɪkrəskəʊp] n.显微镜
- o microscopic [ˌmaɪkrəs'kɒpɪk] adj.极微小的
- o microscopic organism [ˌmaɪkrə'skɒpɪk-'ɔ:gənɪzəm] n.微生物
- o microwave ['maɪkrəweɪv] n.微波
- o microwave oven [ˌmaɪkrəweɪv-'ʌvən] n.微波炉
- o migrate [maɪ'greɪt] v.迁徙
- o Milky Way ['mɪlki-'wei] n.银河
- o mime [maɪm] n.笑剧; 滑稽演员
- o mimed performance [maɪmd-pə'fɔ:məns] n.笑剧、闹剧表演

- o mineral ['minərəl] n. 矿物, 矿石
- o mining ['mainiŋ] n. 采矿
- o minority [mai'nɔ:riti] n. 少数
- o minstrel ['minstrəl] n. 诗歌演唱者
- o mistletoe ['mislteu] n. 槲寄生
- o mite [maɪt] n. 螨类
- o mitochondria [ˌmaɪtə'kɒndriə] n. 线粒体
- o moccasin ['mɒkəsɪn] n. 鹿皮靴
- o moist [mɔɪst] adj. 潮湿的
- o moisten ['mɔɪsn] v. 使潮湿
- o mold [məʊld] v. 塑造
- o molecular cloud [məʊ'lekjələ-klaʊd] n. 分子云
- o molecular hydrogen [mə'lekjələ-'haɪdrədʒən] n. 分子氢
- o molecule ['mɒlikju:l] n. 分子
- o mollusk ['mɒləsk] n. 软体动物
- o molt [məʊlt] v. 换毛
- o molten ['məʊltən] adj. 熔化的; 熔融的
- o molting ['məʊltɪŋ] n. 蜕皮
- o monopolistic [mə'nɒpə'listɪk] adj. 垄断的
- o monopoly [mə'nɒpəli] n. 垄断
- o monsoon [mɒn'su:n] n. 季风
- o Moon [mu:n] n. 月球
- o moose [mu:s] n. 驼鹿
- o moraine [mɔ'reɪn] n. 冰碛
- o mortar ['mɔ:tə] n. 砂浆, 灰浆
- o mortise ['mɔ:tɪs] n. 榫眼
- o mosaic [mɔ'zeɪɪk] n. 镶嵌工艺
- o moth [mɒθ] n. 蛾
- o mother-of-pearl [ˌmʌðər-əv-pɜ:l] n. 珍珠母
- o motion ['məʊʃən] n. 运动
- o motivate ['məʊtɪveɪt] v. 激发
- o motor ['məʊtə] n. 运动肌
- o mound [maʊnd] n. 堆
- o mouth of river [maʊθ-əv-'rɪvə] n. 入海口
- o move over [mu:v-'əʊvə] v. 挪开
- o mud [mʌd] n. 泥
- o multicellular [ˌmʌlti'seljələ] adj. 多细胞的
- o multiply ['mʌltɪplaɪ] v. 增加
- o municipal [mju'nɪsɪpəl] adj. 市政的
- o muscle ['mʌsl] n. 肌肉
- o muscle tone ['mʌsl-təʊn] n. 肌肉弹性, 肌肉张力
- o musculoskeletal [ˌmʌskjələʊ'skelɪtəl] adj. 肌(与)骨骼的

- o musical instrument ['mju:zɪkəl-'ɪnstɹrɪmənt] n. 乐器
- o mutualism ['mju:tʃuəlɪzəm, 'mju:tʃuəlɪzəm] n. 共生
- o Myanmar ['mjænmə:] n. 缅甸
- o myth [miθ] n. 神话

-N-

- o nail [neɪl] n. 指甲
- o National Weather Service ['næʃənəl-'weðə-'sɜ:vɪs] n. 国家天气服务局
- o neck [nek] n. 颈项
- o necklace ['neklɪs] n. 项链
- o nectar ['nektə] n. 花蜜
- o needle ['ni:dl] n. 针
- o negative ['negətɪv] adj. 负的
- o negative image ['negətɪv-'ɪmɪdʒ] n. 底片 (负像)
- o negative mold ['negətɪv-kɑ:st] n. 凹模
- o neighboring ['neɪbərɪŋ] adj. 邻近的
- o nematode ['nemətəʊd] n. 线虫
- o Neptune ['neptjʊ:n] n. 海王星
- o nervous system ['nɜ:vəs-'sɪstəm] n. 神经系统
- o nest [nest] n. 巢
- o net [net] n. 网
- o neutral tissue ['njuərəl-'tɪʃju:] n. 神经组织
- o neutron ['nju:trɒn] n. 中子
- o new [nju:] adj. 新月 (的)
- o newt [nju:t] n. 蝾螈
- o niche [nitʃ] n. 壁橱; 合适的环境
- o nictitating membrane ['nɪktɪ'teɪtɪŋ-'membreɪn] n. 瞬膜: 鸟类、爬行动物和—
些哺乳动物的内生透明眼睑, 闭上可保护和滋润眼睛
- o nitrogen ['naɪtrədʒən] n. 氮
- o nitrogen cycle ['naɪtrədʒən-'saɪkl] n. 氮循环
- o nocturnal [nɒk'tə:nl] adj. 夜间的
- o node [nəʊd] n. 中转站
- o nondegradable ['nɒndɪ'greɪdəbl] adj. 不可以进行分解和还原处理的
- o nonprofit ['nɒn'prɒfɪt] adj. 非赢利的
- o Northern coniferous forest ['nɔ:ð(ə)n-kəʊ'nɪfərəs-'fɒrɪst] n. 北方针叶树林
- o notation [nəʊ'teɪʃən] n. 谱号 [musical notation: 音乐记谱]
- o nucleic acid [nju:'kli:ɪk-'æsɪd] n. 核酸
- o nucleus ['nju:kliəs] n. 细胞核; 原子核; 核子
- o nurse [nɜ:s] v. 护理
- o nut [nʌt] n. 坚果

- o nutrient ['nju:triənt] n. 营养品; 营养物质
- o nutritional ['nju:'triʃənəl] adj. 营养的

-O-

- o oak [əuk] n. 橡树; 橡木
- o oboe ['əubəu, 'əubi] n. 双簧管
- o observational [ˌɒbzə'veiʃənəl] adj. 观测的
- o obtain [əb'tein] v. 获取
- o occupation [ˌɒkjʊ'peɪʃən] n. 职业
- o occupy ['ɒkjʊpaɪ] v. 占用, 填满
- o occur [ə'kɜ:] v. 出现, 发生
- o oceanography [ˌəʊʃə'nɒgrəfi] n. 海洋学
- o octave ['ɒktiv] n. 八度音程 (音阶)
- o octopus ['ɒktəpəs] n. 章鱼
- o offensive [ə'fensiv] adj. 冒犯的
- o oil shale [ɔɪl-ʃeɪl] n. 油母页岩
- o olfactory [ɒ'læktəri] adj. 嗅觉的
- o opaque [əu'peɪk] adj. 不透明的; 不反射光线的
- o operation [ˌɒpə'reɪʃən] n. 运营
- o opposite ['ɒpəzɪt] n. 对立物
- o optic axis ['ɒptɪk-æksɪs] n. 光轴
- o optical phenomena ['ɒptɪkəl-fi'nɒmɪnə] n. 光学现象
- o optics ['ɒptɪks] n. 光学
- o orbit ['ɔ:bit] v. 绕 (吸引中心) 运转 n. 轨道
- o orchestra ['ɔ:kɪstrə] n. 管弦乐
- o orchid ['ɔ:kɪd] n. 兰花
- o order ['ɔ:də] n. 目
- o Ordovician [ˌɔ:də'viʃən] n. 奥陶纪
- o organelle [ˌɔ:gə'nel] n. 细胞器
- o organic remains [ˌɔ:gænɪk-rɪ'meɪns] n. 有机残留物
- o organism ['ɔ:gənɪzəm] n. 有机物
- o origin ['ɒrɪdʒɪn] n. 起源
- o ornamental [ˌɔ:nə'mentl] adj. 装饰性的
- o osteon ['ɒstɪən] n. 骨单位 (密质骨构造的基本单位)
- o ostrich ['ɒstrɪtʃ] n. 鸵鸟
- o otherwise ['ʌðəwaɪz] adj. 别样的, 另外一种的
- o oven ['ʌvən] n. 烤炉
- o overfill ['əʊvə'fɪl] v. 把……装得溢出
- o overlap [ˌəʊvə'læp] v. 与……重叠
- o overlapping [ˌəʊvə'læpɪŋ] n. 重叠

- o overlie [ˌəʊvəˈlaɪ] v. 在……上面
- o overspin wire [ˌəʊvəspin-ˈwaɪə] n. 缠弦
- o ovulation [ˌɒvjʊˈleɪʃən] n. 排卵
- o ovum [ˈəʊvəm] n. 卵子
- o owl [aʊl] n. 猫头鹰
- o oxygen [ˈɒksɪdʒən] n. 氧; 氧气
- o oyster [ˈɔɪstə] n. 牡蛎
- o ozone layer [ˈəʊzəʊn-ˈleɪə] n. 臭氧层

-P-

- o pack into [pæk-ˈɪntu] v. 将……装(塞, 挤)入
- o paint [peɪnt] n. 油漆
- o paleontologist [ˌpæliənˈtɒlədʒɪst] n. 古生物学者
- o paleontology [ˌpæliənˈtɒlədʒi] n. 古生物学
- o Paleozoic [ˌpæliəˈzəʊɪk] n. 古生代
- o palm frond [pɑ:m-frɒnd] n. 棕榈叶
- o pan [pæn] n. 平底锅
- o pancreas [ˈpæŋkriəs] n. 胰, 胰腺
- o pantomime [ˈpæntəmaɪm] n. 哑剧表演
- o papyrus [pəˈpaɪərəs] n. (古代)纸(莎)草纸 (plural: papyrus [pəˈpaɪəri])
- o parallel [ˈpærəlel] adj. 平行的
- o paralysis [pəˈrælɪsɪs] n. 麻痹, 瘫痪
- o parasite [ˈpærəsait] n. 寄生物
- o parasitic [ˌpærəˈsɪtɪk] adj. 寄生的
- o parathyroid [ˌpærəˈθaɪrɔɪd] adj. 副甲状腺的
- o parchment [ˈpɑ:tʃmənt] n. 羊皮纸
- o parlor [ˈpɑ:lə] adj. 客厅的
- o parrot [ˈpærət] n. 鹦鹉
- o partial eclipse [ˈpɑ:ʃəl-ɪˈklɪps] n. 偏食
- o participate [pɑ:ˈtɪsɪpeɪt] v. 参与
- o particle [ˈpɑ:tɪkl] n. 微粒; 颗粒
- o partridge [ˈpɑ:trɪdʒ] n. 鹌鹑
- o pass through [ˈpæs-θru:] v. 经过
- o passage [ˈpæsidʒ] n. 转变
- o passageway [ˈpæsidʒweɪ] n. 出入口
- o passive elongation [ˈpæsɪv-ɪˈlɒŋˈgeɪʃən] n. 被动延伸
- o pastel [pæsˈtel] n. 彩色蜡笔画; adj. 色彩浅而柔和的
- o pastry [ˈpeɪstri] n. 糕点
- o patent [ˈpeɪtənt] v. 获得专利
- o patrilineal [ˌpætrəˈlɪniəl] adj. 父系的

- o pattern ['pætən] n.模式
- o peak [pi:k] n.山顶
- o pebble ['pebl] n.鹅卵石
- o peculiar [pi'kju:ljə] adj.罕见的
- o pedal ['pedl] n.踏板
- o pegbox ['pegbɒks] n. (弦乐器的) 弦轴箱
- o pelvis ['pelvis] n.骨盆
- o penetrate ['penitreit] v.穿透
- o penguin ['peŋɡwɪn] n.企鹅
- o pentatonic [,pentə'tɒnik] adj.五声音阶的
- o perceive [pə'si:v] v.察觉; 感到
- o perch [pə:tʃ] n.栖木 : 供鸟栖息的枝条
- o percussion [pə:'kʌʃən] n.打击乐器
- o perform [pə'fɔ:m] v.演奏; 执行
- o performer [pə'fɔ:mə] n.表演者
- o periodically [,piəri'ɒdikəli] adv.周期性地
- o peripheral nervous system [pə'riferəl-'nə:vəs-'sistəm] n.周围神经系统
- o peristalsis [,peri'stælsis] n.蠕动
- o permanent ['pə:mənənt] adj.持久的
- o permanent writing ['pə:mənənt-'raɪtɪŋ] n. 能够长时间保留的书写
- o perpendicular [,pə:pən'dikjulə] adj.垂直的
- o perpendicularly [,pə:pən'dikjuləli] adv.垂直地
- o personal hygiene practice [pə:sənəl-'haɪdʒi:n-'præktɪs] v.个人卫生习惯
- o persuade [pə'sweɪd] v.说服
- o pest [pest] n.害虫
- o pesticide ['pestisaɪd] n.杀虫剂
- o petal ['petl] n.花瓣
- o petroleum [pi'trəʊliəm] n.石油
- o phagocyte ['fæɡəʊsaɪt] n.噬菌细胞
- o pharynx ['færiŋks] n.咽
- o phase [feɪz] n.位相
- o phosphorus ['fɒsfərəs] n.磷
- o photography [fə'tɒɡrəfi] n.摄影
- o photosynthesis [fəʊtəʊ'sɪnθəsis] n.光合作用
- o physical science ['fɪzɪkəl-'saɪəns] n.物理科学
- o piano [pi'ænəʊ] n.钢琴
- o pianoforte [pi,ænəʊ'fɔ:ti] n.钢琴(piano的旧称)
- o pick up [pɪk-ʌp] v.获得
- o pickaxe ['pi,kæks] n.镐
- o picture frame ['pɪktʃə-freɪm] n.相框
- o pigment ['pɪɡmənt] n.色素
- o pile [paɪl] n.堆积, 堆

- o pile up [pail-ʌp] v. 堆积
- o pilgrim ['pilgrim] n. 移民者：1620年，在新英格兰建立普利茅斯殖民地的英国主张脱离国教者
- o pillar ['pilə] n. 柱状物
- o pine [pain] n. 松树
- o pink [pink] n. 粉红色
- o pipe [paip] n. 风笛
- o pipeline ['paɪpleɪn] n. 管道，输送管道
- o pistil ['pɪstɪl] n. 雌蕊
- o pitch [pɪtʃ] n. 音高
- o pituitary [pi'tju:itəri] n. (脑) 垂体
- o pizzicato [ˌpɪtsɪ'kɑ:təu] n. 拨奏乐曲
- o plait [plæt] v. 把……打成辫
- o plane [pleɪn] n. 刨子；平面
- o planetary system ['plænɪtəri-'sɪstəm] n. 行星系
- o plank [plæŋk] n. 支架 [wrest plank: (校准弦音的) 扭钥架]
- o plant [plɑ:nt] n. 植物
- o plasma ['plæzmə] n. 血浆
- o plaster ['plɑ:stə] n. 石膏；胶泥，灰泥，涂墙泥
- o plastic ['plæ:stɪk] n. 塑料
- o plastic barrel ['plæ:stɪk-'bærəl] n. 塑料管
- o plate tectonics [pleit-tek'toniks] n. 板块构造论
- o plateau ['plætəu] n. 高地
- o pliers ['plaɪəz] n. 老虎钳
- o plow [plau] v. 犁；耕；n. 犁
- o pluck [plʌk] v. 拨（弦）
- o pluck away [plʌk-ə'wei] v. 冰川冲走（岩石）
- o pluck out [plʌk-aʊt] v. 拔掉
- o plumage ['plu:midʒ] n. 鸟类的羽毛
- o Pluto ['plu:təu] n. 冥王星
- o poach [pəʊtʃ] v. 水煮
- o pole [pəʊl] n. 杆；磁极，电极
- o poliomyelitis [ˌpəʊliəumaɪə'laitɪs] n. 小儿麻痹症，急性骨髓灰质炎
- o pollen ['pɒlɪn] n. 花粉
- o pollinate ['pɒlineɪt] v. 授粉，传粉
- o pollutant [pə'lu:tənt] n. 污染物
- o pollution [pə'luʃən] n. 污染（作用）
- o pond [pɒnd] n. 池塘
- o population [ˌpɒpjə'leɪʃən] n. 个数
- o pore [pɔ:] n. 毛孔
- o portrait ['pɔ:trɪt] n. 肖像
- o positive ['pɒzɪtɪv] adj. 正的

- o positive cast ['pɒzitiv-kɑ:st] n. 正像
- o possession [pə'zeʃən] n. 占有, 拥有
- o post [pəʊst] n. 杆
- o postal service ['pəʊstəl-'sə:vɪs] n. 邮电业
- o post-and-lintel [pəʊst-ænd-'lɪntl] n. 连梁柱
- o posterior [pɒ'stiəriə] n. 身体后部; adj. 后面的, 尾部的
- o posterior lobe [pɒ'stiəriə-ləʊb] n. 后叶
- o posture ['pɒstʃə] n. 姿势
- o pot [pɒt] n. 锅
- o potter's wheel ['pɒtəs-wi:l] n. 陶轮
- o pottery ['pɒtəri] n. 陶器
- o poultry ['pəʊltri] n. 家禽
- o pour [pɔ:] v. 浇注
- o pour on [pɔ:r-ɒn] v. 把……倒在……上
- o practice ['præktɪs] n. 做法, 技术
- o prairie ['preəri] n. 大草原
- o praying mantis ['preɪɪŋ - 'məntɪs] n. 螳螂
- o precipitate [pri'sɪpɪteɪt] v. 沉淀
- o precipitation [pri,sɪpi'teɪʃən] n. 降水; 沉淀
- o precise [pri'saɪs] adj. 精确的
- o predator ['predətə] n. 捕食者
- o predict [pri'dɪkt] v. 预测
- o prediction [pri'dɪkʃən] n. 预报
- o preen [pri:n] v. (鸟)用嘴整理(羽毛)
- o pregnancy ['pregnənsɪ] n. 怀孕
- o preserve [pri'zə:v] v. 保存
- o press [pres] v. 压
- o pressure ['preʃə] n. (气) 压
- o prey [preɪ] n. 被掠食者, 战利品; 猎物
- o priest [pri:st] n. 祭司
- o primarily ['praɪməriɪ] adv. 主要地
- o primary ['praɪməri] adj. 初级的
- o primary wave ['praɪməri-weɪv] n. 纵波
- o primitive ['prɪmɪtɪv] adj. 自然的; 原始的
- o principal ['prɪnsəpəl] adj. 主要的
- o principle ['prɪnsəpəl] n. 原则
- o prism ['prɪzəm] n. 棱镜
- o privately owned ['praɪvɪtli-əʊnd] adj. 私有的
- o privatization [,praɪvətə'zeɪʃən] n. 私有化
- o prize [praɪz] v. 珍视
- o process ['prəʊses] n. 过程
- o processing [prəu'sesɪŋ] n. 处理

- o producer [prə'dju:sə] n. 制片人
- o production [prə'dʌkʃən] n. 作品, 成品, 电影制片
- o profusion [prə'fju:ʒən] n. 大量 (a profusion of: 大量的)
- o progressively [prə'gresivli] adj. 逐步地
- o prohibit [prə'hɪbɪt] v. 禁止
- o project [prə'dʒekt] v. 凸出
- o projection [prə'dʒekʃən] n. 凸出物
- o prokaryote cell [prəu'kæriəut-sel] n. 原核生物细胞
- o prolong [prə'lɒŋ] v. 延长
- o pronounced [prə'naʊnst] adj. 显著的
- o propagate ['prɒpəgeɪt] v. 传播
- o propagation [,prɒpə'geɪʃən] n. 传播
- o propel [prə'pel] v. 推动
- o property ['prɒpəti] n. 财产, 所有物, 道具; 性质
- o props [prɒps] n. 小道具
- o propulsive [prə'pʌlsɪv] adj. 有推进力的
- o protein ['prəuti:n] n. 蛋白质
- o proton ['prəʊtɒn] n. 质子
- o protozoan [ˌprəʊtəu'zəʊən] n. 原生动物 (plural: protozoa[ˌprəʊtəu'zəʊə])
- o puberty ['pjʊ:bəti] n. 青春期
- o public utility ['pʌblik-ju'tiliti] n. 公用事业
- o publicly owned ['pʌblikli:-əʊnd] adj. 公有的
- o pullet ['pʌlɪt] n. 小母鸡
- o pump [pʌmp] v. 输送
- o punch [pʌntʃ] n. 打孔器
- o pupa ['pjʊ:pə] n. 蛹 (plural: pupae ['pjʊ:pi:])
- o pupil ['pjʊ:pəl] n. 瞳孔
- o puppet show ['pʌpɪt - ʃəʊ] n. 木偶剧
- o puppeteer [ˌpʌpɪ'tiə] n. 演木偶戏的人
- o purify ['pjʊəɪfaɪ] v. 净化
- o pursue [pə'sju:] v. 追捕, 追击
- o put together [put-tə'geðə] v. 组合
- o putty ['pʌti] n. (嵌装玻璃等用的) 油灰

-Q-

- o quasar ['kweɪzɑ:] n. 类星体
- o quiescent [kwɪ'esənt] adj. 休眠的
- o quill [kwɪl] n. 羽毛笔
- o quilt [kwɪlt] n. 被子
- o quilting ['kwɪltɪŋ] n. 被子面料

-R-

- o rabbit ['ræbit] n.野兔
- o radiant energy ['reɪdiənt-'enədʒi] n.放射性能量
- o radiate ['reɪdieɪt] v.散发
- o radius ['reɪdjəs] n.半径(plural: radii ['reɪdiəi])
- o rail transport [reɪl-træns'pɔ:t] n.铁路运输
- o rain forest [rein-'fɒrəst] n.雨林
- o rainbow ['reinbəu] n.彩虹
- o raindrop ['reɪndrɒp] n.雨滴
- o ranch [rɑ:ntʃ] v.经营牧场
- o range [reɪndʒ] n.火炉
- o rasp [rɑ:sp] n.粗锉：一种有锋利、突出、尖锐凸起的钝锉
- o ratio ['reɪʃiəu] n.比, 比率
- o react [ri'ækt] v.起反应
- o readjust [,ri:'ə'dʒʌst] v. 重新调整
- o rear [riə] adj.后面的
- o recede [ri'si:d] v.后退
- o receptor [ri'septə] n.感受器
- o recipe ['resəpi] n.秘方
- o recognize ['rekəɡnaɪz] v.认出; 识别
- o recognized ['rekəɡnaɪzd] adj.公认的, 经过验证的
- o recommend [,rekə'mend] v.推荐
- o recreation [,rekri'eɪʃn] n. 消遣
- o rectangular [rek'tæŋɡjələ] adj.矩形的
- o rectrix ['rektriks] n.舵羽(鸟尾上的硬毛)
- o redshift ['redʃɪft] n.红移
- o reduce [ri'dju:s] v.减少
- o reed [ri:d] n.芦苇; 簧片
- o reef [ri:f] n.礁, 暗礁
- o refer to [ri'fə:-tu:] v.指的是……
- o refer to sb / sth as ... [ri'fə:-tu:-'sʌmbədi/'sʌmθɪŋ-æz]
v. 称某人/某物为……
- o reflect [ri'flekt] v.反射
- o reflex [ri'fleks] n.反射
- o refract [ri'frækt] v.使折射
- o refraction [ri'frækʃən] n.折射
- o region ['ri:dʒən] n.地区
- o register ['redʒɪstə] n.音域
- o regolith ['regəliθ] n.风化层

- o regulate ['regjuleit] v. 调节, 校准, 控制
- o regulation [,regju'leiʃən] n. 规章制度
- o rehearsal [ri'hæ:səl] n. 彩排
- o reinforce [,ri:in'fɔ:s] v. 加强
- o release [ri'li:s] v. 释放
- o relief [ri'li:f] n. 浮雕
- o rely on [ri'lai-ɒn] v. 依赖, 依靠
- o remains [ri'meins] n. 遗体
- o renewal [ri'njuəl] n. 更新
- o rent [rent] n. 租金
- o repel [ri'pel] v. 排斥
- o replenish [ri'plenɪʃ] v. 补充
- o reproduce [,ri:prə'dju:s] v. 繁殖
- o reproduction [,ri:prə'dʌkʃən] n. 复制品; 繁殖, 生殖
- o reproductive organ ['ri:prə'dʌktiv-ɔ:gən] n. 生殖器官
- o reptile ['reptail] n. 爬行动物
- o resemble [ri'zembəl] v. 类似; 类似于
- o reservoir ['rezəvɔ:] n. 贮水池, 水库
- o resilient [ri'ziliənt] adj. 有弹性的
- o resist [ri'zist] v. 抵抗; 对抗
- o resonance ['rezənəns] n. 共振
- o respectively [ris'pektivli] adv. 分别地
- o respiratory system [ris'piəretəri-'sistəm] n. 呼吸系统
- o retailing ['ri:teiliŋ] n. 零售业
- o retain [ri'tein] v. 保留
- o retina ['retinə] n. 视网膜
- o retreat [ri'tri:t] n. 冰川后退
- o reveal [ri'vi:l] v. 揭示, 显示
- o revival [ri'vaivəl] n. 复兴大会 (尤指旨在促进基督教信仰的布道会)
- o revolve [ri'vɒlv] v. 公转
- o rhythm ['riðəm] n. 节奏
- o rhythmic patterns ['riðmik-'pætən] n. 有节奏的模式
- o rib [rib] n. 琴骨; 肋骨
- o ribonucleic acid [raibənju:'kli:ik-'æsid] n. 核糖核酸 (略作RNA)
- o ribosome ['raibəsəm] n. 核糖体
- o ridge [ridʒ] n. 海脊, 山脊
- o ritual ['ritʃuəl] n. (宗教) 仪式; adj. (宗教) 仪式的
- o rival ['raivəl] v. 相匹敌
- o roast [rəʊst] v. 烘烤
- o rod [rɒd] n. 杆
- o rolling pin ['rəuliŋ-pin] n. 擀面杖
- o rooster ['ru:stə] n. 公鸡

- o root [ru:t] n. 根
- o root drowning [ru:t-'draunɪŋ] n. 根部溺死
- o rope [rəʊp] n. 绳索
- o rotate [rəu'teɪt] v. 自转
- o rotation [rəu'teɪʃn] n. 自转
- o row [rəʊ] n. 一排, 一行
- o rub [rʌb] v. 摩擦
- o rug [rʌg] n. 地毯
- o rum [rʌm] n. 朗姆酒
- o running survey ['rʌnɪŋ-sə:'veɪ] n. 勘测
- o rupture ['rʌptʃə] adj. 断裂的
- o rush [rʌʃ] n. 灯心草

-S-

- o saclike ['sæklaɪk] adj. 囊状的
- o sacred fire ['seɪkrɪd-'faɪə] n. 祭祀的火焰
- o salamander ['sæləməndə] n. 火蜥蜴
- o saliva [sə'laɪvə] n. 唾液
- o salt solution [sɔ:'lt-sə'lu:ʃən] n. 盐溶液
- o saltwater ['sɔ:'lt,wɔ:tə] n. 盐水
- o sandpaper ['sænd,peɪpə] n. 砂纸; v. 用砂纸磨光(某物)
- o sandstone ['sændstəʊn] n. 砂岩
- o sanitation [,sæni'teɪʃən] n. 公共卫生
- o sap [sæp] n. 树液
- o satellite ['sætəlaɪt] n. 人造卫星
- o Saturn ['sætən] n. 土星
- o sauté [sɔ:'teɪ] v. 炒
- o saw [sɔ:] n. 锯
- o sawfly ['sɔ:'flaɪ] n. 叶蜂
- o sawing ['sɔ:ɪŋ] n. 锯
- o saxophone ['sæksəfəʊn] n. 萨克斯管
- o scale [skeɪl] n. 音阶; 鱼鳞; 刻度
- o scatter over ['skætə-əʊvə] v. 分布在……
- o scavenger ['skævɪndʒə] n. 食腐动物
- o scene [si:n] n. 景色
- o scenery ['si:nəri] n. 舞台布景
- o school [sku:l] n. 学派
- o sclera ['skliərə] n. 巩膜
- o scope [skəʊp] n. 范围; 乐谱; v. 打记号于
- o scorpion ['sko:pɪən] n. 蝎子

- o scrape [skreip] n. 浅坑; v. 刮擦
- o scratch [skrætʃ] v. 刮擦; 刻划
- o screen [skri:n] n. 筛子
- o screw [skru:] v. 拧, 转
- o script [skript] n. 剧本
- o scroll [skrɔ:l] n. 涡卷形头: 提琴类乐器上的弯曲状饰头; 书卷
- o sculpt [skʌlpt] v. 雕刻
- o sculpture ['skʌlptʃə] n. 雕刻
- o sea urchin [si:-'ə:tʃɪn] n. 海胆
- o sealer ['si:lə] n. 封底漆
- o seaport ['si:pɔ:t] n. 港口
- o season ['si:zən] n. 适合某种活动的季节/期间
- o secondary ['sekəndəri] adj. 第二等级的
- o secondary wave ['sekəndəri-weiv] n. 横波
- o secrete [si'kri:t] v. 分泌
- o secretion [si'kri:ʃən] n. 分泌物
- o secure to [si'kjʊə-tu] v. 固定在……
- o sediment ['sedimənt] n. 沉积物
- o sedimentary rock [sedɪ'mentəri-rɒk] n. 沉积岩
- o segmented [seg'mentɪd] adj. 分段的
- o seismic wave ['saizmik-weiv] n. 地震波
- o seismologist [saiz'mɒlədʒɪst] n. 地震学家
- o seize [si:z] v. 抓住, 捉住
- o self-reliance ['self-ri'lains] n. 倚靠自己
- o Seminole ['seminəʊl] n. 塞米诺族印第安人
- o sensation [sen'seɪʃən] n. 感觉
- o sensitized paper ['sensitaɪzd-'peɪpə] n. 感光纸
- o sensory system ['sensəri-'sɪstəm] n. 感觉系统
- o sepal ['sepəl] n. 萼片
- o septic tank ['septik-tæŋk] n. 化粪池
- o sequence ['si:kwəns] n. 顺序, 序列
- o serve [sə:v] v. 满足, 服务
- o settler ['setlə] n. 移民者
- o sew [səʊ] v. 缝合
- o sewage ['sjuidʒ] n. 阴沟污物; 污水
- o sewer [sjuə] n. 排水沟
- o shades of gray [ʃeɪdz-əv-grey] n. 灰度梯度 (深浅不同的各种灰色)
- o shake [ʃeɪk] v. 摇晃
- o shaking ['ʃeɪkɪŋ] n. 摇晃
- o shale [ʃeɪl] n. 页岩
- o shear [ʃiə] v. 剪切
- o shears [ʃiəz] n. 剪刀

- o sheath [ʃi:θ] v. 包裹
- o shed [ʃed] v. 脱落, 蜕 (皮), 脱 (毛)
- o sheer [ʃiə] adj. 绝对的
- o sheet [ʃi:t] n. (一) 张
- o shelf [ʃelf] n. 架
- o shell [ʃel] n. 外壳
- o shellac [ʃe'læk] n. 虫胶清漆 : 用于涂饰木料
- o shellfish ['ʃel,fɪʃ] n. 贝类
- o shelter ['ʃeltə] n. 遮蔽处
- o shield [ʃi:ld] v. 防御, 保护
- o shingle ['ʃɪŋɡl] n. 木瓦
- o shock [ʃɒk] v. 震撼
- o shoreline ['ʃɔ:lain] n. 海岸线
- o shortleaf pine [ʃɔ:t'li:f-pain] n. 短叶松
- o shoulder ['ʃəuldə] n. 肩
- o shrimp [ʃrɪmp] n. 虾
- o side to side [said-tu-said] adv. 从一边到另一边
- o siliceous [si'liʃəs] adj. 由硅石组成的
- o silicon ['silikən] n. 硅
- o silken ['silkən] adj. 柔软光滑的
- o silt [sɪlt] n. 沙, 淤泥
- o silver salt ['silvə-sɔ:lt] n. 银盐
- o similarity [,simi'lærɪti] n. 相似
- o simultaneously [,siməl'teinjəsli] adv. 同时地
- o single-bladed ax ['sɪŋɡl-bleɪdɪd-æks] n. 单刃斧
- o sink [sɪŋk] n. 水槽
- o siren ['saɪərɪn] n. 土鳗; 汽笛
- o sisal ['sisəl] n. 波罗麻 : 一种分布在墨西哥和中美洲的植物
- o skeletal muscle ['skelɪtl-'mʌsl] n. 骨骼肌
- o skeletal nervous system ['skelɪtl-'nə:vəs-'sɪstəm] n. 骨骼神经系统
- o skeleton ['skelɪtn] n. 框架结构, 整体框架; 骨骼
- o skillet ['skɪlɪt] n. 煎锅 (= frying pan)
- o skip [skip] v. 滑, 笔触中途不出油
- o skull [skʌl] n. 颅骨
- o skyscraper ['skaɪ,skreɪpə] n. 摩天大楼
- o slash pine [slæʃ-pain] n. 沼泽松
- o sleet [sli:t] n. 雨夹雪, 雨淞
- o sloping ['sləʊpɪŋ] adj. 倾斜的
- o slough off [slau-ɔf] v. 脱落
- o small intestine [smɔ:l-in'testɪn] n. 小肠
- o smallpox ['smɔ:lpɒks] n. 天花
- o smog [smɒɡ] n. 烟雾

- o smooth [smu:ð] v. 使光滑
- o smooth muscle [smu:ð-'mʌsl] n. 平滑肌
- o smooth over [smu:ð-'əʊvə] v. 擦除
- o smudge [smʌdʒ] v. 弄脏, 涂去, 使模糊
- o snare [sneə] n. 陷阱
- o snout [snaut] n. 口鼻部
- o snowflake ['snəʊfleik] n. 雪花
- o snugly ['snʌgli] adv. 紧紧地
- o soar [sɔ:] v. 振翅飞行
- o social status ['səʊəl-'steitəs] n. 社会地位
- o softwood ['sɒftwʊd] n. 针叶树
- o soil [soil] n. 土地, 土壤
- o solar eclipse ['səʊlə-'i'klips] n. 日食
- o solar system ['səʊlə-'sistəm] n. 太阳系
- o solid ['sɒlɪd] adj. 结实的
- o solidify [sə'lɪdaɪ] v. (使)凝固
- o solitary ['sɒlɪtəri] adj. 独居的
- o solution [sə'lu:ʃən] n. 溶液
- o somatic nervous system [sə'mætɪk-'nə:vəs-'sistəm] n. 躯体神经系统
- o songbird ['sɒŋbə:d] n. 鸣禽
- o sonic ['sɒnɪk] adj. 声音的
- o soot particle [sut-'pɑ:tɪkl] n. 煤烟颗粒
- o sophisticated [sə'fɪstɪkeɪtɪd] adj. 复杂精巧的
- o sound [saʊnd] n. 声音
- o spacecraft ['speɪskrɑ:ft] n. 宇宙飞船
- o span [spæn] v. (桥、拱等) 横跨
- o sparsely [spɑ:sli] adv. 稀少地
- o spatula ['spætjələ] n. (调拌用的) 刮铲
- o spatulate ['spætjʊlɪt] adj. 竹片状的
- o spear [spiə] n. 矛
- o species ['spi:ʃi:z] n. 物种
- o spectrum ['spektrəm] n. 光谱
- o speculate ['spekjuleɪt] v. 思索, 推测 (about, on, upon)
- o sperm [spɜ:m] n. 精子
- o sphere [sfɪə] n. 球(体)
- o spherical ['sferikəl] adj. 球形的
- o spider ['spaɪdə] n. 蜘蛛
- o spinal cord ['spainl-kɔ:d] n. 脊髓
- o spiral arm ['spaɪərl-ɑ:m] n. 旋臂
- o spleen [spli:n] n. 脾
- o splint [splɪnt] n. 薄木条
- o split [splɪt] n. 裂缝; adj. 劈开的

- o spongy bone ['spʌndʒi-bəʊn] n. 松质骨
- o spoon [spu:n] n. 匙
- o sprawl [sprɔ:l] n. (城市的) 无计划扩展
- o spread [spred] v. 扩散, 传播
- o spruce [spru:s] n. 云杉质木材
- o spruce budworm [spru:s-'bʌdwə:m] n. 云杉蚜虫
- o square [skwɛə] n. 直角尺; adj. 正方形的
- o squash [skwɔʃ] n. 南瓜属植物
- o squid [skwid] n. 鱿鱼
- o squirrel ['skwɪrəl] n. 松鼠
- o stadium ['steɪdʒəm] n. 体育场
- o staff [stɑ:f] n. 五线谱; 全体职员
- o stain [steɪn] n. 着色剂
- o staircase ['steəkeɪs] n. 楼梯
- o stalk [stɔ:k] n. 茎
- o stamen ['steɪmən] n. 雄蕊
- o stand [stænd] n. 高大树丛
- o statistics [stə'tɪstɪks] n. 统计学
- o steam [sti:m] v. 蒸
- o steep [sti:p] adj. 陡峭的
- o stellar ['stelə] adj. 恒星的
- o stem [stem] n. 茎
- o step [step] n. 音级
- o stew [stju:] v. 炖: 用小火烧或慢慢地煮沸来烹饪 (食物)
- o stiffen ['stɪfn] v. (使)变硬
- o stilt [stɪlt] n. 支撑物
- o stimulus ['stɪmjʊləs] n. 刺激
- o sting [stɪŋ] v. 螫
- o stir up [stɜ:-ʌp] v. 搅动
- o stitch [stɪtʃ] n. & v. 缝合
- o stock [stɒk] n. 浓汤
- o stomach ['stʌmək] n. 胃
- o stove [stəʊv] n. 炉子
- o straight [streɪt] adj. 直的; 笔直的
- o strand [strænd] n. 串, (绳子的) 股; 线、束
- o straw [stro:] n. 麦秆
- o stream [stri:m] n. 河流
- o streamer tail ['stri:mə-teɪl] n. 带状的尾巴
- o streamlined ['stri:mlaɪnd] adj. 流线型的
- o stretch [stretʃ] v. & n. 伸展; adj. 有弹性的
- o stride [straɪd] n. 进步
- o strike [straɪk] v. 敲击; 照在……上; 击打

- o string [strɪŋ] n. 琴弦, 弦乐器
- o strip [stri:p] n. 条
- o stroke [strəʊk] n. 笔划
- o style-defining [stail-di'fainɪŋ] adj. 定义完整的, 仔细定义过的
- o stylus ['stailəs] n. (古人刻写蜡板的) 尖笔
- o subarctic ['sʌb'ɑ:ktɪk] adj. 亚寒带
- o subdivision ['sʌbdɪvɪʒən] n. 部分
- o subject ['sʌbdʒɪkt] n. 对象
- o subject to ['sʌbdʒɪkt-tu:] v. 受……支配; 使遭受
- o submerged [səb'mə:dʒd] adj. 水下的
- o subtend [səb'tend] v. 对向
- o subtle ['sʌtl] adj. 微小的
- o subtropical ['sʌb'trɒpɪkəl] adj. 亚热带的
- o subunit [sʌb'ju:nɪt] n. 子 (分, 亚) 组
- o suburb ['sʌbə:b] n. 城郊
- o sucker-bearing ['sʌkə-'beərɪŋ] adj. 长着吸盘的
- o suction cup ['sʌkʃən-kʌp] n. 吸盘
- o suffocation [sʌfə'keɪʃn] n. 窒息
- o sugar maple ['ʃʊgə-'meɪpl] n. 糖枫
- o sulfur ['sʌlfə] n. 硫 (磺)
- o sulfur-laden ['sʌlfə-'leɪdn] adj. 含有硫黄的
- o sullage ['sʌlɪdʒ] n. (房屋、街道等排出的) 污物
- o sunscald ['sʌnskɔ:ld] n. 日晒病
- o superior [sju'piəriə] adj. 处于上方的
- o supple ['sʌpl] adj. 柔软的
- o supplement ['sʌplɪment] v. 补充
- o survivor [sə'vaɪvə] n. 幸存者
- o suspend [sə'spend] v. 悬挂; 悬浮 (大气、液体中)
- o sustain [sə'steɪn] v. 维持
- o sustaining [səs'teɪnɪŋ] adj. 支持的
- o swallow ['swɒləʊ] v. 吞咽
- o swampy ['swɒmpi] adj. 沼泽的
- o symbiosis [sɪmbaɪ'əʊsɪs] n. 共生 (现象)
- o symbiotic [sɪmbaɪ'ɒtɪk] adj. 共生的
- o sympathetic vibration ['sɪmpə'tetɪk-vai'breɪʃən] n. 共振
- o synthetic [sɪn'tetɪk] adj. 合成的
- o syrup ['sɪrəp] n. 糖浆

-T-

- o tablet ['tæblɪt] n. 写字板
- o tail [teɪl] n. 尾
- o tailpiece ['teɪlpi:s] n. 系弦钮
- o talon ['tælən] n. 爪
- o tap [tæp] n. 轻打
- o tarsus ['tɑ:səs] n. 跗骨
- o taste bud [teɪst-bʌd] n. 味蕾
- o tattoo [tə'tu:] v. 刺花纹
- o tavern ['tævən] n. 酒馆
- o tear gland [tiə-glænd] n. 泪腺
- o tempera ['tempərə] n. 蛋彩画
- o Temperate evergreen forest ['tempərɪt-'evəɡri:n-'fɒrɪst]
n. 温带绿叶林
- o Temperate rain forest ['tempərɪt-reɪn-'fɒrɪst] n. 温带雨林
- o tenant ['tenənt] n. 房客
- o tend [tend] v. 照顾
- o tendon ['tendən] n. 腱
- o tenon ['tenən] n. (木工的) 榫
- o tension ['tɛnʃən] n. 绷紧状态; 压力, 张力
- o tentacle ['tentəkl] n. 触角, 触须
- o terminal ['tə:mi:nəl] n. 终点站
- o terminate ['tə:mineɪt] v. 结束
- o terrace ['terəs] n. 阳台
- o terra-cotta ['terə'kɒtə] n. 赤陶
- o tessera ['tesərə] n. 镶嵌物
- o texture ['tekstʃə] n. 质地
- o thatch [θætʃ] n. 茅草
- o theater ['θiətə] n. 戏剧
- o theatrical text [θi'ætrɪkəl-tekst] n. 戏剧文本
- o theoretical [θi'əretɪkəl] adj. 理论的
- o thick [θɪk] adj. 厚的
- o thicken ['θɪkən] v. 使变厚, 使变浓
- o thin [θɪn] adj. 薄的
- o thong [θɒŋ] v. 给……装上皮带; n. 皮带
- o throw [θrəu] n. 投, 掷, 抛投掷的距离; v. 投; 掷; 扔; 抛
- o thunderstorm ['θʌndəsto:m] n. 雷雨
- o thymus ['θaɪməs] n. 胸腺
- o tick [tɪk] n. (寄生于体大动物的吸血小虫) 壁虱
- o tidal ['taɪdɪ] adj. 潮汐的
- o tie [taɪ] n. 枕木

- o tile [tail] n. 饰面砖, 瓷砖
- o till [til] n. 冰渍
- o timber ['timbə] n. 木材
- o tint [tint] n. 带白的颜色, 淡色; v. 染色; 着色于……
- o tissue ['tisju:] n. 组织
- o toad [təud] n. 蟾蜍
- o toe [təu] n. 脚趾
- o toll [təul] n. 通行费
- o tone [təun] n. 音调
- o tongue [tʌŋ] n. 舌头
- o toothpaste ['tu:θpeist] n. 牙膏
- o topographic [ˌtɒpə'græfɪk] adj. 地形学上的
- o torn [tɔ:n] n. (tear的过去分词) 撕
- o tornado [tɔ:'neɪdəu] n. 龙卷风
- o total eclipse ['təutəl-i'klɪps] n. 日全食
- o totem ['təutəm] n. 图腾
- o touch [tʌtʃ] v. 接触
- o tough [tʌf] adj. 坚硬的
- o toxic ['tɒksɪk] adj. 有毒的
- o trabecula [trə'bekjulə] n. 小梁
- o trace [treɪs] v. 追踪, 探索; 顺着; 追溯
- o trachea [trə'ki:ə] n. 气管
- o tragedy ['trædzɪdɪ] n. 悲剧
- o transformation [ˌtrænsfə'meɪʃən] n. 转变
- o transient ['trænzɪənt] adj. 瞬间变化的
- o translate into [ˌtræns'leɪt-'ɪntu] v. 转化为
- o translucent [ˌtrænz'lu:snt] adj. 半透明的
- o transmission [ˌtrænz'mɪʃən] n. 传播
- o transmit [ˌtrænz'mɪt] v. 传送, 传导; n. 传送
- o transparent [ˌtræns'peərənt] adj. 透明的
- o transversely [ˌtrænzvə'sli] adv. 横向地
- o trap [træp] v. 捕捉; 困住; n. 陷阱
- o treadle loom ['tredl-lu:m] n. 脚踏织机
- o treble ['trebl] n. 高音部分
- o tremendous [tri'mendəs] adj. 可怕的, 惊人的
- o tremolo ['tremələu] n. 颤音
- o trend [trend] n. 趋势
- o Triassic [traɪ'æsɪk] n. 三叠纪
- o tribe [traɪb] n. 部落
- o trisquare ['traɪ,skweə] n. 曲尺
- o tropical ['trɒpɪkəl] adj. 热带的
- o tropical rain forest ['trɒpɪkəl-reɪn-'fɒrɪst] n. 热带雨林

- o Tropical savanna forest ['trɒpɪkl-sə'vænə-'fɒrɪst] n. 热带稀树草原
- o Tropical scrub forest ['trɒpɪkl-skrʌb-'fɒrɪst] n. 热带灌丛林
- o troposphere ['trɒpəʊsfɪə] n. 对流层
- o trough [trɔ:f] n. 深海槽
- o truly social ['tru:li-'səʊʃəl] n. 完全社会性的
- o tsunami [tsju:'nɑ:mi] n. 海啸
- o tulip ['tju:lip] n. 郁金香
- o tuning peg ['tju:niŋ-peg] n. 调音弦轴
- o tuning pin ['tju:niŋ-pin] n. 调音弦轴
- o turkey ['tə:ki] n. 火鸡
- o tussock moth ['tʌsək-mʌθ] n. 毒蛾
- o tweezers ['twi:zəz] n. 镊子
- o twist [twɪst] v. 捻
- o twist [twɪst] n. 扭曲
- o typhoon [tai'fu:n] n. 台风

-U-

- o ultraviolet ray [ʌltrə'vaɪəlit-rei] n. 紫外线
- o unaided [ʌn'eɪdɪd] adj. 独立的
- o unaided eye [ʌn'eɪdɪd-ai] n. 肉眼
- o undergo [ʌndə'gəʊ] v. 经历 (变迁等); 遭受 (苦难等)
- o underlie [ʌndə'laɪ] v. 成为……的基础
- o underneath [ʌndə'ni:θ] prep. 在……的下面
- o undetectable [ʌndɪ'tektəbl] adj. 未被发现的
- o unfold [ʌn'fəʊld] v. 展开
- o unicellular ['ju:ni'seljulə] adj. 单细胞的
- o uniform ['ju:nɪfɔ:m] adj. 相同的
- o union ['ju:njən] n. 结合
- o unique [ju:'ni:k] adj. 独特的
- o unparalleled [ʌn'pærəleɪd] adj. 无可比拟的
- o unveil [ʌn'veɪl] v. 揭去
- o up and down [ʌp-ænd-daʊn] adv. 上上下下
- o updraft ['ʌpdra:ft] n. 上升气流
- o upright ['ʌpraɪt] adj. 立式的
- o upwell [ʌp'wel] v. 涌起
- o Uranus ['ju:ərənəs] n. 天王星
- o urban ['ə:bən] adj. 城市的
- o urine ['juəɪn] n. 尿
- o utensil [ju:'tensl] n. 器具
- o uterus ['ju:tərəs] n. 子宫

-V-

- o vaccine ['væksi:n] n. 疫苗
- o vacuole ['vækjuəl] n. 液泡
- o vagina [və'dʒaɪnə] n. 阴道
- o valley ['væli] n. (山) 谷; 流域, 河谷
- o valve [vælv] n. 瓣膜
- o vapor ['veɪpə] n. 蒸气
- o variation [ˌveəri'eɪʃən] n. 变体
- o various ['veəriəs] adj. 许多的, 多样的; 不同的, 好几个
- o varnish ['vɑ:nɪʃ] n. 清漆: 用于把表面涂成坚硬的、有光泽的透明的膜
- o vary ['veəri] v. 不同, 变化
- o vasopressin [ˌvæsəʊ'presɪn] n. 血管加压素, 血压激素
- o vast [vɑ:st] adj. 大量的
- o vaudeville ['vɔ:dəvɪl] n. 歌舞杂耍
- o vault [vɔ:lt] n. 拱 (不连柱的)
- o vehicle ['vi:ɪkl] n. 交通工具; 调漆料
- o vein [veɪn] n. 静脉
- o vellum ['veləm] n. 牛皮纸
- o velocity [vi'lɒsɪti] n. 速率
- o venae cavae ['vi:ni:-'kɪvi:] n. 腔静脉 (singular: vena ['vi:nə] cava ['ki:və])
- o vent [vent] n. 孔, 口, 裂口
- o ventilation [ˌventɪ'leɪʃən] n. 通风设备
- o venture ['ventʃə] v. 冒险
- o Venus ['vi:nəs] n. 金星
- o Venusian [vi:n'ju:siən] adj. 金星的
- o vertebra ['vɜ:tɪbrə] n. 脊椎骨
- o vertebrate ['vɜ:tɪbrɪt] n. 脊椎动物
- o vessel ['vesəl] n. 血管
- o veto ['vi:təu] v. 否决
- o via ['vaɪə] prep. 通过
- o vibration [vaɪ'breɪʃən] n. 振动
- o vibration consequence [vaɪ'breɪʃən-'kɒnsɪkwəns] n. 振动频率
- o victim ['vɪktɪm] n. 牺牲品
- o vigorously ['vɪgərəsli] adv. 竭力地
- o vine [vaɪn] n. 藤蔓
- o violent [ˌvaɪələnt] adj. 猛烈的
- o violin [ˌvaɪə'lin] n. 小提琴
- o virtual ['vɜ:tʃuəl] adj. 虚的
- o virtually [ˌvɜ:tʃuəli] adv. 实际上

- o virus ['vaɪərəs] n.病毒
- o viscous ['vɪskəs] adj.黏性的
- o vital ['vaɪtəl] adj.极重要的
- o volcanic vent [vɒl'kænik-vent] n.火山口
- o volcano [vɒl'keɪnəʊ] n.火山
- o volume ['vɒljʊ:m] n.体积; 卷
- o voluntary ['vɒləntəri] adj.自主的

-W-

- o wagon ['wæɡən] n.四轮马车
- o waist [weɪst] n.昆虫腹部的较窄部位
- o wall hanging [wɔ:l-'hæŋɪŋ] n.壁毯
- o waning ['weɪnɪŋ] adj. (月亮) 渐亏的
- o warm-blooded [wɔ:m'blʌdɪd] adj.(动物) 恒温的
- o warp [wɔ:p] n.织物上纵的方向的纱或线
- o warped [wɔ:pd] adj. 弯曲的
- o wasp [wɔsp] n.黄蜂
- o waste dumps [weɪst-dʌmps] v.废料堆
- o waste substance [weɪst-'sʌbstəns] n.废物
- o waterfowl ['wɔ:tə'fau] n.水鸟
- o waterproof ['wɔ:təpru:f] adj.防水的
- o watershed ['wɔ:tʃed] n.流域
- o water-soluble ['wɔ:tə-'sɒljubl] n.可溶于水的
- o wattle ['wɒtl] n.柳栅：用嫩枝、芦苇或枝条编成的构架，用以筑墙、篱笆和屋顶
- o wave motion [weɪv-'məʊʃən] n.波动
- o wax [wæks] n.石蜡; 蜡
- o wax-coated [wæks-'kəʊtɪd] adj.涂蜡的
- o waxing ['wæksɪŋ] adj. (月亮) 渐盈的
- o weather ['weðə] n.天气
- o weather forecast ['weðə-'fɔ:kə:st] n.天气预报
- o weathering ['weðərɪŋ] n.侵蚀
- o wedge [wedʒ] v.楔入
- o wedge-shaped ['wedʒ-'ʃeɪpt] adj.楔形的
- o weed [wi:d] n.杂草
- o weft [weft] n.织物上横的方向的纱或线
- o well-defined [weln-'di'faɪnd] adj.明确的
- o well-seasoned [wel-'si:znd] adj.完全风干的
- o whale [hweɪl] n.鲸
- o whisk [hwɪsk] n.搅拌器

- o whistle ['hwisl] n. 口哨
- o whitewash ['(h)waitwɔʃ] v. 用石灰水粉刷
- o widespread ['waɪdspred] adj. 普遍的
- o wildfowl ['waɪldfaʊl] n. 野禽
- o wind [waɪnd] v. 绕, 缠
- o wind instrument [waɪnd-'ɪnstrəmənt] n. 管乐器
- o winding ['waɪndɪŋ] adj. 弯曲的
- o windpipe ['waɪnpaɪp] n. 气管
- o wine [waɪn] n. 葡萄酒
- o withdrawal [wɪð'drɔ:əl] n. & v. 提取, 收回
- o withstand [wɪð'stænd] v. 经受, 承受
- o woody material [wudi-mə'tiəriəl] n. 木质材料
- o woodwind ['wudwaɪnd] n. 木管乐器
- o woodworking ['wud,wə:kɪŋ] n. 木工艺, 木工行业; adj. 木工的
- o work surface [wə:k-'sə:fəs] n. 工作台
- o World Meteorological Organization [wə:ld-,mi:tiəə'lɒdʒɪkəl-,ɔ:ɡənai'zeɪʃən]
n. 世界气象组织
- o worldwide belt ['wə:ldwaɪd-belt] n. 世界范围分布
- o worn [wɔ:n] adj. 用旧了的, 磨破了的
- o wrap [ræp] v. 缠绕
- o wraparound ['ræpə,raʊnd] adj. 裹身的
- o writing implement ['raɪtɪŋ-'ɪmplɪmənt] n. 书写用具
- o wrongdoer ['rɔ:ŋ,du:ə,'rɔ:ŋ] n. 做坏事的人

-X-

- o X ray [eks-reɪ] n. X光
- o xylophone ['zaɪləfəʊn] n. 木琴

-Y-

- o yard [jɑ:d] n. 庭院
- o yarn thread [jɑ:n-θred] n. 纺线丝
- o yearn [jɜ:n] v. 向往
- o yellow poplar ['jeləu-'pɒplə] n. 北美鹅掌楸
(poplar, 相当于中国人说的“杨树”)
- o yield [ji:ld] n. 产量; v. 生出, 出产; 产生

-Z-

- o zither ['ziðə] n. 齐特琴 (与中国的“箏”相似)
- o zone [zəʊn] n. 地带

附录一：托福专题分类 (TOEFL Topic Categories)

BASIC CATEGORY

Arts

Life Science

Physical Science

Social Science

Arts

Architecture

Industrial design/art

City planning

Crafts: weaving, knitting, fabrics, furniture, carving, mosaics, ceramics, etc; folk and tribal art

Cave/rock art

Music and music history

Photography

Literature and authors

Books, newspapers, magazines, journals

Life science

Extinction of or conservation efforts for animals and plants

Fish and other aquatic organisms

Bacteria and other one-celled organisms

Viruses

Medical techniques

Public health

Physiology of sensory organs

Biochemistry

Animal behavior, e.g., migration, food foraging, defensive behavior

Habitats and the adaptation of animals and plants to them

Nutrition and its impact on the body

Animal communication

Physical science

Weather and atmosphere

Oceanography

Glaciers, glacial landforms, ice ages

Deserts and other extreme environments
Pollution, alternative energy, environmental policy
Other planets' atmospheres
Astronomy and cosmology
Properties of light, optics
Properties of sound
Electromagnetic radiation
Particle physics
Technology of TV, radio, radar
Math
Chemistry of inorganic things
Computer science
Seismology (plate structure, earthquakes, tectonics, continental drift,
structure of volcanoes)

Social science
Anthropology of non-industrialized civilizations
Early writing systems
Historical linguistics
Business, management
TV/radio as mass communication
Social behavior of groups, community dynamics, communal behavior
Child development
Education
Modern history (including the history of urbanization and industrialization
and their economic and social effects)

附录二 常见元音的拼法

/æ/

[a] add, average, can, stand

/ɑ:/ /ɑ:r/

[a] answer, ask, chance

[aɪ] calm, half, calf, psalm

[ɑr] arch, archive, dark, yard

/ʌ/

[u] ugly, us, but, fund

[o] become, above, dozen, shove

[ou] country, touch, encourage

/e/

[e] education, every, never, second

[ea] bread, measure, ready, realm

/i:/

[e] economic, evil, immediate, previous, scene

[ea] eagle, appeal, reveal, flea, tea

[ie] achieve, priest, yield

[ei] counterfeit, perceive, seize

[ee] agree, guarantee, knee

/ɪ/

[y] any, country

[e] coyote, simile, psyche, recipe

[i] chilli, semi, fungi, modus operandi

[ie] cookie, movie, prairie

[ey] journey, money, turkey

[ea] guinea

[ee] coffee, jubilee

[ois] chamois

[æ] larvae

[is] precis

/ɪ/ (旧版音标)

[i] if, index, big, build, until

[y] analytic, cynical, symptom, bicycle, pyjamas

[e] economy, ensure, extreme, predict, wallet, women

[a] average, damage, village

/ə:r/

[ear] early, earnest, rehearse, yearn

[ur] urban, urge, hurt, furniture

[er] alert, commercial, defer, transfer

[ir] dirt, confirm

[our] courteous, journey
 [or] word, worthy
 /ə/ /ər/
 [a] about, company, poera, arena
 [o] obey, original, million, political, actor, major
 [i] family, policy
 [e] children, similan
 [ah] hookah, purdah
 /ɔ:/
 [or] orchestra, organize, abortion, divorce
 [al] almost, although, bald, fall, call
 [au] auction, autumn, fault, pause, onslaught
 [aw] bawl, lawyer, spawn, law, raw
 [ou] bought, sought
 [our] course, resource
 [oar] aboard, hoard
 [or] for, metaphor
 [ore] adore, score
 [ar] award, quarter
 [aur] theaurus
 /ɔ/
 [o] obvious, opposition, body, possible
 [en] enfant, entrepreneur * 法语词汇
 [a] equality, squad, swap, warrant
 [au] because, bureaucracy
 /u:/
 [oo] oodles, ooze, mood, cartoon
 [ou] acoustic, soup, boutique, boutquet
 [o] lose, prove, tomb
 [u] crucial, rule, truth
 [ui] fruit, recruit
 [ew] blew, shrew, Hebrew
 /u/
 [oo] foot, mushroom
 [u] bull, guru, Buddhist
 [o] wolf, woman, bosom
 /ju:/
 [u] use, unique, future, produce
 [eu] euphemism, feudal, neutrail
 [ew] lewd, pewter, few, nephew, stew
 [ue] argue, statue, tissue
 /ju:r/
 [ure] endure, pure, secure

/ai/

[i] ice, ivy, guide, private, plight, tight, fungi, nuclei, stimuli

[y] analyze, rhyme, tyrant, rely, supply

[ei] Fahrenheit, neither, seismic

[ie] belie, tie, vie, science

[ia] alliance, biased

[ya] cyanide, hyacinth

[ire/ier/yer/yre] admire, satire; amplifier, qualifier; flyer, dryer; byre, pyre

/ei/

[a] able, atheist, alias, aviator, ancient...

[ai] aid, aim, ailment...

[ay] bay, mayor, say

[ei] eight, beige, rein, obey, survey, veil...

/oi/

[oi] oil, exploit, poison

[oy] annoying, royal, alloy, toy

/au/

[ou] ouch, about, proud

[ow] brown, crowd, towel, cow, sow

/əu/

[o] open, opaque, control, local, ago, jumbo

[oa] oak, coax, goat, throat

[ou] mould, shoulder, soul, though

[ow] arrow, mellow, narrow

[oe] toe, woe

[eau] bureau, plateau

/eər/

[air] air, dairy, prairie, affair, stair

[are] various, wary, careful, share, scare

[aire] billionaire, doctrinaire

[ear] bear, wear

[ere] premiere, there, where

/iər/

[ear] ear, appear, rear, year

[er] arterial, hero, material, serious

[eer] beer, peer, volunteer

[ere] adhere, mere, sphere

[ier] frontier, cahier

/uər/

[ur] assurance, jury, rural

[our] gourmet, tour, velour

[ure] assure, sure

[oor] moor, poor

附录三 常见辅音的拼法

/b/

[b] back, debate, rob

[bb] bubble, rabbit, ribbon

/p/

[p] party, recipe, cup

[pp] apper, apple, snippet

/d/

[d] day, adult, food, need, trade, side

[dd] add, shudder, reddish, wedding

/t/

[t] take, still, political, pant, support

[tt] attack, letter, witty, putt, watt

[tte] cassette, diskette

/g/

[g] go, legal, sugar, dog, rug, zigzag

[gg] struggle, aggravate, egg

[gu] guess, guitar

[gh] ghost, ghoulish

[x] /gz/ auxiliary, example, exorbitant

[gue] vogue, fatigue, ialogue, prologue]

/k/

[c] can, cup, credit, act, vehicle, misic, traffic

[cc] acclaim, occurrence, tobacco, vaccine

[k] key, kite, bike, market, desk, fork

[ch] chaos, chord, ache, scheme, stomach, psych

[ck] pocket, cuckoo, nickel, sick, attack

[que] antique, critique, unique

[qu] quite, queen, quote, conquer, mosquito

[x] fix, axis, toxic

/v/

[v] value, several, university

[vv] navy, savvy

/f/

[f] face, professor, beef, wolf

[ff] affair, suffer, traffic, cliff, sniff

[ph] phone, phrase, cipher, prophet, triumph

[gh] cough, tough

/ð/

[th] than, this, further, together, smooth

[the] bathe, loathe, soothe

/θ/

[th] think, three, author, nothing, both, wreath

/z/

[z] zoo, zip, zero, wizard, zigzag

[zz] blizzard, puzzle

[s] easy, prison, visit, arise, pause, lens, news

/s/

[s] set, state, industry, gas, epidermis

[ss] assail, assist, assume, press

[c] cell, cent, cyclone, ceremony, facet, specify, notice

[sc] scenario, scent, discern

[ps] psyche, psalm

/dʒ/

[j] joke, job, jeopardy, pajamas, trajectory, subject, injure

[g] gender, gene, gym, surgery, strategic, age, urge

[dg] budget, gadget, judge, ridge

[du] educate, individual, arduous

/ʃ/

[sh] she, shoulder, worship, publish

[ch] chauvinist, chef, cliché

[sch] schwa, schmooze

[s] sugar, sure

[ti] confidential, potential, nation, position, direction, section, infectious, initiate

[ci] artificial, special, magician, physician, spacious, conscious, ancient

[si] apprehension, tension

[ssi] admission, passion, recession

[ss] assure, issue

[che] niche, moustache

/tʃ/

[ch] chair, chief, achieve, voucher, reach, torch

[tch] itchy, kitchen, batch, switch, wretch

[tu] actual, sanctuary, perpetuate, situate, statue, virtue, fatuous, impetuous, future, venture

/l/

[l] last, only, particular

[ll] caterpillar, dull, bill

/m/

[m] most, number, home, arm, seem

[mm] ammo, immune, hammer, mammal

[gm/lm/mb/mn] paradigm, palm, bomb, column

/n/

[n] nation, any, won, down

[nn] antenna, tennis, dinner

[gn/kn] gnomonic, knite, know, align, sovereign

/ŋ/

[n] concrete, congress, monkey

[ng] sing, young

/h/

[h] hard, herb, ahead, perhaps

[wh] who, whole

/r/

[r] radio, run, different, serious

[rr] arrange, irritate, warrior, carry

[rh] rhyme, rhythm, rhetoric

[wr] write, wrong

/ʒ/

[g] genre, regime, garage, beige

[j] courgette, bijou

[su] casual, pleasure

[si] decision, occasion

/w/

[w] way, work, world, twin, towards

[wh] what, whim

[u] quality, require, anguish, language, persuade

/j/

[y] yellow, yolk, beyond, lawyer

[u/eu] uniform, use, euphemism, document

[i] behavior, junior, opinion

附录四 TOEFL常见熟词辟义列表

A

address [ə'dres] v. 应付, 处理(问题等)

aging ['eidʒɪŋ] n. 陈酿

appropriate [ə'prəʊpriɪt] v. 擅用, 挪用, 占用, 盗用

article ['ɑ:tɪkl] n. 物品; 制品, 商品

arrested [ə'restɪd] adj. 不良的, 滞留的. (arrested development 发育不良)

articulate [ɑ:'tɪkjʊleɪt] adj. 有关节的; 有节的; articulate structure n. 节体动物

assume [ə'sju:m] v. 承担, 担任. 假装, 装作……的样子, 采取(……态度)

B

bark [bɑ:k] n. 树皮; 三桅帆船

bill [bil] n. 账单, 清, 议案, 法案; (水禽等细长而扁平的)嘴(猛禽的钩状嘴通常叫 beak)

blind [blaɪnd] n. 百叶窗

book [buk] vt. 预定, 定(戏位、车位等); 托运(行李等)

build [bɪld] n. 骨格, 体格, 成形

C

catch [kæʃ] n. 陷阱, 圈套, 诡计; 料不到的困难

champion ['tʃæmpjən] vt. 维护, 拥护, 主张; 为……而奋斗.

champion a cause 维护一项事业.

chest [tʃest] n. 箱, 函, 柜, 匣; 银箱; 金库, 公款, 资金

close [kləʊz] adj. 闷气的, 闷热的

complex ['kɒmpleks] n. 络合物, 复合物, 综合体

concern [kən'sɜ:n] n. 商行, 公司; 财团; 康采恩; 事业, 业务

consume [kən'sju:m] vi. 枯萎; 憔悴 The flowers consumed away. 花枯萎了.

be consumed [away] with (envy, fever, ambition, grief)

count [kaʊnt] n. 起诉理由, 罪状.

critical ['krɪtɪkəl] adj. 危急的; 决定性的, 重大的

coat [kəʊt] v. 涂上一层(例如油漆)

cure [kjʊə] v. (鱼等用腌、熏、晒、烤等的)加工保藏(法).

cut [kʌt] vt. 生长, 出(牙齿)

D

date [deɪt] n. 海枣

deal [di:l] n. (松等的)木板; 木材, 木料; adj. 松木的

dear [diə] adj. 昂贵的, 高价的

deed [di:d] v. 立契转让

default [di'fɔ:lt] n. & v. 不履行; 违约; 拖欠

dock [dɒk] n. 草本植物 vt. 剥夺, 扣去……的应得工资

down [daʊn] n. (美国)沙丘; (蒲公英等的)冠毛; 鸭绒, 绒毛; (鸟的)绒羽; 柔毛.

汗毛,软毛,毳毛

draw [drɔ:] vt.提取(钱款); 使打成平局

drill [dril] vt.(用钢钻)钻(孔);在……上(用钢钻)钻孔

drive [draiv] n.冲力,动力;干劲;努力;魄力;精力

E

eat [i:t] vt.蛀;腐蚀;消磨

exploit [iks'plɔit] n.功绩,功劳,勋绩

exponent [eks'pəunənt] n.典型,样品

F

factor ['fæktə] n.因子,因数; 倍;乘数;商

fair [fɛə] n.(英国)定期集市,庙会.商品展览会,展销会,商品交易会

fashion ['fæʃən] vt.形成,铸成,造,作 (into; to)

fault [fɔ:lt] n.(地质)断层

felt [felt] n.毛毡;毛布;毡制品;油毛毡.

figure ['figə] n.人影,人形;人物

fine [fain] n.罚款;adj.细小,精致

fly [flai] n.(男裤的)拉链

functional ['fʌŋkʃən] adj.从使用的观点设计[构成]的

G

game [geim]n.&adj.(集合词)猎物,野味;(鸽等的)群;野外游戏 [游猎、鹰狩等]

give [giv] n.弹性

H

hide [haid] n.兽皮

hit [hit] vt.偶然碰见,遭遇

hold [həuld] n.(货船)船舱

humor ['hju:mə] n.(眼球的)玻璃状液体;(旧时生理学所说动物的)体液;(植物的)汁液.

I

import ['impɔ:t] n.意义,含义

inviting [in'vaitɪŋ] adj.引人注目的,吸引人的

involved [in'vɒlvɪd] adj.复杂的,难缠的

issue ['ɪʃu:] n.&v.流出,(血、水等的)涌出; [法律]子孙,子女

J

jar [dʒɑ:] vi.(1)给人烦躁[痛苦]的感觉,刺激 (on) (2)(发出刺耳声地)撞击 (on upon against) (3)震动,震荡(不和谐地)反响,回荡.(4)(意见、行动等)不一致,冲突,激烈争吵(with). jar on sb.给某人不快之感.

L

lace [leis] vt.&vi.系牢, 系紧

late [leit] adj.已去世的, 已故的

lay [lei] adj. (1)一般信徒的, 俗人的, 凡俗的 (opp. clerical). (2.)无经验的, 外行(人)的 (opp. professional).

lead [led] n. 铅

leave [li:v] n. (1)许可, 同意. (2)告假, 休假; 假期

letter ['letə] n. 出租人; letters 证书, 许可证

literature ['litəritʃ] n. 文献

lot [lɒt] n. 土地

M

make [meik] n. 构造

means [mi:nz] n. 财力、资产

measure ['meʒə] n. 准绳, 韵律 a measure of = is determined by; ……的体现 : The rate at which a molecule of water passes through the cycle is not random but is a measure of the relative size of the various reservoirs.

meet [mi:t] n. 比赛

minute ['minit] adj. 微小的, 细小的

N

novel ['nɒvəl] adj. 新的, 新颖的; 新奇的, 珍奇的, 异常的

O

observe [əb'zə:v] vi. 陈述意见, 评述, 简评 (on; upon) strange to observe 讲起来虽奇怪. I have very little to observe on what has been said. 关于刚才所听到的我没什么话好讲. vt. observe silence 保持沉默. observe a rule 遵守规则. organ ['ɔ:gən] n. [音乐](教堂用的)管风琴 (= (美国) pipe organ); (足踏)风琴; 手摇风琴; 口琴. 机构; 机关; 机关报[杂志]; 喉舌; 报刊.

outstanding [aʊt'stændɪŋ] adj. 未付的, 未清的; 未解决的; 未完成的

P

partial ['pɑ:ʃəl] adj. [植物; 植物学] 后生的, 再生的.

pen [pen] n. (1) (家畜等的) 围栏, 槛. (2) 一栏[一圈]家畜.

period ['piəriəd] [音乐] 乐段

pile [pail] n. (1) 高大建筑; (2) 痔疮 (3) 软毛, 绒毛; 毛茸. (布、绒的) 软面.

pitch [pitʃ] n. 沥青; 含有沥青的物质; 松脂, 树脂

pool [pu:l] n. [医学] 淤血

pound [paʊnd] n. 兽栏 v. (连续) 猛击; 乱敲; 砰砰地乱弹(钢琴等), 乱奏(曲子)

preserve [pri'zə:v] n. (1) 禁猎区; (2) 蜜饯

produce [prə'dju:s] n. 物产; 产品, 农产品; 制品, 作品

project [prə'dʒekt] v. 使突出, 使凸出; 伸出. The upper storey projects ove

the street 二楼伸出街上.

promise ['prɒmɪs] n.(前途有)希望;(有)指望

pronounced [prə'naʊnst] adj.决然的,断然的,强硬的;明白的,显著的.

provide [prə'vaɪd] v.规定

providing [prə'vaɪdɪŋ] conj.[provided] 倘若……只要,在……条件下.

Q

quality ['kwɒləti] adj.(1)优质的,高级的.(2)上流社会的.

quarters ['kwɔ:təs] n.寓所,住处;[军事]营房,驻地,营盘,宿舍;岗位.

R

rate [reɪt] v.(1)被估价;被评价 The ship rates as a ship of the line.这条船列入战列舰级.(2)申斥,斥责,骂

rear [riə] v.饲养(家畜等);抚养,教养(孩子);栽培(作物).

relief [ri'li:f] n.[雕刻]凸起;浮起,浮雕;浮雕品;[绘画]人物凸现,轮廓鲜明

rent [rent] v.1.(rend 的过去分词) 撕碎;n.[地质学;地理学]断口;(意见等的)分裂,分歧;(关系等的)破裂

retire [ri'taɪə] vi.就寝,去睡觉

run [rʌn] n.丝袜上的洞

S

save [seɪv] conj.除了

say [seɪ] n.发言权

scale [skeɪl] n.阶梯,梯子;v.用梯子上;爬越,攀登;n.天平;n.鳞;(锅垢、锈)v.剥鳞/垢、锈

school [sku:l] n.(鱼、鲸等水族动物的)群;队 a school of dolphins 一群海豚.

score [sko:] n.[音乐]总谱,乐谱;(电影歌舞等的)配乐

scores [sko:z] n.许多 scores of fossil remains

screen [skri:n] n.筛子

season ['si:zən] vt.(1)使熟练;使(习)惯.(2)风干;晒干(木材);晾干,对...进行干燥处理;使陈化.(3)使适应(气候等).(4)给……加味[调味].(5)给……增加趣味.(6)缓和,调和.

secretary ['sekretəri] n.(1)(上部附有书橱的)写字台.(2)书写体大写铅字.

secure [si'kjʊə] vt.搞到;把……拿到手;得到;获得

serve [sə:v] vi.[网球]开球;发球

shower ['ʃaʊə] n.(美)(为新娘等举行的)送礼会;(婚前、产后)的聚会

shrink [ʃrɪŋk] n.精神病医师

sound [saʊnd] vi.(1)测水深;探测(上层空气).(2)试探(别人的意见);调查(可能性).(3)(鱼或鲸鱼)突然潜入海底.

sow [səʊ] n.大母猪

spell [spel] vt.(1)招致,带来;(2)轮班,换班,替班;(3)符咒,咒语.(4)吸引力,诱惑力,魔力,魅力.

spoke [spəuk] n.(车轮的)辐条
 spot [spot] vt.认出,发现,定位
 spring [sprɪŋ] n.弹簧;泉水v.扭伤(腿)
 stand [stænd] v.忍受;n.床头柜
 standard ['stændəd] n.直立支柱;灯台;烛台,电杆,垂直的水管(电管).
 start [stɑ:t] v.(船材、钉等)松动,翘曲,歪,脱落.
 stem [stem] v.起源于,起因于,(由……)发生,来自 (from out of). Correct decisions stem from correct judgments.正确的决心来自于正确的判断.
 still [stil] n.蒸馏锅 v.蒸馏
 strain [streɪn] n.血统,家世;族,种;[生物学]品系,系;菌株;变种,小种.
 subscribe [səb'skraɪb] v.同意、赞成
 stroke [strəuk] n.笔画

T

temper ['tempə] n.(黏土的)黏度;(灰泥的)稠度;
 tender ['tendə] v.正式提出; tender one's resignation 提出辞呈
 till [tɪl] n.[地质学;地理学]冰碛土(物)
 train [treɪn] n.schene, trick ; 敲门,绝技

U

utter ['ʌtə] adj.完全的,十足的

V

vessel ['vesəl] n.船,舰;飞船

W

wage [weɪdʒ] v.实行,进行,发动(战争等) (on against)
 way [wei] adv.(美口)……得多,远为.与 above, ahead, behind, below, down, off, out, over, up 等副词、介词连用,以加强语气. way back 老早以前. way down upon the river Thames 在老远老远的泰晤士河边. way up 还在上面;好得多.
 way out of balance 逆差很大很大.
 weather ['weðə] vt.[地质学;地理学](常用被动语态)使风化
 well [wel] n.井;vt.涌出,喷出(up/out/forth)
 wind [wɪnd] n.肠气,屁;v.嗅出,察觉,嗅到猎物的气味(winded/winded) ;v. (winded/wound) 吹(角笛、喇叭等). wind a call 吹哨子(召唤).v.(wound/wound) 卷绕,缠绕;上发条

附记

本书成书过程中，张艺馨、朱莎莎、赵媛、徐薇等老师做了大量的辅助工作，并通过教学实践为本书提出了很多建设性意见。在此，衷心感谢他们的辛勤工作。另外，还有很多学生，如包括（且不仅包括）华盛顿大学的高雅、康奈尔大学的魏天天、耶鲁大学的曲直、布朗大学的张少鹏、斯沃兹茅的钟灵等同学，在此过程中通过自身的学习经历为编者、老师提供了大量合理而又有效的反馈，虽然无法列出所有同学的姓名，但在此也由衷地感谢他们的努力。

还有许多充满善意和热心的读者。正是因为有他们的帮助，才使得很多原本可能无法注意到的、或者不可能想象的各种纰漏剔除。我的个人网站是 www.lixiaolai.com；我的电子邮箱是 lixiaolai@gmail.com。希望热心的读者在阅读本书的过程中遇到任何可能的纰漏之时，能在我的网站留言，或者直接发送邮件告知，万分感谢！

这本书的出版过程中，还得到了无数意外的帮助。最令人惊讶也最令人感动的来自胖兔子粥粥的插图。粥粥同学决定突破英语难关，而他的方式颇为独特：每天至少背完一篇文章，并为该文章的文字配上插图。大约从2009年的7月份开始，粥粥言出必行，期间只有几次因为工作原因（比如出差）而间断；至10月底，粥粥同学不仅背完这些单词，还一幅不差地画完了插图——使这本原本几乎注定枯燥的词汇书一下子改头换面，成为同类书籍中的异类。而这原本是我根本不可能奢望的事情。

我感激粥粥，并不仅仅是因为他画了这么多可爱的插图，更是因为他为读者树立了一个榜样。粥粥是专业漫画人，对已经离开学校参加工作几年、原本在高校底子就较差（粥粥自我陈述）的他来说，学英语、背单词的难度可能要比大多数在校学生要大很多。然而，他是个言出必行的人，不仅背完，还要画完，还要作为出版人全程参与本书整个出版过程。粥粥背单词的过程，无疑为读者树立了一个活生生的榜样，而这样的榜样对读者来说，几乎是无价的。谢谢粥粥。

最后，真心希望这本书能够给读者带来更大的进步。

李笑来
二〇一〇年初 于北京

补充说

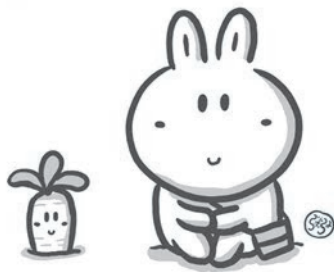
虽然这是一本以词汇为主的书，但是要学好英语，还是要听说读写齐头并进的，这也是这本词汇书为啥有大段的文章、还有老外来通篇朗读的原因：只背单词，肯定不是学英语的全部。不仅要背单词，遇到好句子也要背，遇到漂亮文章也要背——死记硬背的另外一个说法是“博闻强识”。

说个做这本书时的趣事：本书里所有文章的音频都是一个老美念的。这老美还是个前雅思考官，文化水平不可谓不高，托福词汇显然他也都认识，但这书里的音频，他真是频频念错，足足搞了6、7个小时，差点没昏过去……（其实这现象也不难解释。你自己找一篇《南方周末》上的报道、或者是挑一篇科学松鼠会的文章，试试能不能不打磕巴、有声有色地念出来？说明啥？听说读写，不是一码事儿！你肯定不能光靠背单词就把英语学好！）

英语学习肯定没有秘诀，速成没有可能。但英语学习肯定也不需要天赋，只需要一些努力罢了。不要着急。静静地等待梦想长大。

英语门外汉胖兔子粥粥
二〇一〇年九月于北京

不用着急。
静静地等待梦想长大。



ps：胖兔子粥粥号称是拯救地球小分队的队长嘛，所以本书的音频就不给你光盘了，环保！大家去李笑来老师博客上下载吧：www.lixiaolai.com。如果你对学习英语有啥疑惑，也欢迎翻看，学习外语的真知灼见都在里面咯。另外贴个小广告：想学好英语的，不要错过笑来老师的新书——《人人都能用英语》！！很快就要上市啦，敬请围观！还有猛烈推荐那本《把时间当作朋友》！还有这些书的插图都是我画的！哇哈哈哈！